DCAM

Data Management Capability Assessment Model

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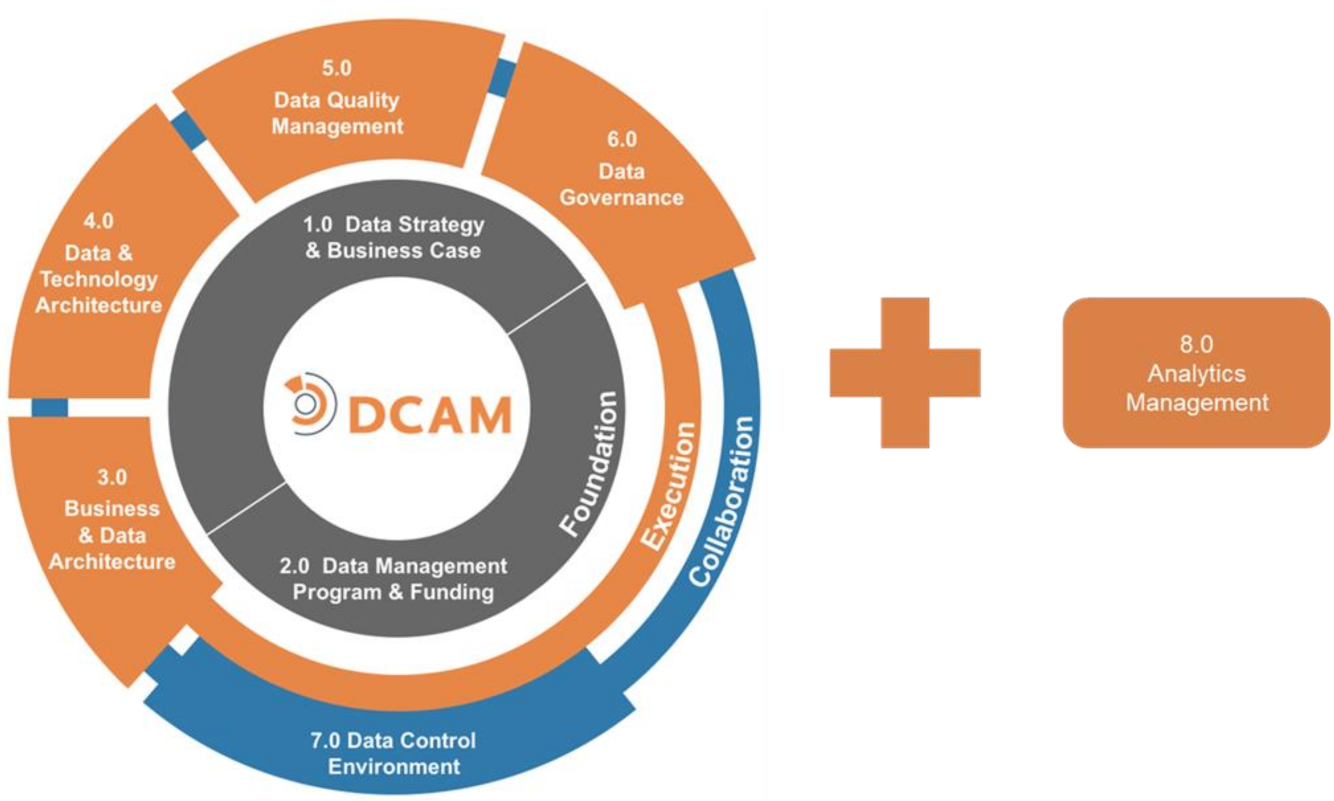
The DCAM® Framework



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Revision History

Date	Description	Version
June 2019	Initial release of DCAM version 2 – Major release	v2.1.1
August 2019	Update of DCAM Framework graphics	v2.1.2
October 2020	Initial release of DCAM+ Analytics Component	v.2.2.1

Foreword

The *Data Management Capability Assessment Model* (DCAM®) is a structured resource that defines and describes the capabilities needed to establish and sustain a successful data management (DM) initiative in any organization. The Enterprise Data Management Council created the Model based on the practical experiences and hard-won lessons of many of the world's leading organizations. The result is the synthesis of a broad range of DM best practices from across the full spectrum of interconnected business *processes*. The DCAM addresses the strategies, organization-wide structures, technology, and operational practices needed to drive DM successfully. It addresses the tenets of DM based on an understanding of business value combined with the hard reality of implementation.

To manage data in today's organizations, we must start by recognizing that proper DM is about managing data as meaning. This relatively new concept for many organizations is not very well understood. Managing data, according to its meaning, is a *process* of defining each piece of data by what it represents or describes in the real world. This *process* results in a direct, readily comprehensible label for that data. By adding descriptive *metadata*, the precise nuanced connection between each piece of data and the real world is established. Data exists everywhere within an organization and must be managed consistently within a well-defined control framework. The DCAM defines the framework and capabilities required to make DM a critical part of an organization's everyday operational fabric.

The challenges of properly managing data are significant. In most organizations, there are numerous legacy data repositories and an overabundance of *functions* to unravel. There are social and political barriers to overcome. There are real technical challenges and execution gaps to address. *Data ownership* and accountability are hard to establish. Historically, funding often has been project-based, making DM an intermittent priority. Data's now critical place in the organization requires a commitment to robust, ongoing funding. Organizations have an additional challenge to build the strong executive support needed to ensure that the organization stays the course in the face of short-term measurement criteria, operational disruption, and conflicting *stakeholder* priorities.

We understand this reality because we've been there, and we have the scars to show for it. Data is foundational. It is the lifeblood of the organization. The bad data tax is a significant expenditure for many organizations though it may remain hidden in accepted inefficiencies and stunted results. Unraveling data silos through the creation of harmonized data is a prerequisite for eliminating redundancy, reducing reconciliation, and automating business *processes* across the organization.

Managing this kind of fully interconnected data is essential if we are to gain insight from analytics, feed our *models* with confidence, enhance our service to clients and capitalize on new, but often fleeting, business opportunities. DCAM provides the guidance needed to assess the current-state of any organization's DM and define the objectives and framework for the target-state of the DM initiative.

The DCAM is organized into seven core components and one optional component:

Core:

- 1.0 Data Management Strategy & Business Case
- 2.0 Data Management Program & Funding Model
- 3.0 Business & Data Architecture
- 4.0 Data & Technology Architecture
- 5.0 Data Quality Management
- 6.0 Data Governance
- 7.0 Data Control Environment

The core components are organized into 31 capabilities and 106 sub-capabilities.

Optional:

- 8.0 Analytics Management

This optional component covers Analytics Management and is relevant where the scope of the DM program or the Chief Data Officer's responsibilities cover the Analytics *functions* of the organization. This component is organized into seven capabilities and 30 sub-capabilities.

The EDM Council is indebted to the dozens of members who contributed to the development of the initial DCAM in 2014 and the subsequent updates. DCAM is quickly gaining adoption across the DM industry. DCAM training has now been delivered to more than 1,000 students around the world. Organizations ranging from banks to brokerages to consultancies to regulatory bodies have successfully adopted the DCAM Framework. Born in the financial service industry, increasingly organizations from other industries, manufacturing to government, are applying DCAM to their environment.

To keep the model relevant and on the leading edge, the EDM Council has committed to the product management discipline to review and update the Framework regularly. For this current version, a small but global team of representatives from organizations and consultancies was assembled into a Work Group to recommend and approve the latest updates to the Framework.

To ensure DCAM addresses the needs of the DM industry in each unique organization, we have and always will accept input and contribution from members who have worked closely with the model.

Acknowledgement

The EDM Council is committed to leveraging the knowledge of the data management practitioners across our membership. For Version 2, a small but global team of representatives from organizations and consultancies was assembled into a Work Group to recommend and approve the latest updates to the Framework. A further group was formed to create the Analytics Management component. A thank you is extended to the groups listed below for their role in advising the enhancements to DCAM.

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Contents

Introduction

The *Data Management Capability Assessment Model* (DCAM®) defines the scope of capabilities required to establish, enable and sustain a mature data management (DM) discipline. It addresses the strategies, organizational structures, technology and operational best practices needed to successfully drive DM. It addresses the tenets of DM based on an understanding of business value combined with the reality of operational implementation.

Overview

The concept of data as a foundational aspect of business operations has arrived. It is now widely understood as one of the core factors of input into the full range of business and organizational *processes*. An organization that is effective in its use of data is one that implements and manages a data control environment. The data control environment yields a wide range of bottom-line benefits to the organization, including reduced operational costs, automated manual *processes*, consolidated redundant systems, minimized reconciliation, and enhanced business opportunities. An organization implements a data control environment to ensure trust and confidence in the data they are relying on for business processing and decision-making. The data control environment eliminates the need for manual reconciliation or reliance on *data transformation* *processes*.

The data control environment concept ensures the data is precisely defined, described using *metadata*, aligned with meaning, and managed across the full *data lifecycle*. The key to first establishing a data control environment, however, is the achievement of unambiguous shared meaning across the organization along with the governance required to ensure precise definitions. Data must be consistently defined by the real thing it represents, such as products, clients, customers, legal entities, transactions, events, and much more. All other *processes* are built upon this foundation.

The opposite of a data control environment is a fragmented data environment. The fragmentation results in application development that produces ad hoc *data models.* The fragmentation exacerbates the problem of common terms that have different meanings, common meanings that use different terms, and vague definitions that don't capture critical nuances. For many organizations, this challenge can be debilitating because there are thousands of *data elements*, delivered by hundreds of internal and external sources, all stored in dozens of unconnected databases. This fragmentation results in a continual challenge of mapping, cross-referencing, and manual reconciliation. The data control environment is dependent on every *data attribute* being understood, at its atomic level, as a fact that is aligned with specific, durable business meaning without duplication or ambiguity. Managing data as meaning is the key to the alignment of data repositories, harmonization of *business glossaries,* and ensuring that application data dictionaries are comparable.

Achieving alignment of business meaning, including the *process* of how terms are created and maintained, can be a daunting task. It is not uncommon to experience resistance from internal business and technology users– particularly when there are multiple existing systems linked to critical business applications. The best strategy for reconciliation in a fragmented environment is to harmonize based on legal, contractual, or business meanings rather than trying to get every system to adopt the same naming convention. Nomenclature represents the structure of data and the unraveling of data structures and *data models* is expensive and not necessary. It is better to focus on precisely defining business concepts, documenting transformation *processes,* and capturing real-world data relationships. Once established, existing systems, glossaries, dictionaries, repositories, etc. can be cross-referenced to the common meaning.

Data as meaning must be managed along with its defining *metadata* to ensure consistency and comparability across the organization. Data meaning and *metadata management* must be understood as the core of your content infrastructure and the baseline for *process* automation, application integration, and alignment across linked *processes*. Some common types of *metadata* include business, operational, technical, descriptive, structural, administrative.

The implementation and management of a data control environment are governed by data *policies*, *standards*, *processes,* and *procedures*. These are the essential mechanisms for establishing a sustainable DM initiative and for ensuring compliance with a data control environment in the face of organizational complexity. Managing meaning is the key to effective DM. Meaning is achieved through the adoption of *semantic standards*. *Standards* are governed by *policy*. The *policy* is established by executive management, supported by *data owners* and enforced by Internal Audit.

Challenges in Creating a Data Control Environment

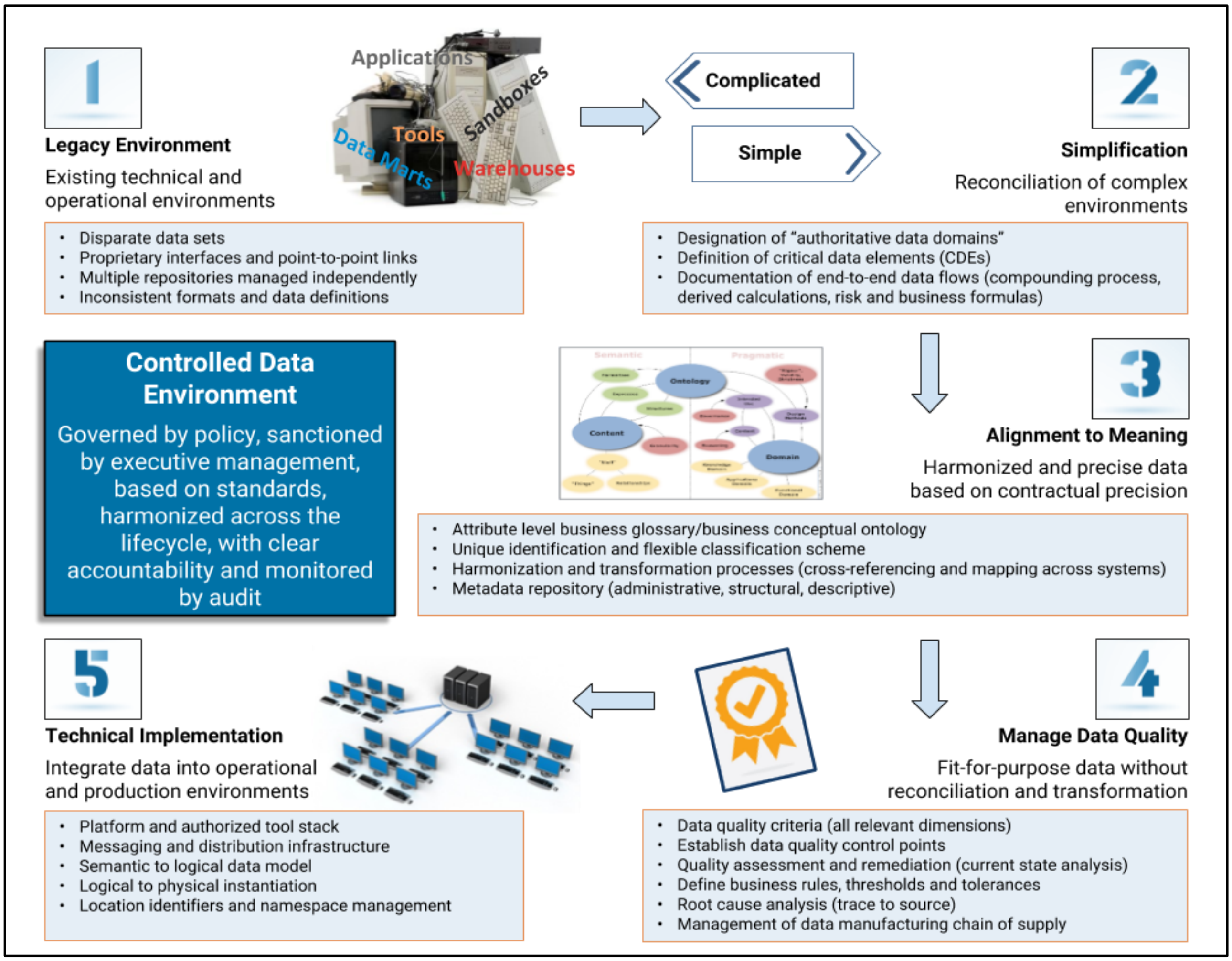

Diagram 0.1: Data Control Environment Challenges

Achieving a data control environment requires the organization to overcome the following challenges.

- Understand the existing legacy data environments, including an inventory of data, point-to-point links, inconsistent definitions, etc.
- Simplify, organize, and categorize the disparate environment into defined *data domains*, with clearly identified *data elements* and documented *data flows*.
- Align *data elements* to unambiguous shared meaning across the organization through the implementation of controls, *policy*, and governance.
- Measure and track data to ensure quality and consistency with minimal reconciliation.
- Align technology to ensure the *principles* and best practices that have been established are enabled across the organization's technology infrastructure.

It is this journey that must be taken to arrive at a data control environment needed to ensure the highest quality of data is delivered to critical *processes* throughout the organization.

Many organizations have made the mistake of trying to solve the DM problem by starting with technology. The DCAM, as a best practice, advocates that the starting point is with the business *process* that defines the requirements for data. Then the business *process* and data can be automated by technology.

Data Management Operating Levels

The DCAM Framework defines the components and capabilities that are required to achieve a data control environment in the organization. However, the Framework is not prescriptive to how the capabilities are executed at the various operating levels of the organization. An organization will need to customize their *operating model* to account for the size, complexity, geography, and culture of the organization.

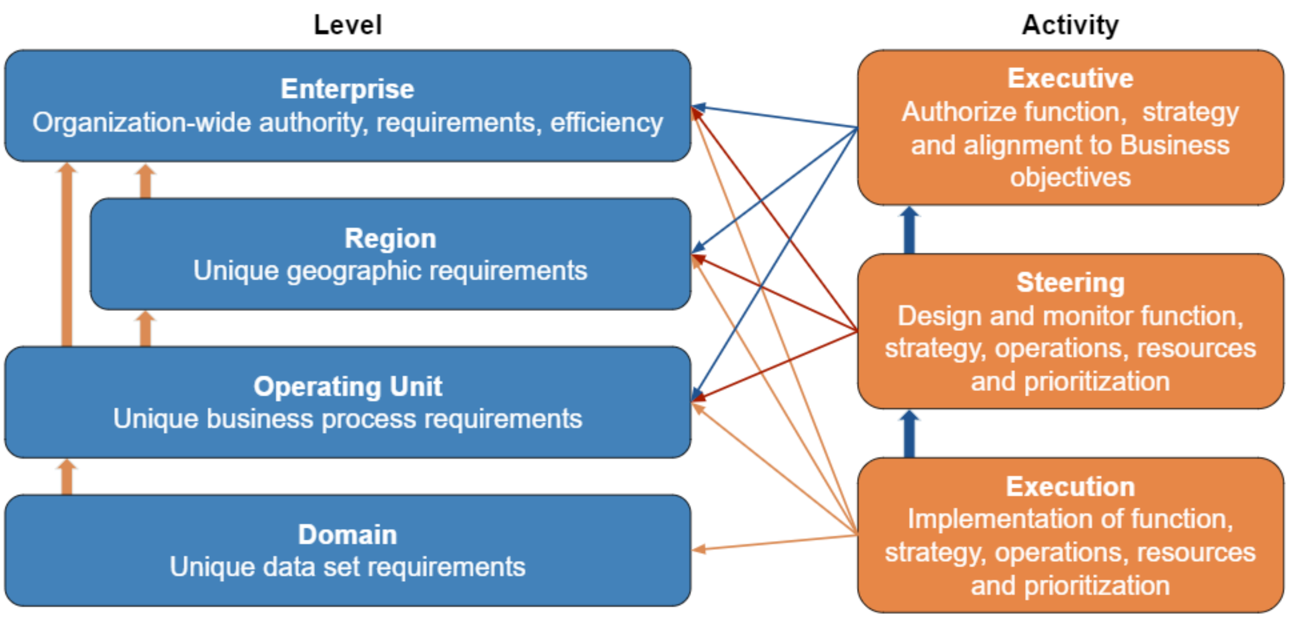

Diagram 0.2: Data Management Operating Levels

When using the DCAM Framework to assess the organization's capabilities it is most informative to evaluate each level as defined in the DM *target operating model* of the organization.

Data Management Stakeholders

The DM *Stakeholder* Accountability Matrix illustrated below is a *construct* for identifying the *stakeholder* types across the *data ecosystem* and the high-level roles and the relationships between the *stakeholders*. Understanding the *stakeholder* types is an important foundation for applying the DCAM Framework in a DM *target operating model* for the organization. It also defines the audience that is targeted for using the DCAM as the capability assessment tool for the organization. Within the Matrix, *data architecture* is the pivotal bridge in the relationship between the business and technology *stakeholders*.

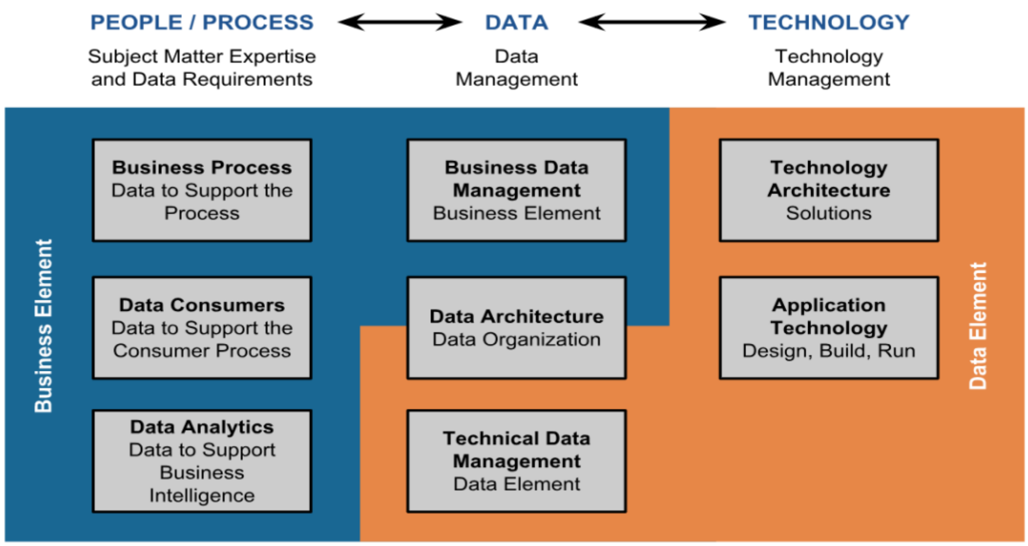

Diagram 0.3: Data Management Stakeholder Accountability Matrix

Emerging Issues

The practice of DM continues to evolve. New technologies and issues introduced into organizations that impact data or DM create additional requirements for the DM initiative. While the CDO may not be directly accountable for executing these technologies or addressing the issues, they must be actively involved in guiding the organization.

Our mission is to continue to evolve DCAM to stay current and support DM professionals to be successful in their roles within their organizations. Based on feedback from our membership, two topics—the rise of Machine Learning (ML) and Artificial Intelligence (AI) in analytics and the issue of data ethics—have emerged as important considerations for DM decision making. The member workgroup supporting the new version of DCAM included key *subject matter experts* on these topics. With their input, we have included considerations for these topics into the existing sub-capabilities in DCAM and introduced new capabilities and sub-capabilities where required.

Analytics & Machine Learning

Many organizations are at various stages of developing and implementing more advanced analytics and machine-learning applications within their operations. Although these applications do not fundamentally alter the required structures of good DM, they do necessitate a variety of changes to quite a few of its elements. These changes are set out under the relevant components and capabilities. However, it is useful to provide some background to the analytics and machine-learning *functions* in an organization so that its requirements will have context.

In broad terms, two main types of data-analytics activities will take place in an organization:

1. Descriptive / Inferential Statistics (D/IS) – This is the summary of key metrics of an organization. Key metrics include income, throughput, sales, and expenses. D/IS may also be estimates of what these metrics are likely to be if they are not directly observable. These summaries may be presented numerically or graphically, such as in the form of dashboards, and may also be generated periodically or on a *live* basis. This type of analysis typically forms part of the business intelligence (BI) or management-intelligence (MI) *function* of an organization.
2. Machine Learning / Artificial Intelligence – As opposed to descriptive and inferential statistics, which aim to present the salient features of a dataset through a direct summary of the data, this type of analysis employs algorithms to discern useful and valuable patterns from the data with little, or no, human input beyond setting up the learning algorithm. These patterns can be derived from both structured and unstructured data. The patterns uncovered are often more valuable than those discerned by humans for several reasons, including:
    a. These types of algorithms can consider many more data dimensions and their interactions than humans. They can also consider many more examples of a phenomenon simultaneously than a human can; and
    b. Machines do not suffer from many of the cognitive biases that result in poor human pattern-recognition outcomes.

A consequence of the complexity of the patterns recognized by these types of algorithms is that, despite the accuracy of the recognized patterns, the underlying pattern is often too complex to be understood by humans and *functions* effectively as a black box. The lack of understanding may not always be acceptable, depending on the application of the algorithm.

Both D/IS and ML/AI may be used as the basis for other types of analysis and forecasting activities.

Regarding the *data flow* that takes place for analytical activities, both classes of analytical activities go through approximately the same steps:

1. Raw input data is sourced from *reference databases* into a single processing workspace;
2. The raw input data is manipulated and transformed into a form that is necessary for the analytical activities;
3. The analytical activities take place:
    a. In the case of D/IS, this would be the generation of summary metrics;
    b. In the case of ML/AI, this would be the training of algorithms to recognize the required patterns;
4. The output of the analysis stage is made available to authorized users in the desired output format:
    a. In the case of D/IS, this would be the output of summary metrics in the form of a numerical or graphical report or dashboard;
    b. In the case of ML/AI, this would be the publication of trained and tested algorithms to those users who are authorized to use them or the publication of the outputs of the trained algorithms as they are applied to current data.

New technologies such as ML/AI introduce unique considerations for DM. That said, the successful implementation of all the DM capabilities is your best approach to addressing these new requirements. DCAM V2 now incorporates these considerations across the spectrum of capabilities and sub-Capabilities as a guide to empowering advanced analytics.

Data Ethics

ML/AI and other technological innovations have increased the sheer volume of data available to organizations and created a greater distance between human judgment and automated decision-making. One of the DM considerations raised by technological innovations in data ethics. In our discussions with DM experts around the world, the topic of data ethics has emerged as a primary concern, especially in light of the record-breaking fines imposed on organizations that trade in personal data and fail to protect user privacy. The concern is compounded by the rapid pace at which companies are pursuing products and services that leverage the *internet of things*, including facial recognition security systems and self-driving cars. DCAM V2 addresses this concern proactively to help DM professionals instill an organizational culture aligned with a Code of Ethics that articulates their organizations' values and priorities and provides *guidelines* for operationalization.

Automated decision making—using algorithms to classify data and make determinations based on these classifications—has the potential to revolutionize DM *processes* that were previously tedious, expensive, and impossible for humans to conduct. For example, convolutional neural networks—the most commonly applied machine learning technique used in medical imaging—are employed in the detection of tumors in diagnostic radiology. However, the reliability and *accuracy* of these algorithms depend on many factors, including the quality of training data, and the volume and how well the data corpus represents what it is supposed to represent. The best algorithms work because they are trained on accurate yet generalizable data so that new information is not forced into inappropriate classifications.

What this means for data ethics is that the complexity of the algorithms employed by an organization must be supported by heavily annotated training data and a broadly representative, very large data corpus. Furthermore, those responsible for data ethics within the organization must understand how the algorithms classify data, and how this classification may have unintended consequences. In some organizations, CDOs are being assigned accountability for data ethics, while others appoint a Chief Ethics Officer. Regardless of the approach, there are responsibilities and requirements for data and DM inherent in the operations of organizations that prioritize data ethics.

The assumptions shape the ability of algorithms to contend with data that varies from its training data that programmers make when contemplating the realm of possible outcomes. Though advances in machine learning have shone a spotlight on data ethics, this issue is not confined to the realm of artificial intelligence. Efforts to uphold privacy commitments to the people represented by the data are undermined often by simple human error or outdated systems protocols. While compliance with legal requirements for DM is helpful, it represents a low bar for privacy protection. Even the most well-intentioned data professionals working with the cleanest datasets may cause harm inadvertently, especially if they are not aware of how algorithms replicate societal inequities. In other words, "Good data can create bad outcomes."

Similarly, those responsible for governing data ethics in an organization must be prepared to consider how both automated and human decision making may affect organizational *stakeholders*. The *stakeholders* may be known entities or unknown populations who may be affected unintentionally by the organization's operations. For example, Amazon faced tremendous scrutiny when it rolled out its same-day Prime delivery service in metropolitan areas across the United States. On the surface, the data-driven decision to prioritize offering same-day service to customers living in ZIP codes where there is a high concentration of Prime members and expand the service to other ZIP codes over time seems logical in terms of cost and efficiency. However, given that the customers who pay to be Prime members are clustered in predominantly white neighborhoods, Amazon's same-day delivery service areas ended up reinforcing a chronic, systematic inequality in access to retail services. Since the news broke about the racial disparity in Amazon's same-day Prime delivery zones, the company has changed its approach. The reputational damage, though, will take longer to mitigate.

Incorporating data ethics into the DCAM framework is designed to help organizations establish *policies* and *procedures* that increase the likelihood that data-driven decisions with potential for such unintended outcomes will be identified and modified accordingly. Of course, it is difficult to anticipate how *data classification* and use will affect people in dynamic contexts. An organizational culture that embraces data ethics can go a long way toward minimizing vulnerability to data breaches and offsetting the biases inherent in programming assumptions. Every member of the organization who interacts with data and data systems has the potential to contribute to a culture of data ethics.

DCAM: A Framework for Sustainable Data Management

A complex set of DM capabilities are required to achieve a data control environment. The *Data Management Capability Assessment Model* is a framework for executing a robust, sustainable DM *function*. It is also an essential tool for ongoing assessment and benchmarking of an organization's DM capabilities.

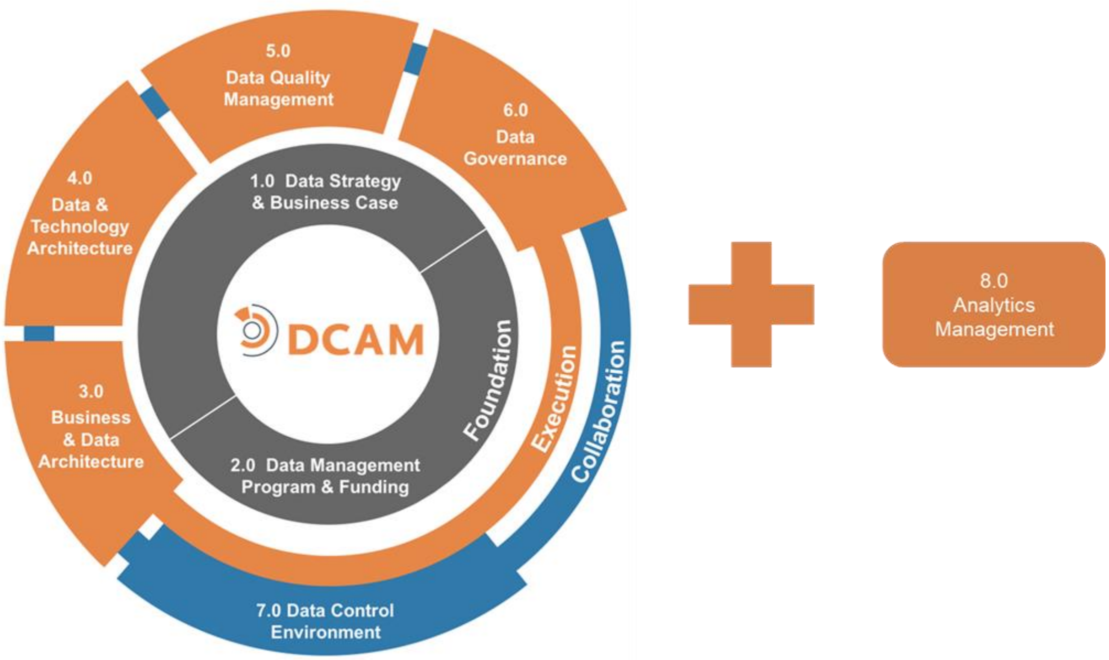

Diagram 0.4: DCAM Framework

The DCAM Framework consists of seven core components and one optional component. The first component, Data Management Strategy & Business Case and the second component, the Data Management Program & Funding Model, are ***foundational*** to the other five core components.

The next four core components of the DCAM framework, Business & Data Architecture, Data & Technology Architecture, Data Quality Management, and Data Governance are the ***execution*** components.

The final core component is the ***collaboration*** activity–Data Control Environment. It is here that the execution components are put into operation by the *data producer* to bring a defined set of data into control and make it available to *data consumers* at a point in time that is either real-time or a period end.

The seven core Components include 31 Capabilities and a total of 106 Sub-capabilities. The definition and scope of each component are presented below.

The eighth component, Analytics Management, is optional and is relevant where the scope of the DM program or the Chief Data Officer's responsibilities cover the Analytics *functions* of the organization. This component has seven capabilities and 30 sub-capabilities.

DCAM: The Scope of the Eight Components

1.0 Data Management Strategy & Business Case

The **Data Management Strategy (DMS) & Business Case** component is a set of capabilities to define, prioritize, organize, fund, and govern DM and how it is embedded into the operations of the organization in alignment with the objectives and priorities of both the *enterprise* and operating units. The DM business case is the justification for creating and funding a DM initiative. The business case articulates the major data and data related issues facing an organization or operational unit and describes the expected outcomes and benefits that can be achieved through the implementation of a successful DM initiative.

- Establish a DMS *function* within the Office of Data Management (ODM).
- Work with DM Program Management Office (PMO) to design and implement sustainable business-as-usual *processes* and tools for the DMS *function*.
- Align the DMS with the business strategy, objectives, and priorities, including prioritization of data based on its criticality to the business.
- Define the rationale and business case for the management of data as an asset through the organization-wide DM initiative.
- Ensure the DMS is aligned with the organization-wide Enterprise Data Management Principles.
- Articulate the DM target and current-state. Then, using DCAM as an assessment tool for gap analysis and prioritized gap closure, create a cohesive execution plan.
- Define the high-level execution roadmap.
- Define strategy execution risks and mitigations.
- Define DM performance metrics.
- Document the DMS with a compelling presentation of the value of an organization-wide DM initiative.
- Ensure that the DMS governance is integrated into the Data Governance (DG) structure.

2.0 Data Management Program & Funding Model

The **Data Management Program (DMP) & Funding Model** component is a set of capabilities to manage the Office of Data Management (ODM). These organizational structures include resource requirements and a full range of Program Management Office (PMO) activities such as the execution of program management, *stakeholder* management, funding management, communications, training, performance measurement. The DM funding model within the DMP is designed to provide the mechanism to ensure the allocation of sufficient capital needed for implementation of the Program. It also defines and describes the methodologies used to measure both the costs and the organization-wide benefits derived from the DM initiative.

- Establish a DMP *function* to implement the Program Management Office (PMO) capabilities within the ODM.
- Facilitate the design and implementation of sustainable business-as-usual DM *processes* and tools across the components and their capabilities.
- Establish roles and responsibilities related to the DM capabilities aligned with an organizational structure and execute in an ODM.
- Define Funding Model, secure and monitor funding, and institute cost and benefits tracking aligned to the Business Case.
- Establish the DM execution roadmap with supporting project plans to build upon the high-level Data Management Strategy (DMS) roadmap.
- Engage each *stakeholder* across the *data ecosystem* as appropriate to their roles in resource alignment, funding, communications, training, and skill development.

- Manage the DM initiative by monitoring and socializing DM performance metrics.
- Ensure that the DMP governance is integrated into Data Governance (DG).

3.0 Business & Data Architecture

The **Business and Data Architecture (DA)** component is a set of capabilities to ensure integration between the business *process* requirements and the execution of the DA *function*. The business *architecture function defines the business process*. DA defines *data models* such as *taxonomies* and ontologies, as well as *data domains*, *metadata*, and business-critical data to execute *processes* across the data control environment. The DA *function* ensures the control of data content, that the meaning of data is precise and unambiguous and that the use of data is consistent and transparent.

- Establish a DA *function* within the Office of Data Management (ODM).
- Work with DM *Project Management Office* (PMO) to design and implement sustainable business-as-usual *processes* and tools for DA, including the required integration with *business architecture*.
- Identify and establish *data domains*, authoritative sources, and *provisioning points*.
- Identify and inventory the data to support the business requirements, including all necessary *metadata,* including a glossary, dictionary, classification, lineage, etc.
- Define and assign business definitions linked to the data inventory.
- Ensure that the DA governance is integrated into Data Governance (DG) and aligned to both business and technology governance activities.

4.0 Data & Technology Architecture

The **Data & Technology Architecture (TA)** component is a set of capabilities to align the architectural requirements of the business, data, and technology across the organization to support the desired business *process* outcomes. These outcomes include DM *processes* and the required technology infrastructure.

- Work with DM *Project Management Office* (PMO) to design and implement sustainable business-as-usual *processes* and tools for the integration of business and *technology architecture* with data architecture.
- Ensure DM *function* alignment with business, data, and *technology architecture* and strategy.
- Ensure that the DM governance is aligned to both business and technology governance activities.

5.0 Data Quality Management

The **Data Quality Management (DQM)** component is a set of capabilities to define *data profiling*, DQ measurement, defect management, root cause analysis, and data remediation. These capabilities allow the organization to execute *processes* across the data control environment, ensuring that data is fit for its intended purpose.

- Establish a DQM *function* within the Office of Data Management (ODM).
- Work with data management (DM) Program Management Office (PMO) to design and implement sustainable business-as-usual *processes* and tools for DQM.
- Execute DQM *processes* against business-critical data. DQM *processes* include profiling & grading, measurement, defect management, root cause fix, remediation.
- Establish DQ metrics and reporting routines.
- Ensure that the DQM governance is integrated into Data Governance (DG).

6.0 Data Governance

The ***Data Governance*** **(DG)** component is a set of capabilities to codify the structure, lines of authority, roles & responsibilities, escalation protocol, *policy* & *standards*, compliance, and routines to execute *processes* across the data control environment. The component ensures authoritative decision making at all levels of the organization.

- Establish a *data governance function* within the Office of Data Management (ODM).
- Work with DM Program Management Office (PMO) to design and implement sustainable business-as-usual *processes* and tools for *data governance*.
- Define clear roles, responsibilities, and accountabilities for DM resources, including those mandated by DM *policy*.
- Define and operate the *data governance* structure with clear lines of authority, responsibility for decision making, engaged *stakeholders*, adequate oversight, issue escalation paths, and tracking of remediation activity.
- Develop and oversee adherence to comprehensive and achievable DM *policies*, *standards,* and *procedures*, including leading the response to audits.
- Ensure the *data governance* function is aligned with other relevant control *function policies*, *procedures*, *standards*, and governance requirements from information security, privacy, *technology architecture,* etc.

7.0 Data Control Environment Scope

The **Data Control Environment (DCE)** component is a set of capabilities that together form the fully operational data control environment. Data operations, supply-chain management, cross-control *function* alignment, and collaborative *technology architecture* must operate cohesively to ensure the objectives of the DM initiative are realized across the organization.

- Work with DM *Program Management Office* (PMO) to design and implement sustainable business-as-usual *processes* and routines to enable a successful data control environment.
- Bring together the DM components as a coherent, end-to-end *data ecosystem*.
- Follow current DM best practices by routinely reviewing and auditing the capabilities and their *processes*.
- Ensure all facets of DM for business-critical data such as *data lifecycle*, end-to-end *data lineage,* and data aggregations are fully operational.
- Ensure DM is aligned with other control *function policies*, *procedures*, *standards*, and governance.

8.0 Analytics Management Scope

The **Analytics Management** (AM) component is a set of capabilities used to structure and manage the Analytics activities of an organization. The capabilities align AM with DM in support of business priorities. They address the culture, skills, platform, and governance required to enable the organization to obtain business value from analytics.

Scope

- Define the Analytics strategy.
- Establish the AM function.
- Ensure that analytics activities are driven by the Business Strategy and supported by the DM strategy.
- Ensure clear accountability for the analytics created and for their uses throughout the organization.
- Work with DM to align Analytics with Data Architecture (DA) and Data Quality Management (DQM).

- Establish an *analytics platform* that provides flexibility and controls to meet the needs of the different stakeholder roles in the Analytics operating model.
- Apply effective governance over the *data analysis lifecycle*. Governance includes tollgates for model reviews, testing, approvals, documentation, release plans, and regular review processes.
- Ensure that Analytics follows established guidelines for privacy, data ethics, *model bias*, and *model explainability* requirements and constraints.
- Manage the cultural change and education activities required to support the Analytics strategy.

DCAM Uses Cases

DCAM has multiple uses within an organization.

- As a well-defined control framework
- As an assessment tool
- As an industry benchmark

DCAM as a Framework

When an organization adopts the standard DCAM framework they introduce a consistent way of understanding and describing DM. DCAM is a framework of the capabilities required for a comprehensive DM initiative presented as a best practice paradigm. DCAM helps to accelerate the development of the DM initiative and make it operational. The DCAM Framework:

- Provides a common and measurable DM framework
- Establishes common language for DM
- Translates industry expertise into operational *standards*
- Documents DM capability requirements
- Proposes evidence-based artifacts

DCAM as an Assessment Tool

To effectively use DCAM as an assessment tool requires the definition of the assessment objectives and strategy, planning for the assessment management, and adequate training of the participants to establish a base understanding of the DCAM Framework.

The assessment results translate the practice of DM into a quantifiable science. The benefits afforded an organization from the assessment outcomes include:

- Baseline measurement of the DM capabilities in the organization compared to an industry *standard*
- Quantifiable measurement of the progress the organization has made to operationalize the required DM capabilities
- Identification of DM capability gaps to inform a prioritized roadmap for future development aligned to the organization’s business requirements for data and DM
- Focused attention to the funding requirements of the DM initiative

DCAM Scoring Guide

The actual scoring guide used throughout DCAM is as follows. It is designed to assess the phase of capability attainment. It is not an assessment of the maturity or scope to which an organization has applied its capabilities.

By design, the scoring is an even number to force a conscious decision by the rater and avoiding the tendency to select the midpoint of the scoring scale.

SCORE	CATEGORY	DESCRIPTION	CHARACTERISTICS
1	Not initiated	Not Performed	Ad hoc activities performed by heroes
2	Conceptual	Initial Planning Stages	Issues being debated; whiteboard sessions
3	Developmental	Engagement Underway	Key functional *stakeholders* identified; workstreams defined; meetings underway; participation growing; *policies*, roles, and operating *procedures* being established; project/annual funding
4	Defined	Defined and Verified	Business users active; LOB management with P&L responsibility engaged; requirements verified; responsibilities defined and assigned; *policy* and *standards* exist; routines in place; lineage underway; *critical data elements* (CDEs) identified; adherence tracked; multi-year/sustainable funding
5	Achieved	Adopted and Enforced	Executive management sanctioned; proactive business engagement; responsibilities coordinated; *policy* and *standards* implemented; lineage verified; *data harmonized* across repositories; adherence audited; strategic/investment funding
6	Enhanced	Integrated	Fully embedded in the operational culture of the organization with the goal of continuous improvement

Table 0.5: DCAM Scoring Guide

DCAM as an Industry Benchmark

The EDM Council conducted a DM industry benchmark study in 2015, 2017, and 2020. The benchmark is based on the capabilities defined in DCAM and thus can be used in comparison analysis for organizations conducting a DCAM assessment.

The industry benchmark is not restricted to organizations that are using DCAM. Input from a broader range of DM industry practitioners affords an enhanced perspective on the state of the DM industry.

DCAM Release Notes Summary

DCAM v2.2 – November 2020

The DCAM v2 release is considered a ***minor*** release in that it includes an enhancement to the DCAM Framework but it did not structurally change the scoring of DCAM.

The objective of the DCAM v2.2 refresh includes the following:

- Addition of Component 8.0: Analytics Management
  - This additional component is considered an *optional* component in relation to the seven *core* components of DCAM.

  - An assessment conducted with all eight Components is now considered a DCAM+ Assessment, and the overall DCAM+ score would be inclusive of the 8.0 Analytics Management score, while the overall DCAM score would continue to be calculated based on the seven core components.
- Introduction of Core Artifacts required to support each Component.
- Enhancement to the content to support clarity, usability, and consistency in language, format, and presentation style.

For more information about how Component 8.0: Analytics Management is scored or included in a DCAM Assessment, please see the DCAM+ Scoring Guidelines FAQ.

DCAM v1.3 Mapping to v2.2 *– a model is available that presents alignment of the Component, Capabilities, and Sub-capabilities in DCAM version 1.3 to version 2.2.*

DCAM v2.1 – August 2019

The DCAM v2 release is considered a ***major*** release in that it has structural change. The intent is that there will not be a structural change to DCAM more often than 24 months, which will align to the two-year cycle on refreshing the industry benchmark data. In between major releases, there is the potential for a ***minor*** release no more often than six months. A minor release could include changes such as language, clarification, glossary alignment, and enhancements to the Component Introduction and reference to related EDMC best practice materials. Very basic grammar or spelling mistakes will be fixed as uncovered.

The EDM Council will always support one version backward from the current major version (v2.x). In other words, v1.3 will be supported until such time that v3.1 is released.

The objective of the DCAM v2.1 refresh includes the following:

- Changes to the framework components to better present a logical flow of the capabilities
- Introduction of new concepts that have requirements for data and DM
- Enhancement to the content to support clarity, usability, and consistency in language, format and presentation style

Changes to Framework Components

1.0 Data Management Strategy & Business Case

- Changed to strategy capability that aggregates the concepts of the other six components into a summary strategy
- Introduced the concept that a data management strategy is comprised of a data content strategy, data use strategy, and data management deployment strategy
- Integrated Business Case with the data management strategy – previously in a standalone component with Funding Model

2.0 Data Management Program & Funding Model

- Integrated Data Management Program with Funding Model – previously in a standalone component with Business Case

3.0 Business & Data Architecture

- Clarified that data management all starts with the business *process* requirements for data; maintained full *Data Architecture* scope

4.0 Data & Technology Architecture

- Clarified the accountability for Technology to meet the business *process* and data requirements

5.0 Data Quality Management

- Clarified a single data quality *process* that supports all data types

6.0 Data Governance

- Aggregated all governance leveraged by the other components into a single view of governance

7.0 Data Control Environment

- Clarified that the data control environment is the operational execution that brings together the data management components as a coherent, end-to-end *data ecosystem*
- Formalized data risk as a sub-capability

Introduction of New Concepts

Machine Learning and Artificial Intelligence – the introduction of challenges and opportunities using innovative analytics tools and methodologies like Machine Learning (ML) and Artificial Intelligence (AI)

Data Ethics – advanced analytics exposes an organization to increased risk from the misuse of data; data use and outcome is not simply a legal question but also an ethical question

> ***Data Ethics/ML/AI Concepts Mapped to DCAM v2.1*** *– a model is available that aligns the concepts of ML/AI and Data Ethics and their data and data management requirements to related sub-capabilities of DCAM.*

Business Process Design – emphasize the role of the business *process* in defining requirements for data and data management and troubleshooting root-cause-fix; introduce the requirement for *standard* end-to-end DCAM DM *processes*

Enhancements to the Content

***Standard* Language –** improved the consistent use of language and the readability

DCAM *Business Glossary* – applied the discipline of aligning to the use of terminology as defined in the DCAM *Business Glossary*

Rating Guidance *Standard* Language – established consistent language to describe the six assessment scores of a sub-capability to simplify the respondent scoring *process*

> The rating guidance *standard* language is intended to be used as the basis for understanding how to score each sub-capability. However, each sub-capability has its bespoke rating

guidance, which is specific to the sub-capability context and should be a blend of *standard* language plus specific information *to help you score the sub-capability.*

How to use this Document – Anatomy of DCAM

The DCAM is organized into seven core components and one optional component. Each component is preceded with a definition of what it is, why it is important, and how it relates to the overall data management *process*. These definitions are written for business and operational executives to demystify the data management *process*. The core components are organized into 31 capabilities and 106 sub-capabilities and the optional component is organized into seven capabilities and 30 sub-capabilities. The capabilities and sub-capabilities are the essences of the DCAM Framework. They define the goals of data management at a practical level and establish the operational requirements that are needed for sustainable data management. Finally, each sub-capability has an associated set of measurement criteria. The measurements are used in an assessment of your data management journey.

- **Component** – a group of capabilities that together deliver a foundational tenet of data management; used as a reference tool by the data practitioners who are accountable for executing the tenet
    - **Introduction** – high-level context for the component; used as a background for developing an understanding of the component by data practitioners
        - **Definition** – formal description of the component; used to support common data management understanding and language
        - **Scope** – a set of statements to establish the guardrails for what is included in the component; used to understand and communicate reasonable boundaries
        - **Value Proposition** – a set of statements to identify the business value of delivering the data management component; used to inform the varied business cases for developing the DM initiative
        - **Overview** – more detailed context and accounting at a practical level to establish an understanding of the operational execution required for sustainable data management; used as a guide by the respective data practitioners
        - **Process, Tools & Constructs** – things that are required to support the execution of the component; used for reference and to link to supporting best practice material when available
        - **Core Questions** – high-level but probing inquiries; used to direct exploration of the data management component
- **Capability** – a group of sub-capabilities that together execute tasks and achieve the stated objectives; used as a reference tool by the data practitioners who are accountable for the execution
    - **Description** – brief aggregate explanation of ***what*** is included in the sub-capabilities required to achieve the capability; used in the assessment process to inform the respondent of the scope of what they are rating
- **Sub-Capability** – more granular activities required to achieve the capability; used as a reference tool by the data practitioners who are accountable for the execution
    - **Description** – a brief explanation of ***what*** is included in the sub-capability and used in the assessment process to inform the respondent of the scope of what they are rating
    - **Objective** – identified goals or desired outcomes from executing the sub-capability; used as a basis for defining requirements for the data management process design
    - **Advice** – more detailed but casual insight on the best practice ***how*** to execute the sub-capability with an audit review perspective; used by the data practitioner

- **Questions** – inquiries to direct interrogation of the capability/sub-capability current-state; used by the data practitioner to inform a perspective of the assessment scoring
- **Artifacts** - Required things or evidence of adherence; used for assessment and audit reference and to link to supporting best practice material when available
- **Rating Guidance** – insight for defining an assessment score; used when completing an assessment survey

DCAM Business Glossary

The EDM Council has developed a DCAM *Business Glossary,* which contains ~130 data management *term* names and definitions. DCAM v2.1 has applied these *terms* consistently across the document. Where there are *terms* defined in the glossary, the word or phrase is *italicized and underlined* in the text.

> DCAM Business Glossary[1] – *all words or phrases throughout the document that are italicized and underlined are contained in the DCAM Business Glossary available by the link above.*

Acronym Glossary

AI Artificial Intelligence

AM Analytics Management (DCAM Component)

BA *Business Architecture* (DCAM Component)

BI Business Intelligence

CDE *Critical Data Elements*

COO Chief Operating Officer

DA *Data Architecture* (DCAM Component)

DaaS Data as a Service

DBMS Database Management System

DCE Data Control Environment (DCAM Component)

DG *Data Governance* (DCAM Component)

DM Data Management

DMP Data Management Program (DCAM Component)

DMS Data Management Strategy (DCAM Component)

DQ *Data Quality*

DQM Data Quality Management (DCAM Component)

DSA *Data Sharing Agreement*

ETL Extract, Transform, Load

KPIs Key Performance Indicators

KRIs Key Risk Indicators

ML Machine Learning

MRA Memo Requiring Attention

NLP Natural Language Processing

ODM Office of Data Management

PMO Program Management Office

QA Quality Assurance

QC Quality Control

RACI Accountability, Responsibility, Consulted and Informed

[1] https://edmcouncil.org/members/group_content_view.asp?group=195841&id=808338

RCA	Root-cause Analysis	**SLA**	Service Level Agreement
ROI	Return-on-Investment	**STP**	Straight Through Processing
SDLC	Software Development Lifecycle	**TCO**	Total Cost of Operation

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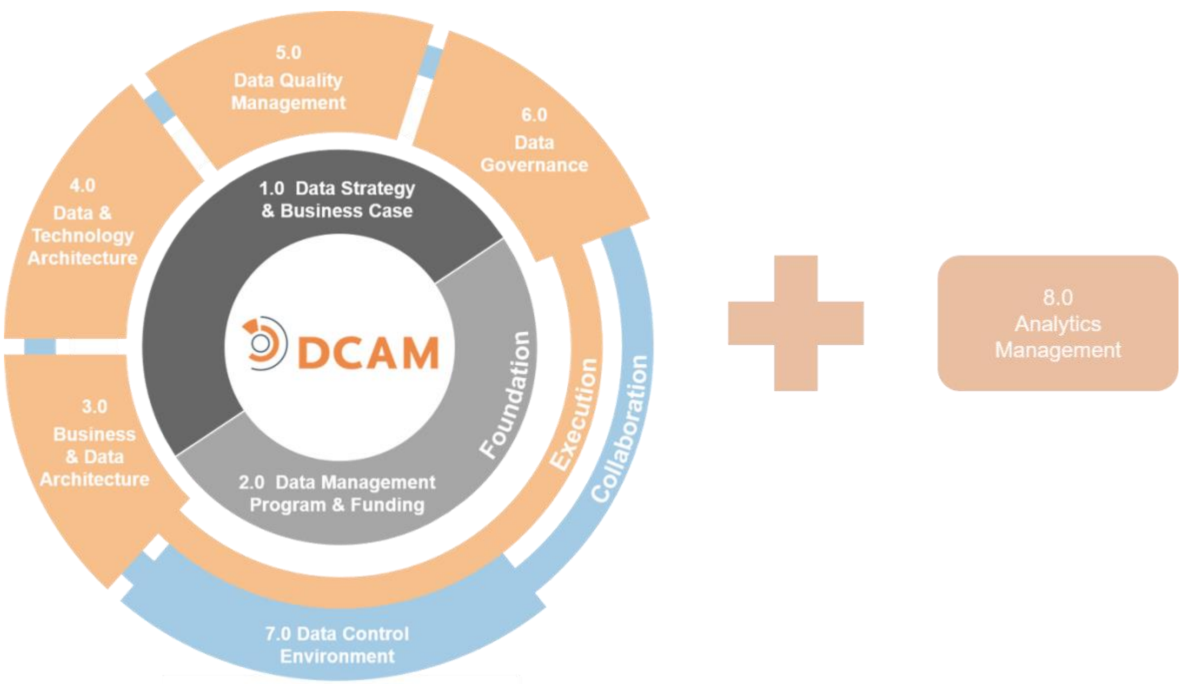
DCAM® Framework

1.0 Data Management Strategy & Business Case



1.0 Data Management Strategy & Business Case

Introduction

The **Data Management Strategy & Business Case** determines how data management (DM) is defined, organized, funded, governed and embedded into the operations of the organization. It defines the long-term vision including a description of *stakeholders* or *stakeholder functions* that must be aligned. Data Management Strategy demonstrates the business value that the program will seek to achieve. It becomes the blueprint for the organization to evaluate, define, plan, measure and execute a successful and mature DM initiative.

The purpose of developing a DM strategy and business case is to articulate the rationale for the DM initiative. The strategy defines *why* the initiative is needed, as well as the goals, and expected benefits. The strategy also describes *how* to mobilize the organization in order to implement a successful DM initiative. The DM business case provides the rationale for the investment in the DM initiative. DM is no different than any other established business *process*. It needs to be justified, funded, measured and evaluated. It provides clarity of purpose, enabling agreement and support of initiative objectives from senior executives as well as program *stakeholders*.

Definition

The **Data Management Strategy (DMS) & Business Case** component is a set of capabilities to define, prioritize, organize, fund and govern DM and how it is embedded into the operations of the organization in alignment with the objectives and priorities of both the *enterprise* and operating units. The DM business case is the justification for creating and funding a DM initiative. The business case articulates the major data and data related issues facing an organization or operational unit and describes the expected outcomes and benefits that can be achieved through the implementation of a successful DM initiative.

Scope

- Establish a DMS *function* within the Office of Data Management (ODM).
- Work with DM Program Management Office (PMO) to design and implement sustainable business-as-usual *processes* and tools for the DMS *function*.
- Align the DMS with the business strategy, objectives, and priorities, including prioritization of data based on its criticality to the business.
- Define the rationale and business case for management of data as an asset through the organization-wide DM initiative.
- Ensure the DMS is aligned with the organization-wide Enterprise Data Management Principles.
- Articulate the DM target and current-state. Then, using DCAM as an assessment tool for gap analysis and prioritized gap closure, create a cohesive execution plan.
- Define the high-level execution roadmap.
- Define strategy execution risks and mitigations.
- Define DM performance metrics.
- Document the DMS with a compelling presentation of the value of an organization-wide DM initiative.
- Ensure that the DMS governance is integrated into the Data Governance (DG) structure.

Value Proposition

Organizations that have *enterprise* and operating unit executives who understand, support, and offer direction for the organization-wide DM initiative lower organizational risk and get better acceptance of the DM Initiative at all levels of staff. Staff engagement in the sustainable management of data for the short and long-term success of the organization is essential. Organizations that implement effective DM get a return on investment from several areas:

- Efficiency and effectiveness of data issue resolution, compliance, and auditable demonstration
- Improved ***enterprise*** risk management
- Efficiency in business ***process*** optimization
- Innovation and differentiation for customers

Overview

The DMS component is one of the three foundational components of the DCAM Framework. The DMS is what integrates the strategies of each of the other components of the Framework into an overall strategy for the execution of the DM initiative. It is important to note that the DMS is a *function*, within the Office of Data Management in which strategic planning capabilities and skills reside. (You may refer to *Overview* in the *Data Management Program* section of this guide for detail on the structure of the Office of Data Management).

Using the DCAM Framework provides a structure for the DM initiative that includes the core *principles* of DM. It also helps *stakeholders* understand the value of DM as it relates to their operating units and strategic initiatives.

The strategy should align with the organization's articulated *target operating model* for executing DM with a roadmap and timeline to achieve the target. It is important for a strategy to compare the target-state to the current-state in order to show the organizational, functional, operational, and technological gaps and inefficiencies. The Strategy can then define, prioritize, and schedule gap closure. DCAM used as a capability assessment tool is fundamental to the analysis of gaps in each operating level and organizational unit.

The DMS integrates the Framework components at each operating level throughout the organization. For detail on the levels and context at which DM operates, refer to *Data Management Operating Levels* in the introductory section of this guide.

A strategy must be documented for each organizational unit at the various operating levels of the organization and work in concert with the other organizational units across the organization. Within a single organizational unit, each DCAM component has a unique input to the strategy that is then integrated with the other component input and prioritized in the final strategy for that organizational unit. These inputs must align with the DM *target operating model* for the organization. The *target operating model* defines the expected component and capability requirements for the operating level of the organizational unit.

Because not all organizational units will be at the same maturity in the design and execution of their DM initiative these strategies are specific to their business objectives, priorities, and identified DM inefficiencies and gaps.

The importance of the physical documentation should not be underestimated because the document is the primary internal marketing tool to drive understanding and support from all *stakeholders* at all levels of the organization.

The DMS includes the business case that describes how value will be realized from the data assets of an organization, through the collaboration of business, data, and technology.

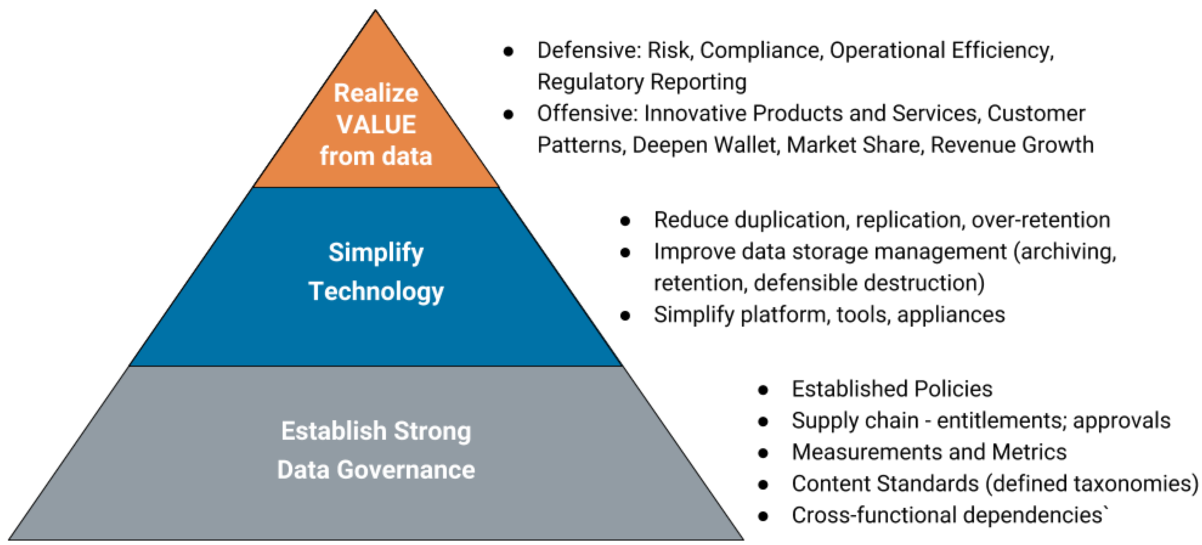


Diagram 1.1: Data Asset Value Model

The DM business case is the cost-benefit realization of the set of activities and deliverables expected from the DM initiative. The DM business case answers the question: Why the firm is focusing on data management? This helps achieve alignment across the *stakeholders*. The business case helps management understand the costs, benefits, and risks associated with the evolution of the DM initiative. It is essential to link the business case with realistic strategic and tactical measurement criteria and align them with the long-term sequence plan for the DM initiative. This enables the organization to understand the total costs associated with implementation as well as maintenance of the DM initiative and helps ensure that it is sufficiently funded to meet both near and long-term objectives.

The DM business case articulates the benefits of DM, in alignment with the objectives defined, communicated and agreed upon in the DMS. It discusses the defensive benefits of the initiative including operational cost reduction, improved regulatory reporting, streamlined risk management, controlled *data governance*, improved *data quality* (DQ). It also highlights the offensive benefits of the initiative which include advanced *analytics*, improved customer service, innovative product development, increased revenues, improved market penetration.

In some cases, the best way to build support for the business case is through a demonstrative proof of concept or pilot project. In these instances, a specific pain point or high-profile business objective would be selected and used to demonstrate the benefits of implementing effective DM. If this approach is used, it is important to select a project that is achievable and can provide quick wins. This approach builds confidence among *stakeholders* on the foundational benefits of DM to ensure sustainability. Regardless of whether you define the business case with or without a proof of concept, all activities must align to the strategic business objectives of the organization.

The DM strategy and the business case are not static and must be able to evolve as the priorities and needs of the organization change. The most effective and successful DM strategies are living artifacts that are visibly endorsed by executive management and are supported by mandatory organizational *policy*.

Core Questions

- → Does the DMS clearly articulate the reason and the importance of implementing the DM initiative at each level of the organization?
- → Is there executive buy-in across business, operations and technology?

→ Do *stakeholders* agree to support and sustain a DMS *function*?
→ Has the DMS sufficiently defined the immediate, medium and long-term goals and objectives of the organization-wide DM initiative?
→ Is the DMS in line with organizational priorities?
→ Has the DMS effectively identified the critical areas of focus, including how priorities are established and verified?
→ Has the DMS identified the *operating model* and required staffing resources needed to establish, lead and maintain the DM initiative?
→ Is the Data Management Business Case aligned with the strategic goals of the organization?

Core Artifacts

The following are the **core artifacts** required to execute an effective Data Management Strategy & Business Case capability. Items with an '*' link to published best practice *guidelines*.

- Data Management and Business Strategy Alignment Matrix
- Data Management Accountability Matrix*
- Data Management Business Case
- DCAM Business Glossary*
- Data Management Framework
- Data Management Operating Model
- Data Management Strategy
- Data Risk Appetite Statement
- *Stakeholder* Analysis

1.1 The Data Management Strategy (DMS) is Specified and Shared

The DMS is a collaboration of business, data and technology stakeholders. The strategy must identify the high-level organizational objectives identified by the organization's senior executives. The strategy will include the objectives of each of the components in the DCAM Framework that results in an integrated implementation roadmap. The authority for the DMS is from approval by the defined governance structure of the DM initiative. The authority is further enhanced by Internal Audit review of the enforceability of the DMS.

1.1.1 The DMS is developed, documented, and consolidated

Description

The DMS is a collaboration with the full spectrum of business, technology and operations management *stakeholders*. Together they document the DMS.

Objectives

- Align DMS with business, technology, and operations.
- Document DMS.
- Publish DMS to all *stakeholders*.

Advice

The DMS is both a statement of approach and a marketing document for presentation to *stakeholders*. Without a formally defined and collaborative statement of its strategy, the organization's approach to DM can become a series of short-term, defensive reactions. Without collaboration, the strategy can be viewed as irrelevant to *stakeholders*.

Questions

→ Have all the aspects of the DMS been defined and presented in terms meaningful to each operating level?
→ Are the business, regulatory and operational rationales for the DM initiative defined and verified?
→ Is the DMS aligned with business requirements, implementation plans, technical capabilities, and operational *processes*?
→ Has the DMS been documented and published?
→ Is the approach to DM clearly defined?
→ Are *stakeholders* aligned on the specified approach?

Artifacts

- Vision statement of the target-state DM initiative and what will be achieved
- Definition of the foundational *principles* and illustration of why they are essential
- Business requirements and priorities; *process* for establishing and approving each
- Benefits – answers why we are doing this; articulates the value proposition and how it aligns to organizational *principles*
- Mapping of strategy to technical and operational capabilities
- List of *stakeholders* and evidence of bi-directional communication
- Distribution lists and approvals from *stakeholders*
- Evidence that the strategy published

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No formal DMS exists.	No formal DMS exists, but the need is recognized, and the development is being discussed.	The formal DMS is being developed.	The formal DMS is defined and validated by directly involved *stakeholders*.	The formal DMS is established and understood across the organization and is being followed by the *stakeholders*.	The formal DMS is established as part of business-as-usual practice with a continuous improvement routine.

1.1.2 The DMS is aligned with high-level organizational objectives

Description

High-level organizational objectives are those identified by executive management as organizational goals (e.g., the organizational objective is to improve customer support and services).

Objectives

- Fully map DMS and align it with the high-level organizational objectives.
- Secure approval of DMS by the executive committee and *stakeholders*.
- Define a *process* to ensure the future alignment of the DMS to organizational objectives.

Advice

The goal is to ensure that the DMS supports the current objectives of executive management. These high-level objectives need to be translated into requirements for data or DM and evaluated against gaps and inefficiencies, often called pain points, that currently exist within the organization. A DM initiative that is not synchronized with the high-level objectives of the organization can result in a misalignment of data priorities. Critically, this misalignment can lead to a perception of the DMS as irrelevant among executive management. Achieving executive buy-in is critical for the organization to adopt the DM initiative. With the right executive support the time required for the organization to embrace a data culture and adopt the DM initiative will reduce exponentially.

Questions

→ Has the DMS been aligned and mapped to organizational objectives?
→ Has the alignment been verified and approved by *stakeholders* and executive management?

Artifacts

- Mapping of organizational objectives to data concepts and from there to strategy, with verification from executive committee and *stakeholders*
- High-level roadmap on how the strategy will be implemented in alignment with business objectives
- List of *stakeholders* and evidence of bi-directional communication
- Distribution lists and approval documents from *stakeholders*

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No formal DMS exists.	No formal DMS exists, but the need is recognized, and the development is being discussed.	The formal DMS is being developed.	The formal DMS is defined and is aligned with high level organizational objectives.	The formal DMS is established and recognized as aligned to high level organizational objectives.	The formal DMS is reviewed against high level organizational objectives at least annually.

1.1.3 The DMS addresses the core strategy concepts from each DCAM component

Description

The DMS is a high-level aggregation of the strategies for implementation of the other Components of the DCAM Framework. It represents a prioritized balancing of business objectives and resources that results in an integrated implementation roadmap.

Objectives

- Include core concepts from each of the DCAM Framework Components in the DMS.
- Prioritize the core DCAM components based on their alignment to the business objectives.
- The defined outcomes reflect advancement of the current-state toward the *target operating model*.

Advice

A DMS may exist at each of the operating levels of an organization: *enterprise*; Region; operating units; and *Domain*. The DMS integrates the strategy for each DCAM Framework Component based on the business objectives of the operating level implementing it. Often, implementation may require alignment with the business objectives of a higher operating level or control *function* of the organization.

Core strategic concepts from each DCAM Framework Component should be identified in the DMS. The DMS should represent the business objectives of the current planning period against the DM *target operating model*. It should prioritize outcomes and resources across each of the DCAM Framework Components.

Questions

→ Have the core strategic concepts of the DM Program & Funding Model been integrated into the DMS?
→ Have the core strategic concepts of the Business & Technology Integration been integrated into the DMS?
→ Have the core strategic concepts of the Business & Data Architecture been integrated into the DMS?
→ Have the core strategic concepts of the Data Quality Management been integrated into the DMS?
→ Have the core strategic concepts of the Data Governance been integrated into the DMS?
→ Have the core strategic concepts of the Data Control Environment been integrated into the DMS?

Artifacts

- Evidence of content management with authoritative *data domains*, *critical data elements*, *taxonomies* and *ontology*, identifiers, *systems of record*
- Verification of DM measured by program, outcome, quality and usage
- Evidence of implementation of DM for architectural *principles*, cross-functional collaboration, operational capabilities, incremental strategy

- Communication and training materials used for *stakeholder* education and socialization
- Evidence of governance Including organizational structure, *policy*, controls, stewardship, accountability, audit and enforcement
- List of *stakeholders* and evidence of bi-directional communication
- Verified mapping of strategy to technical and operational capabilities
- Evidence that the strategy was approved and published

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No formal DMS exists.	No formal DMS exists, but the need is recognized, and the development is being discussed.	The formal DMS is being developed.	The formal DMS is defined and includes core concepts from the other Components.	The formal DMS is established and recognized as coherent and aligned with the other Components.	The DMS is reviewed, updated and realigned with the other Components at least annually.

1.1.4 The DMS includes an established mechanism for approval

Description

Approval of the DMS requires a mechanism and a governance structure that provides authority for the Strategy.

Objectives

- Create a mechanism to capture feedback from *stakeholders*.
- Incorporate *stakeholder* feedback into the DMS.
- Have *stakeholders* review and approve the DMS.

Advice

An effective DMS needs buy-in from all the *stakeholders* within the organization because it has significant implications for the *stakeholders* responsible for operations. *Stakeholders* are more likely to buy into the DM initiative if they see evidence of their influence over the finished DMS. This buy-in typically makes implementation easier. DMS approval is best managed as an iterative *process* focused on the needs of business and balanced against the requirements for implementation.

Questions

→ Is there a mechanism to obtain and verify feedback from *stakeholders* on the aspects of the DMS?
→ Is there a mechanism to obtain and verify feedback from *stakeholders* on the implementation strategy?

Artifacts

- Approval *process* documentation that defines the mechanism of the *process* of approval
- List of *stakeholders* by *function*
- List of *stakeholders* and evidence of bi-directional communication
- Distribution lists and approvals from *stakeholders*

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No formal DMS exists.	No formal DMS exists, but the need is recognized, and the development is being discussed.	The formal DMS is being developed.	The formal DMS is defined and includes an approval mechanism.	The formal DMS is established and recognized as approved.	The DMS is reviewed, updated and re-approved at least annually.

1.1.5 The DMS has been evaluated as being enforceable

Description

Enforceability of the DMS should be evaluated by the Internal Audit *function* of the organization.

Objectives

- Have Internal Audit review and approve the DMS.
- Have Internal Audit determine that implementation of the DMS can be enforced via their existing examinations.

Advice

Engagement with Internal Audit is an important way to ensure that the DM initiative is viable from an organizational point of view. This will typically require education of Internal Audit about DM concepts and *principles*. This early stage activity and the development of a partnership with Internal Audit can also help ensure their engagement in oversight as a priority. If the DMS can be audited, it becomes real in the eyes of both the organization and industry regulators. Internal Audit can be an important supporter of the implementation of a sustainable DM initiative.

Questions

→ Is Internal Audit familiar with the concepts associated with DM?
→ Has Internal Audit reviewed the DM initiative and determined that it can be audited via scheduled exams?

Artifacts

- Evidence of communication with Internal Audit about the concepts in the DMS
- Evidence of internal reviews and approval of the DMS
- Verification that the DMS can be enforced and audited

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No formal DMS exists.	No formal DMS exists, but the need is recognized, and the development is being discussed.	The formal DMS is being developed.	The formal DMS is defined and includes an explanation of how it is enforced.	The formal DMS is established and recognized as enforceable.	The formal DMS is established as part of business-as-usual practice. It is recognized as the normal way of working rather than as a mandate.

1.2 The Data Management Business Case is Defined

The DMS must reflect the high-level business requirements of the organization. These requirements will inform the priorities of the DM initiative and the financial return on investment from managing data. The business stakeholders must validate the quantifiable business outcomes of the business case.

1.2.1 High-level business requirements are documented

Description

High-level business requirements are those identified by the operating units, often reflecting the high-level *enterprise* requirements identified by executive management. It is important that the DMS reflect both the organizational requirements as well as operating units' requirements.

Objectives

- Document the high-level business requirements for the *enterprise* and critical operating units.
- Verify high-level business requirements for the *enterprise* and critical operating units and incorporate them into the DMS.

Advice

High-level business requirements such as objectives, goals, pain points and priorities are derived based on discussions with representatives from the operating units at all levels of the organization. The discovery *process* and verification of these requirements are done on an iterative basis and needs to be balanced against operational reality as well as budgetary requirements. The goal is not only to define objectives and requirements but to prioritize them based on dependency, budget, and implementation reality. The objective of the business case is to achieve *stakeholder* buy-in for how the initiative will operate and how it will deliver against the objectives of the business. It drives home the fact that the benefits outweigh the costs.

A unique consumer of data that cuts across all operating units is the data analytic or data science team. It will be critical to obtain input for business requirements from the practitioners and executives of these teams. The methodologies and tools they deploy and the speed at which they are evolving may call for an ongoing routine more frequent than the annual planning cycle. Ensure that the scope of discussion covers both internal activities and data *analytics* that may be acquired externally.

The investment in a shared understanding of the business objectives that drives the DMS is essential for *stakeholders* to buy into the long-term view of the DM initiative. A DMS cannot be built in a vacuum. Instead, it must reflect the requirements of the *enterprise* and individual operating units.

Questions

→ Have the business requirements been documented?
→ Have the business requirements been captured from all applicable levels of the organization?

Artifacts

- Documentation of the high-level objectives, requirements and verification
- List of *stakeholders* and evidence of bi-directional communication

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No formal definition of business requirements exists.	No formal definition of business requirements exists, but the need is recognized, and the development is being discussed.	The formal business requirement definition capability is being developed.	The formal business requirement definition capability is defined.	The formal business requirement definition capability is established and used by *stakeholders*.	The formal business requirement definition capability is reviewed and updated at least annually.

1.2.2 Business requirements have been prioritized, approved, and incorporated into the DMS

Description

The prioritized *enterprise* and operating units' business requirements will inform the priorities of the DM initiative.

Objectives

- Secure *stakeholders* review, prioritization and approval of the business requirements incorporated into the DMS.
- Establish regular requirements for review cycles.

Advice

Incorporation of business requirements into the DMS is based on defining the priorities of the high-level organizational objectives. Regulators, auditors and *stakeholders* will want to understand how the organization is addressing the correlation of priorities to both funding and operational realities. Discussions about funding can help define *enterprise* and operating units' priorities. This prioritization and approval *process* will clarify what will and will not be done as part of the DM initiative. It is also important to define and illustrate how the organization will deal with new issues as they arise.

While the DM executive is not usually directly accountable for the advanced *Analytics function* of the organization, they can bring valuable insight to the prioritization of the business requirements. This insight is based on highlighting opportunities where the data and the methods and tools of advanced *analytics* can present a competitive advantage. Tools such as Artificial Intelligence (AI) and Machine Learning (ML) can drive improvements in efficiency, client experience and innovative products and services.

In today's marketplace and regulatory environment, it is equally important that the DM executive bring a lens of the legal and ethical use of data to the *process* of prioritizing the business requirements. This early review will avoid investment in opportunities that ultimately carry ethical risk.

Questions

→ Has the *process* to prioritize and approve high-level business requirements been approved?
→ Do the priorities include links and dependencies?
→ Are the priorities verified and aligned with DM initiative priorities, budget, technology and operations?
→ Is there a *process* for review and prioritization of new requirements?

Artifacts

- Prioritization *process* documentation
- Examples of requirements verification and approval

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No formal DMS exists.	No formal DMS exists, but the need is recognized, and the development is being discussed.	The formal DMS is being developed.	The formal DMS is defined and is aligned with prioritized business requirements.	The formal DMS is established and recognized as aligned to prioritized business requirements.	The formal DMS is reviewed against prioritized business requirements at least annually.

1.2.3 The DM business case is mapped to and aligned with the DMS

Description

The DM business case must align with and reflect the *enterprise* and operating units level objectives, drivers and requirements as detailed in the DMS.

Objectives

- Map the DM Business Case and align it with organizational priorities and objectives.
- Align the DM Business Case with the strategic *enterprise* and operating units' priorities and objectives.
- Include a review of the DM Business Case business requirements to ensure they follow ethical data practices.

Advice

The DM Business Case is the justification for creating and funding a DM initiative. The DM Business Case answers the *why* questions and addresses the *so what* challenges. It articulates the major data and data related issues facing the organization and describes the expected outcomes and benefits that can be achieved through the implementation of a successful DM initiative.

The business case can incorporate defensive objectives related to regulation, risk and compliance. This defensive objective must include an ethical review of data practices. The risk of unethical data practices should be evaluated as part of establishing the business case that the potential of reputational risk may outweigh the business value of the proposed use of data. The practices employed to manage data that must hold up to ethical scrutiny.

It should also address offensive objectives such as business enablement, *analytics* and operational efficiencies. Core objectives must be defined. Implementation approaches must be articulated. The value propositions need to be clearly stated in ways that are meaningful to *stakeholders*.

Questions

→ Does the justification of the business case align with the DMS?
→ Are the objectives defined and verified?
→ Are the value propositions clearly specified and addressed to the *stakeholders*?
→ Is the data in scope to the business requirements and ethical use of the data?

Artifacts

- Business case documentation
- Evidence of alignment between business case, strategy, organizational objectives and priorities
- Evidence of ethical data practices review and related risks

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No formal DMS exists.	No formal DMS exists, but the need is recognized, and the development is being discussed.	The formal DMS is being developed.	The formal DMS is defined and aligned to the DM business case.	The formal DMS is established and recognized as aligned to the DM business case.	The formal DMS and DM business case are reviewed and realigned at least annually.

1.2.4 Expected DM outcomes are defined and sequenced

Description

A primary *function* of the business case is to define the challenges of the current-state and to define the roadmap to improvement.

Objectives

- Define and sequence expected outcomes.
- Define and articulate the current-state to target-state.

Advice

In order to develop the roadmap for current-state to target-state advancement, the dependencies between intermediate steps need to be identified so the right sequence can be defined. Because of the scope of DM, issues need to be prioritized and sequenced. This *process* needs to be formal and transparent to avoid confusion and manage expectations. Strong communication about priorities, sequences, and dependencies is essential.

Questions

→ Have data access and delivery dependencies been defined and verified?
→ Have critical DM concepts been prioritized, sequenced and verified to align with business outcomes?
→ Is there a communication strategy in place to provide visibility and transparency to *stakeholders*?

Artifacts

- Definitions of business outcomes
- Documentation of sequence plans and schedules
- Evidence of *stakeholder* communication through feedback on priorities

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No formal DM business case exists.	No formal DM business case exists, but the need is recognized, and the development is being discussed.	The formal DM business case is being developed.	The formal DM business case is defined and includes high level DM-related business outcomes.	The formal DM business case is established and recognized to include relevant high-level DM-related business outcomes.	The formal DM business is reviewed at least annually and realigned to relevant high-level DM-related business outcomes.

1.2.5 The DM business case is socialized and validated by stakeholders

Description

Buy-in to the business case is affirmed by *stakeholder* validation of the quantifiable business outcomes.

Objectives

- Socialize the DM Business Case to *stakeholders*.
- Review and validate the business objectives.
- Review and approve outcomes, benefits, timelines and target thresholds.

Advice

Stakeholders in *enterprise* and operating units level management must review the business case and validate both objectives and approaches. This *process* needs to be formalized and ongoing, as new priorities are introduced, and existing ones are completed.

Questions

→ Has the business case been socialized to the *stakeholders*?
→ Have the target objectives been reviewed and verified?
→ Have outcomes, benefits, timelines, and thresholds been approved?

Artifacts

- Show mechanism for verification and validation by *stakeholders*
- List of *stakeholders* and evidence of bi-directional communication
- Distribution lists and approvals from *stakeholders*

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No formal DM business case exists.	No formal DM business case exists, but the need is recognized, and the development is being discussed.	The formal DM business case is being developed.	The formal DM business case is defined and validated by directly involved *stakeholders*.	The formal DM business case is established and is recognized and used by *stakeholders*. *Stakeholders* understand the need for data management.	The formal DM business case is reviewed at least annually.

1.3 The Data Management Vision is Defined

Defining the data management vision includes: 1) a data content strategy identifying what data is required; 2) a data usage strategy for how the data will be used; and 3) a DM deployment strategy that aligns the business objectives to a prioritized execution road-map.

1.3.1 The Data Content Strategy is defined

Description

The data content strategy identifies the sources and types of data that are required to meet the prioritized objectives of the business. Sources include both internal and external data.

Objectives

- Define the data that is required to achieve the prioritized business objectives.
- Maintain a backlog of lower priority business objectives and required data for future consideration.

Advice

The data content strategy answers the question of what data is required to support the objectives of the business. The data in scope cannot exceed the capacity of the DM initiative responsible for managing the data. This becomes an exercise of right sizing the priorities of the business to the resources that are provided to the DM initiative.

The types of data include reference, transactional, derived and alternative. Types of alternative data may include geospatial, satellite, mobile, sensor, weather, and social media. All data must be viewed through the lens of ethical access and use.

Questions

→ Have you defined the data required to meet the prioritized business objectives?
→ Have you evaluated the potential for externally sourcing the required data?
→ Have you captured a backlog of business objectives and required data for future consideration?

Artifacts

- Data content strategy included in the DM initiative strategy document
- Definition of the in-scope data aligned to business objectives
- Evidence of review of external data sources

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No formal data content strategy exists.	No formal data content strategy exists, but the need is recognized, and the development is being discussed.	The formal data content strategy is being developed.	The formal data content strategy is defined and validated by directly involved *stakeholders*.	The formal data content strategy is established and understood across the organization and is being followed by the *stakeholders*.	The formal data content strategy is established as part of business-as-usual practice with a continuous improvement routine.

1.3.2 The Data Usage Strategy is identified

Description

The data usage strategy defines how the organization intends to use data. The strategy includes the vision for the adoption and use of advanced *analytics* methods and tools like Machine Learning (ML), Artificial Intelligence (AI) and cognitive thinking. The strategy must identify the appropriate use of innovation within the constraints of achieving ethical data outcomes.

Objectives

- Develop a data usage strategy that addresses the vision for the use of advanced *analytics* methods and tools.
- Ensure that the data usage strategy addresses the vision of the organization in the innovative use and maintenance of data.
- Create a data usage strategy that includes a review of the ethical use and outcome of the data being used.

Advice

The data usage strategy is an extension of the data content strategy. The challenge is to combine the basic requirements for data from the business objectives to explore the organization's vision for using advanced *analytics* methods and tools. Advanced *analytics* can lead to the discovery of opportunities for innovation or efficiencies not previously possible. However, applying the automation and efficiency from these innovative tools to the analysis of data exposes an organization to increased risk from the misuse of data. The data access and use are not simply a legal question but also an ethical question.

The DM executive should provide insight to how these evolving methods and tools can be used to advance the objectives of the business. As part of evaluating these new capabilities a review of the ethical use and–the ethical outcome of the use–of the data is required.

Questions

→ Have the application and value of using advanced *analytics* been evaluated by the organization?
→ Have the methods and tools of advanced *analytics* been evaluated for supporting the *process* execution of the DM initiative?
→ Is it ethical to use the defined data?
→ Is the way the data is being used producing an ethical outcome?

Artifacts

- Data usage strategy included in the DM initiative strategy document
- Evidence of the ethical review of the in-scope data

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No formal data usage strategy exists.	No formal data usage strategy exists, but the need is recognized, and the development is being discussed.	The formal data usage strategy is being developed.	The formal data usage strategy is defined and validated by directly involved *stakeholders*.	The formal data usage strategy is established and understood across the organization and is being followed by the *stakeholders*.	The formal data usage strategy is established as part of business-as-usual practice with a continuous improvement routine.

1.3.3 The Data Management Deployment Strategy is communicated

Description

The Data Management Deployment Strategy defines the prioritized business objectives and their requirements for DM. The strategy includes a vision of the DM initiative target-state, current-state gap analysis, prioritized roadmap and execution resource requirements.

Objectives

- Define the business requirements for data and DM capability.
- Evaluate the DM initiative current-state against the target-state with gap analysis and a gap closure roadmap.
- Define the required execution resources.
- Define the business case for the DM initiative.

Advice

The DM deployment strategy answers the question of what DM is going to be applied to the in-scope data defined in the data content strategy. A solid business case is a critical part of the deployment strategy to validate the value delivered from the investment in the DM initiative.

Questions

→ Is the target-state vision of the DM initiative defined?
→ Has an assessment of the current-state been completed?
→ Have current-state-to-target-state gaps been quantified and prioritized?
→ Is there a defined current-state gap closure roadmap in place?

Artifacts

- Data management strategy included in the DM initiative strategy document
- Current-state gap closure roadmap
- Funding and resource analysis
- Presentation of the business case

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No formal data management deployment Strategy exists.	No formal data management deployment strategy exists, but the need is recognized, and the development is being discussed.	The formal data management deployment strategy is being developed.	The formal data management deployment strategy is defined and validated by directly involved *stakeholders*.	The formal data management deployment strategy is established and understood across the organization and is being followed by the *stakeholders*.	The formal data management deployment strategy is established as part of business-as-usual practice with a continuous improvement routine.

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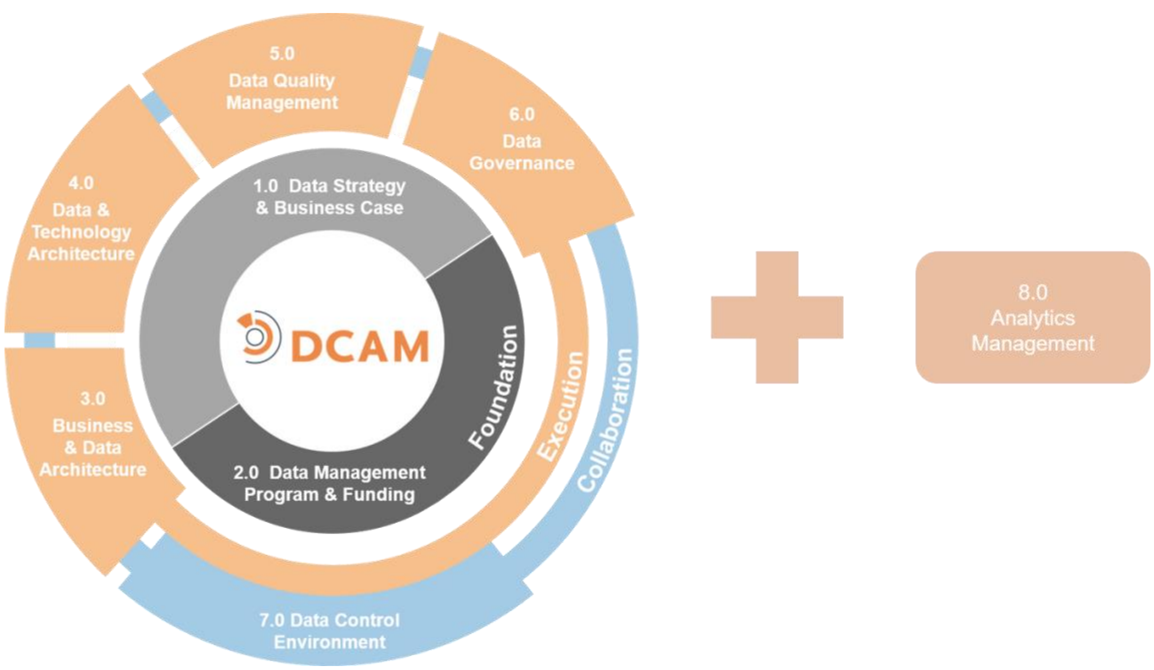
DCAM® Framework

2.0 Data Management Program & Funding Model



2.0 Data Management Program & Funding Model

Introduction

The **Data Management Program** is an organizational *function* dedicated to the management of data as an asset throughout an organization. It illustrates how the management of *data quality* (DQ), its definition and its content support strategic, business and operational objectives. It also reinforces the necessity of orchestration, active collaboration and alignment among diverse *stakeholders* in order to instill confidence in data as a trusted factor of input into business and operational *processes*.

The purpose of a data management (DM) program is to organize and embed the DM concepts into the operational framework of an organization on a sustainable basis. The creation and implementation of the DM program elevates the importance of DM and integrates it as a core aspect of organizational operations. It establishes DM as a sustainable activity by ensuring sustainable funding. It reinforces the importance of managing data across the organization via education, training, and communication.

The **Data Management Funding Model** describes the overall framework and high-level engagement of senior management used to ensure that the objectives and *processes* of DM become a sustainably funded activity within the organization.

Definition

The **Data Management Program (DMP) & Funding Model** component is a set of capabilities to manage the Office of Data Management (ODM). These organizational structures include resource requirements, and full range of Program Management Office (PMO) activity such as execution of program management, *stakeholder* management, funding management, communications, training, performance measurement. The DM funding model within the DMP is designed to provide the mechanism to ensure the allocation of sufficient capital needed for implementation of the Program. It also defines and describes the methodologies used to measure both the costs and the organization-wide benefits derived from the DM initiative.

Scope

- Establish a DMP *function* to implement the Program Management Office (PMO) capabilities within the ODM.
- Facilitate the design and implementation of sustainable business-as-usual DM *processes* and tools across the components and their capabilities.
- Establish roles and responsibilities related to the DM capabilities aligned with an organizational structure and execute in an ODM.
- Define Funding Model, secure and monitor funding, and institute cost and benefits tracking aligned to the Business Case.
- Establish the DM execution roadmap with supporting project plans to build upon the high-level Data Management Strategy (DMS) roadmap.
- Engage each *stakeholder* across the *data ecosystem* as appropriate to their roles in resource alignment, funding, communications, training and skill development.
- Manage the DM initiative by monitoring and socializing DM performance metrics.
- Ensure that the DMP governance is integrated into Data Governance (DG).

Value Proposition

Organizations that ensure the right people are involved in the DM initiative demonstrate the ability to eliminate the replication and misuse of data and improve their ability to integrate data based on *enterprise standards*.

Organizations that define and follow set *processes* and *standard* operating *procedures*, including requesting, sharing, defining, producing, and using data, demonstrate the ability to ensure the sharing of data aligns with *enterprise standards*.

A DM Funding Model with an appropriate, sustainable funding framework is required to ensure the success of the objectives and *processes* of the DM initiative.

Organizations that effectively implement the DMP and achieve an accountable funding model get a return on investment from several areas:

- Establishes an organization-wide data culture
- Efficiencies in operations with repeatable and sustainable *processes*
- Productivity from resource and skill alignment
- Continuity of funding over time
- Sustainability for the DM initiative

Overview

The DMP is one of the two foundational components of the DCAM Framework. It should be established as a formal, independent and sustainable part of the organization, within the ODM. The DMP needs to establish the lines of responsibility and accountability within the ODM organizational structure and beyond into federated organizational structures (e.g. operating units, regions, etc.). The DMP must ensure the ODM has access to the appropriate staff resources and functional capabilities in order to deliver the data needed to support organizational objectives. An effective DMP has the strong support of executive management, appropriate governance authority to ensure the implementation of a data control environment and a well-structured model of how *stakeholders* will engage in data-related issues. An effectively designed DMP, that is flexible enough to accommodate to changing circumstances, will help embed the importance of DM into the culture of the organization. A key goal is to instill a sense of collective respect for the role of data among all *stakeholders*.

It is important to note that the DMP is an organization, or *function*, within the ODM in which typical Program/*Project Management Office* capabilities and skills reside. The DMP *function* organizes DM activity throughout the organization. (Refer to *Data Management Operating Levels* in the introductory section of this guide for detail on the levels and context at which DM operates.) The DMP *function* will operate in a similar way to a typical program or project, but the DMP must be an ongoing, sustainable business-as-usual organization-wide commitment that organizes DM for the long term.

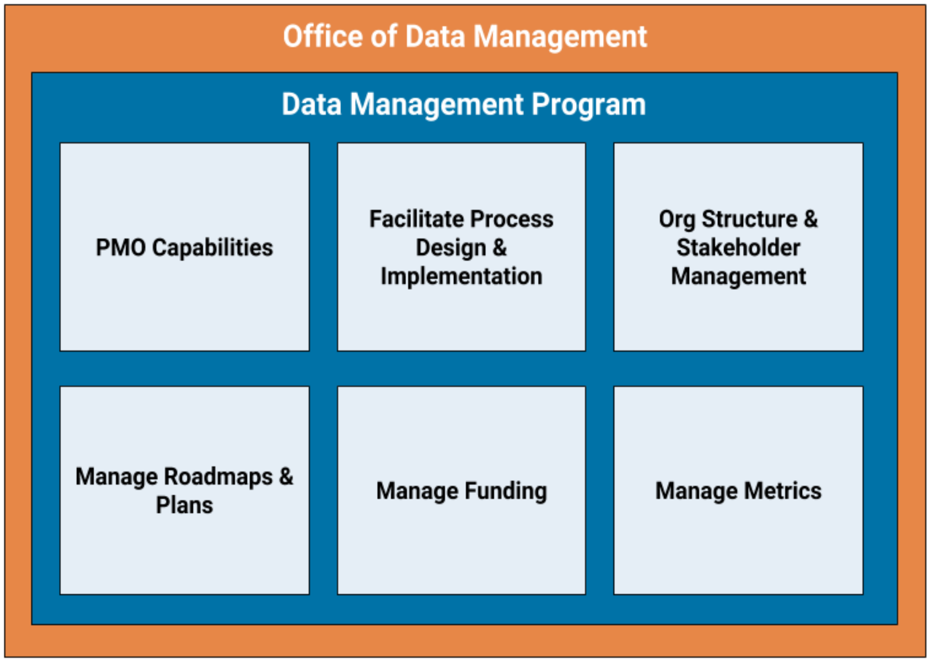


Diagram 2.1: DMP in ODM

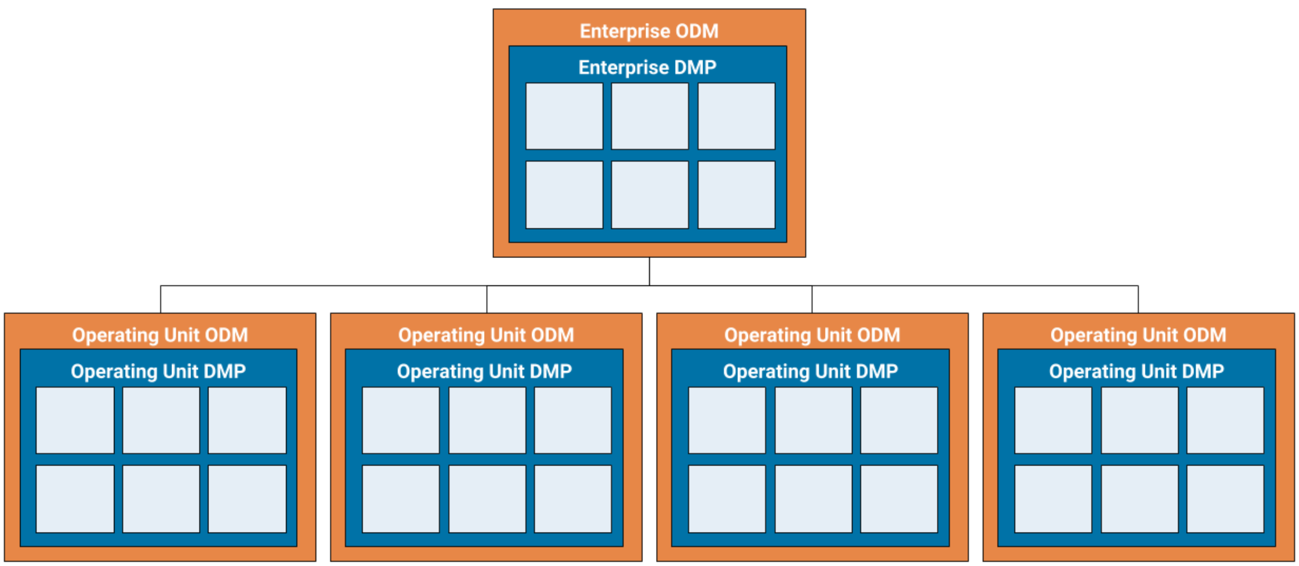


Diagram 2.2: Enterprise DMP with Operating Unit DMPs

The DM funding model defines the mechanism used to generate and maintain the capital needed for the DM initiative throughout its lifecycle. It establishes the methodology used for cost allocation among business lines and can be used to help align *stakeholders* on funding-related issues. In mature organizations, the funding model reflects the individual requirements of the various DM initiative components of the organization and is integrated with governance to ensure that appropriate oversight and accountability is applied to DM. Verifiable metrics are essential and must be aligned with tangible business objectives. A well-structured funding model can help avoid recurring debates over business priorities, mitigate internal competition and facilitate open discussions among *stakeholders*.

Strong consideration should be given to allocating initial funding to the DM initiative as an *enterprise* level expenditure rather than an individual organizational unit approach. This type of grassroots funding can become mired in competition among organizational units, is often aligned with a tactical view of DM and frequently reinforces short-term evaluation cycles. An organization can expect its funding model to evolve along with the maturity of its DM initiative.

There is no single model for funding DM. The specific model implemented will depend on the dynamics and operational culture of the individual firm. Some organizations will fund centrally, others will fund through the organizational units, while still others may take a hybrid approach. There are pros and cons to all of these approaches. Whichever is selected, the fundamental aspects of the funding model, such as investment criteria/priorities, budget management, delivered-versus-expected benefits, allocation methodology and capital needed for ongoing management of the initiative should always be included. Most importantly, the funding model must reflect a multi-year journey, incorporating both initial implementation costs, as well as sustainable ongoing funding. DM must become a day-to-day operation and must be funded accordingly to ensure it becomes part of the fabric of an organization's operation.

Core Questions

- → Is the DMP aligned with the DM strategy and business objectives?
- → Does the DMP have the right mixture of skills, resources, and capabilities to effectively implement and govern the DM *function*?
- → Does the DMP have the appropriate support from executive management?
- → Is the Funding Model sufficient to sustainably support the DM program?
- → Have the funding requirements been translated from the business case and aligned with the objectives, sequence priorities and high-level implementation roadmap of the DM strategy?
- → Does the funding model cover all aspects of DM including tangible, intangible, special requests, urgent/crisis requirements, unique applications?
- → Is there a defined *process* with established criteria for determining and verifying the investment required for DM, and is it aligned with the business structure, priorities and governance *process* organization?

Core Artifacts

The following are the **core artifacts** required to execute an effective Data Management Program & Funding Model capability. Items with an '*' link to published best practice *guidelines*.

- Communications Plan
- Data Management Metrics & Dashboards
- Data Management Organizational Structure*
- Funding Model
- *Stakeholder* Engagement Plan
- Training Plan

2.1 The Data Management Program (DMP) is Established

The DMP provides the required program and project management for the DM initiative. The strategy and approach must be defined and approved by stakeholders. Roles and responsibilities across the stakeholders must be established with operational processes in place.

2.1.1 The DMP strategy and approach are defined and adopted

Description

The strategy and approach must be defined for the DMP. Once established, it must be formally empowered by senior management and its role communicated to all *stakeholders*.

Objectives

- Formally establish the DMP strategy within the organization.
- Obtain executive management support for the DMP strategy.
- Communicate the role of the DMP *function* across the organization through formal, organizational channels.
- Operate the DMP *function* collaboratively with DM initiative *stakeholders*.
- Secure authority to enforce DMP compliance through *policy* and documented *procedure*.

Advice

The ODM, and the DMP within it should be established as an independent entity. Formalization is essential. Be careful about embedding the DM initiative in technology *function*. The creation of a new control *function* needs a clear announcement from executive management to minimize inevitable disruption. Support needs to be broad-based.

Creating the DM initiative without empowerment is useless. As a change *function*, the DMP needs authority to enforce behavioral change. The authority granted must be formal. Support from Internal Audit is a very useful way to ensure compliance with *policy* and *standards*.

Questions

→ Has the DMP *function* been formally established?
→ Is there a DMP strategy and approach in place?
→ Has the ODM been formally communicated to business, technology, operations, finance and risk *stakeholders*?
→ How has executive management demonstrated its support?
→ Has authority been granted to the DMP *function* to implement and enforce best practice via *policy* and *standards*?
→ Has authority been communicated to *stakeholders*?
→ Is there a functional partnership in place with Internal Audit?

Artifacts

- The DMP plan
- ODM charter including strategy and approach
- Description of roles and responsibilities of the DMP *function*
- Communication of specific support from executive management with distribution lists

- *Policies* and *procedures* associated with executing and enforcing DMP
- Bi-directional engagement with *stakeholders* on the DMP *function* authority

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No formal DMP exists.	No formal DMP exists, but the need is recognized, and the development is being discussed.	The DMP is being developed.	The DMP is defined and has been validated by the directly involved *stakeholders*.	The DMP is established and is recognized and used by *stakeholders*.	The DMP is established as part of business-as-usual practice with a continuous improvement routine.

2.1.2 The DMP PMO is established and roles and responsibilities are defined and implemented

Description

The DM initiative will require the coordination of many projects across the organization or division. Resources may be shared. It is important that a PMO is established and appropriately staffed with adequate resources to manage the required workload of the DM initiative. The authority and responsibility of the PMO must be defined and communicated to all *stakeholders*.

Objectives

- Approve and charter the PMO.
- Define and communicate the roles and responsibilities of the DMP *function*.
- Fund and staff the PMO.
- Ensure and enforce alignment of activities and projects to *policy* and *standards* through the authority of the PMO *function*.
- Hold individuals accountable for the PMO performance via annual reviews and compensation considerations.

Advice

DM is no different from any other organizational *function*. It requires coordination. Program coordination must be formalized, appropriately staffed, funded and empowered to ensure alignment among the *stakeholders* and adherence to program deliverables. Without the *function* of the PMO, DM is just another good idea that doesn't get properly implemented on time and within budget. Management of the details associated with the implementation of the DM initiative is one of the real measures of implementation success.

Questions

→ Has the PMO been established?
→ Is the PMO appropriately staffed and funded?
→ Does the PMO have the authority needed to be effective?
→ Have the roles and responsibilities of the PMO been defined, documented and socialized?
→ Have milestones and metrics associated with program delivery been established?

Artifacts

- Evidence of PMO formation such as its charter and approvals
- Evidence of *stakeholder* identification
- *RACI matrix* or other evidence of accountability assignment
- Description of the roles and responsibilities of the PMO
- Staff qualifications and assignments
- Evidence of accountability linked to performance reviews and compensation
- Gap analysis of skills needed and in place
- List of *stakeholders* and evidence of bi-directional communication

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No formal DMP PMO exists.	No formal DMP PMO exists, but the need is recognized, and the development is being discussed.	The DMP PMO is being developed.	The DMP PMO is defined and has been validated by the directly involved *stakeholders*.	The DMP PMO is established and is recognized and used by *stakeholders*.	The DMP PMO is established as part of business-as-usual practice with a continuous improvement routine.

2.1.3 The DMP processes are defined and operational

Description

Formal *processes* must be established for the activities of the DMP. These *processes* align with the DM *policy* and *standards* of the organization and include *procedures*, tools and routines. The routines are required for steady-state operations. Routines include but are not limited to regular *stakeholder* meetings, planning sessions, status reporting, etc.

Objectives

- Establish formal DMP *processes* in alignment with the DM *policy* and *standards*.
- Integrate the DMP *processes* into the overall end-to-end *processes* of the DM initiative.
- Identify, schedule and maintain DMP routines, meetings and working sessions required for operational support.

Advice

Plans and presentations are important, but unless there is evidence of activity being done on a routine basis, the likelihood of a sustained DM initiative is at risk. Routines in the form of standing meetings with high repeatable attendance, planning sessions and regular communications help ensure that DM objectives are taking place.

Questions

→ Have formal *processes* been defined and implemented?
→ Are the *procedures*, tools and routines in place for implementing the *processes*?
→ Are DMP activities part of the normal operational routine of *stakeholders*?
→ Are there standing meetings, planning sessions and regular communications about DMP initiatives?

Artifacts

- *Process* design artifacts, *procedure* guides and published routines
- *Process* performance metrics reports
- Meeting minutes, status reports and DMP announcements

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No formal DMP operational *processes* exist.	No formal DMP operational *processes* exist, but the need is recognized, and the development is being discussed.	The DMP operational *processes* are being developed.	The DMP operational *processes* are defined and have been validated by the directly involved *stakeholders*.	The DMP operational *processes* are established and are recognized and used by *stakeholders*.	The DMP operational *processes* are established as part of business-as-usual practice with a continuous improvement routine.

2.1.4 The DMP has the authority to enforce adherence and compliance

Description

The DMP must be formally empowered by senior management and its role communicated to all *stakeholders*.

Objectives

- Operate the DMP collaboratively with DM initiative *stakeholders*.
- Endow the DMP with the authority to enforce adherence and compliance through *policy* and documented *procedure*.

Advice

Creating the DM initiative without empowerment is useless. As a change function, the DMP needs authority to enforce behavioral change. The authority granted must be formal. Support from Internal Audit is very useful to ensure compliance with *policy* and *standards*.

Questions

→ Has the DMP been established as mandatory?
→ Has authority been granted to implement and enforce best practice via *policy* and *standards*?
→ Has authority been communicated?
→ Is there a functional partnership in place with Internal Audit?

Artifacts

- Communication from executive management with distribution lists
- *Policies* and *procedures* associated with making DM mandatory
- List of *stakeholders* and evidence of bi-directional communication

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No formal DMP exists.	No formal DMP exists, but the need is recognized, and the development is being discussed.	The DMP is being developed.	The DMP is defined and includes an explanation of how it is enforced, and compliance is achieved.	The DMP is established and recognized as enforceable.	The DMP is established as part of business-as-usual practice with a continuous improvement routine.

2.1.5 The DMP concepts are reflected in the DMS

Description

The DMS must include concepts from each of the Framework Components integrated into a coherent strategy. The DMP *function* will incorporate the strategic concepts into a DMP strategy with required detail to support execution.

Objectives

- DMS & Business Case concepts are reflected in the DM Program
- Align *stakeholder* plans and roadmaps with the DMP.
- Obtain approval from *stakeholders* of the DMP.

Advice

The integration of DMP concepts into the DMS is achieved by an agreement between the Operating Level *Data Officer* and the individual responsible for delivering the DMP. The Operating Level *Data Officer* is accountable for establishing priorities across each of the Framework Component requirements. These priorities are in alignment with the roadmap and are essential to achieving the *target operating model*. Inclusion of the DMP concepts in the DMS is critical to the success of the DM initiative.

Questions

→ Does the DMS contain concepts related to the DMP function?
→ Is the DMP strategy and roadmap aligned to the DMS?

Artifacts

- DMS document; evidence of DMP strategic concepts
- DMP strategy and roadmap documents; evidence of DMS strategic objectives

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No DMP concepts exist.	No DMP concepts exist, but the need is recognized, and the development is being discussed.	The DMP concepts are being developed.	The DMP concepts have been defined and validated by directly involved *stakeholders*.	The DMP concepts are established and have been reflected in the DMS (either directly or via an addendum).	The DMP concepts are established as part of business-as-usual practice with a continuous improvement routine.

2.2 The DM Funding Model has been Established, Approved, and Adopted by the Organization

The DM funding model must include resources required for stakeholders to execute the DM initiative. The model must align with the organization-wide funding process and be enforced through the DM governance process.

2.2.1 The DM funding model is matched to business requirements, implementation timelines and operational capabilities

Description

The DM funding model must be in alignment with all resources that are required for the execution of the DMS.

Objectives

- Define and socialize the DM funding model with *stakeholders*.
- Collect and incorporate feedback and into the model.
- Secure funding for key business-driven data requirements.
- Align funding levels to business requirements.
- Ensure funding levels and timing support the appropriate delivery dates of key DM initiatives.
- Obtain approval of funding levels that will sustain data operations.
- Have *stakeholders* review and approve funding commitments.
- Allocate funding to the DMP roadmaps and workstream.

Advice

The funding model is the reality check of the DM initiative. The goal is to ensure that the resources needed to deliver against objectives are available – and to ensure that the business requirements can be satisfied. It is important that all *stakeholders* evaluate the proposed funding and that feedback is captured.

Questions

→ Can the technology infrastructure deliver against requirements?
→ Can the operations function sustain and support the objectives of the DM initiative?
→ Is the funding model appropriate for the initiative?
→ Has the funding model been socialized and approved?

Artifacts

- Alignment of the budget with business requirements and delivery schedules
- Alignment of DM goals with technology and operational capability
- Funding plans and budget allocation
- List of *stakeholders* and evidence of bi-directional communication

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No DM funding model exists.	No DM funding model exists, but the need is recognized, and the development is being discussed.	The DM funding model is being developed.	The DM funding model is defined and has been validated by the directly involved *stakeholders*. The DM funding model is matched to business requirements, implementation timelines and operational capabilities.	The DM funding model is established and operational.	The DM funding model is established as part of business-as-usual practice with a continuous improvement routine. The DM funding model is reviewed for alignment with business requirements, implementation timelines and operational capabilities at least annually, and adjusted accordingly.

2.2.2 The DM funding model is aligned with the funding processes of the organization

Description

The resource funding *process* of the DM initiative must align with the overall funding *process* of the organization.

Objectives

- Ensure that DM funding addresses the current year budget cycle.
- Map the DM funding to a multi-year implementation plan.
- Have the DM PMO review, reconcile and approve resource plans.
- Integrate DM funding as a sustainable corporate *function*.

Advice

The goal is to ensure that the funding model for DM is synchronized with the overall funding approach of the organization (i.e., budget *processes*, cycles, escalations, approvals). Successful DM initiatives leverage existing mechanisms because they are already established and enforceable.

Questions

→ Is the funding *process* being debated with *stakeholders* at the appropriate level of authority?
→ Is the funding model for the current year or does it span multiple years?

Artifacts

- Alignment with budget *processes* and organizational cycles
- A *process* to review, modify, and validate resource plans
- Mapping to current and multi-year implementation plans

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No DM funding model exists.	No DM funding model exists, but the need is recognized, and the development is being discussed.	The DM funding model is being developed.	The DM funding model is being defined and has been validated by the directly involved *stakeholders*. The DM funding model is aligned to the overall funding *process* of the organization.	The DM funding model is established and operational.	The DM funding model is established as part of business-as-usual practice with a continuous improvement routine. The DM funding model is reviewed for alignment with the overall funding *process* of the organization at least annually and adjusted accordingly.

2.2.3 Implementation of the DM funding model is enforced

Description

Enforceability of the DM funding model comes through alignment to the organization's funding *process* and the use of the DM governance structure for approvals.

Objectives

- Obtain *enterprise* and/or operating units' approval and allocate funding.
- Have the DM PMO review all budgets.
- Empower the DM organization to enforce the *enterprise* and operating units funding in accordance with its objectives.

Advice

Funding for the DM initiative is required. Enforcement must come from top-of-the-house. Funding can come from a centralized seed funding approach. It can come from *enterprise* and/or operating units. Regardless–there needs to be evidence of sustainable financial support.

Questions

→ Is the funding enforcement approach documented and verified?
→ Are funding sources identified and confirmed?
→ How will the ODM handle budget reductions or other funding challenges?
→ What is the *process* for prioritizing both discretionary and non-discretionary funding decisions?

Artifacts

- Documented enforcement mechanism
- Communication with *stakeholders* about funding
- Escalation *procedures* to resolve budget shortfalls

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No DM funding model exists.	No DM funding model exists, but the need is recognized, and the development is being discussed.	The DM funding model is being developed.	The DM funding model is defined and is recognized as enforceable by *stakeholders*.	The DM funding model is operational and enforced. DM is best practice is not optional.	The DM funding model is established as part of business-as-usual practice with a continuous improvement routine. The DM funding model is reviewed for enforcement at least annually and adjusted accordingly.

2.3 The Data Management Organizational Structure is Created and Implemented

The ODM includes the organizational team responsible for promoting the DM initiative at the various operating levels of the organization. A senior executive data officer at each organization operating level must be appointed, given authority and allowed to appropriately staff the ODM.

2.3.1 The Office of Data Management (ODM) is created

Description

The ODM includes the organizational team responsible for promoting the DM initiative at the various operating levels of the organization.

Objectives

- Design and plan the ODM.
- Charter and approve the ODM.
- Create the ODM.

Advice

The ODM formalizes and runs the DMP. The ODM needs a visible and strong commitment from executive management. A formal office is necessary to create *policy*, implement *standards*, coordinate governance, and manage organizational collaboration across control *functions*.

Questions

- → Is there a formal and sanctioned ODM?
- → Is it recognized as part of the official corporate structure?
- → Does the ODM have the authority it needs to implement change?
- → Does it have a clear mission and charter?
- → Does the ODM have strong and visible executive support?
- → Does the ODM have sufficient funding and the skill sets needed to accomplish the DM objective?

Artifacts

- DM charter and approvals
- Specific and identifiable organizational structure
- Formal communication from executive management such as any notifications to *stakeholders*
- List of *stakeholders* and evidence of bi-directional communication

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No formal ODM exists.	No formal ODM exists, but the need is recognized, and the development is being discussed.	The ODM is being developed.	The ODM structure is defined and has been validated by the directly involved *stakeholders*.	The ODM is established and is recognized and used by *stakeholders*.	The ODM is established as part of business-as-usual practice with a continuous improvement routine.

2.3.2 The ODM has an executive owner

Description

A senior executive *data officer*, for example a *Chief Data Officer* or an operating unit *data officer* must be appointed and be given full authority to run the ODM. The role and scope of responsibility for this position must be clearly defined and communicated to the organization.

Objectives

- Establish an understanding of the need for an authoritative data executive.
- Define the ownership role and responsibility of the data executive.
- Hire the executive owner or appoint one from within.
- Communicate the duties and authority of the executive owner to *stakeholders*.

Advice

It is essential that a single, high-level executive be responsible for the DM initiative. A committee cannot run the DM initiative. To ensure that the DM initiative is sustainable, a senior executive with authority must be appointed and must have broad support from other executives. The executive in charge needs to be the visible advocate for DM, communicating with vision and passion. This executive is the chief diplomat for collaboration as well as the person who runs the DM initiative. Simply appointing the executive is not sufficient – the role and authority necessary to implement a change of this magnitude must be communicated to all *stakeholders*.

Questions

→ Has the remit of the *data officer* been defined, socialized and documented?
→ Has a senior executive been hired to run the DMP?
→ Has the executive been empowered with the authority necessary to implement the DM initiative?
→ Have the lines of authority for the *data officer* been defined and established?
→ Has the role of the DMP and the *data officer* been sanctioned and communicated to *stakeholders*?

Artifacts

- *Data officer* job definition including skills and expectations
- Named individual performing the DM executive *function*
- Executive management communication to *stakeholders*
- List of *stakeholders* for communication about *data officer*

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
The ODM has no executive owner.	The ODM has no executive owner, but the need is recognized, and ownership is being discussed.	The ODM executive ownership is being developed.	The ODM has an executive owner who has been validated by the directly involved *stakeholders*.	The ODM executive owner is established and recognized by *stakeholders*.	The ODM executive owner is established and recognized as driving continuous improvement.

2.3.3 The ODM is funded and staffed by individuals with the required skill-sets

Description

The ODM must be appropriately funded and staffed with the required DM skill-sets

Objectives

- Get approval for funding the ODM.
- Gain approval to hire skilled personnel.

Advice

Be wary of DMPs that are approved but not given the authority to hire or acquire essential operational talent. It is not necessary for all the tasks associated with DM to be the sole tasks of the personnel responsible for them. Some staff may be shared with other business units where they have unrelated *functions* and responsibilities.

Whether it exists as a stand-alone group or whether many of the operational *functions* are embedded in the business is dependent on the strategy and culture of the organization. Regardless, the DMP needs dedicated resources with appropriate skill sets and at least a small, dedicated, central coordination *function*. Finding the right people requires ramp-up time and contingency plans. Capabilities to manage inevitable turnover should be designed to avoid single points of failure and prevent operational bottlenecks are essential.

The scope of the DM initiative continues to evolve. As new issues, opportunities or regulation emerge be mindful of need for additional skills across the *stakeholder* teams. An example of this would be the skills needed to support the advanced *analytics* tools and methodologies and the inherent challenges of integrating data ethics into the governance and operations of the data.

Questions

→ Is the *operating model* for the ODM established?
→ Are the resources needed to support the DM initiative defined and acquired?
→ Does the ODM have the authority to hire, or approval to acquire, the skill sets needed for implementation?
→ Are all the required skills represented across the *stakeholder* teams?
→ Has ramp-up time for staff onboarding and funding commitment been anticipated?

Artifacts

- *Operating model* with a resource plan for the ODM
- Job descriptions for the defined organizational structure
- Gap analysis of skills needed
- List of required skills
- Confirmation of approved budget and authorization to hire

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No formal funding or requisite skilled staffing of the ODM exists.	No formal funding or requisite skilled staffing of the ODM exists, but the need is recognized, and funding/staffing is being discussed.	The formal funding and requisite skilled staffing of the ODM is being developed.	The funding and staffing requirements of the ODM are in place and has been validated by the directly involved *stakeholders*. Actual hiring and placement of individuals within roles may (still) be in progress.	The funding is established and staffing of the ODM is complete. The critical threshold of requisite skilled staffing has been attained.	The funding and staffing of the ODM is established as part of business-as-usual practice with a continuous improvement routine.

2.4 The Roadmaps for the DMP are Developed, Socialized, and Approved

Program roadmaps must describe the steps required to attain the DM initiative's target-state for its organizational structure and function. Stakeholders must be engaged and validate the roadmaps. Once roadmaps are approved they must be translated into detailed project plans.

2.4.1 Program roadmaps are defined, developed, and aligned with the DMS

Description

Program roadmaps must describe the steps required to attain the DM initiative's target-state for its organizational structure and *function*. Roadmap topics include but are not limited to, governance structure; content management strategy; infrastructure design and data architecture.

Objectives

- Develop DMP roadmaps.
- Align DMP roadmaps with all aspects of the DMS.

Advice

Strategies should include high-level roadmaps. Those roadmaps should be supported by detailed project plans that may include roadmaps for each project plan. Therefore, roadmaps may exist at multiple levels within the overall planning structure.

Defined and detailed roadmaps are needed to establish and communicate the required actions to reach the DM initiative's target-state across all the DM framework components. Roadmaps must be consistent with the strategy, as they illustrate the path for implementation. These plans don't have to be fully fleshed out, but they do need a clear, real-world definition of what will be done and when. Short-term roadmaps (i.e., 30/60/90-day) do need to be comprehensive. More flexibility is acceptable for long-term plans. Questions should be raised about scope, practicality and achievability of each plan. Identify what dependencies exist within or between plans. Dependencies add risk.

Questions

→ Have clearly defined program roadmaps been developed?
→ Are roadmaps and plans tangible (i.e., can they be measured)?
→ Have the dependencies been defined, documented and verified?
→ Are any/all dependencies included in respective budgets?

Artifacts

- Program roadmaps, including evidence on how they align to data the management strategy
- Maps of dependencies associated with implementation
- Outcomes and projected deliverables
- Budget alignment with strategy, roadmaps and dependencies

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No formal DMP roadmaps exist.	No formal DMP roadmaps exists, but the need is recognized, and the development is being discussed.	Formal DMP roadmaps are being developed and aligned to the DMS.	Formal DMP roadmaps are defined and aligned to the DMS.	Formal DMP roadmaps are established and recognized as aligned to the DMS by directly involved *stakeholders*.	Formal DMP roadmaps are established as part of business-as-usual practice with a continuous improvement routine. DMP roadmaps are realigned to the DMS at least annually.

2.4.2 Program roadmaps are socialized and agreed to by *stakeholders*

Description

It is essential that roadmaps be shared with *stakeholders*. Working with *stakeholders* during the development phases invites collaborative feedback and buy-in.

Objectives

- Align DMP roadmaps with the DM *stakeholder* roadmaps.
- Verify and approve DMP roadmap alignment with *stakeholders*.

Advice

Sharing the program plans with *stakeholders* helps ensure support. This will require discussion and likely a modification of plans. The back-and-forth collaboration is essential if you want *stakeholders* to own the outcomes, deliverables and commitments.

Questions

→ Have the roadmaps been shared with key *stakeholders*?
→ Has feedback, including suggestions and concerns, been captured and addressed?
→ Have final, agreed to roadmaps been developed?

Artifacts

- Alignment of roadmaps with DM strategy
- List of *stakeholders* and evidence of bi-directional communication with organizational units
- Verification and approval of roadmaps

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No formal DMP roadmaps exist.	No formal DMP roadmaps exists, but the need is recognized, and the development is being discussed.	Formal DMP roadmaps are being developed, in collaboration with directly involved *stakeholders*.	Formal DMP roadmaps are defined and validated by the directly involved *stakeholders*.	Formal DMP roadmaps are established and being followed by directly involved *stakeholders*.	Formal DMP roadmaps are established as part of business-as-usual practice with a continuous improvement routine. DMP roadmap continuous improvement is in collaboration with directly involved *stakeholders*.

2.4.3 Project plans are developed detailing deliverables, timelines, and milestones

Description

Once roadmaps are agreed to and approved, they must be translated into tangible mechanisms for delivery. The DMP office is responsible for the creation, coordination and management of the DM project plans.

Objectives

- Develop project plans.
- Align project plans to the programs' implementation roadmaps.
- Routine program review *procedures* are in place to track the progress of development plans.

Advice

Program roadmaps need to be translated into detailed project plans. The management of these project plans should be centralized via the established PMO to ensure adherence and delivery. The project plans need to contain practical deliverables and reflect the priorities that were negotiated with *stakeholders*. They must be in alignment with approved budgets.

Questions

→ Do practical project plans exist?
→ Are they aligned with program roadmaps and budgets?
→ Is there a centralized mechanism (PMO) in place to oversee implementation?
→ Are routine project review *procedures* in place to track progress?

Artifacts

- Evidence of completed project plans with defined deliverables
- Timeframes and milestones in line with implementation roadmaps
- Evidence of a centralized PMO
- Program review *procedures* to track progress

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No DMP project plans exist.	No DMP project plans exist, but the need is recognized, and the development is being discussed.	DMP project plans are being developed.	DMP project plans are defined and have been validated by the directly involved *stakeholders*.	DMP project plans are established and being followed by directly involved *stakeholders*.	DMP project plans are established as part of business-as-usual practice with a continuous improvement routine.

2.5 Data Management Process Excellence Program is Established

The success of the DM initiative requires organization-wide standard DM processes that are repeatable, sustainable and measurable. The organization should leverage existing industry standards and best practice. The use of the standard processes must be required by policy.

2.5.1 DM process standards are defined and implemented organization-wide

Description

The success of the DM initiative depends on repeatable execution of *standard*, sustainable and measurable DM *processes*. The *enterprise* DMP, working with the operating units DMP, is responsible for developing *standard* DM *processes* across the organization.

Objectives

- Create a centralized capability to achieve *process* excellence and continuous improvement within the ODM.
- Build a mechanism to create *standard* DM *processes* across the DM Framework components.
- Adopt *standard* DM *processes* across the DM Framework components.

Advice

Standard DM *processes* across the organization are critical to collaboration that is required between *data domains* and operating units. The relationships among *data producers* and *data consumers* are many-to-many across the organization. Variation in the *processes* results in an inability to achieve interoperability in the DM initiative. The *standard processes* also are required to leverage *standard* DM technology across the organization. Beware the trap of deploying DM technology without the *process* in place to determine how that technology will be used.

Standardizing DM *processes* helps to ensure that core *principles*, such as data ethics are embedded into the operations of the organization. The *process* discipline reinforces cultural alignment to the organization's defined *principles*.

Questions

→ Is there a *process* optimization framework adopted for the DM initiative?
→ Are there *process* management tools in place?
→ Does the data *policy* and *standards* of the organization require *standard process* execution?

Artifacts

- *Process* Optimization Framework
- *Process* management toolset
- *Policy* requirement for *standard* *processes*

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No *standardized* DM *processes* exist.	No *standardized* DM *processes* exist, but the need is recognized and the development is being discussed.	*Standardized* DM *processes* are being developed.	*Standardized* DM *processes* are defined and have been validated by the directly involved *stakeholders*.	*Standardized* DM *processes* are established and being followed by directly involved *stakeholders*.	*Standardized* DM *processes* are established as part of business-as-usual practice with a continuous improvement routine. *Processes* are reviewed for relevance and accuracy at least annually and adjusted accordingly.

2.5.2 DM processes are informed by industry standards and best practices

Description

Engagement with industry trade groups, research organizations and *standards* bodies ensure that the organization is aware of and aligned with the latest DM trends and new developments related to DM best practices.

Objectives

- Create a formal *function* with dedicated resources to maintain participation in DM education, industry activities and events.
- Keep *stakeholders* informed about changes and events in the DM industry.

Advice

The organization should participate in industry trade groups to stay informed of new developments in DM. Likewise, engagement with *standards* bodies need to be formalized. Internal owners and facilitation agents are a good way of ensuring the flow of information across the organization. Embedding engagement with external organizations and *standards* bodies into job descriptions is useful to ensure participation.

As new issues emerge – like advanced *analytics* and data ethics – leveraging early best practices can be extremely valuable to the organization in terms of increased effectiveness and adoption speed.

Questions

→ Do you have a strategy for engagement with the DM industry outside of your organization?

→ Are the appropriate owners and facilitation agents identified and engaged?
→ Does executive management understand the value proposition and buy into the engagement activity?

Artifacts

- Evidence of participation in industry trade groups
- Evidence of attending training and DM education classes and forums

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No *standardized* DM *processes* exist.	No *standardized* DM *processes* exist, but the need is recognized and the development is being discussed.	*Standardized* DM *processes*, informed by industry *standards* and best practice, are being developed	*Standardized* DM *processes* are defined and have been validated by the directly involved *stakeholders*. *Processes* are informed by relevant industry *standards* and best practice.	*Standardized* DM *processes* are established and being followed by directly involved *stakeholders*. *Stakeholders* recognize that the *processes* are informed by industry *standards* and best practice.	*Standardized* DM *processes* are established as part of business-as-usual practice with a continuous improvement routine. *Processes* are reviewed for relevance and accuracy to industry *standards* and best practice at least annually and adjusted accordingly.

2.5.3 DM processes are supported by policy and auditable

Description

The DM *policy* must require the adoption of *standard* DM *processes* and tools across the organization. Routines should be established that include quality control (QC) self-assessments within the *process*, quality assurance (QA) monitoring from outside the *process* and formal audit performed by Internal Audit.

Objectives

- Perform QC self-assessments of the operating units DM *processes*.
- Perform QA monitoring of the operating units DM *processes* by the *enterprise* level ODM.
- Empower the *enterprise* level ODM to require remediation of gaps found in operational *processes*.
- Charge Internal Audit to perform routine examinations of the DM *processes*.
- Generate formal audit issues if operational gaps are uncovered.

Advice

DM *processes* should be routinely monitored. Monitoring occurs on three levels. First, QC self-attestation where the *stakeholders* evaluate and assert they are following the DM *processes*. Second, QA by ODM, where the DM initiative works with *stakeholders* to validate compliance. Third, internal review where Internal Audit has formally validated that *processes* are being followed.

Questions

→ What are the mechanisms to ensure QC and QA of the management *processes*?
→ What is the schedule for Internal Audit review of the DM *processes*?

Artifacts

- Evidence of self-attestation and *enterprise* ODM review
- Evidence of Internal Audit engagement and review

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No *standardized* DM *processes* exist.	No *standardized* DM *processes* exist, but the need is recognized and the development is being discussed.	*Standardized* DM *processes*, supported by *policy*, are being developed	*Standardized* DM *processes* are defined and have been validated by the directly involved *stakeholders*. *Processes* are supported by *policy* and auditable.	*Standardized* DM *processes* are established and being followed by directly involved *stakeholders*. *Stakeholders* recognize that the *processes* are supported by *policy* and auditable.	*Standardized* DM *processes* are established as part of business-as-usual practice with a continuous improvement routine. *Processes* are reviewed for relevance and accuracy to *policy* and audit requirements at least annually and adjusted accordingly.

2.6 Stakeholder Engagement is Established and Confirmed

A broad set of stakeholders are required to effectively manage the data. Accountable engagement is required and should be reinforced through active monitoring and individual performance review. The ODM must have visibility and ideally, authority to review and approve stakeholder committed budgets.

2.6.1 Stakeholders commit and are held accountable for the DMP deliverables

Description

DM requires participation and cooperation from staff and resources outside the DMP organizational structure, as well as staff and resources from other organization-wide control *functions*. Those identified as *stakeholders* must be held accountable for on time and on budget program delivery. To strengthen that commitment, performance in support of the DMP should be reflected in *stakeholder* reviews and compensation.

Objectives

- Communicate roadmap and program milestones to *stakeholders*.
- Initiate a review of program deliverables by *stakeholders*.
- Document *stakeholder* agreement with program deliverables.

- Formalize *stakeholder* accountability to the program deliverables through job description modification and compensation modification.

Advice

DM is a collaborative activity with implications across the organization affecting multiple *stakeholders*. Performance commitments from *stakeholders* are an essential aspect of a successful DMP and comes in many forms. It involves a financial commitment as well as an operational commitment that is often a daily necessity. It requires accountability. Ensuring that data is properly curated, secure and accessible is a shared responsibility. *Stakeholders* should be accountable as part of the risk framework of the organization to evaluate the appropriate and ethical use of data. All of these accountabilities should be formally included in performance review and compensation.

The expanding importance of advanced *analytics* within an organization brings with it a diverse set of *stakeholders* and unique requirements for data. Make sure the *stakeholder* analysis aggressively identifies all data *stakeholders*.

Questions

→ Have *stakeholders* been identified and verified?
→ Have *stakeholders* demonstrated a commitment to the objectives of the DMP?
→ Is funding in place to verify commitment to DMP deliverables?
→ Is there a mechanism to ensure accountability such as alignment with performance review and compensation?

Artifacts

- Roster of *stakeholders*
- Documentation of commitment plus incremental, milestone and final deliverables
- Evidence of bi-directional feedback and approvals
- Mechanisms to ensure accountability (i.e., modification of job descriptions or performance review criteria)

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No formal *stakeholder* accountability for DMP deliverables exists.	No formal *stakeholder* accountability for DMP deliverables exists, but the need is recognized, and the development is being discussed.	The formal *stakeholder* accountability for DMP deliverables is being developed.	The *stakeholder* accountability for DMP deliverables is defined and has been validated by the directly involved *stakeholders*.	The formal *stakeholder* accountability for DMP deliverables is established and recognized by other *stakeholders*.	The formal *stakeholder* accountability for DMP deliverables is established as part of business-as-usual practice with a continuous improvement routine. Such maturity may include the definition of accountabilities via job description modification and be reflected in compensation.

2.6.2 Resource plans are aligned with and verified against initiative requirements

Description

Proper resource levels with appropriate skill sets must be secured by *stakeholders*.

Objectives

- Complete resource planning.
- Review and reconcile resource plans and get them approved by the PMO.
- Put the approved resources in place.

Advice

The goal is to ensure that resource plans are sufficient to support the program deliverables, timelines and commitments. Resource allocation plans must be verified and approved. Be aware of potential budget cuts. DM is collaborative–cuts can have a cascading effect. Make sure the commitments are strong.

Questions

→ Have *stakeholders* pledged sufficient resources to implement project plans and meet program roadmaps?
→ Do the resources exist, or do they need to be acquired?
→ If they need to be acquired, has sufficient ramp-up time been incorporated into deliverable timeframes?
→ Does the ODM have authority to review and modify resource plans of *stakeholders*?

Artifacts

- Resource plans and documents
- Evidence of *processes* to review, modify and validate resource plans

- Bi-directional feedback of reviews, reconciliation and approval

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No resource plans for the DMP exist.	No resource plans for the DMP exist, but the need is recognized, and the development is being discussed.	The resource plans for the DMP are being developed.	The resource plans for the DMP are defined and have been validated by the directly involved *stakeholders*.	The resource plans for the DMP are established and resources are in place.	The resource planning for the DMP is established as part of business-as-usual practice with a continuous improvement routine. Plans are reviewed at least annually, and resources realigned accordingly.

2.6.3 Funds are allocated and aligned to program roadmaps and workstreams

Description

Sufficient funding dedicated to the DMP must be committed to by business, technology and operations. In a mature DMP, the ODM is granted authority to review and approve committed budgets.

Objectives

- Align funding to the program roadmaps and workstreams.
- Discuss and mitigate funding challenges.
- Obtain completed reviews of funding allocations by the ODM PMO.
- Approve and allocate funding levels.

Advice

Funding plans have dependencies and interrelationships. The goal is to ensure that all *stakeholder* plans are approved and aligned with the objectives of the DMP. There is no single strategy for funding DM initiatives. The strategy will depend on the culture of the organization. Some fund centrally. Some require operating units allocations. Some provide seed funding for early-stage activity. Some mix and match. Regardless of the funding mechanism, accountability and predictability are required and the ODM needs a reliable and realistic mechanism to ensure funding commitment.

Questions

→ Have budgets been prioritized to ensure adequate funding for the DMP?
→ Are budgets aligned to DM initiative deliverables?
→ Does the ODM have the authority to challenge *stakeholders* about budget commitments?

Artifacts

- Funding plans and budget allocation

- Funding approval and authorization to spend
- Escalation *procedures* for budget shortfalls

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No DMP roadmap or workstream funding exists.	No DMP roadmap or workstream funding exists, but the need is recognized, and the development is being discussed.	The DMP roadmap and workstream funding and allocation is being developed.	The DMP roadmap and workstream funding and allocation is defined and has been validated by the directly involved *stakeholders*.	The DMP roadmap and workstream funding and allocation is established and funding is in place.	The DMP roadmap and workstream funding and allocation is established as part of business-as-usual practice with a continuous improvement routine. Funding is reviewed at least annually and reallocated accordingly.

2.7 Communications and Training Programs are Designed and Operational

Internal communications and formal training are needed to affect the required organization-wide behavior and cultural change. For organizations subject to regulatory oversight, active communication with the authorities is essential.

2.7.1 Internal communication plans have been defined and approved

Description

Plans for internal communications are needed to drive awareness and adherence to the DMP. The communication strategy must be designed and implemented according to the culture of the organization and incorporated into the DMS.

Objectives

- Have *stakeholders* develop, socialize and approve internal communication.
- Establish communication channels.
- Implement an operational communication program.
- Incorporate the communication strategy goals and objectives into the DMS.

Advice

Communications is not a sideline activity. It would benefit from a dedicated professional. Introduce continual communications to reinforce DM concepts and emphasize that it is not a one-and-done *process*. Think about a variety of communications channels and mechanisms to keep the content fresh. The full spectrum of communications channels (i.e., websites, access portals, reference libraries, documents, training materials, town hall meetings, etc.) is needed to ensure that *stakeholders* understand the goals, objectives and *processes* associated with the DMP.

Also, think about ways to involve the full spectrum of participants including PR, HR, executive management, Internal Audit in the communications program. There are lots of subtle but critical concepts like the difference between correcting bad data and fixing data problems at the source. Focus on communication as a two-way street. Maintain an active mechanism for discussions about value derived vs. the inevitability of operational disruption. Take advantage of the existing internal communications infrastructure to develop and implement an organization-wide communications strategy. It is important to evaluate the degree to which executive management is participating in these activities. Having and highlighting their engagement denotes importance and sends a clear, positive message of support for the DM initiative.

The goal of the communications plan is to establish an organization-wide data culture that understands the importance of data and the appropriate and ethical use of the data. Every individual should know their role in meeting the data objectives of the organization.

Questions

- → Has the importance of communication and training been defined as part of the DMS?
- → Does the communications plan include a track for bringing new employees onboard?
- → Does the communications strategy define the core goals and objectives of the DM initiative?
- → Have the plans been created, published and approved?
- → Are the communications channels defined and established?
- → Is the internal communications program operational?

Artifacts

- Communications plan and records of the channels to be used
- Evidence illustrative of content and methods used
- Definition of mechanisms for engagement
- List of *stakeholders* and evidence of bi-directional communication

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No internal communication plans exist.	No internal communication plans exist, but the need is recognized, and the development is being discussed.	Internal communication plans are being developed.	Internal communication plans are defined and have been validated by the directly involved *stakeholders*.	Internal communication plans are established and operational.	Internal communications are established as part of business-as-usual practice with a continuous improvement routine.

2.7.2 Plans for communication with external regulatory bodies are defined and approved

Description

For organizations subject to regulatory oversight, communication with regulators and government authorities are essential as well as functional. The regulatory mandate for DM can be an important aspect of the overall DM initiative.

Objectives

- Plan and get *stakeholder* approval for a proactive strategy for communications with regulatory bodies.
- Establish *procedures* for communications with regulators. These *procedures* are usually carried out through the organization's compliance department.
- Maintain routine communications with regulatory bodies.

Advice

Communication with regulators about data challenges and objectives should be proactive and transparent. Regulators are aware of the challenges of implementing a data control environment. They have a stake in the matter as well, through both market structure oversight and linked risk analysis. It is much wiser to identify your own data challenges and create plans for remediation. Too often, the alternative is to have them discovered by regulators during an audit, which results in a Memo Requiring Attention (MRA) written against the organization.

Questions

→ Does the organization have a regulatory communication strategy?
→ Has the strategy been developed in conjunction with your compliance and risk teams as the first line of engagement?
→ Are routine communications with regulatory bodies taking place?

Artifacts

- Regulatory roster and evidence of communication
- An internal *procedure* for approval of regulatory engagement and communication
- Self-identified audit reports

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No formal communication plans for external regulatory engagement exist.	No formal communication plans for external regulatory engagement exist, but the need is recognized, and the development is being discussed.	Formal communication plans for external regulatory engagement are being developed.	Formal communication plans for external regulatory engagement are defined and have been validated by the directly involved *stakeholders*.	Formal communication plans for external regulatory engagement are established and operational.	Formal communication plans for external regulatory engagement are established as part of business-as-usual practice with a continuous improvement routine. Engaged bodies are reviewed for relevance and accuracy at least annually and adjusted accordingly.

2.7.3 Formal training programs have been defined and implemented

Description

Behavior and culture change are required for effective DM. Formal training is needed to ensure those with data responsibility are operating in accordance with established data *policy* and *standards*.

Objectives

- Design and make operational DM training programs.
- Provide training and require it by *policy* where appropriate.
- Incorporate the training program scope, goals and objectives into the DMS.

Advice

Training has a broad scope and all aspects need to be included in the DM training program. Consider areas such as ***functional training***, as with the role of the data steward; ***operational training*** including the implications of the DM initiative and where *stakeholders* go for support; ***concept training***, for example why DM is critical and what is meant by adopting a DM culture; and ***dependency training*** which includes how to collaborate to effectively manage data assets. The desired outcome is the continual reinforcement of the objectives of the DM initiative. In order to achieve an organization-wide data culture, every individual in the organization needs to understand their accountability to data and data ethics.

Training across all four areas is what will empower every individual to understand their accountability to ensure the organization has the high-quality data it needs and that the use of data is appropriate and ethical. Senior management support of the training program resonates across the organization and helps establish and maintain sufficient funding. Look for opportunities to partner with human resources and ways to tie DM, and its training, into compensation and retention.

Questions

→ Is there a formal training program for the data roles and broader *stakeholders*?
→ Have the training curricula been developed in collaboration with the organizational units and other control *functions*?
→ Is participation mandatory and part of the control *process*?
→ Have the aspects of the DM training program been evaluated for completeness and value?
→ Are the training program goals and objectives as well as the scope and core aspects incorporated into the DMS?

Artifacts

- Training program definition
- Approaches and methodologies used
- Training curricula and materials
- Class rosters, certificates of accomplishment and other communication about participation
- Training and testing results
- List of *stakeholders* and evidence of bi-directional communication

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No formal training programs exist.	No formal training programs exist, but the need is recognized, and the development is being discussed.	Formal training programs are being developed.	Formal training programs are defined and have been validated by the directly involved *stakeholders*.	Formal training programs are established and operational.	Formal training programs are established as part of business-as-usual practice with a continuous improvement routine. Training programs reviewed for relevance and accuracy at least annually and adjusted accordingly.

2.8 The DMP is Measured and Evaluated Against Business Objectives

Measuring the DMP includes: 1) program metrics; 2) outcome metrics; 3) process metrics; and 4) financial metrics.

2.8.1 Program metrics are defined and used to track progress

Description

Develop program metrics for Key Performance Indicators (KPIs) and Key Risk Indicators (KRIs) to track the progress of the implementation and adoption of the DM initiative. Include other critical metrics that demonstrate the health and wellness of the DM initiative. The metrics include program capabilities such as organizational structure rollout, skill-set hiring, leader appointments, *policy* adoption and *standards* implementation.

Objectives

- Obtain review and approval of the program metric plans and approach from *stakeholders*.
- Include program metrics in the DMS.
- Establish program metrics and thresholds.
- Track and report program metrics to *stakeholders*, including executive management.
- Use program metrics to inform and drive program decisions and modifications.

Advice

Program metrics are designed to track progress against the implementation plans for the DM initiative. This is the implementation of metrics at the end-user level. This is designed to provide evidence of progress against the DMP roadmap. When you can validate that you are on track, that is a valuable tool for communication across *stakeholders* including executive management.

Questions

- → Has the concept of metrics related to the DM initiative itself been defined?
- → Have the program-related metric plans been socialized and verified?
- → Is there a method for monitoring progress against the strategy and roadmaps of the DMP?
- → Have metrics been defined across the core DM capabilities?
- → Do *stakeholders* support the metrics program?
- → What reporting mechanisms are being used?

Artifacts

- Definition of program metrics
- Evidence of the program metrics tracking and reporting within the DMS
- List of *stakeholders* and evidence of bi-directional communication
- Approvals and sign-off Reports, dashboards, heat maps and other forms of output

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No formal DM program metrics exist.	No formal DM program metrics exist, but the need is recognized, and the development is being discussed.	Formal DM program metrics are being developed.	Formal DM program metrics are defined and have been validated by the directly involved *stakeholders*.	Formal DM program metrics are established and operational.	Formal DM program metrics are established as part of business-as-usual practice with a continuous improvement routine. Metrics are reviewed for relevance at least annually and adjusted accordingly.

2.8.2 Outcome metrics are defined and used to track against business objectives

Description

Outcome metrics are measurements of the net effect of the DM initiative. Define metrics that cover the stated business objectives that align to the data, data usage and DM deployment strategies in the DM strategy. Examples of outcome metrics include lower operational failures, streamlined reporting, reduced number of data reconciliations, improved data discovery, improved access to critical data and the use of data is appropriate and ethical.

Objectives

- Obtain review and approval of the plans and approach for the outcome metrics from *stakeholders*.
- Include outcome metrics in the DMS.
- Establish outcome metrics and thresholds.
- Track and report outcome metrics to *stakeholders*, including executive management.

- Use outcome metrics to inform and drive program decisions and modifications

Advice

Stakeholders need to understand the concepts of factor of input and data interoperability, harmonization and *process* automation. It is important to measure value, but it must be remembered that data is only one input. It is quite possible to have good data, but not to achieve the desired outcome because of some operational deficiency. Measuring areas such as Straight Through Processing (STP), reduction of repairs, improved discovery, consolidation of technology is multidimensional and not always easy to trace back to DM.

Questions

→ What are the expected operational outcomes associated with DM and proper data hygiene?
→ How will the organization measure both the defensive and offensive value of DM?
→ Do *stakeholders* understand the concepts associated with data as a trusted factor of input?
→ Do *stakeholders* understand the concepts of linked analysis and causality?
→ Do *stakeholders* understand the concepts of appropriate and ethical use of data?

Artifacts

- Definition of outcome metrics
- Evidence of the outcome metrics tracking and reporting within the DMS
- Approvals and sign-off reports, dashboards, heat maps and other forms of output
- List of *stakeholders* and evidence of bi-directional communication

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No formal DM outcome metrics exist.	No formal DM outcome metrics exist, but the need is recognized, and the development is being discussed.	Formal DM outcome metrics are being developed.	Formal DM outcome metrics are defined and have been validated by the directly involved *stakeholders*.	Formal DM outcome metrics are established and operational.	Formal DM outcome metrics are established as part of business-as-usual practice with a continuous improvement routine. Metrics are reviewed for relevance at least annually and adjusted accordingly.

2.8.3 Process metrics are defined and used to drive continuous improvement

Description

A core aspect of business *process* optimization is establishing metrics to measure the performance of that *process*. When the *process* is not performing as designed there is an opportunity for *process* improvement.

Objectives

- Obtain review and approval of the *process* metric plans and approach from *stakeholders*.
- Include *process* metrics in the DMS.
- Establish *process* metrics and thresholds.
- Track and report *process* metrics to *stakeholders*, including executive management.
- Use *process* metrics to inform and drive program decisions and modifications.

Advice

Without *process* metrics it cannot be demonstrated that the *processes* are performing as designed. The metrics are intended to be an early alert system to identify *process* failures. Additionally, the metrics often support assessing the impact and priority of an underperforming *process*.

Questions

→ Are there methods of measuring the performance of the DM *processes*?
→ Have metrics been defined across the DM *processes*?
→ Do *stakeholders* support the metrics program?
→ What reporting mechanisms are being used?

Artifacts

- Definition of *process* metrics
- Evidence of the *process* metrics tracking and reporting within the DMS
- Approvals and sign-off Reports, dashboards, heat maps and other forms of output
- List of *stakeholders* and evidence of bi-directional communication

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No formal *process* metrics exist.	No formal *process* metrics exist, but the need is recognized, and the development is being discussed.	Formal *process* metrics are being developed.	Formal *process* metrics are defined and have been validated by the directly involved *stakeholders*.	Formal *process* metrics are established and operational.	Formal *process* metrics are established as part of business-as-usual practice with a continuous improvement routine. Metrics are reviewed for relevance at least annually and adjusted accordingly.

2.8.4 Financial metrics for total program costs and benefits (ROI) are tracked and reported

Description

Data and DM expenses occur throughout an organization and need to be assessed in the context of the overall DM initiative. Determining the current cost at the *enterprise* level as well as the operating unit level establishes a benchmark that can be referred to as the organization-wide DM initiative is established and deployed. The metrics should include cost-benefit analysis that results in an accountable return-on-investment (ROI) measurement.

Objectives

- Define what the DM benefits are for your organization.
- Capture current-state data and DM expenses at the *enterprise* and operating unit levels.
- Analyze total expense to establish a cost benchmark for comparison to future costs as the DM initiative is implemented.
- Establish a sustainable methodology for calculating the financial benefits of the DM initiative at the *enterprise* and organizational unit levels. Use either an established organizational *standard* or a newly created method.
- Obtain review and approval of the cost-benefit methodology from *stakeholders*.
- Measure and monitor financial benefits and use these results for *enterprise* and organizational unit level decision-making.

Advice

It is important to establish a cost baseline for the DM initiative. This baseline is an essential metric that repeatedly will prove to be valuable. Prepare for discussions on what constitutes data expenses as well as on the appropriate methodology to use to capture spending by data category.

Capturing the benefits of DM is required to ensure continued buy-in to the DM initiative. Benefits should be understood in the context of the entire organization. This understanding rests on the evaluation of all the organization-wide dependencies associated with trusted data and the impact of bad data. This usually will not fit into *standard* criteria for the calculation of project-based ROI. Determine the current methodology for these calculations. Examine the value proposition from four dimensions:

1) operational efficiency (cost)
2) trust (model-based strategies)
3) insight (upselling and predictive analysis)
4) flexibility (ability to adapt to changing circumstances)

Positive cost-benefit analysis of the DM initiative is necessary to ensure organizational buy-in. DM affects many systems and *processes* across the organization and may need to be evaluated beyond *standard* project-based return-on-investment (ROI) methodologies.

Determining ROI should include a defensive and offensive view of the benefits. Evaluating the benefits of maintaining ethical usage of data is a defense-based analysis measuring cost avoidance. Whereas, the benefits tied to advanced *analytics* is an offense-based analysis measuring increased opportunity and the resulting revenue.

Ensure that captured metrics are being properly used for decision-making, resource allocation, task prioritization, and other similar business objectives. Graphic expressions using tools such as heat maps help put DM into context. The ability to "name and shame" is useful in loosening purse strings.

Questions

→ What methodology is used to capture total DM spend (i.e., acquire, cleanse, store, manipulate, transform, integrate, distribute), as well as on soft metrics like reconciliation, lack of capability, missed opportunity, capital charges, inefficient operations and collateral calculations?
→ What is the organizational view of the benefits associated with DM?
→ What are the methodologies used to calculate financial and operational benefits?
→ How have these metrics been used to establish and remediate priorities?

Artifacts

- Expense categories with evidence of agreement
- Cost allocation methodology
- Total Cost of Operations (TCO) calculation with worksheets, approvals, reporting, ROI criteria
- Documentation of methodologies and illustrations of how applied
- Evidence of use of metrics to evaluate, adjust and enhance the DM initiative
- List of *stakeholders* and evidence of bi-directional communication

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No formal financial metrics exist.	No formal financial metrics exist, but the need is recognized, and the development is being discussed.	Formal financial metrics are being developed.	Formal financial metrics are defined and have been validated by the directly involved *stakeholders*.	Formal financial metrics are established and operational.	Formal financial metrics are established as part of business-as-usual practice with a continuous improvement routine.

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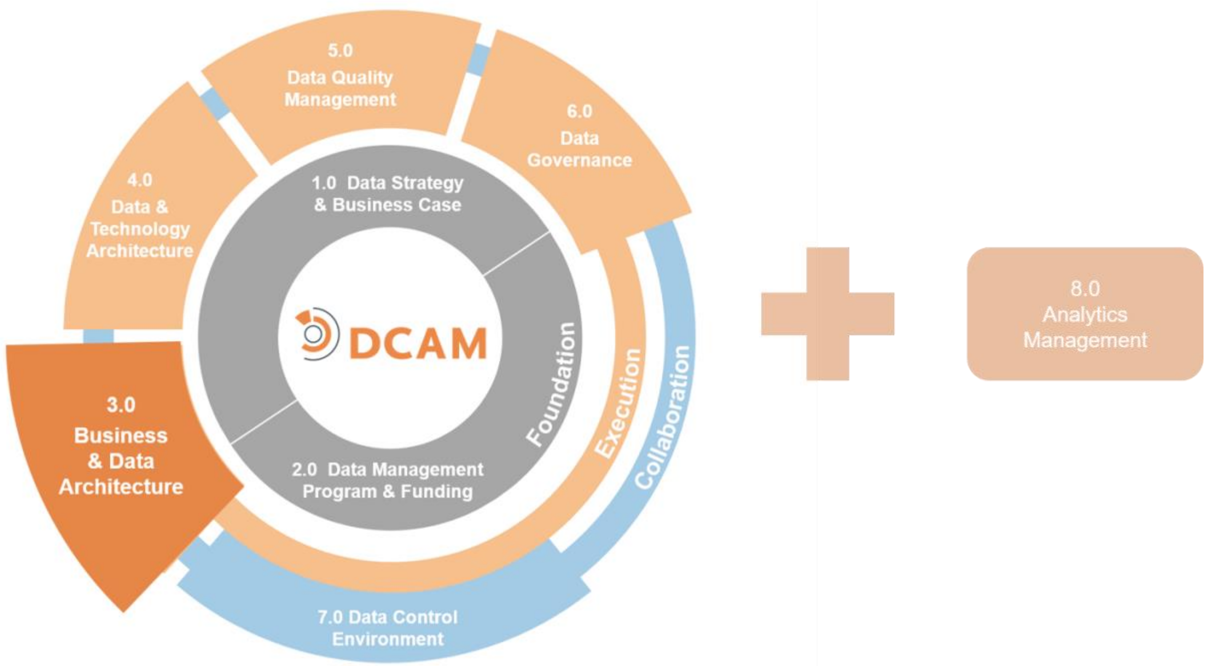
DCAM® Framework:

3.0 Business & Data Architecture



3.0 Business & Data Architecture

Introduction

The path to integrated architecture across the organization begins with *business architecture* and how it defines requirements for *data architecture*.

Business Architecture *is the strategy, design and execution of the capabilities needed to support the organization business* functions.

Data Architecture *is the strategy and execution of how data is designed (identified and described) to support the business objectives.*

Business architecture defines the *processes* required to meet the objectives of the business. The *processes* have requirements for data and data management (DM). Those requirements must be defined as input and output of the business *process*.

Data architecture speaks to the design, definition, management and control of information content. Data architecture identifies *data domains*, documents *metadata*, defines *critical data elements*, establishes *taxonomies* and ontologies that are critical to ensuring that the meaning of data is precise and unambiguous, and that the usage of data is consistent and transparent.

A *data architecture function* establishes consistency in definition and use of data throughout an organization. Adhering to a prescribed data architecture forces business and technology resources to take the necessary steps to define and document data meaning, define the appropriate use of the data, and to ensure that proper governance is in place to manage data as meaning on a sustainable basis.

Definition

The **Business and Data Architecture (DA)** component is a set of capabilities to ensure integration between the business *process* requirements and the execution of the DA *function*. The business *process* is defined by the *business architecture function*. DA defines *data models* such as *taxonomies* and ontologies as well as *data domains*, *metadata*, and business critical data to execute *processes* across the data control environment. The DA *function* ensures the control of data content, that the meaning of data is precise and unambiguous and that the use of data is consistent and transparent.

Scope

- Establish a DA *function* within the Office of Data Management (ODM).
- Work with DM *Project Management Office* (PMO) to design and implement sustainable business-as-usual *processes* and tools for DA including the required integration with *business architecture.*
- Identify and establish *data domains*, authoritative sources, and *provisioning points*.
- Identify and inventory the data to support the business requirements, including all necessary *metadata* including glossary, dictionary, classification, lineage, etc.
- Define and assign business definitions, linked to the data inventory.
- Ensure that the DA governance is integrated into Data Governance (DG) and aligned to both business and technology governance activities.

Value Proposition

Organizations that identify, record and make available information about the internal constituencies that define, produce, and use specific data demonstrate efficient and effective coordination, cooperation, and communications around this data.

Organizations that document information about highly valued *data elements* demonstrate improved understanding and business use of these data.

Organizations that effectively implement DA to understand their data and *data ecosystem* get a return on investment from several areas:

- Operational excellence in business *processes* creates efficiencies and lowers operating cost
- Creation of straight-through, fit-for-purpose data, reduced data debt and remediation costs and increased value derived from advanced *analytics*
- Greater understanding of your data leads to data simplification and reduces the cost of DM and maintenance
- Understanding your data also reduces operational, financial and reputational risks associated with using the wrong data for *analytics*, decision-making, and regulatory reporting

Overview

The DA component establishes the unambiguous definition and use of data throughout an organization. Adhering to a prescribed data architecture forces business and technology to take the necessary steps to define and document data meaning, define the appropriate use of the data, and to ensure that proper governance is in place to manage data as meaning on a sustainable basis.

Data exists throughout an organization across all facets of business operations. The design of an organization's data architecture is based on a comprehensive understanding of business requirements and their impact on what data is needed. Unraveling the business *process* informs how data should be identified, defined, modeled and related. *Technology architecture* then dictates how the data architecture design is organized and placed into physical repositories in order to provide optimized access, security, efficient storage management and speed of processing.

To establish a successful DA *function*, the following sequence of activities is required.

First, the following two activities will allow an organization to understand what data is needed to satisfy the business requirements.

- **Identification of Logical *Data Domains***
  Logical *data domains* are the logical groupings of data, not the databases themselves, that are needed to satisfy the business requirements.
- **Identification of Physical Repositories**
  Underlying the logical *data domains* are multitudes of physical, often overlapping repositories of data that will map into the logical *data domains*. Identification of these underlying physical repositories is a critical step towards minimizing the complexity of legacy environments, reducing replication, better understanding *data lineage*, assigning *data ownership*, and assessing *data quality* (DQ).

Once the *data domains* and their underlying physical sources of data have been identified, precise business definitions using common *semantic* language for the identified data must be assigned and agreed upon by *stakeholders*. DA is about managing the meaning of data. The importance of assigning precise definitions in the context of business reality, the creation of a shared data dictionary and getting the agreement from both *data* *producers* and *data consumers* cannot be minimized. Without this common understanding of *data attributes*, aligned to business meaning, data architecture will struggle to succeed. The risk of inappropriate use of data will increase and the ability to share data across an organization with confidence will be hindered.

The next step in addressing data architecture is to define data *taxonomies* and business ontologies. Data *taxonomies* define how data entities are structurally aligned and related. For each officially designated *data domain* that is identified, inventoried and deemed critical, a *taxonomy* must be defined and maintained. The *taxonomy* is then mandated for all systems using this data as input into their business *processes*. With critical business *function* *taxonomies* defined and in place, the organization needs to model the relationships between *taxonomies* into a business *ontology*. Ontologies are the relationships and knowledge of multiple related *taxonomies* across *data domains*.

A comprehensive data architecture *process* may include the following; however, this level of complexity is not always required.

- The business *function* defines the data in a business model based on the requirements for data as an input and output of the business *process*.
- These business *function* *data models* are then consolidated into an organization-wide business data model.
- For each *data domain*, a *taxonomy* is defined, maintained, and mandated for all systems using this data as input to the business *process*. Data *taxonomies* define how data entities are structurally aligned and related.
- With business *function* *taxonomies* defined and in place, the organization *models* the relationships between *taxonomies* into a business *ontology*. These ontologies represent the relationships and knowledge of multiple related *taxonomies* across *data domains*.

Taxonomies and ontologies define and relate the content of data to enable the organization to realize the maximum value of its data in a consistent and controlled manner. Once the content is defined, it needs to be precisely described using *metadata*. Some of the types of *metadata* may include, but not limited to, business, operational, technical, descriptive, structural and administrative.

Core Questions

→ Are business *stakeholders* driving requirements for data?
→ Are *policies* in place to govern the creation and maintenance of *data attributes* and relationships?
→ Are governance procedures in place to ensure adherence to established data architecture *standards*?
→ Are design reviews in place and required to ensure enhancements and new development are utilizing *standard* data architecture definitions?
→ Is adherence to data architecture *standards* auditable?

Core Artifacts

The following are the **core artifacts** required to execute an effective Business & Data Architecture capability. Items with an '*' link to published best practice *guidelines*.

- Business & *Data Architecture* *Policy* Alignment Matrix
- *Business Element – Data Element Model**
- DCAM *Business Glossary* – Data Dictionary Model
- *Data Asset* Inventory
- Data Domain Inventory Registry
- *Data Lineage* – *Data Flow* Model
- *Data Model* Inventory
- Data Domain Identification Criteria
- Data Classification Catalog

3.1 Data Architecture (DA) function is established

The DA function strategy and approach must be defined and approved by stakeholders. Roles and responsibilities across the stakeholders must be established with operational processes in place.

3.1.1 The DA strategy and approach are defined and adopted

Description

The strategy and approach for the DA *function* must be defined and reflect the related vision and objectives of the Data Management Strategy (DMS). Once established, it must be formally empowered by senior management and its role communicated to all *stakeholders*.

Objectives

- Formally establish the DA strategy and approach within the organization.
- Get approval of the DA strategy and approach from *stakeholders*.
- Ensure alignment of *stakeholder* plans and roadmaps with the DA strategy and approach.
- Obtain executive management support for the DA strategy.
- Communicate the role of the DA *function* across the organization through formal organizational channels.
- Operate the DA *function* collaboratively with DM initiative *stakeholders*.
- Secure authority to enforce DA compliance through *policy* and documented procedure.

Advice

The DA *function* is a critical bridge between the business and technology *stakeholders* in the DM initiative. Irrespective of whether the DA *function* aligns organizationally to the business or to the technology *function*, the pivotal role as the bridge between these two *stakeholder* groups must be recognized. Sometimes data architecture is mistakenly viewed as a subset of *technology architecture*. Successful data architecture requires the integration of subject matter expertise from both *business architecture* and *technology Architecture*.

Alignment of the DA strategy and roadmap to the DMS vision and objectives is achieved by agreement between the Operating Level *Data Officer* and the individual responsible for delivering the *data governance function*. The Operating Level *Data Officer* is accountable for establishing priorities across each of the Framework Component requirements.

Questions

→ Has the DA *function* been formally established?
→ Is there a DA strategy and approach in place?
→ Is the DA strategy and roadmap aligned to the DMS?
→ Has the DA *function* been formally communicated to business, technology, operations, finance and risk *stakeholders*?
→ How has executive management demonstrated its support?
→ Has authority been granted to the DA *function* to implement and enforce best practice via *policy* and *standards*?
→ Has authority been communicated to *stakeholders*?
→ Is there a functional partnership in place with Internal Audit?

Artifacts

- The DA Plan
- Description of the roles and responsibilities of the DA *function*
- Communication of specific support from executive management with distribution lists
- *Policies* and *procedures* associated with executing and enforcing DA
- Bi-directional engagement with *stakeholders* on the DA *function* authority

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No formal DA strategy exists.	No formal DA strategy exists, but the need is recognized and the development is being discussed.	The formal DA strategy is being developed.	The formal DA strategy is defined and has been validated by the directly involved *stakeholders*.	The formal DA strategy is established and understood across the organization and is being followed by the *stakeholders*.	The formal DA strategy is established as part of business-as-usual practice with a continuous improvement routine. The strategy and approach are reviewed and updated at least annually.

3.1.2 The DA stakeholder roles and responsibilities are defined and communicated

Description

The DA *function* will require the coordination between the roles of the DA, *business architecture* and *technology architecture*. The integration activities between these disciplines must be defined by specific role descriptions within the different *stakeholder* teams. It is critical that each team be staffed at an optimal level for the scope and volume of work.

Objectives

- Define and communicate the roles and responsibilities of the DA *function*.
- Fund and staff the DA *function*.
- Ensure and enforce alignment of activities and projects to *policy* and *standards* through the authority of the DA *function*.
- Hold individuals accountable for the data control environment performance via annual reviews and compensation considerations.

Advice

Effective DA is critical to the DM initiative. The data architect needs an understanding of the business *process* and the technical environment to excel as the bridge between these two disciplines. The data architect will partner with the *business data steward* and *technical data steward*. The *business data steward* is accountable for the *business element,* which defines all the requirements for data. The technical data steward is accountable for the data element, which is the physical execution of the concept defined by the *business element*.

Questions

→ Has the DA *function* been established?
→ Is the DA *function* appropriately staffed and funded?
→ Does the DA *function* have the authority needed to be effective?
→ Have the roles and responsibilities of the DA *function* been defined, documented and socialized?
→ Have the skills for data ethics review and execution of Machine Learning (ML) and Artificial Intelligence (AI) tools been added or developed within the *stakeholders*?
→ Have milestones and metrics associated with DA execution been established?

Artifacts

- Evidence of *stakeholder* identification
- *RACI matrix* or other evidence of accountability assignment
- Description of the roles and responsibilities of the DA *function*
- Staff qualifications and assignments
- Evidence of accountability linked to reviews and compensation
- Gap analysis of skills needed and those in place
- List of *stakeholders* and evidence of bi-directional communication

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No formal DA roles & responsibilities exist.	No formal DA roles & responsibilities exist, but the need is recognized and the development is being discussed.	The formal DA roles & responsibilities are being developed.	The DA roles & responsibilities are defined and have been validated by the directly involved *stakeholders*.	The DA roles & responsibilities are established and are recognized and used by *stakeholders*.	The DA roles & responsibilities are established as part of business-as-usual practice with a continuous improvement routine. The roles & responsibilities are reviewed and updated at least annually.

3.1.3 The DA processes are defined and operational

Description

Formal *processes* have been established for the activities of the DA *function*. These *processes* align with the DM *policy* and *standards* of the organization and include *procedures*, tools and routines. The routines are required for steady-state operations.

Objectives

- Establish formal DA *processes* in alignment with the DM *policy* and *standards*.
- Integrate the DA *processes* into the overall end-to-end *processes* of the DM initiative.
- Identify, schedule and maintain DA routines, meetings and working sessions required for operational support.

Advice

The DA *subject matter experts* should work with the business *process* design and optimization service within the Data Management Program (DMP) *function*. Together they will create and monitor the implementation of the DA *processes* in alignment to the end-to-end *process* across the full DM initiative.

Questions

→ Have formal *processes* been defined and implemented?
→ Are the *procedures*, tools and routines in place for implementing the *processes*?
→ Have innovative technologies such as AI and ML been considered as part of the DCE *process* and infrastructure?
→ Has the review of data ethics been included in the Data Quality Management (DQM) strategy and approach?
→ Are DA activities part of the normal operational routine of *stakeholders*?
→ Are there standing meetings, planning sessions and regular communications about DA initiatives?

Artifacts

- *Process* design artifacts, *procedure* guides and published routines
- *Process* performance metrics reports
- Meeting minutes, status reports and DA announcements

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No formal DA operational *processes* exist.	No formal DA operational *processes* exist, but the need is recognized and the development is being discussed.	The DA operational *processes* are being developed.	The DA operational *processes* are defined and have been validated by the directly involved *stakeholders*.	The DA operational *processes* are established and are recognized and used by *stakeholders*.	The DA operational *processes* are established as part of business-as-usual practice with a continuous improvement routine.

3.2 Business Architecture (BA) is Integrated with Data Architecture (DA)

The DM initiative must be engaged with the business architecture function for three activities: 1) the business process must define data as an input and output; 2) manage restrictions on data access and use by the business process; and 3) root-cause-fix for data issues that involve people or process deficiency. This engagement must be supported by mutual governance and policy.

3.2.1 BA defines process input and output data requirements

Description

The DM initiative must be engaged with the *business architecture function* of the organization to support the identification of requirements for data as input and output of the business *process* design and optimization activity.

Objectives

- Define a *process* optimization framework for the business architecture *function*. Include the identification of requirements for data as input and output of the *process*.
- Engage the DA *function* in business *process* design and optimization. Integrate requirements for data into the *standard data models* of the organization.

Advice

The BA *function* should engage with business *subject matter experts* to define requirements for data. The DA *function* then will engage with the *business data stewards* to interpret the requirements for data into a complete set of *business element* requirements. They should then record these requirements as *metadata*. The DA *function* can then engage with the technology *function* to translate *business element* requirements into the requirements for the physical *data elements*. In this way these physical *data elements* are brought into the real world.

Questions

→ Are the BA *function* and the DA *function* integrated into the business *process* design and optimization activities?
→ Does the business *process* design and optimization activity generate requirements for data as input and output of the *process* steps?

Artifacts

- Business & Data Architecture *Policy* Mapping
- Business & Data Architecture Target-state
- Business & Data Architecture Execution Roadmaps
- *Process* Optimization Framework
- *Business Element* – *Data Element Construct*

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No BA *processes* for data requirement definition in business *processes* exist.	No BA *processes* for data requirement definition in business *processes* exist, but the need is recognized and the development is being discussed.	BA *processes* for data requirement definition in business *processes* are being developed.	BA *processes* for data requirement definition in business *processes* are defined and validated by directly involved *stakeholders*.	BA *processes* for data requirement definition in business *processes* are established and are recognized and used by *stakeholders*.	BA *processes* for data requirement definition in business *processes* are established as part of business-as-usual practice with a continuous improvement routine.

3.2.2 Business data requirements must include data usage, data restrictions and data ethics considerations

Description

The business *process* design and optimization effort must define requirements for data as an input and output of the business *process* activities. To *process* these requirements a review of any restrictions on the data usage is required. These restrictions may be due to data-subject owner consent, privacy, ethics of accessing the data and ethics of the data usage output of the business *process*.

Objectives

- Define a *process* to evaluate the data usage restrictions of the data.
- Define data restrictions that include the ethics of accessing the data and the ethics of the data usage output of the business *process*.

Advice

As the BA *function* delivers requirements for data to the DA *function*, a review of the data usage restrictions must be completed. This review should involve the business and technical data stewards as required. The data usage restriction review must be exhaustive across all regulatory, internal *policy* and ethical considerations.

Questions

→ Have all the usage restrictions for the data been defined and reviewed?
→ Is ethical data access and data usage output defined in the data usage restrictions?

Artifacts

- *Process* documentation for data usage restrictions
- Data usage restriction *metadata*

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
When *processing* business requirements, there is no review of data usage and data restrictions from an ethical perspective.	When *processing* business requirements, there is no review of data usage and data restrictions from an ethical perspective, but the need is recognized and the development is being discussed.	The capability to *process* business requirements, with a review of data usage and data restrictions from an ethical perspective, is being developed.	The capability to *process* business requirements, with a review of data usage and data restrictions from an ethical perspective, is defined and validated by directly involved *stakeholders*.	The capability to *process* business requirements, with a review of data usage and data restrictions from an ethical perspective, is established, recognized and used by *stakeholders*.	The capability to *process* business requirements, with a review of data usage and data restrictions from an ethical perspective, is established as part of business-as-usual practice with a continuous improvement routine.

3.2.3 BA processes incorporate root cause fix of people or process

Description

When data fit-for-use issues are encountered, the BA *function* must be engaged if it is determined that the root cause fix required involves a people or *process* deficiency.

Objectives

- Define a *process* to engage the BA *function* to conduct root cause analysis and execute root cause fix solutions when the deficiency is related to people or business *process*.

Advice

When data fit-for-use issues are discovered the first step is to triage the potential cause. The triage review may determine it is a data issue, a technology issue, a business *process* issue or a people issue. The type of issue will define the subject matter expertise required to determine the root cause fix. In the case of either business *process* or people issues, the *business architecture function* should be engaged in the root cause analysis. Direct business *subject matter experts* should be engaged as needed.

Questions

- → Is there a triage *process* in place to identify suspected root cause of data defects?
- → Does the data defect issues log include categorizing issues that align to people or *process*?
- → Is the business architecture *function* involved in the analysis and root cause fix of data defects aligned to people and *process*?

Artifacts

- *Process* for engagement of business architecture in root cause fix activities
- Issues log that includes defects aligned to people and *process*

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
BA *processes* do not include root cause fix of people or *process*.	BA *processes* do not include root cause fix of people or *process*, but the need is recognized and the development is being discussed.	BA *processes* to include root cause fix of people or *process*, are being developed.	BA *processes* to include root cause fix of people or *process*, are defined and validated by directly involved *stakeholders*.	BA *processes* to include root cause fix of people or *process*, are established, recognized and followed by *stakeholders*. Root cause fixes of people and *process* are evident.	BA *processes* to include root cause fix of people or *process*, are established as part of business-as-usual practice with a continuous improvement routine. Root cause fixes of people and *process* are the natural way of working.

3.2.4 DA governance is aligned with BA governance

Description

Alignment of DA and BA governance includes: a definition of the business *process*; data requirements as an input and output of the business *process*; and integration of *data consumer* requirements including third party data contract specifications. DA governance is a part of the overall DM governance structure and routines.

Objectives

- Align DA governance with BA governance to ensure *semantic* definitions, *taxonomies* and *critical data elements* (CDEs) are properly assigned and maintained.
- Utilize DA governance to monitor the capture of appropriate *business metadata* as defined by DM *policies*.

Advice

The goal is to ensure that the management of data meaning is aligned with defined business *processes*. *Business elements* including their definitions and relationships to one another need to be properly assigned and maintained to capture and align with business reality. Data meaning needs to be aligned with operational *processes* and third-party data requirements. Collaboration is required to manage data vendor, *data producer* and *data consumer* relationships and entitlement control needed to maintain the flow of data.

Questions

→ Have business *processes* been defined and verified?
→ Are governance *procedures* in place to ensure unambiguous shared meaning across the organization?
→ Have the business *processes* defined the priority data (CDEs) that is critical to the *process*?
→ Are there mechanisms to ensure collaboration between *data producers* and *data consumers*?
→ Are third party data requirements and restrictions defined and accessible?

Artifacts

- Business *process* flow diagrams with data as an input and output
- Bi-directional communication on data definitions and relationships
- CDEs defined by business *processes*
- Security and privacy classifications

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
DA governance is not aligned with BA governance.	DA governance is not aligned with BA governance, but the need for alignment is recognized and the development is being discussed.	Aligned DA governance and BA governance is being developed.	Aligned DA governance and BA governance is defined and validated by directly involved *stakeholders*.	Aligned DA governance and BA governance is established, recognized and used by *stakeholders*.	Aligned DA governance and BA governance is are established as part of business-as-usual practice with a continuous improvement routine.

3.3 Identify the Data

Identifying the data includes: 1) defining the logical data domains; 2) mapping of physical data repositories to the logical data domains; and 3) cataloging the physical data in the repositories.

3.3.1 Logical data domains have been identified, documented, inventoried and authorized

Description

Identification of logical *data domains* must be driven by the business from the perspective of what data is needed to perform the required business *functions*. A logical *data domain* is the representation of a category of data that has been designated and named. Logical *data domains* represent the data, not the databases, which are needed to satisfy the business *process* requirements.

Objectives

- Involve business *process* *subject matter experts* in the identification of the logical *data domains*.

- Identify and prioritize logical *data domains*.
- Structure logical *data domains* to contain the data of the *domains* irrespective of the various organizational structures where the data may be produced organization-wide.

Advice

The overall goal is to ensure the proper use of data and to get *stakeholders* to think about DM in terms of data content concepts and not the physical database repositories. All this needs to be based on an understanding of how the business *functions* operate in reality. Once the logical *data domains* are defined, they must be mapped to their physical locations and associated with *authoritative provisioning points*. The first step, however, is to define the *domains*.

Data domains include both internally generated data as well as externally acquired data. It is imperative that these strategic data assets are identified and inventoried to ensure their proper use in all *data consumer* critical business *processes*.

Questions

→ Have *data domain* owners who are responsible for the quality and availability of the data been identified?
→ Has the business domain owner, as well as the DA *function*, been involved in the designation of the *authoritative data domains*?
→ Have *data domain* *taxonomies* and conceptual/logical *models* been verified by business subject experts?
→ Are all critical business *functions* represented in the discussion?
→ Is the distinction between *data domains* and databases clear?

Artifacts

- *Policy* indicating what *authoritative data domains* are and how they are to be used
- Criteria for determination of *authoritative data domains*
- Inventory of *authoritative data domains*
- List of *stakeholders* and evidence of bi-directional communication

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No logical *data domains* exist.	No logical *data domains* exist, but the need is recognized and the development is being discussed.	Logical *data domains* are being developed.	Logical *data domains* are being defined and validated by directly involved *stakeholders*.	Logical *data domains* are established, recognized and used by *stakeholders*.	Logical *data domains* are established as part of business-as-usual practice with a continuous improvement routine.

3.3.2 Physical repositories of data have been located, documented and inventoried

Description

Underlying the logical *data domains* are physical repositories of data that will map to those logical *data domains*. The physical repositories may include application data, databases, spreadsheets, data warehouses, data lakes, streaming data or data stored in cloud services.

Objectives

- Link underlying physical repositories to the logical *data domains* and record those links.
- Identify repositories that have been inventoried and document that the inventory is actively maintained.

Advice

The *data elements* in any logical *data domain* need to be mapped to where the data physically resides. The first step is the creation of the inventory of data repositories. It doesn't really matter where the content resides. This may be external, streaming, master/slave and cloud-based. What is important is that it is linked to the *authoritative data domains* and enforced. This is not about centralizing the data in a warehouse. It is adequate if the data has a unique repository identified as the known source of the data.

Questions

→ Have the inventories of data repositories been compiled and verified?
→ Have the *authoritative data domains* been mapped to their physical location?
→ Are controls implemented to ensure namespace integrity and accessibility?
→ Has *policy* been drafted, verified and sanctioned on the use of *authoritative provisioning points*?

Artifacts

- Inventory of data repositories and authoritative distribution points
- Mapping of *authoritative data domains* to the physical location
- *Policy* statements on the use of *authoritative provisioning points*

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
There is no inventory of physical repositories of data.	There is no inventory of physical repositories of data, but the need is recognized and the development is being discussed.	The inventory of physical repositories of data is being developed.	The inventory of physical repositories of data is defined and validated by directly involved *stakeholders*.	The inventory of physical repositories of data is established, recognized and used by *stakeholders*.	The inventory of physical repositories of data is established as part of business-as-usual practice with a continuous improvement routine.

3.3.3 Physical data has been cataloged

Description

After the physical repositories of data aligned to the *data domains* are established, the next step is to catalog the physical data in the repositories.

Objectives

- Establish a catalog of *data elements* aligned to the *data domain*.
- Capture basic *metadata* on the *data elements*.
- Make the data catalog accessible to *stakeholders*.

Advice

The *data elements* aligned to a *data domain* must have basic *metadata* captured including but not limited to source, term name, term definition, field name and field location. The basic *metadata* is required to make the data accessible. Any *data consumer* and particularly the data *analytics* consumers will need this *metadata* as part of their discovery *process* in advance of defining data for production usage. As data is prioritized based on business needs and business *process* criticality the data *policy* will require a more comprehensive set of business and *technical metadata* to be captured.

Questions

→ Have the *data elements* aligned to a *data domain* been cataloged?
→ Has basic *metadata* been captured on the *data elements*?
→ Is the *metadata* accessible to *stakeholders* in a data catalog?

Artifacts

- Data *policy* for basic *metadata* capture on *data elements* aligned to a *data domain*
- Data Catalog

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
There is no catalog of physical data.	There is no catalog of physical data, but the need is recognized and the development is being discussed.	The catalog of physical data is being developed.	The catalog of physical data is defined and validated by directly involved *stakeholders*.	The catalog of physical data is established, recognized and used by *stakeholders*.	The catalog of physical data is established as part of business-as-usual practice with a continuous improvement routine.

3.4 Define the Data

Defining the data includes: 1) defining and documenting the conceptual and logical models; 2) establishing the business process definition of the data; and 3) use of taxonomies to establish relationships between the data.

3.4.1 Enterprise entities are identified, defined, modeled and standardized

Description

Conceptual and logical *models* for all *enterprise data domains must be* defined and documented. Alignment to the models must be required by policy and integrated into the *enterprise* change management policies.

Objectives

- Define and document conceptual and *logical data models*.
- Verify conceptual and *logical data models* with key *stakeholders*.
- Capture data object relationships and document them into domain ontologies.
- Verify *authoritative data domains* with business *subject matter experts*.
- Publish *authoritative data domain taxonomies* and demonstrate use by upstream/downstream systems.
- Align and cross-reference internal *taxonomies* to global *standards*.

Advice

Conceptual *models* define how business *processes* work in the real world. Logical *models* organize the data in into manageable *domains*. *Taxonomies* define hierarchical relationships. *Taxonomies* are critical to establishing a common definition and language of data organization-wide and are required to ensure the data's proper use. Establishing these models should be done independent of the future physical instantiation of the data, however, issues related to relational data design vs. *semantic* (i.e.: graph or OWL design) should be considered strategically in designing the *enterprise* entity data.

Once the models are designated, they need to be managed and required by *policy* to ensure that they are implemented, maintained and used. Alignment to *data domain taxonomies* and conceptual *models* should be formally mandated by the organization's change management *policies* including change approvals, impact analysis and controlled implementation.

Data repositories contain data that represents real concepts required in the business *processes*. The data have terms, define characteristics, express conditions, define triggers, specify requirements and translate activities in the business *process*. The goal is the creation of a single conceptual view of data that defines how business concepts and *processes* work in the real world. The conceptual model is used to express the requirements for data in *business terms*, based on the business *process* activities at the most granular level needed in the business *process*. The logical *data model* is used to define a logical set of data to align to a *data domain*. The scope of the logical *data set* can be validated by whether there is a natural subject matter expertise to support the *data set*. If the scope of the logical *data set* is to broad it will require desperate subject matter expertise and be inefficient to manage.

Granular data is often used to manufacture derived concepts to manage the objectives of the business *processes*. Derived concepts can sometimes be very complex and be manufactured from many *data elements* that may have variation if the data comes from multiple sources. Ensuring that the data is aligned with common meaning is an essential requirement for achieving automation, performing advanced *analytics* and generating trusted reports. This is one of the essential goals of the DM initiative and the building block of most business *processes*. Without shared meaning and transparency about how derived concepts are created, the organization will have challenges unraveling interconnections, managing complexity and most importantly using data to drive innovation.

Questions

- Have conceptual and *logical data models* defining the organization-wide *data domains* been verified by business subject experts?
- Are data *taxonomies* and *models* documented, made accessible and used in existing and new systems?
- Have *policies* and *standards* for managing *data models* and *taxonomies* been defined, verified, sanctioned and made accessible?
- Has governance over *taxonomies* been aligned with existing change management *policies*?
- Has the model of terms, definitions, and relationships been verified by business *stakeholders* and stored as *metadata*?
- Are there mechanisms for access such as glossaries that can be used as reference points for implementation?

Artifacts

- *Policy* and *standards* on use and maintenance of *data models* and *taxonomies*
- Mapping and transformation to ensure implementation by upstream and downstream systems
- Business conceptual terms, definitions and relationships

- Agreement on business meaning verified by *stakeholders*
- *Data models* and *taxonomies* recorded in *metadata* repository
- *Metadata* repository content accessible to *stakeholders*
- List of *stakeholders* and evidence of bi-directional communication

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
Enterprise data entities are not defined, modeled or *standardized*.	*Enterprise* data entities are not defined, modeled or *standardized*, but the need is recognized and the development is being discussed.	*Enterprise* data entities are being defined, modeled and *standardized*.	*Enterprise* data entities are defined, modeled and *standardized*, and validated by directly involved *stakeholders*.	*Enterprise* data entities are established, recognized and used by *stakeholders*. *Enterprise*-level *data models* are recognized and used by *stakeholders*.	*Enterprise* data entities and data modelling are established as part of business-as-usual practice with a continuous improvement routine.

3.4.2 Business definitions are composed, documented and approved

Description

Business definitions must be developed as non-technical descriptions of *data attributes* that are based on contractual, legal and/or business facts.

Objectives

- Document business definitions and verify them with *stakeholders*.
- Assign approved business definitions to defined *taxonomies*, which are fully attributed conceptual *models*.

Advice

The precise meaning of data gets convoluted as data is moved around, copied and renamed. This is a problem because most organizations are run by business applications. Applications are driven by software– each with their own unique data model. And all these *models* use glossaries as core factors of input. Meaning is often aligned with the specific software to make sure it works. It is not aligned across all applications and not harmonized across the organization. This creates circumstances where organizations use the same terms to mean different things and refer to identical things using different terms. These problems can be exacerbated when aligning front office to back-office *processes* because terms used in the front office don't always communicate critical nuances that are needed to meet legal obligations in the back-office. These definitional differences create problems with integration and make it difficult to unravel complex business calculations or reuse data across new applications. The goal is the agreement on the meaning of data terms in the context of how they are used.

Questions

→ Has the business meaning of atomic and derived terms been defined and verified?

→ Have legal and compliance representatives been involved in the legal language used to define business concepts?

Artifacts

- *Business glossaries*
- Complete front-to-back *stakeholder* engagement
- List of *stakeholders* and evidence of bi-directional communication

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
Business definitions are not documented.	Business definitions are not documented, but the need is recognized and the development is being discussed.	Business definitions are being developed.	Business definitions are defined and validated by directly involved *stakeholders*.	Business definitions are established, recognized and used by *stakeholders*.	Business definitions are established as part of business-as-usual practice with a continuous improvement routine. Business definitions are reviewed and updated at least annually.

3.4.3 Unique identification and classification are defined, applied and in use

Description

Data identification, classification and *taxonomy* schemes and methodologies must be used to ensure precise organization of data. Establishing these schemes and methodologies are critical to defining how things relate, establishing *standard* treatment of data organization-wide and for aggregating data for analytical purposes. Unique and precise identifications, classifications and *taxonomies* are a foundational concept and is a required aspect for regulatory reporting, risk analysis and other internal *analytics*.

Objectives

- Define identifiers for *critical business elements*.
- Assign and publish internal entity IDs and use them across business *processes*.
- Align and cross-reference internal IDs to industry *standard* identifiers.

Advice

Data identification schemes are required for data factors of input such as Customer ID, Legal Entity ID and Product ID.

Data classification schemes are required for data factors such as privacy treatment, info-security treatment, masking, encryption and risk analysis.

Data *taxonomy* schemes define how things relate. *Taxonomies* define the relationship of *Business Elements* within a *data domain*. *Taxonomies* are critical to establishing a common definition and language of data organization-wide and are required to ensure the data's proper use.

Standard identification, classification and *taxonomy* schemes need to be mapped to any proprietary identifiers used in consuming applications. Unique identification is a core foundational tenet of DM that is governed by *policy* and enforced by *standards*.

Policies, *procedures*, and *standards* are needed to ensure the appropriate assignment, use, and maintenance of identification, classification and *taxonomy* schemes. The creation of these schemes requires participation from business, data, technology and legal and compliance *stakeholders*. In many cases, compliance *policies* may already exist, but they may not be integrated into the appropriate use or Software Development Lifecycle (SDLC) and/or change management *processes* within an organization.

Establishing these schemes and methodologies are critical to defining how things relate, establishing *standard* treatment of data organization-wide and for aggregating data for analytical purposes. Unique identifications, classifications and *taxonomies* are a foundational concept and is a required aspect for regulatory reporting, risk analysis and other internal *analytics*.

Identification schema for instruments, entities, customers, and products need to be unique and precise. *Standard* identifiers need to be mapped to any proprietary identifiers used in applications consuming data. Unique identification is a core foundational tenet of DM that must be governed by *policy* and enforced by *standards*. The scope and value of advanced *analytics* is very dependent on the *standard* identification schema.

Questions

→ Have unique and precise identification schema been established for all instruments, entities, customers, and products?
→ Has *policy* been developed and approved to ensure these identifiers are used in business applications?
→ Have *standard* identifiers been published and cross-referenced to any proprietary identifiers?

Artifacts

- *Policy* about *standard* identifiers
- Inventory of identification *standards* being used
- Cross-referencing and transformation documentation

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
There are no identification, classification and *taxonomy* schemes.	There are no identification, classification and *taxonomy* schemes, but the need is recognized and the development is being discussed.	Identification, classification and *taxonomy* schemes are being developed.	Identification, classification and *taxonomy* schemes are defined and validated by directly involved *stakeholders*.	Identification, classification and *taxonomy* schemes are established, recognized and used by *stakeholders*.	Identification, classification and *taxonomy* schemes are established as part of business-as-usual practice with a continuous improvement routine. The schemes are reviewed and updated at least annually.

3.4.4 Metadata is defined, modeled and standardized

Description

An organization-wide *standard* *metadata* *model* is required. The *metadata* is the *data domain* that is owned and managed by the DM initiative. The same DM initiative requirements for managing any *data domain* are applied to the DM *data domain*.

Objectives

- Define and implement the *metadata* *model* for the *data domain* owned and managed by the DM initiative.
- Include all data required as input and output of the DM business *process* in the *metadata* *model*.
- Manage the DM *data domain* according to the data *policy* and *standards*.

Advice

The DM *data domain* includes all *metadata* defined as requirements for data as input and output of the DM initiative business *processes*. To execute *standard* DM *processes* the *metadata* in the *processes* must align to a *standard* *metadata* *model*. An additional benefit of *standard* *metadata* is the interoperability across all the *data domains* as they interact. And finally, the *standard* *metadata* creates the ability to support, as appropriate, the implementation of DM *processes* centrally in the organization.

Another significant application of *metadata* is the *data classification* schema that allows identification of data flagged for issues such as privacy, sensitive data, consent requirements, data-subject rights and the ethical use, access and outcome of data.

The types of *metadata* are many and may include business, operational, technical, descriptive, structural, administrative.

Questions

→ Is the DM *data domain* defined with a *metadata* *model*?
→ Does the *metadata* *model* include all data required by the DM business *processes*?
→ Does the *metadata* *model* include a comprehensive *data classification* schema?

Artifacts

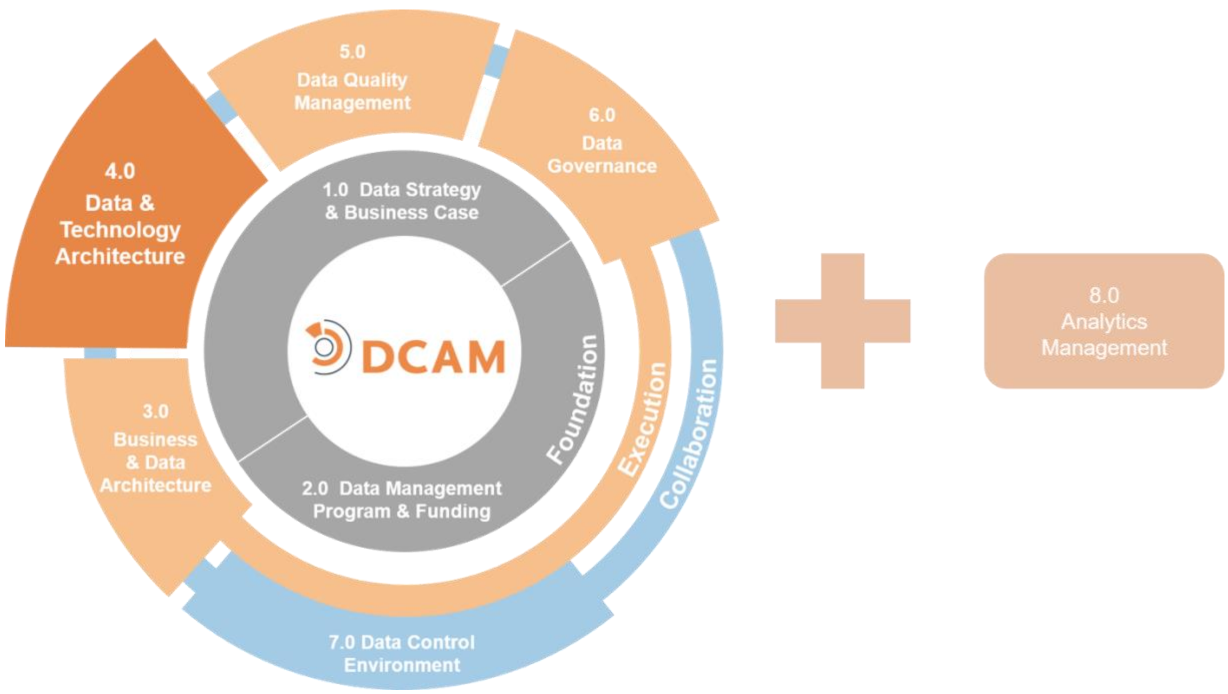
- DM *data domain* included in the organization *data domain* structure
- *Metadata* *model* defined for the DM *data domain*
- Data classification schema

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
There is no *metadata model*.	There is no *metadata model*, but the need is recognized and the development is being discussed.	The *metadata model* is being developed.	The *metadata model* is defined and validated by directly involved *stakeholders*.	The *metadata model* is established, recognized and used by *stakeholders*.	The *metadata model* is established as part of business-as-usual practice with a continuous improvement routine. The *metadata model* is reviewed and updated at least annually.

DCAM® Framework:

4.0 Data & Technology Architecture



4.0 Data & Technology Architecture

Introduction

The requirements for data as well as for data management (DM), are interpreted through data architecture. This interpretation defines the requirements to design the physical data consumed, produced and provisioned by the business *process*. Data architecture is a bridge between the requirements for data of the business *process* and the physical execution of that data in technology infrastructure.

> ***Technology Architecture*** *is the strategy and execution of how the physical infrastructure is designed to support the business and data needs of the organization.*

Technology architecture refers to the strategy, design and implementation of the technology infrastructure which supports the defined business and data architecture. *Technology architecture* defines the platforms and the tools and how they need to be designed for maximum efficiency to support the requirements of the business *process*, data and DM. The purpose of *technology architecture* is to support the business *process* and define how data is physically acquired, moved, persisted and distributed in a streamlined and efficient manner. Physical data proximity, bandwidth, *processing* time, backup, recovery and archiving are some of the important elements of a mature *technology architecture*.

The efficient and effective movement of data is critical to business operations. *Technology architecture* determines how data, tools and platforms operate in collaboration to satisfy business requirements. The proper alignment of these components dictates application efficiency and system *processing* speed. This enables organizations to control costs and achieve infrastructure scalability and elasticity which are characteristic of an organization that is designed for long-term implementation success.

The roadmaps to achieve architecture integration across business, data and technology need to be aligned on a path to the target-state. The roadmaps define the governance and controls that are needed to ensure compliance across the organization.

Definition

The **Data & Technology Architecture (TA)** component is a set of capabilities to align the architectural requirements of the business, data, and technology across the organization to support the desired business *process* outcomes. These outcomes include the DM *processes* and required technology infrastructure.

Scope

- Work with DM *Project Management Office* (PMO) to design and implement sustainable business-as-usual *processes* and tools for the integration of business and *technology architecture* with data architecture.
- Ensure DM *function* alignment with business, data and *technology architecture* and strategy.
- Ensure that the DM governance is aligned to both business and technology governance activities.

Value Proposition

Organizations that effectively deliver *technology architecture* aligned to business and data architecture get a return on investment from several areas:

- A simplified environment allows an organization to be nimble to market, agile in response to regulatory changes and faster to innovate
- Tool selection and implementation are simplified, aligned to business *process* and data requirements which reduces complexity and cost
- A data storage strategy is developed consistent with the objectives of business while it controls cost and risk
- Operational risk architecture ensures the continuous flow of data to critical business *functions* in the event of an outage incident

Overview

Information architecture is the combination of data architecture and *technology architecture*. *Business architecture* or data architecture should not dictate technology. The technology infrastructure of an organization is the responsibility of the technology *function*. However, the requirements of business architecture and data architecture inform technology. Data architecture captures the information requirements of the business *process* and translates them into the ***what, where and when*** of data: ***what*** data is needed; ***where*** is it to be delivered, and by ***when***. *Technology architecture* is the enabler and defines the plan and roadmap for implementation.

There are four areas of *technology architecture* that are critical to a successful DM initiative.

- **Database Platforms:** *Technology architecture* defines acceptable data platforms for *enterprise* use. *Enterprise*-class database platforms, appliance technologies, distributed computing, and in-memory solutions all need to be defined, communicated and governed by *technology architecture*.
- **Tools:** Often one of the biggest expenses and source of inconsistent handling of data is the proliferation of multiple, disparate DM technology tools within an organization. *Technology architecture* must define the allowable tool stacks – what Business Intelligence (BI) tools, extract, transform & load (ETL) tools, and various discovery tools are permitted for use within the organization.
- **Storage Strategy:** *Technology architecture* must define how organizations will store and maintain its data. Several aspects of the target-state storage strategy are the determination of how to maintain data and control data storage. This includes decisions about the use of internal versus external cloud technology, how data will be archived and retained, and how data will be defensibly removed and destroyed from the organization's infrastructure.
- **Operational Risk Planning:** A sound *technology architecture* addresses operational risk, business continuity, and disaster recovery strategies. Data is the lifeblood of an organization and needs proper planning to ensure that *data flows* to all parts of the organization even in the face of events that interrupt business continuity.

Finally, all the above aspects of a sound *technology architecture* must be supported by a strong technology governance *operating model* integrated with the governance of business and data architecture. *Policies* must be in place, agreed to by business, data and technology *stakeholders*, supported by executive management, and subject to internal audit scrutiny and adherence. Without integrated governance over business, data and *technology architecture*, infrastructure and tools will grow independently of each other. This uncontrolled infrastructure will result in inefficiencies and security issues putting *data quality* (DQ) and the organization at risk.

Core Questions

→ Is *technology architecture* driven by business *process* and data requirements?
→ Are *policies* in place to integrate the governance of business, data and *technology architecture*s?

→ Are governance *procedures* in place to ensure adherence?

→ Are design reviews in place and required to ensure enhancements and new development utilize *standard* business, data and *technology architecture* definitions?

→ Is adherence to data architecture *standards* auditable?

Core Artifacts

The following are the **core artifacts** required to execute an effective Data & Technology Architecture capability. Items with an '*' link to published best practice *guidelines*.

- Data & *Technology Architecture* *Policy* Alignment Matrix
- Data Management Tool Stack Roadmap
- Storage Criteria, Strategy, & Roadmap
- Tool Selection Strategy Document

4.1 Technology Architecture (TA) is defined in support of the data management initiative

The Technology *function* must work in collaboration with the DM initiative. Together they define the organization-wide data platform, storage and distribution infrastructures. This collaboration must be supported by mutual governance and policy.

4.1.1 DM is engaged in the Technology vision and strategy

Description

The role of the Technology *function* is to define and design the architecture needed to accommodate data requirements in collaboration with business *process* requirements. The Technology *function* must work in collaboration with the DM initiative. Together they define the database strategies, *analytics platform* approaches, middleware solutions, storage and retention technologies, information security considerations, and all other aspects of the holistic technology infrastructure needed to support the DM goals and objectives of the organization.

Objectives

- Design an integrated TA strategy. Obtain agreement from business, data and technology senior executive *stakeholders* and socialize the strategy.
- Develop the TA strategy to include the review of business objective requirements that can be supported by innovative technology tools and methods. Key among these are Machine Learning (ML) and Artificial Intelligence (AI).
- Support and enforce the integrated architecture strategy using the Internal Audit *policy*.

Advice

The technology vision must ensure that the DM initiative can be supported by the Technology *function*. The Technology *function* runs the technology infrastructure. It does not define data functionality or requirements. Evaluate the TA strategy with DM objectives at the forefront.

An obligation of the data architecture (DA) and TA *functions* is to evaluate the business objectives and inform the business senior executives of opportunities to deploy innovative technology tools and methods like ML/AI. The review should include opportunities to support the DM initiative objectives and *processes* using these tools and methods.

Questions

→ What are the mechanisms to ensure a formal, ongoing partnership between the Office of Data Management (ODM) and the Technology *function*?

→ Have the DA and TA *functions* evaluated and communicated opportunities to deploy innovative technical tools and methods in support of the defined business objectives?

Artifacts

- TA strategy alignment with DM strategy, inclusive of the DA strategy
- List of *stakeholders* and evidence of bi-directional communication

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
DM is not engaged in the Technology vision and strategy.	DM is not engaged in the Technology vision and strategy, but the need for engagement is recognized and the development is being discussed.	The approach to engaging DM in the Technology vision and strategy is being developed.	The approach to engaging DM in the Technology vision and strategy is defined and has been validated by the directly involved *stakeholders*.	DM is engaged in the Technology vision and strategy, and this is recognized by *stakeholders*.	DM engagement in the Technology vision and strategy is established. The nature of engagement is reviewed and updated at least annually.

4.1.2 DM is engaged in the definition and development of the organization-wide platform infrastructure

Description

The DM initiative must be engaged with TA in the platform infrastructure to define and execute a strategy, an execution roadmap and a governance structure. For a technology strategy to be sustainable, it must have a budget commitment over the life of the designed roadmap.

Objectives

- Design an integrated platform infrastructure strategy. Obtain agreement from business, data, and technology senior executive *stakeholders*.
- Develop an integrated platform infrastructure roadmap.
- Develop budgets and gain approval. Integrate the budget into the organization's budget *processes*.
- Build and make operational an integrated platform infrastructure governance system and *policies*.
- Ensure that all enhancements and new developments are subject to a review and approval consistent with the defined platform infrastructure strategy and roadmap.
- Obtain the backing of Internal Audit for the platform infrastructure strategy.

Advice

TA is accountable for the platform infrastructure strategy, the execution roadmap and. The strategy must be coordinated with business requirements for the Database Management Systems (DBMS) organization-wide. The ODM should be a facilitator, providing coordination among the *stakeholders* to represent these business considerations. Prioritization is critical.

Some opportunities for innovation in the organization-wide platform infrastructure involve new technology tools and methods. The infrastructure may benefit from the addition of AI, ML, Natural Language Processing (NLP), knowledge graphs and big data.

Questions

→ What are the mechanisms to ensure coordination among business, data, and technology at the proper levels of the organization?
→ Are legal and compliance involved in these decisions?
→ Are additional legal requirements considered?
→ What are the mechanisms for coordination and prioritization?

Artifacts

- Platform infrastructure requirements, strategy, and roadmap
- Evidence of alignment with the DM strategy
- Evidence of alignment, socialization, and approval
- DBMS consideration represented in *policy* and *standards*
- Evidence of DBMS consideration in Software Development Lifecycle (SDLC) tollgates at the onset of technology platform enhancements or new development

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
DM is not engaged in organization-wide platform infrastructure matters.	DM is not engaged in organization-wide platform infrastructure matters, but the need for engagement is recognized and the development is being discussed.	The approach to engaging DM in organization-wide platform infrastructure matters is being developed.	The approach to engaging DM in organization-wide platform infrastructure matters is defined and has been validated by the directly involved *stakeholders*.	DM is engaged in organization-wide platform infrastructure matters, including definition, roadmap development and approval, and this is recognized by *stakeholders*.	DM engagement in organization-wide platform infrastructure matters is established. The nature of engagement is reviewed and updated at least annually.

4.1.3 DM is engaged in the definition and development of the organization-wide data storage infrastructure

Description

The DM initiative must be engaged with TA in the data storage infrastructure to define and execute a strategy, an execution roadmap and a governance structure. For a technology strategy to be sustainable, it must have a budget commitment over the life of the designed roadmap.

Objectives

- Design an integrated storage management strategy. Obtain agreement from business, data, and technology senior executive *stakeholders*.
- Develop an integrated data storage roadmap.
- Develop budgets and gain approval. Integrate the budget into the organization's budget *processes*.
- Build and make operational a storage management infrastructure governance system and *policies*.
- Ensure that all enhancements and new developments are subject to a review and approval consistent with the defined storage management strategy and roadmap
- Obtain the backing of Internal Audit for the storage management strategy.

Advice

Data storage management strategy, execution roadmap and governance are each a TA *function*. The strategy must be coordinated with business requirements for areas like archiving capability, legal and compliance considerations, retention and defensible destruction of data. It is important to extend the storage strategy to incorporate *sandbox environments*. The ODM should be a facilitator, providing coordination among the *stakeholders* to represent these business considerations. Additionally, data storage, archive, and retrieval plans must be coordinated across the various *stakeholders* (business, data, technology, legal, and compliance). Prioritization is critical.

Some opportunities for innovation in the organization-wide data storage infrastructure involve new technology tools and methods. Consider the roles Hadoop, cloud and on-prem/off-prem-Data might play in building new infrastructure. There are also, new opportunities for data to be encrypted, tokenized, anonymized, or pseudonymized at rest, in transit and in storage.

Questions

→ What are the mechanisms to ensure coordination among business, data, and technology at the proper levels of the organization?
→ Are legal and compliance involved in these decisions?
→ Are additional legal requirements considered in the strategy including but not limited to masking, anonymization, and the full complement of personal privacy rights?
→ What is the organization's appetite for cloud data storage services?
→ How will the organization manage data reconstruction from an archive?
→ What are the mechanisms for coordination and prioritization?

Artifacts

- Storage requirements, strategy, and roadmap
- Evidence of alignment with the DM strategy
- Evidence of alignment, socialization, and approval
- Storage consideration represented in *policy* and *standards*
- Evidence of storage consideration in SDLC tollgates at the onset of technology platform enhancements or new development

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
DM is not engaged in organization-wide data storage infrastructure matters.	DM is not engaged in organization-wide data storage infrastructure matters, but the need for engagement is recognized and the development is being discussed.	The approach to engaging DM in organization-wide data storage infrastructure matters is being developed.	The approach to engaging DM in organization-wide data storage infrastructure matters is defined and has been validated by the directly involved *stakeholders*.	DM is engaged in organization-wide data storage infrastructure matters, including definition, roadmap development and approval, and this is recognized by *stakeholders*.	DM engagement in organization-wide data storage infrastructure matters is established. The nature of engagement is reviewed and updated at least annually.

4.1.4 DM is engaged in the definition and development of the organization-wide data distribution infrastructure

Description

The DM initiative must be engaged with TA in the data distribution infrastructure to define and execute a strategy, an execution roadmap and a governance structure. For a technology strategy to be sustainable, it must have a budget commitment over the life of the designed roadmap.

Objectives

- Design an integrated data distribution strategy. Obtain agreement from business, data, and technology senior executive *stakeholders*.
- Develop a data distribution infrastructure roadmap.
- Develop budgets and gain approval. Integrate the budget into the organization's budget *processes*.
- Build and make operational a data distribution infrastructure governance system and *policies*.
- Ensure that all enhancements and new developments are subject to a review and approval consistent with the defined data distribution strategy and roadmap.
- Obtain the backing of Internal Audit for the data distribution strategy.

Advice

TA is accountable for a data distribution infrastructure strategy, execution roadmap and governance structure. The strategy must be coordinated with business requirements for areas like data access controls, information security restrictions, authoritative data *provisioning points* and data as a service (DaaS). The ODM should be a facilitator – providing coordination among the *stakeholders* to represent these business considerations. Prioritization is critical.

Some opportunities for innovation in the organization-wide data distribution involve new technology tools and methods. Consider the roles DaaS–API/Technology; web-based access; macros for power users and various BI tools might play in building new infrastructure.

Questions

→ What are the mechanisms to ensure coordination among business, data, and technology at the proper levels of the organization?
→ Are legal and compliance involved in these decisions?
→ What is the organization's appetite for data distribution as a service?
→ What are the mechanisms for coordination and prioritization?

Artifacts

- Platform infrastructure requirements, strategy, and roadmap
- Evidence of alignment with the DM strategy
- Evidence of alignment, socialization, and approval
- Data distribution infrastructure represented in *policy* and *standards*
- Evidence of data distribution infrastructure consideration in SDLC tollgates at the onset of technology platform enhancements or new development

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
DM is not engaged in organization-wide data distribution infrastructure matters.	DM is not engaged in organization-wide data distribution infrastructure matters, but the need for engagement is recognized and the development is being discussed.	The approach to engaging DM in organization-wide data distribution infrastructure matters is being developed.	The approach to engaging DM in organization-wide data distribution infrastructure matters is defined and has been validated by the directly involved *stakeholders*.	DM is engaged in organization-wide data distribution infrastructure matters, including definition, roadmap development and approval, and this is recognized by *stakeholders*.	DM engagement in organization-wide data distribution infrastructure matters is established. The nature of engagement is reviewed and updated at least annually.

4.1.5 DM governance is aligned with TA governance

Description

The DM governance must be engaged with TA governance to align with the overall technology vision and strategy along with the specific strategies for organization-wide platform, data storage and data distribution infrastructures. Alignment to TA governance *processes* may include design reviews, permit to build, permit to use and permit to send approvals.

Objectives

- Develop an integrated governance structure and *policies* in alignment with the TA and DA strategies.
- Ensure all technology development meets the TA *policy* requirement to follow DA *policy* and *standards*.
- Submit all enhancements and new developments for architectural platform design review and approval.

Advice

Technology governance is the responsibility of the technology group. Because of the close relationship between the DM initiative and technology implementation, it is imperative that the DM *function* collaborate with the technology group. Together they create, support and enforce technology *policy* and *standards* that impact the DM initiative.

This collaboration with technology is designed to ensure alignment with business *processes*, compliance with data restrictions and harmonization with both technical and architectural *standards*. This will include design reviews to ensure that technology implementation follows DA *policy* and *standards* and that toll gates are in place to review the technical implementation.

TA governance guarantees adherence to declared platform and tool *standards*. The objective is to ensure that existing TA governance *policies* reflect the goals of the DM initiative.

The emerging AI and ML tools also have potential application to enhancing the DM *processes* and should be considered in the overall DM technology tool stack planning.

Questions

→ Is TA governance aligned with DA governance?
→ Is DA involved in the technical design review *process*?

→ Are authorizations to build, use and send data in place?
→ Are legal and compliance involved in this effort?
→ Is TA *policy* aligned with the DM strategy?

Artifacts

- Evidence of governance collaboration such as *stakeholder* meeting agendas and results, escalation *procedures*
- Evidence of toll gate review *process*, for example, minutes and meeting outcomes
- TA *policy* and DA *policy* reference the other without duplication
- List of *stakeholders* and evidence of bi-directional communication on technical review and authorizations

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
DM governance is not aligned with TA governance.	DM governance is not aligned with TA governance, but the need for alignment is recognized and the development is being discussed.	Aligned DM governance and TA governance is being developed.	Aligned DM governance and TA governance is defined and validated by directly involved *stakeholders*.	Aligned DM governance and TA governance is established, recognized and used by *stakeholders*.	Aligned DM governance and TA governance is are established as part of business-as-usual practice with a continuous improvement routine.

4.2 DM Technology Tool Stack is Identified and Governed

The DM business process must define requirements for the DM technology tool stack. Execution roadmaps and governance ensure the tools are adopted.

4.2.1 DM technology tool selection strategy is defined and verified by stakeholders

Description

The DM initiative is just like any other business *process*. Technology requirements must be defined by the DM business *process* and the requirements for data from the business *process*. The DM executive is accountable for the requirements for technology and technology is responsible for a DM technology strategy that delivers against those requirements.

Objectives

- Design an integrated DM technology tool strategy. Obtain agreement on the strategy from business, data and technology senior executive *stakeholders*. Socialize the DM technology tool strategy.
- Demonstrate support for the DM technology tool strategy by corporate *policy*. Engage with Internal Audit for enforcement.

Advice

Ensure that there is a defined DM technology tool selection strategy and that it is aligned with the DM initiative. The ODM must ensure that the DM initiative can support the use of various tools (i.e., modeling, ETL, *metadata*, glossary, DQ, *analytics*). This support capability requires coordination between business, data, and technology *stakeholders* to define the DM initiative business *process* and data requirements for the DM tool selection.

Questions

→ What are the mechanisms for coordination between the technology tool selection *process* and DM?
→ Are the *policies*, *procedures*, and *processes* that govern this relationship defined and verified?

Artifacts

- Tool selection strategy document
- List of *stakeholders* and evidence of bi-directional communication

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
There is no DM technology tool selection strategy.	There is no DM technology tool selection strategy, but the need is recognized and the development is being discussed.	The DM technology tool selection strategy is being developed.	The DM technology tool selection strategy is defined and validated by directly involved *stakeholders*.	The DM technology tool selection strategy is established, recognized and followed by *stakeholders*.	The DM technology tool selection strategy is established as part of business-as-usual practice with a continuous improvement routine. The strategy is reviewed and updated at least annually.

4.2.2 DM Technology tool roadmap is developed and implemented

Description

A technology tool roadmap must be developed to execute the DM technology infrastructure defined in the tool selection strategy.

Objectives

- Develop an integrated technology tool roadmap in adherence to the technology tool strategy. Include *guidelines* for new development as well as decommission plans for non-*standardized* legacy tool implementations.
- Develop budgets aligned to the organization's budget cycle *processes*. Move the budget through the *process* to approval.

Advice

Once the tool selection strategy is defined, it needs to be converted into an actionable implementation roadmap. Make sure the technology roadmap is practical, aligned to business priorities and harmonized with internal procurement *processes*.

Questions

→ Is the DM technology tool roadmap aligned with internal procurement *processes*?

→ Is the DM technology tool roadmap shared with the ODM?

Artifacts

- Technology tool roadmap document
- Evidence of alignment with the DM strategy
- Documented approvals

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
There is no technology tool roadmap.	There is no technology tool roadmap, but the need is recognized and the development is being discussed.	The technology tool roadmap is being developed.	The technology tool roadmap is defined and validated by directly involved *stakeholders*.	The technology tool roadmap is established, recognized and followed by *stakeholders*.	The technology tool roadmap is established as part of business-as-usual practice with a continuous improvement routine. The roadmap is reviewed and updated at least annually.

4.2.3 DM technology tool governance is integrated into Data Governance (DG)

Description

Technology, in partnership with the DM initiative, must define and govern the DM related technology stack. Data tools include but are not limited to data discovery tools, DQ tools, *data profiling* tools, *metadata* tools, lineage tools, BI tools and *data governance* tools.

Objectives

- Put in place an operational, integrated technology tool governance structure with a complete set of associated *policies*. Confirm that the structure is in alignment with the DM strategy.
- Get a review and approval of all enhancements and new development projects to ensure alignment with technology tool selection. Technology governs the permissible technology stack for related data tools.
- Implement data tool governance structure and complete operational roll-out across all technology development teams.

Advice

Ensure the governance structure for DM technology tools is in place and operational, and that the governance *policies* and practices incorporate the DM initiative requirements. Different tools that perform the same *functions* can produce disparate data. Beyond this, the proliferation of tools can increase complexity, add cost and inhibit systems integration. However, the business needs the flexibility to acquire the best tools for their objectives. The challenge is, the ODM must manage the disparate data.

Questions

→ What are the mechanisms for coordination between the technology tool selection governance, and DM?
→ How is the organization working with innovation teams to keep abreast of new technology capabilities?

→ Is there a mechanism for collaboration between DM and data tool selection?
→ Is there an agreement between technology and business regarding the scope and controls associated with technology tools?

Artifacts

- Technology tool governance structure document
- Evidence of governance operation such as meetings and *procedures*, documents
- *Policy* and *standards* for data tool governance which are documented and verified
- List of authorized tools
- Bi-directional communication between business, data and technology on tool selection and criteria for approval

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
DM technology tool governance is not integrated into DG.	DM technology tool governance is not integrated into DG, but the need is recognized and the development is being discussed.	The approach to integrating DM technology tool governance into DG is being developed.	The approach to integrating DM technology tool governance into DG is defined and validated by directly involved *stakeholders*.	DM technology tool governance is integrated into DG, and this is recognized by *stakeholders*.	DM technology tool governance integration into DG is established. The nature of the integration is reviewed and updated at least annually.

4.3 Operational Risk Planning is in Place

The DM governance structure must be in alignment with operational risk governance and engaged in the contingency planning and testing for data access and maintenance in the event of an operational disruption.

4.3.1 Operational risk governance structure and processes are in place and implemented

Description

The DM initiative governance structure must be in alignment with operational risk governance. The data *policy* must align with the operational risk *policy* without creating duplication in the *policies*.

Objectives

- Develop an integrated operational risk governance structure and *policies* and make them operational.
- Approve operational risk plans for all enhancements and new development through project tollgates.
- Bring Internal Audit in to approve the operational risk planning.

Advice

Operational risk routines such as disaster recovery, Business Continuity Planning, cyber-threats, degrees of interconnectedness, testing and dependency planning must be formalized for all data systems. Changes to existing systems and new development must be evaluated against the operational risk *guidelines*. Operational risk will also be driven by regulation. Governance oversight should ensure compliance in advance of an internal audit or regulatory examinations.

Questions

→ Is operational risk governance formalized and aligned with the organization's risk and escalation plans?
→ Is operational risk governance aligned with the DM governance mechanisms?

Artifacts

- Disaster recovery governance mechanisms
- Evidence of disaster recovery planning collaboration with compliance

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
There are no operational risk governance structure and *processes*.	There are no operational risk governance structure and *processes*, but the need is recognized and the development is being discussed.	Operational risk governance structure and *processes* are being developed.	Operational risk governance structure and *processes* are defined and validated by directly involved *stakeholders*.	Operational risk governance structure and *processes* are established and are recognized and followed by *stakeholders*.	Operational risk governance structure and *processes* are established as part of business-as-usual practice with a continuous improvement routine. *Processes* are reviewed and updated at least annually.

4.3.2 Data infrastructure contingency planning is defined and in place

Description

The DM initiative, while not directly accountable, must be engaged in the contingency planning and testing for data access and maintenance in the event of an operational disruption.

Objectives

- Develop an integrated technology operational risk management strategy. Obtain agreement from business, data, and technology senior executive *stakeholders*.
- Engage Internal Audit to review and support the integrated technology operational risk management strategy.

Advice

Operational risk requirements for data must be considered early in the SDLC. These requirements will define the data architecture's choice of data storage applications and data recovery strategies. When evaluating a data storage application, keep in mind applications may hold data from multiple *data domains*. The most stringent data recovery requirements might be applied to all data in the application. The more stringent data recovery has a higher cost so this may inflate storage cost for all the data.

Questions

→ What are the mechanisms to ensure collaboration between DM and operational risk contingency planning?

→ Are all data dependencies defined and understood for recovery?

Artifacts

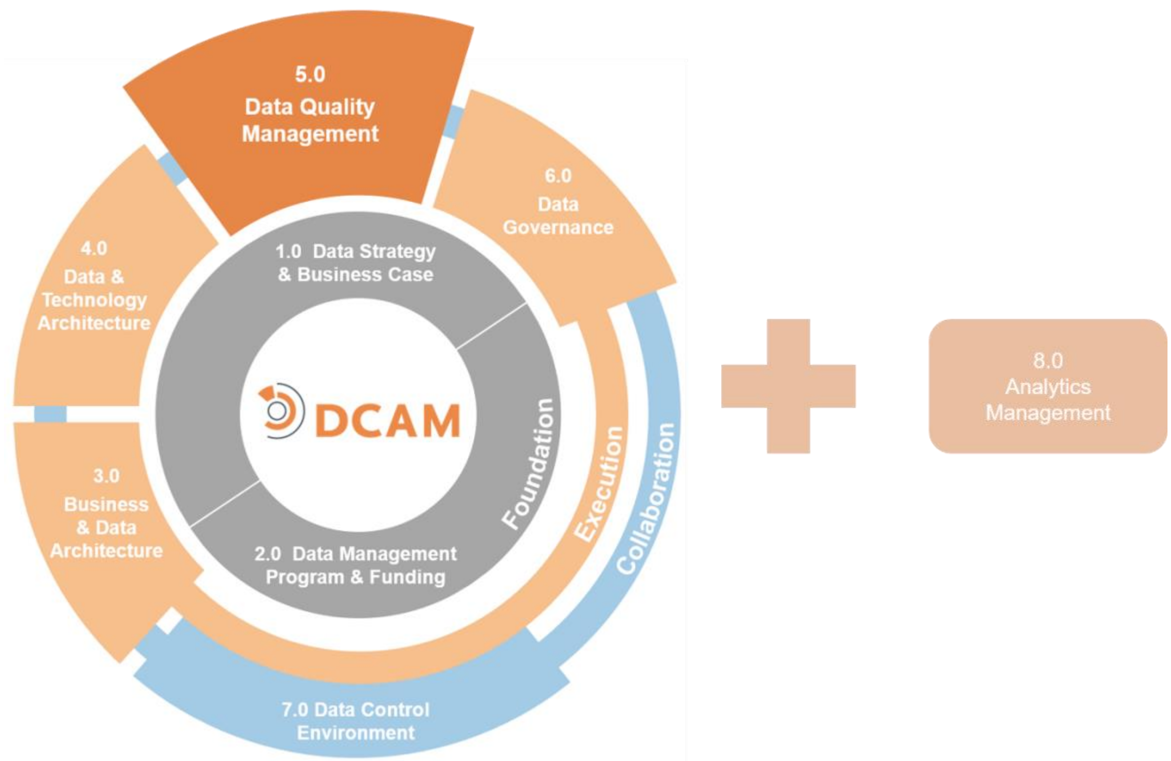
- Disaster Recovery Plan
- Tests and results of Disaster Recovery Plan

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
There is no data infrastructure contingency planning.	There is no data infrastructure contingency planning, but the need is recognized and the development is being discussed.	Data infrastructure contingency planning is being developed.	Data infrastructure contingency planning is defined and validated by directly involved *stakeholders*.	Data infrastructure contingency planning is established and is recognized and followed by *stakeholders*.	Data infrastructure contingency planning is established as part of business-as-usual practice with a continuous improvement routine. Data infrastructure contingency planning is tested, reviewed and updated at least annually.

DCAM® Framework:

5.0 Data Quality Management



5.0 Data Quality Management

Introduction

The **Data Quality Management** *function* defines the goals, approaches and plans of action that ensure data content is of sufficient quality to support defined business and strategic objectives of the organization. The *function* should be developed in alignment with business objectives, measured against defined *data quality* (DQ) *dimensions* and based on an analysis of the current state of DQ. Data Quality Management is a series of *processes* across the full data supply chain to ensure that the data provisioned meets the needs of its intended consumers.

DQ requires an understanding of how data is sourced, defined, transformed, provisioned and consumed. DQ is not a *process* itself but describes the degree in which data is fit-for-purpose for a given business *process* or operation.

Definition

The **Data Quality Management (DQM)** component is a set of capabilities to define *data profiling*, DQ measurement, defect management, root cause analysis and data remediation. These capabilities allow the organization to execute *processes* across the data control environment ensuring that data is fit for its intended purpose.

Scope

- Establish a DQM *function* within the Office of Data Management (ODM).
- Work with data management (DM) Program Management Office (PMO) to design and implement sustainable business-as-usual *processes* and tools for DQM.
- Execute DQM *processes* against business-critical data. DQM *processes* include profiling & grading, measurement, defect management, root cause fix, remediation.
- Establish DQ metrics and reporting routines.
- Ensure that the DQM governance is integrated into the Data Governance (DG).

Value Proposition

Organizations that build, formalize and assign DQ responsibilities into daily routine and methodology achieve a sustainable organization-wide data culture.

Organizations that effectively implement Data Quality Management and achieve the appropriate level of DQ across the *data ecosystem* get a return on investment from several areas:

- Better risk management
- Enhanced *analytics*
- Better client service and product innovation
- Improved operational efficiencies

Overview

DQ is a broad conceptual term that needs to be understood in the context of how data is intended to be used. Perfect data is not always a viable objective. The quality of the data needs to be defined in terms that are relevant to the *data consumers* to ensure that it is fit for its intended purpose. The overall goal of DM is to ensure that *data consumers* have confidence in the data they receive. These consumers are using this data to support their business *functions*. For them to make accurate decisions the data must reflect the facts the data is designed to represent without the need for reconciliation or manual transformation.

The organization needs to develop a DQM strategy and establish the overall plans for managing the integrity and relevance of its data. One of the essential objectives is to create a shared culture of DQ stemming from executive management and integrated throughout the operations of the organization. To achieve this cultural shift, the organization must agree on both requirements and the measurement of DQ that can be applied across multiple business *functions* and applications. This will enable business sponsors, *data producers*, *data consumers* and technology stakeholders to link DQ management *processes* with objectives.

DQ can be segmented into dimensions:

- ***Accuracy***: the relationship of the content with original intent
- ***Completeness***: the availability of required *data attributes*
- ***Coverage***: the availability of required data records
- ***Conformity***: alignment of data content with required *standards*
- ***Consistency***: how well the data complies with the required formats/definitions
- ***Timeliness***: the currency of content representation as well as whether the data is available/can be used when needed
- ***Uniqueness***: the degree that no record or attribute is recorded more than once

The identification and prioritization of *data quality dimensions* foster effective communication about DQ expectations and are an essential prerequisite of the DM initiative.

Creating a profile of the current state of DQ is an important aspect of the overall DQM *function*. A new profile should be created periodically when data is transformed. The goal is to assess patterns in the data as well as to identify anomalies and commonalities as a baseline of what is currently stored in databases and how actual values may differ from expected values. Once the data profile is established, the organization needs to evaluate the data against the quality tolerances and thresholds defined by the DQ requirements. The evaluation also examines business requirements to validate that the data is fit-for-purpose.

The purpose of this evaluation *process* is to measure the quality of the most important business attributes of the existing data and to determine what content needs remediation. A responsibility of the *data producer* and *data consumer* is to identify the data that is critical to the *data consumer's* business *process*. Prioritizing the data based on criticality then informs the DQM *function* which attributes require a heightened level of control and quality review. The designation of criticality requires that the highest level of *accuracy* and DQ treatment is applied. The assessment *process* identifies the data that needs to be cleansed to meet *data consumer* requirements. Data cleansing should be performed against a predefined set of *business rules* to identify defects that can be linked to operational *processes*.

Data cleansing should be performed as close to the point of capture as possible. There should be clear accountability and a defined strategy for data cleansing to ensure that cleansing rules are known and to avoid duplicate cleansing *processes* at multiple points in the *data lifecycle*. The overall goal is to clean data once at the point of data capture based on verifiable documentation and *business rules* as well as to fix the *processes* that allowed defective data into the system at the root cause. Data corrections must be communicated to, and aligned with, all downstream repositories and upstream systems. It is important to have a consistent and documented *process* for issue escalation and change verification for both *data producers* and data vendors.

It is also important to ensure that data meets quality *standards* throughout the lifecycle so that it can be integrated into operational data stores. This aspect of the DQ management *process* is about the identification of data that is missing, determination of data that needs to be enriched and the validation of data against internal *standards* to prevent data errors before data is propagated into production environments.

For DQ to be sustained, a strong governance structure with the highest level of organizational support from senior executive management must be in place. This supports the DQM activities and ensures compliance to DQ *processes*. DQ *processes* need to be documented, operationalized and routinely validated via DM reviews and formal audit *processes*.

DQ cannot be achieved through central control. Organization-wide DQ requires the commitment and participation of a broad set of *stakeholders*. DQ is the result of a series of business *processes* creating a data supply chain. Therefore, *stakeholders*, along that chain must be in place, authorized and held responsible for the quality of data as it flows through their respective areas. DQ requires coordinated organizational support. DQM *processes* and objectives must be part of the operational culture of an organization for it to be sustained and successful.

Core Questions

→ Is it understood that poor quality data is an indication of a broken business *process* or technology?
→ Is it understood that instituting a DQ system is a cultural shift that touches all aspects of business, operations and technology *processes*?
→ Is the required training in place to sustain the DQM *function*?
→ Are the necessary people and funding resources earmarked to implement and operate the DQM *function*?
→ Are the necessary resources in place to provide organization-wide training to support a sustainable, DQ cultural change?

Core Artifacts

The following are the **core artifacts** required to execute an effective Data Quality Management capability. Items with an '*' link to published best practice *guidelines*.

- *Critical Data Element* Criteria
- *Data Profiling* Methodology
- Data Quality Dimensions Framework*
- Data Quality Metrics & Dashboards
- *Data Quality Rules* Inventory
- Defect Management Methodology
- Root Cause Analysis Methodology

5.1 Data Quality Management (DQM) is Established

The DQM function strategy and approach must be defined and approved by stakeholders. Roles and responsibilities across the stakeholders must be established with operational processes in place and auditable.

5.1.1 The DQM strategy and approach are defined and adopted

Description

The strategy and approach must be defined for the DQM *function* and reflect the related vision and objectives of the Data Management Strategy (DMS). Once established, it must be formally empowered by senior management and its role communicated to all *stakeholders*.

Objectives

- Formally establish the DQM strategy and approach within the organization.
- Get approval of the DQM strategy and approach from *stakeholders*.
- Ensure alignment of *stakeholder* plans and roadmaps with the DQM strategy and approach.
- Obtain executive management support for the DQM strategy.
- Communicate the role of the DQM *function* across the organization through formal channels.
- Operate the DQM *function* collaboratively with DM initiative *stakeholders*.
- Secure authority to enforce DQM compliance through *policy* and documented procedure.

Advice

The DQM strategy and approach encompasses the ***what, how and who*** of DQ. It needs to address ***what*** scope of data is to be scrutinized and reviewed; ***how*** the DQ assessments will be performed with metrics defined; and ***who*** will be responsible with defined roles. DQM needs to be closely aligned with the organization's business objectives to ensure that the most important data is properly maintained and monitored. DQM involves cultural change. It is critical that a documented DQM strategy and approach is socialized with business, data and technology *stakeholders* to ensure awareness, support and commitment.

The rapidly evolving focus on data ethics is introducing new requirements for the DQM *function*. These requirements include an ethical review as part of determining that data produced is fit-for-purpose. Additionally, DQM is one of the areas where the use of Machine Learning (ML) and Artificial Intelligence (AI) may assist in the *processes* used to achieve quality data. These requirements and opportunities should be evaluated in the strategy and approach of the DQM *function*.

Alignment of the DQM strategy and roadmap to the DMS vision and objectives is achieved by agreement between the operating level *data officer* and the individual responsible for delivering the DG *function*. The operating level *data officer* is accountable for establishing priorities across each of the Framework Component requirements.

Questions

→ Has the DQM *function* been formally established?
→ Is there a DQM strategy and approach in place?
→ Is the DQM strategy and roadmap aligned to the DMS?
→ Have innovative technologies such as ML and AI been considered as part of the DQM *process* and infrastructure?
→ Has the review of data ethics been included in the DQM strategy and approach?

→ Has the DQM *function* been formally communicated to business, technology, operations, finance and risk *stakeholders*?
→ How has executive management demonstrated its support?
→ Has authority been granted to the DQM *function* to implement and enforce best practice via *policy* and *standards*?
→ Has authority been communicated to *stakeholders*?
→ Is there a functional partnership in place with Internal Audit?

Artifacts

- The DQM plan
- Description of the roles and responsibilities of the DQM *function*
- Communication of specific support from executive management with distribution lists
- *Policies* and *procedures* associated with executing and enforcing DQM
- Bi-directional engagement with *stakeholders* on the DQM *function* authority

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No formal DQM strategy exists.	No formal DQM strategy exists, but the need is recognized and the development is being discussed.	The formal DQM strategy is being developed.	The formal DQM strategy is defined and has been validated by the directly involved *stakeholders*.	The formal DQM strategy is established and understood across the organization and is being followed by the *stakeholders*.	The formal DQM strategy is established as part of business-as-usual practice with a continuous improvement routine. The strategy and approach are reviewed and updated at least annually.

5.1.2 The DQM stakeholder roles and responsibilities are defined and implemented

Description

DQM requires a network of data stewards and *subject matter experts* to ensure data is properly captured, processed and delivered. Accountable parties must be identified and the roles and responsibilities must be clearly communicated.

Objectives

- Define and communicate the roles and responsibilities of the DQM *function*.
- Fund and staff the DQM *function*.
- Ensure and enforce alignment of activities and projects to *policy* and *standards* through the authority of the DQM *function*.
- Hold individuals accountable for the DQM performance via annual reviews and compensation considerations.

Advice

DQM involves numerous *stakeholders* who are responsible for data requirement capture, *data profiling*, remediation, definitions, *metadata*, transformation, root cause analysis, entitlement control and coordination across the full *data ecosystem*. These efforts involve the assignment and empowerment of owners, stewards, curators and custodians. These accountable parties need to be at the right levels of seniority as well as understand all the internal *processes* associated with DQM.

With the addition of a data ethics review in the DQM *process*, subject matter expertise will be required either through the addition of experts or appropriate training of the data stewards. Similarly, to the extent that ML and AI are used to support the DQM *process*, additional skills will need to be added or developed within the *stakeholders*.

Questions

→ Has the DQM *function* been established?
→ Is the DQM *function* appropriately staffed and funded?
→ Does the DQM *function* have the authority needed to be effective?
→ Have the roles and responsibilities of the DQM *function* been defined, documented and socialized?
→ Have the skills for data ethics review and execution of ML and AI tools been added or developed within the *stakeholders*?
→ Have milestones and metrics associated with the DQM *function* been established?

Artifacts

- Evidence of *stakeholder* identification
- *RACI matrix* or other evidence of accountability assignment
- Description of the roles and responsibilities of the DQM *function*
- Staff assignments and qualifications
- Evidence of accountability linked to performance reviews and compensation
- Gap analysis of skills needed and in place
- List of *stakeholders* and evidence of bi-directional communication

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No formal DQM roles & responsibilities exist.	No formal DQM roles & responsibilities exist, but the need is recognized and the development is being discussed.	The formal DQM roles & responsibilities are being developed.	The DQM roles & responsibilities are defined and have been validated by the directly involved *stakeholders*.	The DQM roles & responsibilities are established and are recognized and used by *stakeholders*.	The DQM roles & responsibilities are established as part of business-as-usual practice with a continuous improvement routine. The roles & responsibilities are reviewed and updated at least annually.

5.1.3 The DQM processes are defined and operational

Description

Formal *processes* must be established for the activities of the DQM *function*. These *processes* align with the DM *policy* and *standards* of the organization and include *procedures*, tools and routines. The routines are required for steady-state operations.

Objectives

- Establish formal DQM *processes* in alignment with the DM *policy* and *standards*.
- Integrate the DQM *processes* into the overall end-to-end *processes* of the DM initiative.
- Identify, schedule and maintain DCE routines, meetings and working sessions required for operational support.

Advice

The DQM *subject matter experts* should work with the business *process* design and optimization service within the Data Management Program (DMP) team. Together they will create and monitor the implementation of the DQM *processes* in alignment to the end-to-end *process* across the full DM initiative.

The DQM *process* design should include the requirements for ethical review as part of determining the data is fit-for-purpose. The design should also incorporate ML and AI into the *process* if included in the DQM strategy and approach.

Questions

→ Have formal *processes* been defined and implemented?
→ Are the *procedures*, tools and routines in place for implementing the *processes*?
→ Have innovative technologies such as AI and ML been considered as part of the DQM *process* and infrastructure?
→ Has the review of data ethics been included in the DQM strategy and approach?
→ Are DQM activities part of the normal operational routine of *stakeholders*?
→ Are there standing meetings, planning sessions and regular communications about data initiatives?

Artifacts

- *Process* design artifacts, procedure guides and published routines
- *Process* performance metrics reports
- Meeting minutes, status reports and DMP announcements

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No formal DQM operational *processes* exist.	No formal DQM operational *processes* exist, but the need is recognized and the development is being discussed.	The DQM operational *processes* are being developed.	The DQM operational *processes* are defined and have been validated by the directly involved *stakeholders*.	The DQM operational *processes* are established and are recognized and used by *stakeholders*.	The DQM operational *processes* are established as part of business-as-usual practice with a continuous improvement routine.

5.1.4 The DQM processes are auditable

Description

DQ auditing must occur on three levels:

> **Quality Assurance (QA):** the accountable business performs self-assessments based on defined DQ thresholds, *processes*, and objectives.
>
> **Quality Control (QC):** the DM initiative performs a facilitated audit of the accountable business' DQ and *processes* and is empowered to force the business to remediate any gaps found to ensure adherence to DQ thresholds and *standards*.
>
> **Internal Audit:** the accountable business' DQ and *processes* are subject to audits. Failure to satisfy this review may result in formal escalated audit findings written against the business.

Objectives

- Data Stewards have performed self-assessment of the accountable DQ and processes (QA).
- The DM initiative has performed facilitated assessments of the accountable business DQ and *processes* (QC).
- The DM initiative is empowered to force business teams to remediate gaps found in the operational DQ *processes*.
- Internal Audit performs routine examinations of the accountable business DQ and *processes*.
- Formal audit Issues are generated if operational gaps are uncovered.

Advice

DQ *processes*, validation, root cause analysis, remediation, etc., should be routinely audited. Audit occurs on three levels. First, self-attestation – where the *stakeholders* evaluate and assert they are following the *data quality rules*. Second, through the Office of Data Management (ODM) – where the DM initiative works with *stakeholders* to validate compliance. Third, through internal review where Internal Audit has formally validated that *processes* are being followed.

Questions

→ What are the mechanisms to ensure validation, root cause analysis, and remediation?
→ Is Internal Audit involved in DQM?

Artifacts

- Evidence of self-attestation and *enterprise* ODM review
- Evidence of Internal Audit engagement and review

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
There is no oversight of the DQM *processes*.	DQM oversight strategies and approaches are being discussed.	Three levels of *data quality* review are being defined.	Three levels of *data quality* review have been identified and are being shared with *stakeholders* for review and approval.	Three levels of *data quality* review have been implemented.	The *process* to ensure the auditability of DQM *processes* has a routine in place to identify opportunities for continuous improvement.

5.2 Data is Profiled and Measured

Profiling and measuring the data includes: 1) prioritizing the data in scope based on criticality and materiality; 2) defining and testing data quality rules based on business rules; and 3) measuring that the data is fit-for-purpose.

5.2.1 Data has been identified and prioritized

Description

The data in scope as defined by the business objectives must be prioritized based on its criticality and *materiality* to the *data consumer* business *process*.

Objectives

- Define a *process* for prioritizing data.
- Identify the scope of data subject to DQM, both current and historical.
- Prioritize the scope of data in alignment with the DMS and business priorities.

Advice

An organization may establish data prioritization tiers. The DM *policy* and *standards* should define levels of *data control* to apply to each prioritization tier. The highest-level tier is a *critical data element* (CDE). Designated CDEs receive the highest-level of control to ensure the quality of these attributes is maintained. CDE designation is a controlled *process* to achieve agreement between the *data producer* and *data consumer*.

Questions

→ Has the *process* to prioritize data been defined?
→ Has the scope of data subject to DQM been identified, prioritized and verified?

Artifacts

- Prioritized *data domain* inventories
- Prioritized CDE inventory
- Bi-directional communication about the inventories

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
Data subject to DQM has not been identified or prioritized.	The scope of data subject to DQM is being discussed. The concept of CDEs is being debated.	The scope of data subject to DQM is being identified and shared with *stakeholders*. CDEs are being defined.	The scope of data subject to DQM is prioritized and aligned with both strategy and business priorities. CDEs are verified.	The scope of data subject to DQM is approved. CDEs are designated and actively maintained.	The *process* to identify and prioritize all relevant data has a routine in place to identify opportunities for continuous improvement.

5.2.2 Data quality rules are defined and tested

Description

Data quality rules based on *business rules* must be defined and tested to confidently validate the data is *fit-for-use*.

Objectives

- Define a *process* for the development of *data quality rules*.
- Define *business rules* which can be interpreted into *data quality rules* and used to measure the quality of data.
- Define a *process* for the testing of *data quality rules*.
- Establish an environment and tool set for the running and testing of rules.
- Socialize DQ rules and test output to *stakeholders*.

Advice

Business rules are the basis for developing *data quality rules* necessary to profile the quality of the data. The *data quality dimensions* establish a range of potential rules that may be needed to determine DQ. A critical part of defining quality rules is to test the rule outcome. Testing is an iterative activity during the design of an individual rule. Testing and re-testing with each rule refinement is an essential activity for identification of the range of rules across the *data quality dimensions*. These dimensions are required for the *data quality rule*-set to accurately measure the DQ.

Data quality rules should be developed to test the various dimensions of quality. Not every *data element* can be tested for each dimension. There is an art and a science to writing quality rules. The rules and the range of rules will evolve over time. As new quality defects surface they will guide the design of new rules to detect those issues in the future. A mature *data quality rule* set is a coveted asset of an organization.

The ability to test *data quality rules* requires a testing environment. This may be a data sandbox where the data stewards can play with the data and experiment with various rules and their output. Additionally, the proper technical infrastructure to write, store, run and analyze rules is needed.

The skill set required to test rules may include business *process* subject matter expertise, DQ management expertise and technical infrastructure and coding expertise. Make sure you assess the skills and available resources carefully as this range of skill may not be available through a single individual.

Questions

→ Is there a defined *process* for the design and testing of *data quality rules*?

→ Are the *data quality rules* based on defined *business rules*?
→ Is there a data sandbox and appropriate tools for running *data quality rule* tests?
→ Have the *stakeholders* validated that the range of *data quality dimensions* applied in the rule set is adequate to determine the DQ?

Artifacts

- *Process* design artifacts, *procedure* guides and published routines
- Documented *data quality dimensions*
- Criteria used to evaluate DQ
- *Data quality rules* repository
- *Data quality rules* recorded as *metadata*
- Testing result reports and dashboards
- Testing tools
- Testing environment
- List of *stakeholders* and evidence of bi-directional communication

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No DQ rules or testing capability exist.	No DQ rules or testing capability exist, but the need is recognized and the development is being discussed.	DQ rules and testing capability are being developed.	DQ rules and testing capability have been defined and validated by directly involved *stakeholders*.	DQ rules and testing capability are established and are recognized and used by *stakeholders*.	DQ rules and testing capability are established as part of business-as-usual practice with a continuous improvement routine.

5.2.3 The data is profiled, analyzed and graded

Description

The in-scope data must be profiled to determine the full spectrum of *data quality dimensions* (i.e., *accuracy*, *completeness*, *coverage*, *conformity*, *consistency*, *timeliness*, *uniqueness*). This analysis must include both a row-based analysis examining the *accuracy* of the record and a column-based, statistical analysis. *Metadata* must also be reviewed to ensure the description and intended use of data is properly defined.

Objectives

- Define a *process* for profiling, analyzing and grading data.
- Profile and statistically analyze in-scope data.
- Review the *metadata* and perform gap analysis.
- Measure, monitor and grade in-scope data.
- Capture DQ metrics on a routine basis.
- Report DQ metrics to business, data and technology *stakeholders*.

Advice

The DQM *function* is to establish that the data is fit-for-purpose and can be trusted. *Data profiling* creates a quality benchmark for the organization. Evidence of *data profiling* will be expected in any Internal Audit review or regulatory examination. Data needs to be assessed against both fit-for-purpose criteria and the *data quality dimensions*. DQ *business rules* need to be defined and memorialized. A statistical and columnar analysis should be included to ensure that data is reasonable. Certain *data domain* types, such as time-series data, need to be evaluated against additional criteria like gaps, spikes and abnormalities.

The primary stakeholders involved in this *process* are the *data producer* and *data consumer*. Ultimately quality is defined by the business *process* requirements of the *data consumer* and should be formally agreed to by the *data producer*. Creating a *standard* and automated *process* for routinely executing the quality metrics and reporting the results is critical to meet the time constraints of the data supply chain.

Metrics are used to track DQ and drive data remediation efforts. Control points along the data supply chain capture DQ metrics that are used to produce DQ dashboards. The requirements of the *data consumer* are used to establish quality thresholds for the data. These thresholds permit the grading of the data as to the defined levels of acceptable DQ based on the minimal requirements of specific *data consumer*.

The *data profiling*, analyzing and grading *process* should include a periodic review of the ethical use and outcome of the data as part of determining fit-for-purpose of the data. Particular attention should be paid to the use of proxy values in a *data set*.

A mechanism for executing DQ rules and generating outcome reports are required to support the *data profiling*, analyzing and grading *process*. The use of AI and ML may assist in the *process*.

Questions

- → Has the in-scope data been profiled, analyzed and graded?
- → Is DQ profiled against business logic rules as well as for reasonableness against statistical expectations?
- → Are the right business, operational, analytical, data and technical *stakeholders* involved in the *process*?
- → Have innovative technologies such as ML and AI been considered as part of the *process* infrastructure?
- → Has the review of data ethics been included in the *process*?
- → Are *standard* criteria for measuring DQ defined and verified?
- → Are metrics being collected and reported on a routine basis?
- → Are the results of data measurement and grading captured as *metadata*?

Artifacts

- *Business rules* and *data profiling* and measurement criteria
- Statistical analysis results
- A mechanism for assigning and reporting grades for DQ
- DQ metric reports, dashboards, heat maps and other forms of output
- List of *stakeholders* and evidence of bi-directional communication

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
Data is not profiled, analyzed or graded for the purpose of assessing DQ.	Data is not profiled, analyzed or graded for the purpose of assessing DQ, but the need is recognized and the development is being discussed.	*Data profiling*, analysis and grading, for the purpose of assessing DQ, is being developed.	*Data profiling*, analysis and grading, for the purpose of assessing DQ, has been defined and validated by directly involved *stakeholders*.	*Data profiling*, analysis and grading, for the purpose of assessing DQ, is established and conducted by *stakeholders*.	*Data profiling*, analysis and grading, for the purpose of assessing DQ, is established as part of business-as-usual practice with a continuous improvement routine. It is recognized as the normal way of working.

5.3 DQ Issues are Remediated

Data remediation plans must be developed and executed to resolve the most pressing DQ issues. The remediation must include both correcting the existing data and performing root-cause-fix to eliminate future data defects.

5.3.1 Data remediation has been prioritized, planned and actioned

Description

Based on the current state analysis, remediation plans must be developed to address the most pressing DQ issues. Ongoing DQ evaluation and maintenance and timelines must also be established.

Objectives

- Define a *process* for prioritizing and executing data remediation.
- Develop and prioritize data remediation plans.
- Prepare for immediate action to deal with high priority data remediation.
- Establish timelines for ongoing remediation.

Advice

Data remediation is about correcting the defective data that has been identified. This data should be corrected as close to the source of data capture as possible. Make sure the remediation activities are not one-off *processes*, but rather established as part of the DQM routine. Data remediation needs to be implemented for both *data-at-rest* and *data-in-motion*.

Questions

→ Is a DQ issue prioritization *process* in place?
→ Have data remediation plans been developed, verified and prioritized?
→ Has appropriate funding been allocated?
→ Is there a communications *process* related to data remediation?

Artifacts

- DQ defect reports
- Data remediation plan
- Evidence of issue prioritization
- Evidence of remediation being accomplished
- List of *stakeholders* and evidence of bi-directional communication

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
Data remediation is not prioritized, planned and actioned.	Data remediation is not prioritized, planned and actioned, but the need is recognized and the development is being discussed.	Data remediation prioritization, planning and actioning is being developed.	Data remediation prioritization, planning and actioning has been defined and validated by directly involved *stakeholders*.	Data remediation prioritization, planning and actioning is established, recognized and used by *stakeholders*.	Data remediation prioritization, planning and actioning is established as part of business-as-usual practice with a continuous improvement routine. It is recognized as the normal way of working.

5.3.2 Root-cause analysis (RCA) process is defined

Description

Data remediation must include both correcting the existing data that is defective and determining the root-cause of the DQ deterioration to avoid the reoccurrence of defective data in the future.

Objectives

- Define a *process* for conducting root-cause analysis and fixes.
- Determine the data defect root cause.
- Identify and implement corrective measures to business, data and/or technology *processes*.

Advice

Remediating DQ issues is not merely an exercise in data correction. DQ issues can be systemic. Evaluate the depth and breadth of DQ to determine if the organization is focused more on tactical repair versus the upstream remediation of a root-cause fix. A strong reporting structure is needed to ensure that upstream systems are aware of repetitive or continuing DQ problems.

Data defects may have a people, *process*, data or technical source. Having the right subject matter expertise from each of these areas will be important to the analysis of the root-cause.

Questions

→ Are root cause analysis problems defined?
→ Are corrective measures linked to root cause analysis?

Artifacts

- Evidence of DQ defect reporting across the data supply chain
- Evidence of root cause analysis and remediation being performed

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No Root-cause analysis (RCA) *process* is defined.	No RCA *process* is defined, but the need is recognized and the development is being discussed.	The RCA *process* is being developed.	The RCA *process* is been defined and validated by directly involved *stakeholders*.	The RCA *process* is established, recognized and used by *stakeholders*.	The RCA *process* is established as part of business-as-usual practice with a continuous improvement routine. The *process* is reviewed and updated at least annually.

5.4 DQ is Monitored and Maintained

Monitoring and maintaining the data includes: 1) implementing data quality control points; 2) capturing DQ metrics to identify defective data, and 3) continuous monitoring of the data.

5.4.1 DQ control points are in place

Description

Data control points must be developed to quantitatively measure the quality of data as it flows through business and technology *processes*.

Objectives

- Define a *process* to define DQ control points.
- Put DQ control points in place and bring them to a fully operational state along the data supply chain.
- Record DQ controls as *metadata*.

Advice

DQ is governed by developing control points along the data supply chain. DQ control points need to be applied at both the point of data entry into the organization and at the point of entry into the consuming application as well as when data moves and transforms along the supply chain. DQ controls include the implementation of *business rules*, establishing workflows, setting DQ tolerances and monitoring data movement.

Questions

→ Are control points defined, verified and documented?
→ Are *business rules* defined, verified, documented and approved?
→ Are business *process* flows defined and the way they handle exceptions verified?
→ Are control points, *business rules* and *process* flows operational?

Artifacts

- Documentation on control points, *business rules* and *process* flows
- Control *process* review and sign-off

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No DQ control points are defined.	No DQ control points are defined, but the need is recognized and the development is being discussed.	DQ control points are being developed.	DQ control points are defined and validated by directly involved *stakeholders*.	DQ control points are established and recognized by *stakeholders*.	DQ control points are established as part of business-as-usual practice with a continuous improvement routine. Control points are reviewed for relevance and accuracy at least annually and adjusted accordingly.

5.4.2 Data issues are managed

Description

Control points along the data supply chain capture DQ metrics that are used to produce DQ dashboards which are used to identify defective data. The DQ defects must be part of the issue management routine of the DM initiative. The DQ issue management *process* must track an issue to resolution and provide continuous *stakeholder* communication.

Objectives

- Define a *process* for managing data issues to resolution.
- Drive and prioritize remediation efforts using DQ metric reports.
- Establish an issue management reporting routine and infrastructure.

Advice

Stakeholder engagement, inclusive of the *data consumer*, is critical for successful DQ issue management. The issues need to be managed through all stages of resolution. These stages include defect triage, prioritization, root-cause analysis, root-cause fix and remediation of defective data. *Stakeholder* communication throughout this *process* is critical and must include communication with the *data consumer*. The *data consumer* must be made aware of the defective data and its impact on their business *process*. They may need to participate in the analysis and determination of an acceptable resolution.

Important tools to support the resolution *process* are an issue log and a status tracking system. The link to the issue record should be part of the *metadata* for all instances of defective data. This record will communicate to all users across an organization and can be used to help minimize duplication of effort when an issue is uncovered at multiple points along the data supply chain.

Often, particularly in the early stages of the DM initiative, the volume of defective data may be greater than the resources required to resolve the issues. Documenting the prioritization *process* in the issue log even when it results in a backlog of issues is evidence that the issue was known and evaluated rather than new issues being uncovered as part of an audit.

Questions

→ Are DQ metric reports and dashboards distributed on a routine basis?
→ Are metrics used to identify DQ issues and drive remediation?
→ Are the DQ issues captured as *metadata*?
→ Are the right business, operational, analytical, data and technology resources involved in defining DQ requirements?

Artifacts

- DQ dimension metrics
- DQ metric reports, dashboards, heat maps and other forms of output
- List of *stakeholders* and evidence of bi-directional communication

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
Data issues are not managed.	Data issues are not managed, but the need is recognized and the development is being discussed.	Data issue management is being developed.	Data issue management has been defined and validated by directly involved *stakeholders*.	Data issue management is established, recognized and used by *stakeholders*.	Data issue management is established as part of business-as-usual practice with a continuous improvement routine. It is recognized as the normal way of working.

5.4.3 Continuous monitoring is performed

Description

Data is monitored at control points. Control points must be established where data enters a business *process* or when it enters a consuming application. To achieve continuous monitoring the data must be checked anytime there is data entering either type of control point. This monitoring may be real-time, a batch *process* or on demand.

Objectives

- Define a *process* for continuous monitoring of DQ.
- Establish an infrastructure for continuous monitoring of DQ.

Advice

The *process* of continuous monitoring has costs, benefits and operational challenges. Some form of automation is required to achieve continuous monitoring. Legacy systems often are not capable of continuous monitoring. It would be cost prohibitive to add these quality checks at the point of data capture or data use. The challenge is to then define a technical solution that allows execution of the quality checks as close to the point of data capture or load that is an acceptable cost.

Questions

→ Is there a *process* for continuous monitoring of DQ?
→ Is the infrastructure in place to support continuous monitoring of DQ?
→ Are the defined control points and respective *data quality rules* being monitored?

Artifacts

- Schedule of DQ monitoring
- DQ defect reports

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
Continuous monitoring at DQ control points is not performed.	Continuous monitoring at DQ control points is not performed, but the need is recognized and the development is being discussed.	Continuous monitoring at DQ control points is being developed.	Continuous monitoring at DQ control points is defined and validated by directly involved *stakeholders*.	Continuous monitoring at DQ control points is established and recognized by *stakeholders*.	Continuous monitoring at DQ control points is established as part of business-as-usual practice with a continuous improvement routine. It is recognized as the normal way of working.

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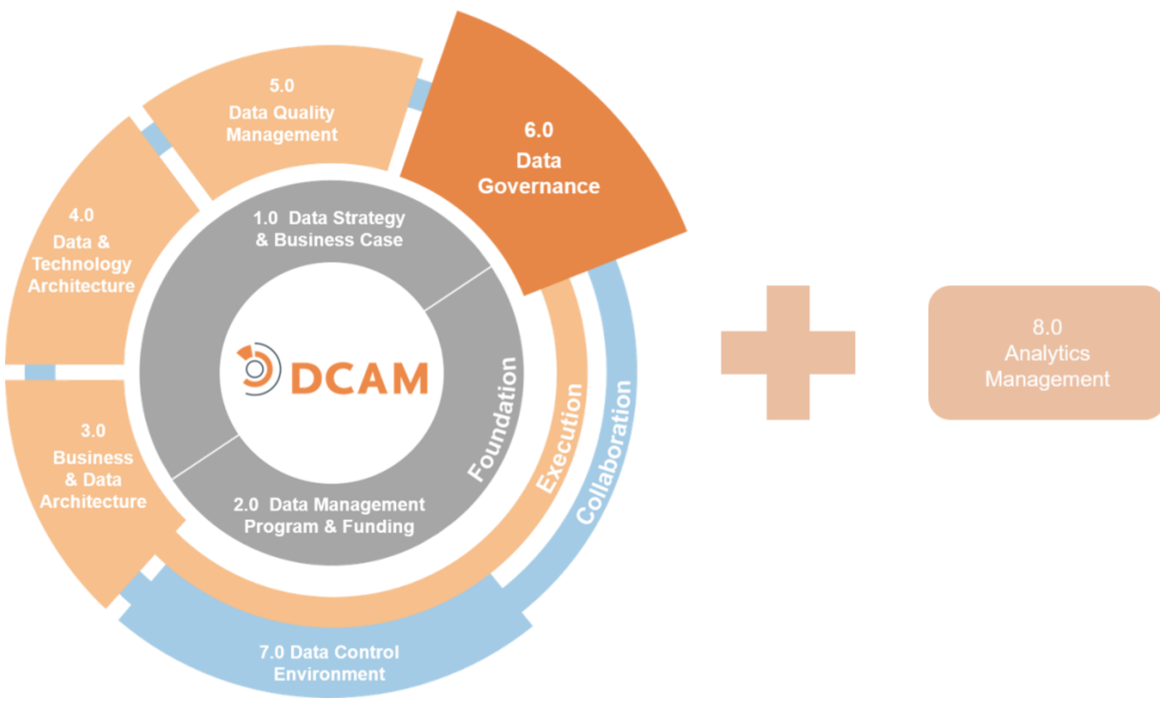
DCAM® Framework:

6.0 Data Governance



6.0 Data Governance

Introduction

Data governance function is the backbone of a successful data management (DM) initiative. *Data governance* is the *process* of setting *standards*, defining rules, establishing *policy* and implementing oversight. It is these steps that ensure adherence to DM best practices. Governance formalizes and empowers the DM initiative to ensure propagation and sustainability throughout the organization.

The purpose of *data governance* is to formalize DM as an established business *function*. *Data governance* establishes the rules of engagement, drives the prioritization of funding and enforces compliance. *Data governance* delineates the *guidelines* for data movement. These movement *guidelines* prescribe how data will be acquired, persisted, distributed, appropriately used, archived and/or defensibly destroyed. *Data governance* formalizes oversight by establishing control *guidelines*, approval *processes* and evaluation of adherence to *policies* and *procedures*. It identifies *stakeholders* and empowers them. *Data governance* ensures that DM *principles* are fully detailed and adoption is achieved. Business, data and technology *functions* are held responsible for the maintenance, quality and proper use of data throughout the organization as part of the *Data governance function*.

Definition

The ***Data Governance*** **(DG)** component is a set of capabilities to codify the structure, lines of authority, roles & responsibilities, escalation protocol, *policy* & *standards*, compliance, and routines to execute *processes* across the data control environment. This ensures authoritative decision making at all levels of the organization.

Scope

- Establish a *data governance function* within the Office of Data Management (ODM).
- Work with DM Program Management Office (PMO) to design and implement sustainable business-as-usual *processes* and tools for *data governance*.
- Define clear roles, responsibilities, and accountabilities for DM resources including those mandated by DM *policy*.
- Define and operate the *data governance* structure with clear lines of authority, responsibility for decision making, engaged *stakeholders*, adequate oversight, issue escalation paths and tracking of remediation activity.
- Develop and oversee adherence to comprehensive and achievable DM *policies*, *standards* and *procedures*, including leading the response to audits.
- Ensure *data governance* function is aligned with other relevant control *function policies*, *procedures*, *standards*, and governance requirements from information security, privacy, *technology architecture* etc.

Value Proposition

Organizations that build, effectively communicate, and enforce DM *policies* assure themselves of lower levels of *enterprise* risk when it comes to DM and data compliance assessments.

Overview

Governance is the key to successful DM. It establishes lines of authority and ensures that the *principles* of DM can and will be implemented. It establishes the mechanisms for *stakeholder* collaboration and defines the organizational structure by which the DM initiative will be governed. The governance infrastructure determines where the initiative resides in the corporate hierarchy, helps manage *stakeholder* expectations, aligns *policies* and *standards* to the organization's mission and values, ensures the adoption of *policies* and *standards*, articulates the mechanism for conflict resolution, ensures adequate funding and sets the methodology for measuring DM progress.

Governance over the DM initiative is multidimensional and includes activities related to each of the DCAM components. And while the most appropriate structure for any individual organization will vary, a clear mission with links to tangible business objectives, as well as a mechanism for realignment, are essential for long-term success. For example, *domain* councils might exist to oversee the intersection of business, data and technology *functions*. Governing boards might be created to establish business data priorities and resolve conflicts. Tactical groups might exist to manage workflow, perform data reconciliation, address quality of *critical data elements*, perform business analysis and provide triage to resolve issues with defective data or outcomes that violate the organization's ethical *standards*. All these aspects need to be linked into an overall framework if governance is going to embed DM concepts into the culture of the organization successfully.

The *data governance* framework establishes the mechanism by which authoritative decisions for data and DM are made across each of the DCAM components. To implement governance, the organization must ensure that the deployment plan will be effective within their business environment. The governance structure should define the in-scope data that is required by the business objectives and establish the DM initiative strategy and approach. It should give authority and required funding for the ODM, establish *policies* and *standards*, make authoritative decisions about DM and data. The governance structure should provide an issues management and escalation procedure. After the initial implementation, the governance framework itself needs to be evaluated, measured and adjusted based on business reality and to ensure that it is fully integrated into business-as-usual *processes*.

Core Questions

- → Have the DM *policies* been defined, developed and validated with key *stakeholders*?
- → Has a governance structure been established with the *stakeholders* identified, charters written and responsibilities assigned?
- → Are there mechanisms in place for issue management, escalation and resolution?
- → Are there mechanisms in place for establishing and resolving prioritization issues among *stakeholders*?
- → Are the appropriate executives identified and engaged?
- → Has the methodology to ensure compliance with established *policies*, *standards* and *processes* across the full *data lifecycle* been defined?
- → Have the metrics been validated by *stakeholder* criteria, aligned with business objectives and collected in a timely manner?
- → Are the metrics actionable and achievable within the organization to improve DM and meet business objectives?

A Note on Data Ethics

Our society generates more than 2.5 quintillion bytes of data per day, which are collected and analyzed using sophisticated algorithms to empower automated decision making. These technologies can improve education, energy efficiency, health, security, and many other aspects of daily life. However, decisions made by algorithms and artificial intelligence cannot account for the complexity and diversity of individual values, such as privacy and equity. Given this reality—and the vulnerability it presents for both organizations and individuals—*enterprise* data management must entail a commitment to ethical data practice that extends beyond routine legal compliance.

Data Ethics involves the study and evaluation of problems related to data, algorithms, and information practices to formulate and support morally good solutions. In other words, Data Ethics answers the question: How *should* we leverage and manage data? Three premises underlie the governance of Data Ethics:

- Algorithms are biased by virtue of their heuristic nature
- Critical reflection is required to interrogate the data
- Each context is unique

As regulation lags behind innovation, organizations must anticipate how lawmakers may address ethical concerns, and proactively establish the governance structures to sustain operations as new legislation is enacted. By aligning governance with the organization's values and initiating the discourse needed to effect changes in organizational culture, *enterprise* data management firms will insulate themselves from ethical liability and be poised to lead the development of ethically sound *standards* and practices.

Core Artifacts

The following are the **core artifacts** required to execute an effective Data Governance capability. Items with an '*' link to published best practice *guidelines*.

- Data Code of Ethics
- Data Governance Operating Model
- Data Management *Policy*
- Data Management Principles
- Data Management Standards
- Data Sharing Agreement

6.1 Data Governance (DG) Function is Established

The DG function strategy and approach, inclusive of the governance organization structure, must be defined and approved by stakeholders. Roles and responsibilities across the stakeholders must be established with operational processes in place.

6.1.1 The DG strategy and approach are defined and adopted

Description

The strategy and approach for the DG *function* must be defined and reflect the related vision and objectives of the Data Management Strategy (DMS). Once established, it must be formally empowered by senior management and its role communicated to all *stakeholders*.

Objectives

- Formally establish the DG strategy and approach within the organization.
- Get approval of the DG strategy and approach from *stakeholders*.
- Ensure alignment of *stakeholder* plans and roadmaps with the DG strategy and approach.
- Obtain executive management support for the DG strategy.
- Communicate the role of the DG *function* across the organization through formal, organizational channels.
- Operate the DG *function* collaboratively with DM initiative *stakeholders*.
- Secure authority to enforce DG compliance through *policy* and documented *procedures*.

Advice

The *data governance* structure creates a mechanism for authoritative decision making about the data and DM initiative. There should be a single structure to govern all aspects of the DM initiative at all of the various levels of the organization. It is important to align decisions to the appropriate level for an authoritative decision while maintaining the required level of subject matter expertise. Resist over-engineering. Position governance as a mission control activity not a judge.

When establishing any governing bodies, limit the scope of their oversight to the expertise of the participants. This limitation will maintain engagement because of the native interest in the topics covered. The DG activity must also align to other governance control *functions* throughout the organization to ensure compliance. As appropriate, *stakeholders* from these other governing areas should be engaged in the DG activity.

DG focuses on the organizational requirements necessary to ensure that the objectives of the DM initiative can and will be implemented. It is critical that the organization understands what they are governing as well as the practical aspects of getting *stakeholders* to alter behavior before seeking to implement a governance structure.

Don't lead with governance details too early in the DM initiative development cycle. DG follows the establishment of the DM initiative and the strategic engagement with the organization's data objectives and challenges. At the strategy level, the primary goal is buy-in to the fact that *data governance* is a mandatory activity – and that it will change the way people operate. Early and interactive engagement with *stakeholders* will help reinforce buy-in. Think of this as *crafting the governance deal* with an appropriate balance between the concepts of governance (clarity on need), the value of governance (coordination and predictability) and the impact of governance (operational and cultural implications).

As an organization formalizes data ethics activities as part of their culture and governance imperative, if it is not fully integrated with the DM initiative, at the very least it must be aligned with the *data governance* *function*.

Alignment of the DG strategy and roadmap to the DMS vision and objectives is achieved by agreement between the operating level *data officer* and the individual responsible for delivering the DG *function*. The operating level *data officer* is accountable for establishing priorities across each of the Framework Component requirements.

Questions

→ Has the DG *function* been formally established?
→ Is there a DG strategy and approach in place?
→ Is the DG strategy and roadmap aligned to the DMS?
→ Has the concept of data ethics been included in the DG strategy and approach?
→ Has the DG *function* been formally communicated to business, technology, operations, finance and risk?
→ How has executive management demonstrated its support?
→ Has authority been granted to the DG *function* to implement and enforce best practice via *policy* and *standards*?
→ Has authority been communicated to *stakeholders*?
→ Is there a functional partnership in place with Internal Audit?

Artifacts

- The *data governance* plan
- *Data governance* structure documentation
- Charter for required governing bodies
- Description of the roles and responsibilities of the DG *function*
- Communication of specific support from executive management with distribution lists
- *Policies* and *procedures* associated with executing and enforcing DG
- Bi-directional engagement with *stakeholders* on the DG *function* authority

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No formal DG strategy exists.	No formal DG strategy exists, but the need is recognized and the development is being discussed.	The formal DG strategy is being developed.	The formal DG strategy is defined and has been validated by the directly involved *stakeholders*.	The formal DG strategy is established and understood across the organization and is being followed by the *stakeholders*.	The formal DG strategy is established as part of business-as-usual practice with a continuous improvement routine.

6.1.2 The governance structure is designed and implemented

Description

The governance structure must align to the operating levels of the organization. In addition to governance at the organization-wide level, Individuals must be appointed in business lines and control *functions* and given the responsibility of DM within those verticals, preferably, reporting into the Chief Operating Officer (COO) or business leader within that group.

Objectives

- The governance structure has been defined, documented and shared with relevant *stakeholders*.

- Operating governance structures have been implemented.
- Working committees are established with written and approved charters.
- *Stakeholders* have been appointed.
- *Stakeholder* roles and responsibilities have been communicated.
- *Stakeholders* are held accountable for their participation in the DM initiative via performance reviews and compensation considerations.

Advice

This is how the DG *processes* will work in reality, including the operating structure, roles, responsibilities and coordination mechanisms. There is no single correct way to define a governance structure. It is dependent on the size of the organization, the scope of the activity, the skill of the staff and the culture of the organization. Developing a new DG mechanism will likely require new skill sets. Collaboration with senior business *stakeholders* for appointment of stewards and with HR for recruiting will help facilitate implementation. Formal training such as DM strategic concepts will help with onboarding.

Questions

→ Has the governance structure been defined and socialized to make sure it is appropriate for the organization?
→ Have the roles, *functions*, and responsibilities been defined and verified?
→ Have potential stewards been identified in collaboration with business *stakeholders*?
→ Is there a succession plan in place?
→ Is there an onboarding and training mechanism to support acclimation to new DM *functions*?

Artifacts

- Governance structure such as organization charts, operating structure, roles, and responsibilities
- *RACI matrix*, or equivalent, denoting accountability
- Operating *procedures* such as how appointments are determined, onboarding and training requirement evidence
- Working groups and committee designations, their charters and participant rosters, minutes and directives
- Bi-directional communication such as *stakeholder* rosters, internal memos, and distribution lists

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No governance structure exists.	The concepts associated with establishing a governance structure are being discussed.	Organization-wide governance structure is being developed. Representatives from involved business lines and control *functions* are participating in the planning *process*.	Organization-wide governance structure has been defined and staffed. Individuals have been informed of their role and responsibilities.	Organization-wide governance structures are implemented. Working committees are operational. *Stakeholders* are held accountable for their participation in the DM initiative.	*Stakeholders* performance reviews and compensation are aligned with the organization governance objectives.

6.1.3 The DG stakeholder roles and responsibilities are defined and implemented

Description

High-level structure and the roles and responsibilities of the DM organization must be established. The roles and responsibilities of the operating units' data executives and data stewards must be addressed in the DMS.

Objectives

- Define and communicate the roles and responsibilities of the DG *function*.
- Fund and staff the DG *function*.
- Ensure and enforce alignment of activities and projects to *policy* and *standards* through the authority of the DG *function*.
- Hold individuals accountable and reward excellence within the DG performance roles through annual reviews.

Advice

Think carefully about how the governance *process* will work in the real world. It is important to evaluate roles and *functions* from all perspectives including that of sponsors with executive authority, owners and other accountable parties. Both business stewards who manage data content and technology stewards who manage technical implementation also have roles that deserve evaluation.

With the addition of a data ethics review in the DG *process*, subject matter expertise will be required either through the addition of experts or appropriate training of the *data governance* team.

Questions

→ Has the DG *function* been established?
→ Is the DG *function* appropriately staffed and funded?
→ Does the DG *function* have the authority needed to be effective?
→ Have the roles and responsibilities of the DG *function* been defined, documented and socialized?
→ Have the skills for data ethics review and execution of Machine Learning (ML) and Artificial Intelligence (AI) tools been added or developed within the *stakeholders*?
→ Have milestones and metrics associated with DG execution been established?

Artifacts

- Evidence of *stakeholder* identification
- *RACI matrix* or other evidence of accountability assignment
- Description of the roles and responsibilities of the DG *function*
- Staff qualifications and assignments
- Evidence of accountability linked to reviews and compensation
- Gap analysis of skills needed and those in place
- List of *stakeholders* and evidence of bi-directional communication

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No formal DG roles & responsibilities exist.	No formal DG roles & responsibilities exist, but the need is recognized and the development is being discussed.	The formal DG roles & responsibilities are being developed.	The DG roles & responsibilities are defined and have been validated by the directly involved *stakeholders*.	The DG roles & responsibilities are established and are recognized and used by *stakeholders*.	The DG roles & responsibilities are established as part of business-as-usual practice with a continuous improvement routine.

6.1.4 The DG processes are defined and operational

Description

Formal *processes* must be established for the activities of the DG *function*. These *processes* align with the DM *policy* and *standards* of the organization and include *procedures*, tools and routines. The routines are required for steady-state operations.

Objectives

- Establish formal DG *processes* in alignment with the DM *policy* and *standards*.
- Integrate the DG *processes* into the overall end-to-end *processes* of the DM initiative.
- Identify, schedule and maintain DG routines, meetings and working sessions required for operational support.

Advice

The DG *subject matter experts* should work with the business *process* design and optimization service within the Data Management Program (DMP)

team. Together they will create and monitor the implementation of the DG *processes* in alignment to the end-to-end *process* across the full DM initiative.

Address up-front some of the more challenging organizational issues about how *data governance* will affect *stakeholders*. Don't underestimate the difficulties associated with, or minimize the importance of, getting agreement on essential concepts like authority, *policy*, and control.

DG is not a project but part of a sustainable program of work that becomes part of the organizational DNA. A smoothly functioning DG *function* is defined by the routines that support it. The goal is to ensure that DM becomes adopted as business-as-usual across the organization.

Whether the data ethics activities are a direct accountability of the DM *function* or simply an aligned responsibility, additional steps in the DM *processes* will be required to execute the ethical management of the data.

Questions

→ Have formal *processes* been defined and implemented?
→ Are the *procedures*, tools and routines in place for implementing the *processes*?
→ Has the review of data ethics been included in the DG strategy and approach?
→ Are DG activities part of the normal operational routine of *stakeholders*?

→ Are there standing meetings, planning sessions and regular communications about DG initiatives?

Artifacts

- *Process* design artifacts, *procedure* guides and published routines
- *Process* performance metric reports
- Meeting minutes, status reports and DG announcements

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No formal DG operational *processes* exist.	No formal DG operational *processes* exist, but the need is recognized and the development is being discussed.	The DG operational *processes* are being developed.	The DG operational *processes* are defined and have been validated by the directly involved *stakeholders*.	The DG operational *processes* are established and are recognized and used by *stakeholders*.	The DG operational *processes* are established as part of business-as-usual practice with a continuous improvement routine.

6.2 Policy and Standards are Written and Approved

DM policy and standards must be established for the organization and approved by stakeholders and executive governing bodies. The policy and standards must align with cross-control function policy and standards and be auditable.

6.2.1 Policy and Standards are written and complete

Description

Policy and *standards* must reflect the basic *principles* of DM. *Policy* and *standards* must define how business, technology, and operations *functions* manage and control data. They address how data is acquired, managed, maintained and delivered throughout an organization.

Objectives

- Develop *policy* and *standards* in collaboration with business, technology, and operations *stakeholders*.
- Complete and verify *policy* and *standards*.
- Align *policy* and *standards* with the DMS.

Advice

The development and implementation of *policy* and *standards* take the DM initiative from conceptual to functional. These are the rules for the data with a rationale to ensure that data is trusted and managed. They need to be both practical and stringent enough to change the way the organization operates. They are to be implemented via data *standards* and based on core *principles*. They must be linked to strategy and integrated into the Software Development Lifecycle (SDLC) *process*. The development and implementation of DM *policy* should be viewed as the bedrock of the DG program.

Although they can vary, most *policy* and *standards* will contain rules and *guidelines* pertaining to *data ownership*, data definition, *data lineage*, *metadata*, *data quality* (DQ), data access, permissible use, data sourcing and controls.

Questions

- → Have the DM *policies* and *standards* been created and published?
- → Are the *policies* and *standards* complete and linked to control *functions* (e.g., cross-border issues, security, privacy), data acquisition *processes* (e.g., legal contracts, entitlements), data usage (e.g., authorizations, redistribution), data retention (e.g., Create, Read, Update, Delete–CRUD), quality control (e.g., *business rules*, logic checks, transformations), data meaning (e.g., identifiers, definitions, classifications), formats and messaging (e.g., schemas, *metadata*, ISO *standards*)?
- → Are they linked to, and aligned with the DMS?
- → Have they been developed and verified in collaboration with *stakeholders*?
- → Are they aligned with the SDLC *process*?
- → Have they been reviewed and approved by both Internal Audit and executive management?
- → Is the organization able to comply with the DM *policy*, or is a defined burn-in period required?

Artifacts

- Definition of the areas that are covered by *policy* and *standards*
- Documented and approved *policies* and *standards*
- Approvals from Executive Committee and Board
- Evidence that *policies* and *standards* have been communicated
- List of *stakeholders* and evidence of bi-directional communication

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No *policy* and *standards* exist.	No *policy* and *standards* exist, but the need is recognized and the development is being discussed.	*Policy* and *standards* are being developed.	*Policy* and *standards* have been defined.	*Policy* and *standards* are established.	*Policy* and *standards* are established as part of business-as-usual practice with a continuous improvement routine.

6.2.2 Policy and Standards have been reviewed and approved by organizational stakeholders

Description

Policy and *standards* must be shared and reviewed by *stakeholders* to ensure agreement, alignment, and buy-in. *Policy* and *standards* are vital for effective DG and should be subjected to a rigorous challenge *process* by *stakeholders*. DG must be aligned with and become integrated in the existing governance structures of the organization.

Objectives

- Develop *policy* and *standards* in collaboration with business, technology, and operations *stakeholders*.
- Submit *policy* and *standards* to the organizational governance mechanism for evaluation.
- Obtain approval of *policy* and *standards*.

Advice

The *policy* and *standards* need to be grounded in the real world. They must be developed collaboratively with *stakeholders* and approved through the authority of executive management. Without this verification and approval *process*, support and adherence will be difficult to achieve.

Questions

→ Have the right *stakeholders* at the right levels of seniority been involved in the development *process*?
→ Have *policies* and *standards* been verified and approved by *stakeholders*?

Artifacts

- A roster of *stakeholders* and communication trail
- Formal approval and associated communications

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No *policy* and *standards* exist.	No *policy* and *standards* exist, but the need is recognized and the development is being discussed.	*Policy* and *standards* are being developed.	*Policy* and *standards* have been defined and validated by directly involved *stakeholders*. The *policy* and *standards* are recognized by these *stakeholders* as being practical and usable.	*Policy* and *standards* are established and recognized and followed by *stakeholders*. The *policy* and *standards* are recognized by *stakeholders* as being practical and usable.	*Policy* and *standards* are established as part of business-as-usual practice with a continuous improvement routine. The *policy* and *standards* are reviewed for practicality and usability at least annually and updated accordingly.

6.2.3 Policy and Standards have been reviewed and approved by executive governing bodies

Description

Policy and *standards* must be recognized and supported by senior executive management. DG must be aligned with and become integrated in the existing governance structures of the organization.

Objectives

- *Policy* and *Standards* have been submitted to the organizational governance mechanism for evaluation.
- *Policy* and *Standards* have been approved.

Advice

Policy needs to carry the authority of executive management.

Questions

→ Do those involved in corporate-level review fully understand the DM imperative and challenges?

→ Was the approval *process* formal with the right executives involved in the *process* and through established organizational approval *processes*?

Artifacts

- Distribution roster
- Evidence of evaluation like a Board of Director agenda and minutes
- Formal approval and associated communications

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No *policy* and *standards* exist.	No *policy* and *standards* exist, but the need is recognized and the development is being discussed.	*Policy* and *standards* are being developed.	*Policy* and *standards* have been reviewed by the organization's governance control *function* and executive management.	*Policy* and *standards* have been formally approved by the organization's governance control *function* and executive management.	*Policy* and *standards* are established as part of business-as-usual practice with a continuous improvement routine. The *policy* and *standards* are reviewed for practicality and usability at least annually and updated accordingly.

6.2.4 Policy and Standards are aligned with the organization-wide control function policies and standards

Description

All data introduced into or delivered out of the *data ecosystem* must be subject to cross-organizational control *standards* to ensure organization-wide compliance. The types of control *functions* that may have *policies* impacting data are legal and compliance, information security, privacy, data usage and cross-border.

Objectives

- Formally recognize cross-control *function* dependencies.
- Ensure these dependencies are reflected in each groups' *policy* and *standards*.
- Establish regular engagement between cross-control *functions* and the DM initiative.
- Apply cross-organizational *data control policies* and *standards* to all data introduced into or delivered out of the ecosystem.

Advice

The DM initiative is not accountable for legal and compliance, information security, privacy, data usage and cross-border control *functions*. However, for these control *functions* to be executed there are requirements for data and DM issues such as identification, classification and access control that are critical to ensuring their success. DM should be working with all control *functions* across the organization, identifying the touch points that impact the other control *function* objectives.

The goal is to ensure that the *policies* and *standards* of DM are aligned with those of the other, organization-wide control *functions*. Take advantage of existing, parallel rules and control *function policies* to integrate them into the DM *policies*, *standards*, and *processes*. Other control *functions* should also reference the *policies*, *standards* and *processes* of the DM initiative.

Questions

- → Are the mechanisms in place to support cross-control *function* collaboration?
- → Is there *policy*, *standards* and *process* alignment between organizational control *functions*?
- → Is cross-control *function* coordination operational and being reviewed by Internal Audit?
- → Are the mechanisms to support coordination with regulators defined and operational?
- → Are formal meetings across control *functions* taking place?

Artifacts

- DM *policies*, *standards* and *processes*
- Other, intersecting control *function policies*, *standards* and *processes*
- Evidence that DM *policies*, *standards* and *processes* align with those of the other control *functions*
- Evidence of collaboration with cross-control *functions* such as communication, joint meetings, minutes, and agendas

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No *policy* and *standards* exist.	No *policy* and *standards* exist, but the need is recognized and the development is being discussed.	*Policy* and *standards* are being developed.	*Policy* and *standards* have been defined and validated by directly involved *stakeholders*. The *policy* and *standards* are recognized by these *stakeholders* as being aligned with the organization-wide business control framework.	*Policy* and *standards* are established and are recognized and followed by *stakeholders*. The *policy* and *standards* are recognized by all relevant *stakeholders* as being aligned with the organization-wide business control framework.	*Policy* and *standards* are established as part of business-as-usual practice with a continuous improvement routine. The *policy* and *standards* are reviewed for cross-organization control *function* alignment at least annually and updated accordingly.

6.2.5 Policy and Standards are enforceable and auditable

Description

Policy and *standards* must be supported by established audit *processes* and routines in partnership with Internal Audit. Lack of adherence to *policy* and *standards* must be elevated as a formal audit issue that requires resolution.

Objectives

- Authorize the ODM to examine and enforce adherence to DM *policy* and *standards*.
- Adhere to the DM *policy* and *standards* examined and enforced by Internal Audit.

Advice

Putting tollgates into production is a balancing act. They must be strong and effective in validating the access and use of data, while at the same time avoid being burdensome and bureaucratic. Establish the necessary review and approval *processes* along the DM lifecycle to ensure that decisions about the acquisition, use and distribution of data adhere to the DM *policy* and *standards*. Project review and approval *processes* will typically include such things as formal data design reviews, formal approvals to build, approvals to access and approvals to distribute.

Questions

→ Are the appropriate first and second lines of defense in place to monitor controls?
→ Are the criteria for tollgates transparent and easy to understand?
→ How are *stakeholders* informed of expectations and reasons for any denials?
→ How is the ODM collaborating with other control *functions* on tollgates?
→ Are new *processes* incorporated into the SDLC *process*?

Artifacts

- Review and approval *process* documentation
- Evidence and evaluation criteria of alignment with existing application development *processes*
- Evidence of alignment with other control *processes*
- Compliance records and illustrations of consequences of non-compliance
- Evidence of the audit *process*

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No policy and standards exist.	No policy and standards exist, but the need is recognized and the development is being discussed.	Policy and standards are being developed.	Policy and standards have been defined and validated by directly involved stakeholders. The policy and standards are recognized by these stakeholders as being enforceable and auditable.	Policy and standards are established and are recognized and followed by stakeholders. The policy and standards are recognized by all stakeholders as being enforceable and auditable.	Policy and standards are established as part of business-as-usual practice with a continuous improvement routine. The policy and standards are reviewed for enforceability and auditability at least annually and updated accordingly.

6.3 Govern the DM Program

Governing the DM program includes: 1) administering program funding; 2) approving program and project DM adherence; 3) enforcing standard DM business process adoption; and 4) implementing issue management.

6.3.1 Program funding governance is established and operational

Description

In order to achieve budget authority, alignment must exist between the DM initiative funding governance, the DM governance structure and the organization-wide funding *model* and *process*.

Objectives

- Get the funding *model* operational.
- Identify and empower the parties accountable for the DM initiative budget.
- Integrate governance of the funding *model* into the governance structure of the DM initiative.
- Align governance of the funding *model* to the overall funding governance of the organization.

Advice

An operational funding *model* means that budgets are secured and aligned to expected deliverables. It means that DM executives are empowered to support the funding commitments. Implicit in an operational funding *model* is that the DM initiative is included in the funding cycle of the organization to secure appropriate levels of funding moving forward. The funding approach must be formalized, ideally as a stand-alone budget.

Questions

→ Does the ODM have the authority to spend?
→ Is the funding *model* governance *process* and governing body authorized by the organization?
→ Is the funding *model* incorporated into the organizational funding cycle and *process*?

Artifacts

- Funding *model*
- Formal approvals from *stakeholders* and budget owners
- Records of spending on DM expenses.

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No governance for program funding exists.	No governance for program funding exists, but the need is recognized and the development is being discussed.	Governance for program funding is being developed.	Governance for program funding is defined and validated by directly involved *stakeholders*.	Governance for program funding is established and is recognized and followed by *stakeholders*.	Governance for program funding is established as part of business-as-usual practice with a continuous improvement routine. It is recognized as the normal way of working.

6.3.2 Program and project review and approval processes are established

Description

Change management *policy* and *standards* must exist in a controlled manner via checkpoints, formal review mechanisms and organizational approval boards. Data and DM requirements must be included in the change *process* to ensure that all new development as well as data access, usage and transmission of data adhere to established DM *policy* and *standards*.

Objectives

- Communicate to *stakeholders* the review and approval *processes* as well as responsibilities for data-related projects.
- Get review and approval *processes* operational such as approval to build, approval to access, approval to use, approval to send.
- Integrate review and approval of data, and the ethical management of data, into the organization's technology development and SDLC *process*.
- Align review and approval *processes* with the control mechanisms of other cross-control *functions* including change management.

Advice

Establish review and approval *processes* as checkpoints along the DM lifecycle to ensure that decisions about acquisition, use, sharing, and distribution adhere to *policies* and *standards*. The implementation of program or project tollgates requires balance. They must be strong enough to be effective without being bureaucratic and burdensome. The objective is to facilitate business and enable data hygiene. If an approval request is denied, it is in the best interest of the DM initiative to help resolve the reason for denial.

Questions

→ Are the appropriate tollgates in place at critical decision points?
→ Are the review and approval *processes* structured to support business *processes* without being overburdensome?
→ Are the criteria for tollgates transparent and easy to understand?
→ Are project review and approval *processes* done collaboratively with other control *functions*?
→ Have *data control* reviews been incorporated into the SDLC *process*?
→ Is the review of data ethics part of the tollgate *process*?

Artifacts

- Documented review and approval *processes*
- Completed review and approval *processes* showing alignment between existing application development and other control *processes*
- Bi-directional communication with *stakeholders*

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No formal project review and approval *processes* exist.	No formal project review and approval *processes* exist, but the need is recognized and the development is being discussed.	Formal project review and approval *processes* are being developed.	Formal project review and approval *processes* are defined and validated by directly involved *stakeholders*.	Formal project review and approval *processes* are established and are recognized and followed by *stakeholders*.	Formal project review and approval *processes* are established as part of business-as-usual practice with a continuous improvement routine. It is recognized as the normal way of working.

6.3.3 Business process optimization for DM is enforced

Description

The DM governance structure must be applied to governing the *standard* DM *processes* in alignment with the organization's *standards* and governance for business *process* operational excellence.

Objectives

- Integrate governance of the DM business *process* optimization into the governance structure of the DM initiative.
- Align governance of the DM business *process* optimization with the organization's overall governance of the business *process* optimization function.

Advice

Effectively governing the *standard* DM *processes* of an organization is critical to maintaining repeatable, sustainable and measurable *processes*. The data *policy* should require the use of *standard* *processes*, making use of the *process* subject to audit. The use of *standard* *processes* across the organization will enhance the data and DM interoperability between *data domains* throughout the *data ecosystem*.

Questions

→ Is the governance of the *standard* DM initiative *processes* integrated into the overall governance of the DM initiative?
→ Is the use of the DM *standard* *processes* enforceable by audit?
→ Is the operational excellence *function* in the ODM aligned with the organization-wide operational excellence *standards* and governance?

Artifacts

- Evidence of use of a *standard* *process* optimization framework
- Evidence of use of *standard* *process* design tools
- Evidence of governance body approval

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No business *process* optimization for DM exists.	No business *process* optimization for DM exists, but the need is recognized and the development is being discussed.	No business *process* optimization for DM is being developed.	Business *process* optimization for DM is defined and validated by directly involved *stakeholders*.	Business *process* optimization for DM is established and is recognized and used by *stakeholders*.	Business *process* optimization for DM is established as part of business-as-usual practice with a continuous improvement routine. It is recognized as the normal way of working.

6.3.4 Issue management process is defined and operational

Description

The issue management *process* includes issue identification, prioritization, resolution tracking and escalation as required. The *process* must support resolution of both DM initiative issues and data issues and leverage the established *data governance* structure. A critical aspect of the issue management *process* is the escalation *procedure* required when agreement cannot be achieved, and conflict resolution is required.

Objectives

- Get the issue management *processes* operational and documented.
- Define issue management routines.
- Align escalation *procedures* with the organizational governance structure.

Advice

The issue management *process* should be integrated with the DM initiative *data governance* structure. Issue management is required for both issues from the practice of DM and the governance of the data. Formality of how issues are managed is essential for both operational sanity and audit requirements. Make sure escalation *procedures* are reviewed by Internal Audit as well as endorsed by executive management.

An established escalation *process* is necessary to resolve conflicts, reconcile priorities and ensure efficient operations. These escalation *procedures* need to be formalized with clearly established roles and responsibilities as well as a defined escalation path.

Questions

→ Is there a defined *process* for issue management?
→ Have you made the distinction between DM initiative issues and data issues?
→ Do escalation *procedures* exist for DM and data issues?
→ Are the right people with the appropriate levels of authority involved in the decision-making *process*?
→ Have issue management and escalation *policies* and *procedures* been reviewed and accepted by audit and senior management?

Artifacts

- *Process* design artifacts, *procedure* guides and published routines
- Issue log, Key Risk Indicators (KRIs) and other performance metrics
- Escalation criteria and *policies*
- Escalation *procedures* and communication about conflict resolution
- Defined escalation path

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No formal issue management *process* exists.	No formal issue management *process* exists, but the need is recognized and the development is being discussed.	The formal issue management *process* is being developed.	The formal issue management *process* is defined and has been validated by the directly involved *stakeholders*.	The formal issue management *process* is established and is recognized and used by *stakeholders*.	The formal issue management *process* is established as part of business-as-usual practice with a continuous improvement routine. It is recognized as the normal way of working.

6.4 Govern the Data Structure

Governing the data structure includes: 1) identifying and using authoritative data domains; and 2) enforcing the definition, approval, publishing and use of *standard data models* and definitions along with identification, classification and *taxonomy* schemes.

6.4.1 Govern the Authoritative Data Domains identification and use

Description

Governance is required to enforce the identification, definition and ultimate use of the organization-wide *authoritative data domains.*

Objectives

- Apply governance to identification and verification of *authoritative data domains* by business *subject matter experts*.
- Apply governance to the use of *authoritative data domains* by upstream/downstream *data consumers*.

Advice

Governance must be in place to ensure *data consumer* use of the *authoritative data domains* that were created as part of the data architecture (DA) activities. The goal is to ensure the authoritative data is consumed consistently by all *data consumers* organization-wide.

Questions

→ Are governance *processes* in place to ensure the identification, maintenance and use of *authoritative data domains*?

→ Are domain owners in place who are responsible for the quality and availability of the data?
→ Has the business domain owner, as well as the DA *function*, been involved in the designation of the *authoritative data domains*?
→ Have *data domain* *taxonomies* and conceptual and logical *models* been verified by business subject experts?
→ Are all critical business *functions* represented in the discussion?

Artifacts

- Criteria for determination of *authoritative data domains*
- Inventory and declaration of *authoritative data domains*
- Listing of domain owners and responsibilities
- Business *process* definition and documentation
- List of *stakeholders* and evidence of bi-directional communication

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No *data domains* exist.	No *data domains* exist, but the need is recognized and the development is being discussed.	*Data domains* are being developed.	*Data domains* are defined and validated by directly involved *stakeholders*.	*Data domains* are established and are recognized and used by *stakeholders*.	*Data domains* are established as part of business-as-usual practice with a continuous improvement routine. The *data domains* are reviewed for accuracy and relevance at least annually and updated accordingly.

6.4.2 Govern the models, glossaries, identifiers, classifications and relationships

Description

Governance is required to enforce the definition, approval, publishing and use of *standard* *data models* and definitions along with identification, classification and *taxonomy* schemes.

Objectives

- Apply governance to the coordinated *process* to define, approve and publish *data models*, definitions, identifiers, classifications and *taxonomies*.
- Apply governance to the use of *data models*, definitions, identifiers, classifications and *taxonomies* for ***Business Elements*** and *data elements*.
- Apply governance to the alignment and cross reference of *models*, definitions, identifiers, classifications and *taxonomies* to industry *standards*.

Advice

Governance must be in place to ensure the creation, maintenance and use of *standard data models*, definitions, identifiers, classifications and *taxonomies* across the organization. The goal is to achieve a precise organization and access to the data by *data consumers*.

Questions

→ Have *standard data models*, definitions, identifiers, classifications and *taxonomies* been defined, maintained and used across the organization?
→ Have *policies* and *standards* been developed and approved to ensure *standard data models*, definitions, identifiers, classifications and *taxonomies* are used?
→ Have *standard data models*, definitions, identifiers, classifications and *taxonomies* been published and cross-referenced to those used in any proprietary applications?
→ Have business, data, technology, legal and compliance *stakeholders* been involved in the definition and verification *process*?
→ Are *standard data models*, definitions, identifiers, classifications and *taxonomies* aligned with other cross-organization control *functions* requirements?
→ Has governance over *standard data models*, definitions, identifiers, classifications and *taxonomies* been aligned with change management *policies*?

Artifacts

- Documentation on the *process* for defining, assigning and maintaining *standard data models*, definitions, identifiers, classifications and *taxonomies*
- *Policy* and *standards* on definition, maintenance and use
- Records of *standard data models*, definitions, identifiers, classifications and *taxonomies* as *metadata*
- Cross-references to other cross-organization control *function* requirements
- Alignment to change management *policies*
- Inventory of *standards* being used
- List of *stakeholders* and evidence of bi-directional communication

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No governance for the listed items exists.	No governance for the listed items exists, but the need is recognized and the development is being discussed.	Governance for the listed items is being developed.	Governance for the listed items is defined and validated by directly involved *stakeholders*.	Governance for the listed items is established and are recognized and followed by *stakeholders*.	Governance for the listed items is established as part of business-as-usual practice with a continuous improvement routine. It is recognized as the normal way of working.

6.5 Govern that the Data is Fit-for-Purpose

Governing that the data is fit-for-purpose includes: 1) controlling the access and use of data; 2) enforcing the contractual restrictions of third-party data; and 3) establishing and monitoring adherence to the *Data Sharing Agreement* (DSA).

6.5.1 Govern the data access and use

Description

Once the *authoritative data domains* have been established, governance is required to control the access and use of the data.

Objectives

- Apply governance to the controlled access and use of data from the *authoritative data domains*.

Advice

Once the *authoritative data domains* have been defined, declared and sanctioned, governance *processes* need to be established to ensure control over the identification, definition, and usage of data. Appropriate usage is based on an understanding of who is using data and for what purpose.

Questions

→ Have *authoritative data domains* been established and sanctioned?
→ Has intended use been defined and verified?
→ Have business meaning and relationships been defined and verified?
→ Are processes in place to control access to the *authoritative data domains*?
→ Are processes in place to ensure appropriate usage?

Artifacts

- *Authoritative Data Domain* designations
- *Metadata* repository with required structural, descriptive and administrative attributes
- List of *stakeholders* and evidence of bi-directional communication

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No governance for data access and use exists.	No governance for data access and use exists, but the need is recognized and the development is being discussed.	Governance for data access and use is being developed.	Governance for data access and use is defined and validated by directly involved *stakeholders*.	Governance for data access and use is established and is recognized and followed by *stakeholders*.	Governance for data access and use is established as part of business-as-usual practice with a continuous improvement routine. It is recognized as the normal way of working.

6.5.2 Govern the adherence to contractual terms & regulatory policy

Description

Governance is required to monitor and enforce the contractual restrictions of third-party data entering the organization.

Objectives

- Apply governance to monitoring the restrictions applied by purchase contract to the use of third-party data.
- Apply governance to the record of third-party data use restrictions as part of the *metadata* of the data.

Advice

How third-party data is managed should be no different than other data in the organization except for the addition of the contractual restrictions that are defined in the purchase contract. The *metadata* of the purchased data should indicate the restrictions on its use, ideally linked to a description of the restrictions. The data should be assigned to a *data domain* and access to the data in compliance to the contract should be controlled.

The third-party contracting *process* should include review of the DM practices applied to their data and the establishment of requirements on the vendor. To introduce external data into the organization's ecosystem, the data should have had similar rigor applied to that of the internally created data. This rigor includes all the cross-organization controls and such topics as privacy and data ethics.

Questions

→ Is the third-party data assigned to an authoritative *data domain*?
→ Are the restrictions of data use being recorded as flags in the *metadata*?
→ Are the data use restrictions centrally captured for reference?

Artifacts

- Third-party data alignment to an Authoritative Data Domain
- *Metadata* model
- Central repository of third-party data restrictions

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No governance for contractual restrictions of use exists.	No governance for contractual restrictions of use exists, but the need is recognized and the development is being discussed.	Governance for contractual restrictions of use is being developed.	Governance for contractual restrictions of use is defined and validated by directly involved *stakeholders*.	Governance for contractual restrictions of use is established and is recognized and followed by *stakeholders*.	Governance for contractual restrictions of use is established as part of business-as-usual practice with a continuous improvement routine. It is recognized as the normal way of working.

6.5.3 Govern the data use according to established Data Sharing Agreements

Description

Governance is required to approve and monitor adherence to the *Data Sharing Agreement* (DSA) established between a *data producer* and *data consumer* and authorize data as fit-for-purpose for the *data consumer's* intended use.

Objectives

- Apply governance to approve and monitor adherence to a DSA.
- Establish a *process* to routinely review and update a DSA.
- Create a routine for validation by the *data producer* that the data is fit-for-purpose for the *data consumer's* intended use.

Advice

A DSA defines the *data consumer* business *process* requirements for data and the *data producer's* terms and restrictions for use of the data. Formal governance is required to approve the agreement. Additionally, a method must be in place to monitor that the terms and restrictions are followed.

The DM *policy* and *standards* should require the DSA to include criteria for determining that the data is fit for purpose such as a defined *data consumer's* intended use and an established minimum threshold of quality.

The governance activity should include the validation and authorization of data as fit-for-purpose. The formality of this will be determined by the governance culture of the organization. Data fit-for-purpose is not only achieving an established threshold of quality but includes the *data producer* understanding the *data consumers* intended use of the data and an agreement that it is the correct data for the use and for how it is being used.

Questions

→ Are DSAs and adherence to the agreement required by DM *policy* and *standards*?
→ Are DSAs in place?
→ Are aspects of the agreement recorded as *metadata*?
→ Are the agreements maintained in a centralized and accessible repository?
→ Are the agreements reviewed routinely and updated to current-state data use?
→ Is there a *data producer* and *data consumer* agreed upon threshold of DQ?
→ Is there a documented *data consumer* intended use of the data?
→ Is there a defined validation by the *data producer* that the data is fit-for-purpose for the *data consumer's* intended use?

Artifacts

- DSAs between *data domains*
- *Metadata* record that the data is covered by a DSA
- *Metadata* record identifying the *data consumer(s)* of a *data element*
- *Metadata* record of quality threshold and intended use
- Evidence that the *data producer* routinely validates and authorizes that the data is fit-for-purpose

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No governance for data sharing exists.	No governance for data sharing exists, but the need is recognized and the development is being discussed.	Governance for data sharing is being developed.	Governance for data sharing, using DSAs, is defined and validated by directly involved *stakeholders*.	Governance for data sharing, using DSAs, is established, and is recognized and followed by *stakeholders*.	Governance for data sharing, using DSAs, is established as part of business-as-usual practice with a continuous improvement routine. It is recognized as the normal way of working.

6.6 Govern the Data Ethics

Governing the data ethics includes: 1) establishing a formal data ethics oversight function; 2) adhering to the ethical access and appropriate use of data; and 3) monitoring whether the outcomes of data access and use are ethical.

6.6.1 Establish a formal data ethics oversight function

Description

A formal data ethics oversight activity is required that includes establishing a governing body, a Code of Data Ethics and senior executive accountability with defined roles and processes for ensuring data ethics.

Objectives

- Create a formal body to oversee both the ethical use of data and the ethical outcomes of data use.
- Craft the Code of Data Ethics for the organization as mandated by the enterprise level senior management team.
- Identify stakeholders and form working groups to make the Code of Data Ethics operational.
- Define and communicate the roles and responsibilities of the data ethics senior officer at all levels of the organization.
- Invest in data ethics training at all levels of the organization.
- Communicate through training and reporting requirements that responsibility for enacting the Code of Data Ethics applies to each individual and is shared by all members of the organization.
- Authorize the executive responsible for the organization's ethical DM to ensure and enforce adherence to the Code of Data Ethics.
- Establish a process to remediate ethical issues.

Advice

Collaborative, routine and transparent information practices lead to ethical *data governance*. The DCAM framework guides organizations through data practices that catalyze the contextual discourse needed to change organizational culture from data-driven to data-ethics-driven. The first step in this endeavor is to establish compliance with a Code of Data Ethics, mandated by the *enterprise*-level senior management team, and implemented at every level of the organization. To implement the Code of Data Ethics with accountability distributed throughout the organization requires operating governance structures that align with the organization's overall governance structure. The data ethics senior officer at all levels of the organization is accountable for ethical *data governance*.

The role of legal and compliance in the DM initiative is to mitigate legal risk. However, as always, law lags behind technological change, and the vast and rapidly expanding data-industrial complex is facing civil and criminal challenges that must be addressed. Let's also recall that legal compliance represents the least common denominator, an outdated and minimal level of compliance. There is a vast difference between following existing regulations and being a leader in ethical DM.

As described in the Introduction, data has meaning. Understanding how that meaning changes in different contexts is the crux of ethical DM. A Code of Data Ethics should articulate how the organization understands the meaning of the data it stewards—now, and in the future.

Codes of Data Ethics typically incorporate some variant of the following general principles:

1. Data elements are entwined with a real person – first do no harm.
2. The effects of DM are inequitable, including some and excluding others.
3. Ensure future data use is consistent with expectations & intentions of its originators.
4. Provenance & tools for analysis must be documented, because they affect how data is presented & used.
5. Be explicit about how the organization's data practices do and do not align with context-dependent expectations.
6. Collect only the data required for the task at hand; less data is less open to misuse.
7. Data subjects have a right to know what data is collected and how that data is used and/or shared. Be open and transparent.
8. Aim to be a leader in data ethics for long-term success.
9. Prioritize design practices that promote and enhance transparency, configurability, accountability, and proactive interrogation of patterns in both training data and outcomes.
10. Welcome internal and external ethical review.

Much of what we hear from many organizations with respect to social responsibility is considered mere lip service. An authentic commitment to data ethics starts with a top-down mandate, which is supported throughout the organization by specific practices and empowered accountability. In other words, employees must be empowered to insist on data practices that are aligned with the data ethics mandate. Without empowered employee actions and established routines, data ethics will not permeate the organizational culture.

Early efforts to inculcate data ethics throughout the organization usually begin with the formation of a steering committee or working group. The group socializes the concept of data ethics among all stakeholders through activities such as large forums that bring stakeholders together to examine the importance of data ethics through the lens of the organization's mission and values. Eventually, organizations committed to data ethics develop formal chartered governance aligned to the governance structure of the organization. Alternatively, if data ethics governance remains separate from *data governance*, alignment routines are required.

Questions

- → Has the organization established and socialized a Code of Data Ethics?
- → Has the organization allocated resources to appropriately staff, train, and socialize Code of Data Ethics governance?
- → Have milestones, metrics, and measurements associated with adherence to the Code of Data Ethics been established?

→ Is the concept of ethical DM understood by all stakeholders at all levels of the organization?
→ Is ethical DM embraced across the full organizational ecosystem?
→ Does the data ethics senior officer at all levels of the organization have the authority needed to be effective?
→ Have the roles and responsibilities of the Chief Data Officer or Chief Ethics Officer been defined, documented and socialized?

Artifacts

- Evidence of enacted Code of Data Ethics
- Staff assignments and qualifications
- Gap analysis of skills needed and in place
- List of stakeholders and evidence of bi-directional communication
- Description of the roles and responsibilities of the DG function for data ethics
- Regular schedule of training and reporting on data ethics implementation
- Written and approved charters of working data ethics committees
- Individuals are held accountable for their adherence to the Code of Data Ethics via annual reviews and compensation considerations.
- Demonstrated hierarchy of responsibility and reporting for ethical DM
- Evidence of events to socialize the data ethics commitment organization-wide

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No formal data ethics oversight function exists.	No formal data ethics oversight function exists, but the need is recognized and the development is being discussed.	The formal data ethics oversight function is being developed.	The formal data ethics oversight function is defined and has been validated by the directly involved stakeholders.	The formal data ethics oversight function is established and is recognized and used by stakeholders.	The formal data ethics oversight function is established as part of business as usual practice with a continuous improvement routine. This is recognized as the normal way of working.

6.6.2 Govern the ethical access and appropriate use of data

Description

Explicit governance of ethical data access and use is required. The ethical evaluation of access and use must include both the actual access and use as well as the data subject's perception of how their data is being accessed and used. Furthermore, this governance must provide processes and protections for personnel who raise concerns about data access and/or use that runs counter to this perception.

Objectives

- Ensure the access and use of data adheres to the Code of Data Ethics.
- Provide a mechanism for personnel to raise concerns about ethical data access and use.

Advice

While 6.5.3 mandates adherence to the Data Sharing Agreement established between a data producer and data consumer and authorizes data as fit for purpose for the data consumer's intended use, 6.6.2 establishes a higher bar for *data governance*. Again, this is the difference between legal compliance and ethical governance of DM. A significant aspect of data ethics governance is the empowerment of personnel at all levels of the organization to question the legal agreements that pertain to data access and use. For example, a data sharing agreement may contain boilerplate stipulations that don't do enough to preserve the trust placed in the organization. Personnel should have a channel through which they may raise concerns about suspected violations of and/or potentially risky data access and/or use.

Some best practices that support the ethical access and appropriate use of data include:

- Embedding provenance information in the metadata to enhance transparency of data collection and practices
- Providing individuals the on-demand (or periodic) opportunity to access to the corpus of data the organization holds about them
- Revisiting consent
- Communicating both legal and perceived consent to third parties
- Proactive disclosure of the organization's DM plan with specific provisions for data disposition
- Resolving to collect only that data which are necessary for the task at hand

Questions

→ Are the Code of Data Ethics used as a guide for evaluating ethical data access and use?
→ Is there a mechanism for personnel to raise concerns about ethical data access and use?

Artifacts

- Metadata schema that includes fields for provenance and consent history
- Survey and/or interview responses demonstrating perceived access and use of data
- Training and internal promotional materials describing a channel for reporting concerns about data access and use
- Communications disclosing the organization's DM plan
- Communications providing people with their data corpuses
- Statistics relating to reduced extraneous data collection over time

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No governance of ethical access and use of data exists.	No governance of ethical access and use of data exists, but the need is recognized and the development is being discussed.	The governance of ethical access and use of data is being developed.	The governance of ethical access and use of data is defined and has been validated by the directly involved stakeholders.	The governance of ethical access and use of data is established and is recognized and used by stakeholders.	The governance of ethical access and use of data is established as part of business as usual practice with a continuous improvement routine. This is recognized as the normal way of working.

6.6.3 Govern the ethical outcomes of data access and use

Description

Explicit governance of ethical outcomes of data access and use is required. The evaluation of ethical derived outcomes must exist within the operating governance structures of the organization in alignment to a socially acceptable code of ethics.

Objectives

- Ensure the derived outcomes from the use of data align with a socially acceptable code of ethics.
- Define and implement operating governance structures which guide and enforce adherence to the Code of Data Ethics.
- Align the data ethics governance structure to both the *data governance* and the organization-wide governance structures.

Advice

Ethical outcomes are those results of data access and use that meet the business needs of the organization without infringing on the human dignity of others. Human dignity can be thought of as what a society tolerates—what is considered, generally, "fair." Organizations have a moral imperative to interrogate the patterns not only in the data used to train their models, but also in their outcomes. To ignore such patterns is unethical. To govern the ethical outcomes of data access and use requires long-term scenario planning and research into the social effects of these practices. This accountability is more complex and involved than legal compliance, but also more directly tied to the organization's values as captured in its Code of Data Ethics.

There are two types of unethical DM: nefarious and unintentional. Nefarious unethical data practices are deployed by people who aim to manipulate decision making to benefit some and disadvantage others. This type of activity is relatively rare in comparison with unintended unethical data practices. Even the most well-intentioned data professionals may inadvertently cause harm, especially if they are not aware of the ways in which algorithms replicate societal inequities. In other words, "Good data can create bad outcomes." For example, in the AI context, it's unethical to train your machine on data without conducting ongoing scenario planning and research on potential negative unintended consequences. By embracing the Code of Data Ethics as an instrumental part of the organizational culture, ethical DM becomes a priority for employees at all levels of functional responsibility. Typically, it is the responsibility of the Chief Data Officer or Chief Ethics Officer to ensure the Code of Data Ethics is embraced, prioritized, and operationalized throughout the organization.

Questions

→ What is the relationship between data ethics and broader ethics activities in the organization?
→ What are the long-term implications of data access and use decisions?
→ What stakeholders should be consulted to better understand the implications of data decisions?

Artifacts

- Records of consultations with stakeholder groups pertaining to the potential for unintended consequences of data and access decisions
- Training curricula that reinforce the organization's approach to ethical DM with case studies highlighting unethical outcomes

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No governance of ethical outcomes of data access and use exists.	No governance of ethical outcomes of data access and use exists, but the need is recognized and the development is being discussed.	The governance of ethical outcomes of data access and use is being developed.	The governance of ethical outcomes of data access and use is defined and has been validated by the directly involved stakeholders.	The governance of ethical outcomes of data access and use is established and is recognized and used by stakeholders.	The governance of ethical outcomes of data access and use is established as part of business as usual practice with a continuous improvement routine. This is recognized as the normal way of working.

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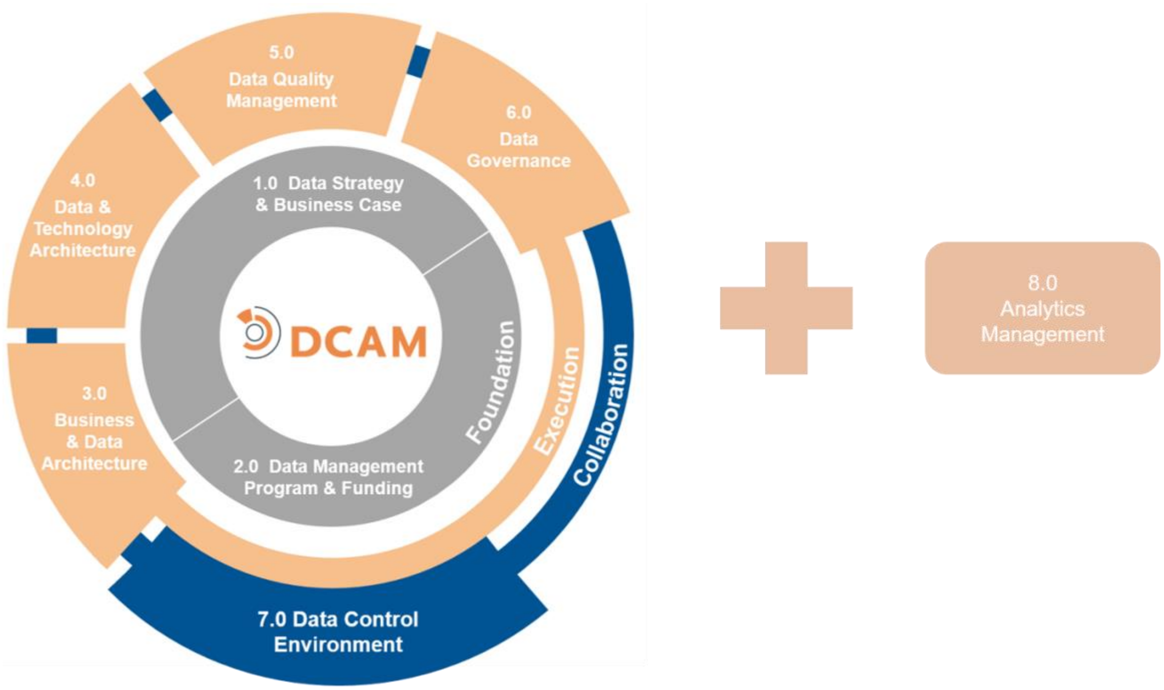
DCAM® Framework:

7.0 Data Control Environment



7.0 Data Control Environment

Introduction

The **Data Control Environment** refers to the state of operation in which the data assets of an organization are holistically managed throughout the organization. There are three elements of a successful data control environment.

1. The data management (DM) objectives and capabilities described within this document have been embraced and adopted throughout the organization.
2. The *data lifecycle* is fully supported by all *stakeholders*. These stakeholders ensure understanding, awareness and control of data throughout the data supply chain–from source to consumption to disposition.
3. DM is part of the organization's *data ecosystem*. It is integrated and coordinated with all other control *functions* organization-wide.

The purpose of the data control environment is to coordinate the people, *process* and technology of DM into a cohesive operational model. The data control environment defines the mechanisms used to capture data requirements, unravel *data flows* and linked *processes* and determine how data is to be delivered to the *data consumer*. The data control environment supports the *data lifecycle*. It ensures that proper resources and controls are in place as data moves throughout its journey. Also, the data control environment ensures collaboration and alignment to cross-organizational control *functions*. Areas such as Information Security, Data Privacy and Change Management must operate in sync with DM to ensure data is properly managed across all business *functions*.

To the extent that the data control environment is not achieved it results in potential data risk. Data risk should be managed in alignment with the overall risk management framework of the organization. Data risk scope includes areas such as data architecture risks, metadata risks, data quality risks, *data governance* risks and Master Data risks.

Definition

The **Data Control Environment (DCE)** component is a set of capabilities that together form the fully operational data control environment. Data operations, supply-chain management, cross-control *function* alignment and collaborative *technology architecture* must operate cohesively to ensure the objectives of the DM initiative are realized across the organization.

Scope

- Work with DM *Program Management Office* (PMO) to design and implement sustainable business-as-usual *processes* and routines to enable a successful data control environment.
- Bring together the DM components as a coherent, end-to-end *data ecosystem*.
- Follow current DM best practices by routinely reviewing and auditing the capabilities and their *processes*.
- Ensure all facets of DM for business-critical data such as *data lifecycle*, end-to-end *data lineage* and data aggregations are fully operational.
- Ensure DM is aligned with other control *function* *policies*, *procedures*, *standards*, and governance.

Value Proposition

Organizations that improve their ability to reconcile requirements for data and accurately share data across the organization's ecosystem are able to better respond to changes in business *process*, regulatory, and audit requirements.

Organizations that deliver data through a controlled environment respond more rapidly to market opportunities and provide innovation to customers.

Overview

The DCE is where the execution components of Business & Data Architecture, Data & Technology Architecture, Data Quality Management and Data Governance are made operational in the data supply chain by the *data producer*. This operationalization brings a defined set of data into control and makes it available to *data consumers* at a point in time, either real-time or period end.

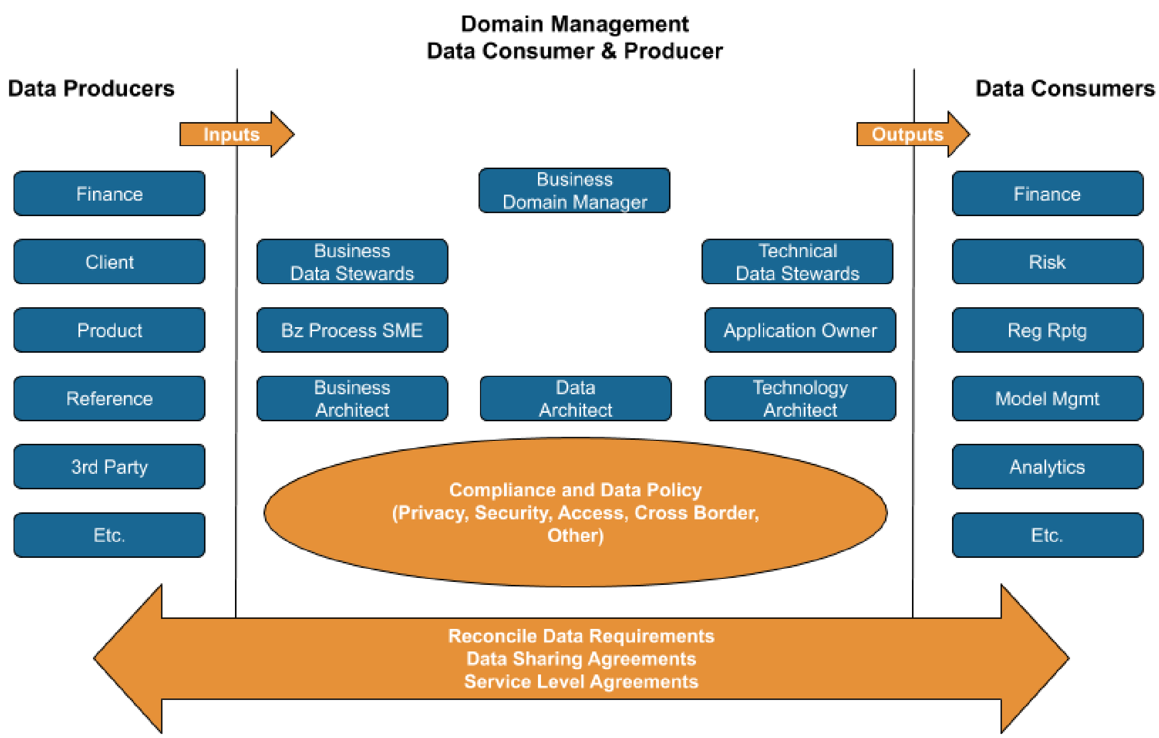


Diagram 7.1: Data Supply Chain

One of the first *functions* within the DCE is the orchestration of the DM component disciplines. These disciplines must be aligned to effectively manage data across the organization. The DCE forces the alignment of all the capabilities discussed in this model into a consistent operational flow. Each capability must be properly resourced and prioritized as well as supported by business, data and technology senior management.

The successful coordination of these components is a determining factor in the success of the DM initiative. It is the responsibility of the DM organization and the senior *data officer* at each level of the organization to structure and coordinate the DM *operating model*. This properly defines data meaning, ensures *data quality* (DQ), and delivers data in a timely and efficient manner. Evidence of the processes must be compiled through demonstration of organizational structures, charters, *policies*, and senior management directives.

Data is a core factor of input into business *functions* and operational *processes*. The *data lifecycle* tracks the progress of data from source to storage, to maintenance, to distribution and to consumption. From this point the data may be reused, sent to the archive and finally to defensible destruction. The mechanisms used to identify, align, and validate the data as factors of input into business *functions* are derived by reverse engineering existing business *processes* into their individual *data elements* and by unraveling the data assembly *processes* used to create the required *data sets*.

This reverse engineering *process* to define data requirements needs to be managed with precision. Only precision will avoid confusion and miscommunication between what the business users truly need for their business *function* and what technology professionals need for technical implementation. Data requirements should be modeled, aligned with business meaning, prioritized in terms of how critical it is to the business *process* and verified by all *stakeholders*. These steps ensure that essential concepts are not lost in translation. This is particularly critical for data that is shared among multiple *data consumers* and for core *data attributes* that are used as a baseline for onward expression in operational calculations or business formulas.

For complex applications and for all data aggregation-related *processes*, it is essential to understand and document how the data moves from system-to-system; how the data is transformed or mapped; and how the data is aligned to business definition and *standard* meaning. Gaining agreement on this *data lineage process* is fundamental for ensuring that the results of decentralized or linked processing can be trusted to be consistent and comparable.

The final aspect of an effective DCE is the integration of DM into the *data ecosystem* of an organization. The *data ecosystem* is a concept that describes how data is managed collaboratively with all cross-organization control *functions*. Control *functions* such as information security, storage management, legal and compliance, privacy, and vendor management all have responsibilities on how data is managed. It is imperative that the *policies* of DM are integrated and aligned with the *policies* of the cross-organization control *functions* to ensure data is being managed consistently and holistically organization-wide.

Finally, a DCE ensures technology's alignment with DM *policies* and best practices. DM capabilities such as architecture, governance, and DQ should be integrated into the organization's Software Development Lifecycle (SDLC) *processes* to ensure that DM considerations are being adequately addressed at the appropriate stages of the development cycle. Nothing should operate in a silo. Operating within an ecosystem recognizes interdependencies and ensures collaboration.

Core Questions

- → Is the concept of a DCE understood by *stakeholders*?
- → Are the aspects of *data control* like terms, definitions, relationships, integration and precedence established on a consistent basis?
- → Are control *processes* applied across the full *data lifecycle*?
- → Are the concepts of *data control* aligned across the full organizational ecosystem?

Core Artifacts

The following are the **core artifacts** required to execute an effective Data Control Environment capability. Items with an '*' link to published best practice *guidelines*.

- Cross-Control Function Alignment Matrix
- Data Risk Framework
- Data Supply Chain Model
- Regulatory Alignment Matrix

7.1 Data Control Environment (DCE) is Evidenced

Evidence of the data control environment is the result of effectively integrating and executing the other six components defined in DCAM. Active engagement by *stakeholders* is required to ensure the DM capabilities are working collaboratively across the organization.

7.1.1 The Data Control Environment is established

Description

To establish the DCE the holistic execution of DM initiative is required at each operating level of the organization.

Objectives

- Charter governing bodies and make them operational.
- Design and fill leadership roles. Document that they are functioning according to prescribed mandates.
- Deliver DM initiative capabilities.

Advice

The data control environment is the result of effectively integrating and executing the other six components defined in the Data Management Capability Assessment Model (DCAM). The integrated execution of these capabilities will create an environment where the people, *process*, data and technology can produce an environment where the data is in control.

Questions

→ Are the governance structures operational to effectively control the data?
→ Are the DM leadership roles at each operating level of the organization defined and operating in harmony as designed?
→ Are the DM initiative capabilities integrated and operational in alignment to the DM *target operating model*?
→ Does the DM initiative have senior management support inclusive of the senior executive suite?

Artifacts

- Evidence of operations in alignment to the DM *target operating model*
- Dashboard summarizing the DM initiative program, outcomes, *process* and DQ metrics
- Evidence of DM support top-down – e.g., formally stated support from the executive suite to the organization

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No formal DCE exists.	No formal DCE exists, but the need is recognized and the development is being discussed.	The formal DCE is being developed.	The formal DCE is defined and has been validated by the directly involved stakeholders.	The formal DCE is established and understood across the organization, and is being followed by the stakeholders.	The formal DCE is established as part of business as usual practice with a continuous improvement routine. The DCE is reviewed and updated at least annually.

7.1.2 The stakeholder roles and responsibilities are defined and implemented

Description

All *stakeholders* critical to the success of the DM initiative must be identified. Roles and responsibilities must be communicated. Active engagement by critical *stakeholders* is operational and evidenced.

Objectives

- Define and communicate the roles and responsibilities of stakeholders who will support the DM initiative.
- Ensure support and provide authority for the DM initiative through stakeholder engagement.

Advice

The DM initiative success demands the engagement of support-*stakeholders*. Management must support the initiative at each operating level of the organization. Also, Internal Audit must be involved throughout to ensure that the DM initiative is auditable and the *policy* and *standards* are enforced. The organization's risk *function* must build an integrated risk framework for managing data risk. Senior leadership, perhaps including the board of directors must be informed, engaged and officially support the DM initiative.

Questions

→ Have the support-*stakeholders* been informed of their roles and responsibilities in supporting the DM initiative?
→ Has the required level of support-*stakeholder* engagement been achieved to provide sustainability of the DM initiative?

Artifacts

- Internal Audit schedules for DM initiative audits
- Integration of data risk into the risk management framework
- Senior management meeting minutes, including DM initiative strategy and performance reporting
- Board of Directors meeting minutes, including DM initiative strategy and performance reporting

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No formal DM roles & responsibilities exist.	No formal DM roles & responsibilities exist, but the need is recognized and the development is being discussed.	The formal DM roles & responsibilities are being developed.	The DM roles & responsibilities are defined and have been validated by the directly involved stakeholders.	The DM roles & responsibilities are established and are recognized and used by stakeholders.	The DM roles & responsibilities are established as part of business as usual practice with a continuous improvement routine. The roles & responsibilities are reviewed and updated at least annually.

7.1.3 DM capabilities are aligned and working collaboratively across the organization

Description

To achieve the DCE the full set of DM capabilities must be operational and applied against a *data domain*. It is important to understand the differences between planned and operational initiatives. Only operating capabilities contribute to the functioning DCE.

Objectives

- Bring DM capabilities into operation and align them with the DM strategy.
- Bring DM capabilities into operation and align them with governance *policy* and *standards*.

Advice

Successful creation of a true DCE happens when all of the defined DM *processes* are operating in concert and all are aligned with the stated DM strategy.

Questions

→ Have the DM *processes* been integrated into a single end-to-end operational *process*?
→ Are the *procedures*, tools and routines in place for implementing the *processes*?
→ Have innovative technologies such as AI and ML been considered as part of the integrated *processes* and infrastructure?
→ Has the review of data ethics been included in the integrated DM *processes*?
→ Are there standing meetings, planning sessions and regular communications validating the *data control*?

Artifacts

- *Process* documentation that shows integration of the DM *processes* into an end-to-end execution
- Documentation of data ethics review as part of the *data controls*
- Governing body minutes and directives related to *data control* monitoring

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No cross-organization DM capability alignment exists.	No cross-organization DM capability alignment exists, but the need is recognized and the alignment is being discussed.	Cross organization DM capabilities are being aligned.	Cross organization DM capabilities are aligned and have been validated as such by the directly involved stakeholders.	Cross organization DM capabilities are aligned and stakeholders are working collaboratively across the organization.	Cross organization DM capability alignment and collaboration are established as part of business as usual practice with a continuous improvement routine. This is recognized as the normal way of working.

7.2 Cross-organization Control Function Collaboration

Cross-control function collaboration includes: 1) aligning DM and other control function policy and standards; 2) establishing engagement routines; and 3) applying cross-organization controls to the data.

7.2.1 Control function and data management policies and standards are aligned

Description

DM controls and best practices must be formally included in cross-organization control *function* *policies* and *standards* to ensure collaboration and alignment.

Objectives

- Create *enterprise* *policy* and *standards* which formally include cross-organization references in each.
- Ensure formal coordination of each groups' *policy* and *standards* through control teams which are held accountable and subject to Internal Audit.

Advice

The goal here is to ensure that the *policies* and *standards* of the DM initiative are aligned with those of the other control *functions*. Take advantage of existing rules and integrate them into the DM *policies*, *procedures*, and *standards*. Other control *functions* should also reference the *standards* and *processes* of the DM initiative.

Questions

→ Are the mechanisms in place to support cross-organization control *function* collaboration?
→ Is there alignment between cross-control *function* *policies*, *standards* and *processes*?
→ Is cross-organization control *function* coordination operational and being reviewed by Internal Audit?

Artifacts

- DM *policies* and *standards*
- Other control *functions*' *policies* and *standards*
- *Policy* and *standards* cross-referencing mechanisms

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No DM and control function policy/standards alignment exists.	No DM and control function policy/standards alignment exists, but the need is recognized and the alignment is being discussed.	DM and control function policy/standards are being aligned.	DM and control function policy/standards are aligned and have been validated as such by the directly involved stakeholders.	DM and control function policy/standards alignment is established and recognized by stakeholders.	DM and control function policy/standards alignment is established as part of business as usual practice with a continuous improvement routine. It is recognized as the normal way of working

7.2.2 Regular routines are established with cross-organization control functions

Description

In order to achieve collaboration among all the control *functions* that have requirements for data or DM a structure of regular interaction is required.

Objectives

- Formally coordinate cross-control *functions* with the DM initiative via regular engagements, meetings and routines.

Advice

Here is where the CDO becomes the *Chief Diplomacy Officer*. There must be an engagement strategy and plan to meet and collaborate with the other control *functions*. This collaboration is required of the *data officer* at each operating level of the organization with their respective cross-organization control *function* peer.

Questions

→ Are the mechanisms to support regular coordination defined and operational?
→ Are formal meetings across control *functions* taking place?

Artifacts

- Engagement plan
- List of *stakeholders* and evidence of bi-directional communication
- Evidence of meetings including minutes and follow-up actions

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
Regular cross-organization control function routines do not exist.	Regular cross-organization control function routines do not exist, but the need is recognized and the development is being discussed.	Regular cross-organization control function routines are being developed.	Regular cross-organization control function routines are defined and have been validated by the directly involved stakeholders.	Regular cross-organization control function routines are established and are recognized and used by stakeholders.	Regular cross-organization control function routines are established as part of business as usual practice with a continuous improvement routine. It is recognized as the normal way of working.

7.2.3 Data entering the ecosystem is subject to cross-organization controls

Description

All new data introduced into or delivered out of the *data ecosystem* must be subject to cross-organization control *standards* to ensure organization-wide compliance.

Objectives

- Subject design review and approval to data introduced into or delivered out of the ecosystem.
- Subject cross-organization *data control policy* and *standards* to data introduced into or delivered out of the ecosystem.

Advice

The goal of cross-organization data control is to ensure that all data entering the ecosystem through any channel is subject to the same restrictions, tollgates, authorizations, and evaluations. The challenge will be to ensure that all of the cross-organization control *functions* understand and recognize the role and authority of the ODM.

Questions

→ Have the cross-organization control *functions' policies* and *standards* been widely communicated?
→ Have the cross-organization control *functions'* requirements for data or DM been inventoried and presented to the DM initiative *stakeholders*?
→ Have the *stakeholders* been informed of their role and responsibility with respect to the onboarding of data into the organizational ecosystem?

Artifacts

- Evidence of cross-referenced rules from other control *functions' policy* and *standards* that demonstrate alignment and collaboration with the DM initiative

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
Cross-organization controls are not applied to all data.	Cross-organization controls are not applied to all data, but the need is recognized and the development is being discussed.	Cross-organization controls are being developed to apply to all data.	Cross-organization controls are defined to apply to all data and have been validated by the directly involved stakeholders.	Cross-organization controls are established on all data and are recognized and used by stakeholders.	Cross-organization controls are established on all data as part of business as usual practice with a continuous improvement routine. It is recognized as the normal way of working.

7.3 Data Risk is Managed

The formal *process* of identifying data risk must be integrated into the DM initiative. Once identified, the risks must be tracked, prioritized and mitigated. These activities should be *standardized* and integrated into the overall risk management framework and *processes* of the organization (e.g.; three lines-of-defense: operating unit, risk function and audit function).

7.3.1 Organizational Unit compliance

Description

All organizational units must be accountable for managing data risk in their unit. The organizational unit is the first-line-of-defense in the organization's risk model.

Objectives

- Define and operationalize the *processes* for self-identification of data risks.
- Manage the data risk *process* to actively track, prioritize and mitigate data risks.
- Engage executive management in data risk management.

Advice

The organizational units are accountable for their data and thus are accountable for the risk associated with their data. The formal *process* of identifying data risk must be integrated into the DM initiative. Once identified, the risks must be tracked, prioritized and mitigated. These activities should be *standardized* and integrated into the overall risk management framework and *processes* of the organization.

The *Chief Data Officer* and in some cases the Chief Operating Officer has a role in supporting the organizational units in the data risk management *process*. This role may even be seen as a review point between the organizational unit and the second-line-of-defense.

Questions

→ Is there a *process* for identifying and managing data risk?

→ Are the *procedures*, tools and routines in place for implementing the *processes*?
→ Does the CDO, the Office of Data Management and/or the COO provide data risk management support to the organizational units?

Artifacts

- Documentation on the *process* to identify and manage data risk
- Evidence of self-attestation and *enterprise* ODM review
- Evidence of CDO or COO data risk management support

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No organizational unit compliance on data risk exists.	No organizational unit compliance on data risk exists, but the need is recognized and the development is being discussed.	Organizational unit compliance on data risk is being developed.	Organizational unit compliance on data risk is defined and has been validated by the directly involved stakeholders.	Organizational unit compliance on data risk is established and is recognized by stakeholders.	Organizational unit compliance on data risk is established as part of business as usual practice with a continuous improvement routine. It is recognized as the normal way of working.

7.3.2 Data Risk function oversight

Description

The data risk *function* must be accountable for establishing the data risk appetite and framework for the organization. The data risk *function* is the second-line-of-defense in the organization's risk model. The CDO must collaborate with the risk *function* to integrate data risk into the overall risk framework.

Objectives

- Establish a data risk appetite statement and *standard* data risk categories and metrics.
- Provide organizational guidance and oversight of the data risk management *process*.

Advice

Data risk must be managed organization-wide in concert with other sources of risk. *Standards* must be set to define data risk appetite along with risk identification, classification and measurement. Because most data has many consumers across the organization a comprehensive view of data risk is critical.

Questions

→ Is there a defined data risk appetite for the organization?
→ Are there *standard* data risk categories and metrics?
→ Has the Risk *function* provided guidance and oversight of the data risk management *process*?

Artifacts

- Data Risk Appetite Statement
- *Standard* data risk categories and metrics
- Risk *function* guidance for the data risk management *process*

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No risk function oversight on data risk exists.	No risk function oversight on data risk exists, but the need is recognized and the development is being discussed.	Risk function oversight on data risk is being developed.	Risk function oversight on data risk is defined and has been validated by the directly involved stakeholders.	Risk function oversight on data risk is established and is recognized by stakeholders.	Risk function oversight on data risk is established as part of business as usual practice with a continuous improvement routine. It is recognized as the normal way of working.

7.3.3 Internal Audit review

Description

Internal Audit must be accountable for periodic review of the DM initiative against the defined *policy* and *standards* of the organization. Internal Audit is the third-line-of-defense in the organization's risk model.

Objectives

- Validate that DM *policy* and *standards* and *processes* are auditable.
- Perform periodic audit according to organization audit *policies*.

Advice

Collaborating with Internal Audit is a valuable exercise to ensure the auditability of the DM initiative. It will also foster a shared understanding of the DM strategy, its objectives, *processes*, and *data controls* that will be valuable to both the data practitioner and the auditor.

Questions

→ Is there a functional partnership in place with Internal Audit?
→ Is Internal Audit familiar with the concepts associated with DM?
→ Has Internal Audit reviewed the DM initiative and determined that it can be audited via scheduled exams?

Artifacts

- Evidence of communication with Internal Audit about the DM concepts
- Verification by Internal Audit that the DM initiative can be enforced and audited
- Internal Audit schedule for DM initiative audits

- Evidence of Internal Audit engagement and review

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No internal audit oversight on data risk exists.	No internal audit oversight on data risk exists, but the need is recognized and the development is being discussed.	Internal audit oversight on data risk is being developed.	Internal audit oversight on data risk is defined and has been validated by the directly involved stakeholders.	Internal audit oversight on data risk is established and is recognized by stakeholders.	Internal audit oversight on data risk is established as part of business as usual practice with a continuous improvement routine. It is recognized as the normal way of working.

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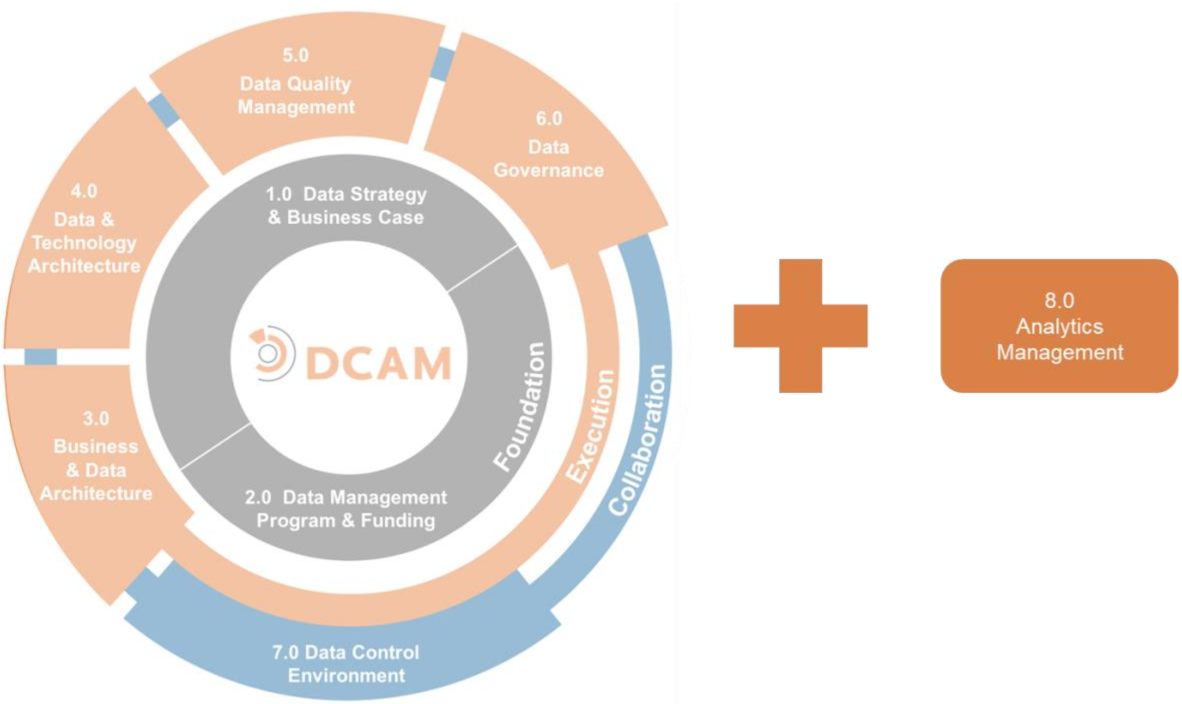
DCAM® Framework

8.0 Analytics Management



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8.0 Analytics Management

Introduction

The first seven components of DCAM define the capabilities required for best practice Data Management (DM). These DCAM best practices guide us regardless of the purpose to which the data is subsequently applied. The issues inherent in Artificial Intelligence/Machine Learning (AI/ML) and an organization's Code of Data Ethics in these components are particularly relevant where Analytics consumes the data. Consumption of data for analytical purposes is increasingly important for many organizations. *Analytics* is dependent on high-quality, well-understood data. *Analytics function*s are, in general, dependent on data produced by areas outside *Analytics* or that is sourced from outside the organization.

The purpose of *Analytics* Management (AM) is to formalize how the *Analytics* activities of an organization are structured, executed, and managed and to ensure they are aligned with the DM activities. The degree to which *Analytics* teams are either centralized or distributed in an organization will depend on the structure and culture of the organization. However, synergies, efficiencies, and effectiveness will be maximized if the teams operate within a well-understood framework as part of a coherent *Analytics* strategy.

Definition

The AM component is a set of capabilities used to structure and manage the *Analytics* activities of an organization. The capabilities align AM with DM in support of business priorities. They address the culture, skills, platform, and governance required to enable the organization to obtain business value from *analytics*.

Scope

- Define the *Analytics* strategy.
- Establish the AM *function*.
- Ensure that *Analytics* activities are driven by the Business Strategy and supported by the DM strategy.
- Ensure clear accountability for the *analytics* created and for their uses throughout the organization.
- Work with DM to align *Analytics* with *Data Architecture* (DA) and Data Quality Management (DQM).
- Establish an *analytics platform* that provides flexibility and controls to meet the needs of the different *stakeholder* roles in the *Analytics* *operating model*.
- Apply effective governance over the *data analysis lifecycle*. Governance includes tollgates for *model* reviews, testing, approvals, documentation, release plans, and regular review *processes*.
- Ensure that *Analytics* follows established *guidelines* for privacy, data ethics, *model bias,* and *model explainability* requirements and constraints.
- Manage the cultural change and education activities required to support the *Analytics* strategy.

Value Proposition

Organizations that best manage their *Analytics function*s and resources effectively maximize the benefits of combining advanced algorithms and high-quality data.

Overview

In simplest terms, *analytics* support decision making by analyzing data. They are not new. Statistical analysis of manually collected data pre-dates the age of computing. Advances in technology have increased the speed, variety, and sophistication of analysis, as well as the quantity and variety of data that can be analyzed. *Analytics*-based, real-time, sophisticated decision making is accelerating and has become a key business differentiator.

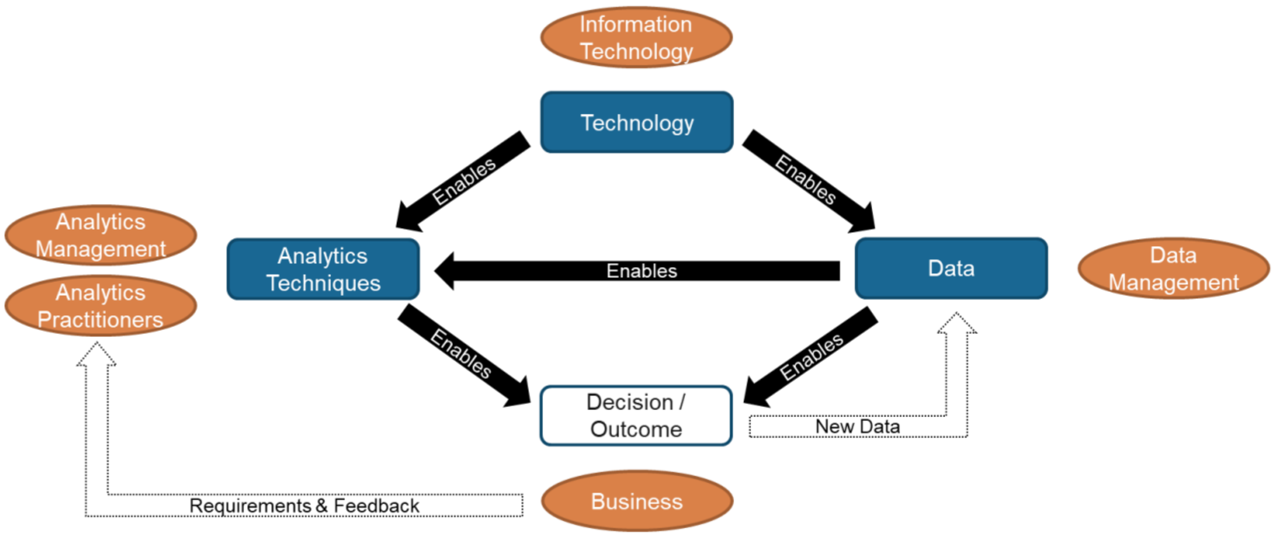

Diagram 8.1: Analytics Enablers and Stakeholders

Advances in technology and data include:

- Lower data storage costs, increasing the quantity of data that can be made available to *analytics*.
- New sources of data such as sensors, telematics, and satellite imagery, enabling new *data sets* and new combinations of *data sets* to be analyzed.
- NoSQL and graph database technologies are enabling greater varieties and quantities of both structured and unstructured data to be accessed and processed efficiently.
- Falling costs of processing and the ease with which cloud computing capacity can be scaled up and down, increasing the affordability of more sophisticated *analytics* techniques.
- Advanced data visualization simplifying how people explore and interpret very large volumes of *analytics* results.
- The evolution of ML, enabling *models* to make decisions with minimal or no human involvement.

Data scientists have the skills to understand a business need, bring together *data sets,* and apply advanced *analytics* techniques to address that need. In many organizations, they may be aligned more directly with the business units they support than with the technology teams that provide more traditional business intelligence *analytics*. In both circumstances, the organization must have a clear *operating model* that provides consistency and efficiency in how the activities are performed. Creators of this *model* must take care to retain the business alignment and agility that the organization demands.

The ever-increasing volume and variety of data available to an organization means the universe of issues that advanced *analytics* can be tasked with is boundless. The organization must have a clear strategy for how *analytics* will be used to support the businesses' objectives. The *operating model* should ensure this *Analytics* strategy can be delivered, and a funding *model* must be established to sustain this effort.

In the absence of effective data management, a significant amount of an *Analytics practitioner's* time is spent manipulating, cleansing, and transforming data in preparation for the *analytics*. Analytics must be aligned with the DM initiative. In particular, DA should ensure that data is understood and that authoritative sources are used where these are available. DQM will provide measures of *data quality* that *Analytics* should reference, and

DQM should be used to manage *data quality* issues uncovered by *Analytics practitioners* as they prepare data for their *models*.

Not all *Analytics* activities will involve the creation of *models* that, once successfully tested, will run as production *processes* or services. Some *analytics* will be one-time exercises to investigate a historical issue or answer a specific question at the moment. There may also be experiments with new data sources or new analytical techniques. The platform that supports *Analytics* must be designed to accommodate these different types of activity and the specific needs of different *stakeholders*.

Most organizations will have restrictions on access to production data, whether driven by commercial sensitivity or personal privacy regulations. *Analytics practitioners* will often need to work with data that has been obfuscated in some way. Requirements for this must be understood and must be facilitated by the *analytics platform*.

Model governance and *model* transparency are critical to the controlled, auditable development and deployment of advanced *analytics*. Compliance with privacy regulations is vital when *models* are released as production assets or services. However, with advanced *analytics*, ML and AI, other governance and control issues also come into play. The need to explain decisions made by *models* may be a regulatory requirement. As with most considerations that need to be confirmed at the point of release into production, requirements for explainability need to be understood as early as possible in the *model* development. This need for early understanding also applies to *model bias*, particularly if this bias could result in prejudice against groups of individuals. In this case, the bias in the *data sets* that are used to train *models* needs to be controlled. These controls may need to be extended to data that the *model* encounters over time in production. Finally, the *models,* and the business issues they are addressing, need to be governed against the Code of Data Ethics of the organization. It is essential to determine if the way the *model* is using the data is ethical and appropriate. This determination then must guide the organization as it decides whether and how it should act on the decisions and recommendations the *model* makes.

Overcoming the challenges an organization will face in delivering its *Analytics* strategy will require cultural and behavioral changes. Analytics practitioners will require specific analytic skill*s*. However, an awareness of both the value that *analytics* can bring and the potential for undesirable impacts created through *analytics* will be needed throughout the organization. Awareness of *model bias* and data ethics considerations will be required by those creating *models* as well as those approving them.

Core Questions

- → Does the *Analytics* strategy clearly articulate the role that *Analytics* will play in delivering the business strategy?
- → Is there buy-in to the *Analytics* strategy, *operating model,* and funding *model* across the organization?
- → Are *Analytics* activities prioritized according to business priorities, and is the value delivered understood?
- → Are AM and DM fully aligned?
- → Does the *analytics platform* support the needs of *Analytics* initiatives and the *Analytics* *operating model*?
- → Are *processes* in place to control the release of *analytics* *models* into production?
- → Is approval and release of *models* aligned with privacy and data ethics governance?

→ Are there initiatives in place to address the cultural change and *Analytics practitioner* education to enable *Analytics* success?

Core Artifacts

The following are the **core artifacts** required to execute an effective *Analytics* Management capability. Items with an '*' link to published best practice *guidelines*.

- *Analytics* Strategy
- *Analytics* Classification System
- *Analytics* *Operating Model*
- *Analytics Methodology*
- *Model Documentation Standard*
- *Data Obfuscation* Strategies
- *Model* Testing, Approval, Release, and Regular Review *Procedures*
- Cultural Behaviors Gap Analysis and Initiatives Backlog
- *Analytics* Skills Gap Analysis

8.1 The Analytics Function is Established

For an *Analytics function* to be sustainable, it must be formally established and recognized. It must be approved by management and supported by an approved funding *model* and an effective governance structure. Roles and responsibilities must be established, and an *analytics methodology* must be adopted.

8.1.1 The Analytics strategy and approach are defined and adopted

Description

The *Analytics* strategy must be defined and formally empowered by senior management, and the role of the *Analytics function* must be communicated to *stakeholders*

Objectives

- Formally establish the organization's *Analytics* strategy.
- Obtain approval for the *Analytics* strategy from *stakeholders*.
- Obtain executive management support for the *Analytics* strategy.
- Communicate the role of *Analytics* across the organization through formal organizational channels.
- Secure the authority to enforce alignment with the *Analytics* strategy through *policy* and documented procedure.

Advice

The *Analytics* strategy must be aligned to the business and operational objectives of the organization and be aligned to the overarching Data Management Strategy (DMS).

The strategy presents the vision for *Analytics*, covering organization, leadership, techniques, platform, *processes,* and culture. It shows how to address the gaps and realize the vision. The goal of developing the Analtyics strategy is to capture the high-level objectives and translate them into achievable *Analytics* solutions.

Achieving *stakeholder* and executive buy-in is critical to the success of the strategy. A well-documented *Analytics* strategy is both a statement of approach and a marketing document for presentation to *stakeholders* and executive management. Effective communication of the strategy is key to empowering a coordinated approach across the organization and avoiding inconsistencies and inefficiencies of different business areas, taking different approaches.

Questions

→ Is there a formal, documented strategy and approach for *Analytics*?
→ Have *stakeholders* been identified and involved in the creation and approval of the strategy?
→ Are the *Analytics* strategy and the vision of the DMS aligned?
→ Does the *Analytics* strategy support the high-level objectives of the organization?
→ Has executive management support for the strategy been obtained?
→ Has the role of *Analytics* been formally communicated throughout the organization?
→ Do *policies* and *procedures* exist with the authority to enforce alignment with the strategy?

Artifacts

- A Vision statement of the target state for *Analytics* and how it supports the organization's objectives
- A prioritized approach to the initiatives required to realize the *Analytics* vision

- Documented alignment of the *Analytics* strategy and vision of the DMS
- List of *stakeholders* and evidence of bi-directional communication
- Evidence of formal communication of the strategy and approach
- Evidence of formal approval of the strategy
- *Policies* and *procedures* enforcing alignment with the strategy

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No formal *Analytics* strategy exists.	No formal *Analytics* strategy exists, but the need is recognized, and the development is being discussed.	The formal *Analytics* strategy is being developed.	The formal *Analytics* strategy is defined and has been validated by the *stakeholders*.	The formal *Analytics* strategy is established and understood across the organization and is being followed by the *stakeholders*.	The formal *Analytics* strategy is established as part of business-as-usual practice with a continuous improvement routine. The strategy and approach are reviewed and updated at least annually.

8.1.2 A categorization system for levels of analytics is defined and adopted

Description

The spectrum of *analytics* (e.g., MI (Management Information), BI (Business Intelligence), ML, or Descriptive, Diagnostic, Predictive, Prescriptive) employed by the organization is defined. Names and descriptions of the different types of *analytics* relevant to the organization are formalized in a categorization system. This system is used in statements of the scope of responsibility in the *Analytics* *operating model*. It provides a common language for the organization to refer to *analytics* and helps avoid confusion and misunderstanding.

Objectives

- Define the different types of *analytics* relevant to the organization.
- Validate the categorization system with *stakeholders*.
- Communicate the categorization system to *Analytics practitioners*.
- Adopt the categorization system in the *Analytics* *operating model*.

Advice

In defining the spectrum of *analytics*, examples will bring these to life. The boundaries of the *analytics* spectrum determine the activities to which the *Analytics* governance and best-practice frameworks apply. For example, an organization may define its lowest boundary of *analytics* such that static MI reports are out-of-scope, whereas self-service BI is considered in-scope.

In setting the categorization system, it is important to consider how to identify the *stakeholders* and to solicit their input. It is also important to identify who should receive communications regarding the outcomes of the *process* of setting the categorization system.

The categorization system for levels of *analytics* will not be static. It will work best by allowing for emerging levels of sophistication as *Analytics* matures and evolves.

Questions

→ What types of *analytics* has the organization developed historically?
→ What types of *analytics* does the organization foresee itself using in the future?
→ Which *stakeholders* should be approached for input into the categorization system for levels of *analytics*?
→ Who should receive the published categorization system, both within and outside the organization?

Artifacts

- Documentation of the categorization system
- Lists of *stakeholders* involved with the categorization system
- Distribution lists for publishable versions of the categorization system

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No categorization system for levels of *analytics* exists.	No categorization system for levels of *analytics* exists, but the need is recognized, and the development is being discussed.	A categorization system for levels of *analytics* is being developed.	A categorization system for levels of *analytics* has been defined, reviewed, and approved.	A categorization system for levels of *analytics* has been adopted.	The relevance and effectiveness of the categorization system for levels of *analytics* are established as part of business-as-usual practice with a continuous improvement routine.

8.1.3 The Analytics operating model is defined, and its structure is implemented

Description

An *operating model* is defined to describe how *Analytics* will be structured within the organization. It establishes the scope and coverage of the different parts of *Analytics* and designs how they should work together to serve the needs of the business.

Objectives

- Define the *Analytics* team's structure within the organization, delineating the scope and coverage of each.
- Define the roles and responsibilities within each type of *Analytics* team.
- Create the organizational structure described in the *operating model*.
- Obtain approval from *stakeholders* for the *Analytics* *operating model*.
- Define the interrelationships between the *Analytics* teams, between *Analytics* teams and the Data Management initiative and data teams, and between *Analytics* teams and the business *function*s.

Advice

The *Analytics* *operating model* encompasses disciplines from different areas of IT (Information Technology), business, and *Analytics*. It provides clarity on how these areas work together to provision the data required by analytical *models* and to deliver the *models* into production.

The *Analytics* *operating model* clarifies the ownership of line-of-business *Analytics*, cross-business *function*s, and any overarching *Analytics* program. It is aligned with other *operating models* within the organization. If there is a strong culture of federated *operating models,* this should be embraced. Conversely, if there is a strong centralized *operating model* culture, carefully consider the benefits before deviating from this approach.

Best practices suggest that the ideal make-up of an *Analytics* team includes people with mathematical, analytical, and technical skills and strong business acumen. The creation of centers of excellence should be considered. Establishing centers of excellence can focus discipline around *Analytics* and instill a level of confidence in the user community. Centers of excellence will bring together the IT, business, and analytical skills that better leverage the organization's investment in technology. These centers transition *Analytics* from technical data gathering to providing solutions through organization-wide awareness, sharing, and best practice.

Having buy-in from *stakeholders* is key if the *Analytics* organization is to be effective. *Stakeholders* should be involved throughout the creation of the *operating model* to avoid misaligned expectations.

Questions

→ Has the structure of the *Analytics* teams within the organization been defined?
→ Have the roles and responsibilities of each type of *Analytics* team been defined?
→ Does the team structure delineate the scope and coverage of each?
→ Are the interactions of *Analytics* teams, the Data Management initiative, and the business defined?
→ Have *stakeholders* approved the *operating model*?
→ Has the functional structure described in the *operating model* been implemented?

Artifacts

- *Operating Model* Terms of Reference
- Organization structure of *Analytics* teams
- Role definitions and RACIs (both between *Analytics* teams and between *Analytics* and other *functions*)
- *Process* *models* for *Analytics*, DM and business interactions
- List of *stakeholders* and evidence of bi-directional communication
- Documented formal approval of the *operating model*

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
The *operating model* for *Analytics* does not exist.	The *operating model* for *Analytics* does not exist, but the need is recognized, and the development is being discussed.	The *operating model* for *Analytics* is being developed.	The *operating model* for *Analytics* has been defined, reviewed, and approved.	The *operating model* for *Analytics* has been implemented, and the organization structure was established.	The *operating model* and organization structure for *Analytics* is established as part of business-as-usual practice with a continuous improvement routine.

8.1.4 The funding model for Analytics has been established, approved and adopted

Description

The *Analytics* funding *model* addresses the resources required for *stakeholders* to implement and sustain the *Analytics function*. The funding *model* aligns with the organization-wide funding *process*. It is enforced through the *Analytics* governance *process*.

Objectives

- Define and socialize the *Analytics* funding *model* with *stakeholders*.
- Collect and incorporate feedback into the *model*.
- Have *stakeholders* review and approve funding commitments for a sustainable *function*.
- Align the funding *model* with the overall funding approach of the organization.
- Measure benefits that help support the funding case.

Advice

The funding *model* must address all aspects of funding *Analytics*. Types of work beyond business-driven *Analytics* projects should be addressed. These include experimentation, foundational work, creation of re-usable *data assets* and ad-hoc, high-priority analysis, and more. Address the funding of software licenses and platform and infrastructure costs. When addressing people costs, take account of differentials in costs of *Analytics* specialists. Alignment with the overall funding approach of the organization may require the *model* to distinguish between *Analytics* initiatives aligned with *operating units* and aspects of the *operating model* that are not *operating unit*-specific.

Ensure that *Analytics* is funded as a sustainable *function*. Use the funding *model* to distinguish aspects of funding that are discretionary from those that are non-discretionary. *Stakeholder* involvement in the creation of the funding *model* is critical to obtaining a financial commitment that can be sustained.

Questions

→ Has a funding *model* for *Analytics* been established?
→ Is the *model* consistent with the established funding approaches and budget *processes* of the organization?

→ Does the *model* address how *Analytics* will be funded as an ongoing, sustainable *function*?
→ Are measurable benefits from *Analytics* initiatives used as support for the funding requirements?
→ Has the *model* been socialized with *stakeholders,* and has their feedback been addressed?
→ Has the *model* been formally approved?

Artifacts

- Documented funding *model*
- A written *process* to review, modify and validate resource plans
- Evidence of alignment with budget *processes* and organizational cycles
- Documentation of measured benefits
- List of *stakeholders* and evidence of bi-directional communication
- Funding plans and budget allocation
- Evidence of formal approval

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No funding *model* for *Analytics* exists.	No funding *model* for *Analytics* exists, but the need is recognized, and the development is being discussed.	The funding *model* for *Analytics* is being developed.	The funding *model* for *Analytics* has been defined, reviewed, and approved.	The funding *model* for *Analytics* has been implemented.	The funding *model* for *Analytics* is established as part of business-as-usual practice with a continuous improvement routine.

8.1.5 Analytics governance structures are in place

Description

Explicit governance of *Analytics* is in place across the organization. The governance structure oversees implementation and sustains the *operating model* for *Analytics*, implementation of the *analytics platform,* and ongoing initiatives to shape the *Analytics* culture of the organization. It ensures alignment of *Analytics* with business strategy, data ethics, and the DMP (Data Management Program).

Objectives

- Define the governance structure for *Analytics*.
- Document and share the governance structure with *stakeholders*.
- Create *policies* to enforce *Analytics* governance implementation.
- Establish governance forums with written and approved charters.
- Implement operating governance structures.
- Appoint *stakeholders*.
- Communicate *stakeholder* roles and responsibilities.
- Hold *stakeholders* accountable for their participation in *Analytics* via performance reviews and compensation considerations.

Advice

The governance structure should complement and align with the *Analytics* *operating model*. Care should be taken to ensure the governance structure does not constrain timely decision making. Members of the *Analytics* governance structure need to be clear about their roles and empowered to effect change. Ensure there is representation from all the disciplines required to resolve issues arising from identification of opportunities, data collection and provisioning, and *model* deployment.

Analytics will have dependencies on other areas of the organization for deliverables such as data collection and *model* deployment. It is particularly important to agree in advance to clear hand-offs and escalation paths. Consider having members of the governance organization reporting to senior management, such as the COO, to gain this empowerment.

Tool selection should be aligned with the business strategy and not driven just by what tooling or expertise exists in the organization.

Ethics and privacy are central themes in the governance of *Analytics*. Start by building on the ethics and privacy data governance *processes* and structures already in place. Governance *processes* need to demonstrate the fairness, transparency, and security of data usage.

Questions

- → Do terms of reference for *Analytics* governance exist?
- → Do *policies* exist that enforce *Analytics* governance implementation?
- → Are there metrics for what is governed and measures of successful compliance?
- → Is there a register of *Analytics* projects with up-to-date status available?
- → Is there a *model* register with review dates and update history?
- → Is there a tooling inventory with usage recommendations available?
- → Is there RACI (Responsible, Accountable, Consulted, Informed) documentation for all *stakeholders* and participants in the *Analytics* governance *process*?
- → Is the governance *process* demonstrating that ethics and privacy are governed?

Artifacts

- *Analytics* governance terms of reference
- Evidence of *policies* written, implemented and enforced to show that *Analytics* is properly governed
- *Analytics* governance metrics and measurements
- *Analytics* project tracking
- *Model* inventory, including *model* reviews and updates
- Tooling inventory and recommendations
- *Analytics* governance RACI

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No governance structures for *Analytics* exist.	No governance structures for *Analytics* exist, but the need is recognized, and the development is being discussed.	*Analytics* governance structures are being developed.	*Analytics* governance structures have been defined, reviewed, and approved.	*Analytics* governance structures are established and operational.	*Analytics* governance structures are established as part of business-as-usual practice with a continuous improvement routine.

8.1.6 An analytics methodology and model documentation standard have been adopted

Description

An *analytics methodology* provides a framework for the activities performed through the *analytics* lifecycle. The lifecycle begins with understanding the business problem and runs through to deployment, operation, and review of the solution. The framework provides a common language and structure for *stakeholders* to refer to the different stages and aspects of *Analytics* activities. A *standard* for *model* documentation ensures consistency across the organization in the way that *model* provenance, assumptions, inputs, outputs, parameters, and limitations are captured and communicated.

Objectives

- Define and adopt a *standard* for *model* documentation.
- Select or develop an *analytics methodology* that defines the *analytics* lifecycle for the organization.
- Obtain approval and support for the methodology and *model documentation standard* from *stakeholders*.
- Ensure the methodology and *model documentation standards* are understood and adopted by *Analytics practitioners*.
- Formally document the *analytics methodology* and *model documentation standard*.
- Provide feedback mechanisms for ongoing refinement and improvement of the methodology and *model documentation standard*.

Advice

Whether the organization should buy or outsource *analytics* solutions as opposed to internally developing them will depend on the maturity and culture of the organization. There will be more flexibility to define and specify the *analytics methodology* for internally developed *analytics* than for externally developed *analytics*. However, in both cases, it may be worthwhile to consider external best practice methodologies and adapting them to the organization.

To ensure that the *analytics methodology* is appropriate to the organization and to get *stakeholder* buy-in, key influencers and *stakeholders* should be involved in the selection or development of the *analytics methodology*. Form a community of “champions” to act as owners of the methods and drivers of improvement.

The *analytics methodology* should focus on significant steps in the end-to-end *process* rather than specific analytical or modeling techniques. *Analytics* tools come and go, while *analytics methodology* should be sustainable and stable. However, to reach a long-term stable *analytics methodology*, the organization must be open to methodology improvements based on experience. The *analytics methodology* should accommodate innovation as well as business use-case-driven *analytics*.

Questions

- → Has an *analytics methodology* that defines the *analytics* lifecycle of the organization been selected or developed?
- → Has a *model documentation standard* been created?
- → Has approval and support for the methodology and *model documentation standard* from *stakeholders* been obtained?
- → Has it been confirmed that the *analytics methodology* and *model documentation standard* have been understood and adopted by *Analytics practitioners*?
- → Have the *analytics methodology* and *model documentation standard* been formally documented?
- → Have feedback mechanisms for ongoing refinement and improvement of the *analytics methodology* and *model documentation standard* been developed and made available to *stakeholders*?

Artifacts

- Records of *analytics methodologies* considered by the organization
- *Analytics methodology* and terminology document
- *Model documentation standard*
- Records of approval by *stakeholders* of the *analytics methodology* and *model documentation standard*
- Evidence of adoption of the *analytics methodology* and *model documentation standard* by *Analytics practitioners*
- Feedback on refinements to the *analytics methodology* and *model documentation standard*
- Feedback from *Analytics practitioners* and other *stakeholders*
- Documentation of mechanisms for ongoing refinement of the *analytics methodology* and *model documentation standard*

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No *analytics methodology* or *model documentation standard* exists.	No *analytics methodology* or *model documentation standard* exists, but the need is recognized, and their development is being discussed.	An *analytics methodology* and a *model documentation standard* are being developed.	An *analytics methodology* and a *model documentation standard* have been defined, reviewed, and approved.	The *analytics methodology* and *model documentation standards* are being followed.	The *analytics methodology* and *model documentation standard* are established as part of business-as-usual practice with a continuous improvement routine.

8.2 Analytics is Aligned with Business and Data Management Strategy

The *Analytics* and DM *functions* must be aligned and together must support the business goals. *Analytics* must be prioritized to meet the needs of business strategy and drive business value.

8.2.1 Dependencies between Analytics and Business Architecture are understood and addressed

Description

Each organization adopts a unique *business architecture*. This architecture defines its structure, governance, *processes,* and information for decision making. The *business architecture* of the organization should inform the organizations' analytical requirements. Those accountable for delivering *analytics* solutions are also accountable for engaging the business *stakeholders*, understanding the impact of business requirements on *analytics*, aligning analytical requirements, and ensuring that these requirements are delivered.

Objectives

- Review the *business architecture* (i.e., strategy, structure, governance, *processes*) – regardless of the degree of formalization.
- Understand and document key analytical requirements appropriate for the business.
- Align the *Analytics* capabilities and *processes* with business requirements.
- Communicate and anchor analytical requirements to *stakeholders*.

Advice

Not all businesses operate the same way. Priorities for decision making may vary across different parts of the same organization. *Analytics* leaders must engage with business leaders to understand their requirements and consider how *Analytics* should best align with the *business architecture*. Note that the *business architecture* is composed of a broad array of elements and not merely the IT architecture.

The *operating model* designers for *Analytics* must ensure it is informed by the business requirements and *business architecture* and is kept current with developing input from business leaders. The proximity of *analytics* to senior decision making may require parts of the *Analytics* organization to be more decentralized and embedded within multiple business units or functional areas. However, some *analytics* may best be enabled by a center of excellence.

The most effective *Analytics* organizations work on projects aligned to the business objectives and *business architecture*. Full engagement with business objectives ensures the *Analytics* portfolio is balanced: strategic vs. tactical; compliance vs. performance; BAU support vs. innovation; balanced scorecard vs. single metric.

Documenting the analytical requirements will achieve a clear and consistent understanding. Ongoing interaction with the evolving business needs keeps *Analytics* focused on immediate priorities.

Questions

→ Is the *Analytics* *operating model* and plan documented?
→ Have key dependencies been taken into account in the design of the *Analytics* organization and its connectivity to decision making?
→ Does the *Analytics* plan include specific reference to business requirements?
→ Have the senior executive and line-of-business executive teams endorsed the *Analytics* plan?
→ Is there a defined *process* for ongoing updates of the *Analytics* plan?

Artifacts

- Evidence of alignment of *Analytics* with business strategy and plan
- *Analytics operating model*
- Documented alignment of *Analytics* and *Business Architecture*
- Details of business *analytics* requirements
- Evidence of senior executive input and approval

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
Dependencies between *Analytics* and *Business Architecture* are not understood.	Dependencies between *Analytics* and *Business Architecture* are not understood, but the need is recognized, and the development is being discussed.	Dependencies between *Analytics* and *Business Architecture* are being determined.	Dependencies between *Analytics* and *Business Architecture* have been defined, reviewed, and approved.	Dependencies between *Analytics* and *Business Architecture* are understood and are being addressed.	Understanding and addressing the dependencies between *Analytics* and *Business Architecture* is established as part of business-as-usual practice with a continuous improvement routine.

8.2.2 The prioritization of Analytics is driven by business strategy

Description

The business strategy drives the *Analytics* roadmap and activities. This fact keeps the focus on *Analytics* investments and how they can maximize business outcomes.

Objectives

- Understand leadership expectations and explore how *Analytics* can support strategic value creation.
- Work collaboratively with leadership to build a comprehensive list of current business opportunities and *analytics* use cases.
- Co-create an *Analytics* vision and roadmap that helps the alignment with business priorities, recognize short- and long-term focus areas, and creates senior management buy-in.
- Create a benefits case for each *analytics* use case and prioritize them according to business value through the *Analytics* prioritization *process*.
- Communicate the *Analytics* vision and the prioritization of *analytics* use cases to *stakeholders*.

Advice

Align the *Analytics* roadmap with the business strategy establishing a forum across the business to supervise prioritization of *analytics* use cases.

A framework to assess the benefits of *analytics* use cases should have clear dimensions for scoring and grading them. Scoring and grading prioritization should include, but not be limited to, the complexity, feasibility, regulatory requirements and business value of the *analytics* solution, the number of data sources involved, and the size of the user community impacted. The framework should be transparent, allowing the fair assessment of *analytics* use cases and the communication of use case prioritization with the business *stakeholders*.

The organization should ensure that *Analytics* activities include both those that drive business outcomes as well as those that develop the foundational and future state *Analytics* capabilities. Create a roadmap that contains a detailed list of initiatives that are sequenced to help the organization deliver on its strategic objectives, and establish *processes* to revisit sequencing as business priorities and market conditions change.

Questions

- → Is there a formal, documented, and prioritized roadmap for *Analytics*?
- → Does a *process* exist to ensure prioritizations are revisited and modified by accountable *stakeholders*?
- → Does the *Analytics* roadmap articulate its support of the organization in its strategic business objectives?
- → Has the *Analytics* vision and roadmap received appropriate high-level sponsorship?
- → Do the *analytics* use cases document the assessments of their business value as well as their feasibility?
- → Are there clear grading criteria for assessing or scoring the potential business value and feasibility of *analytics* use cases?
- → Has communication support been established so business *stakeholders* can prioritize the use cases?

Artifacts

- *Analytics* vision statement or plan
- Prioritized roadmap of *Analytics* initiatives
- Details of use case benefits
- Evidence of high-level sponsorship and support

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
The prioritization of *Analytics* is not driven by business strategy.	The prioritization of *Analytics* is not driven by business strategy, but the need is recognized, and the development is being discussed.	The approach to business strategy driving the prioritization of *Analytics* is being developed.	The approach to business strategy driving the prioritization of *Analytics* has been defined, reviewed, and approved.	The prioritization of *Analytics* is driven by business strategy.	Prioritization of *Analytics* driven by business strategy is established as part of business-as-usual practice with a continuous improvement routine.

8.2.3 Analytics support and influence business needs and is actionable where required

Description

Analytics properly aligned to business objectives can deliver value in multiple ways. Examples include informing decisions, validating understanding, identifying *process* effectiveness, or suggesting actions. *Analytics* teams drive innovation, taking new ideas to the business as well as responding to business-led use cases. Appropriate *analytics* insights should be embedded in business information and *processes*. In many cases, this will include automation of *analytics* with technology for greater speed and efficiency.

Objectives

- Clarify which business questions, decisions, actions or *processes* would be impacted by the analytical solution in the formulation of a new *analytics* use case.
- Early in the design of any *analytics* solution, establish which user community will benefit from the analytical solution.
- Co-develop *analytics* solutions with relevant business teams.
- Ensure access to *analytics* outputs and that visualizations of them are readily available.
- Communicate and deploy outputs in ways the business users can easily consume them, including the development of good visualizations and user interfaces to ensure the explainability of the *analytics* outputs.

Advice

Analytics should both respond to business needs and proactively generate new insights. In each case, there is a responsibility to ensure the business can use the *analytics* outputs to support its decisions and act on the new insight.

Regular communication between the business and *Analytics* teams is critical for success. Interaction allows the analytic solutions to be co-developed so that they are focused on the right needs and delivered in a way that is most effective for the users to act. During this *process*, *Analytics* can add value by considering a broader array of data inputs and insights that may be derived from a mix of internal and external sources.

When designing new *analytics* use cases or initiatives, it is important to consider the actionability of insights. In some instances, limits on actionability may arise from ethical, legal, or commercial considerations. For example, while it may be technically feasible to predict which *customers* are most likely to churn, the business must be extremely careful when deciding whether to act on this insight, to avoid unintended consequences of contacting them. The use of personally identifiable information often requires special care for ethical and regulatory reasons.

All *Analytics* activities must have a purpose. Some analysis is likely to be investigative and so not directly actionable in a business *process*. However, these analyses should fit into a decision *process* where the exploratory analysis helps the business choose the next course of action.

Analytics teams have multiple options for the deployment of insights depending on the context and time-sensitivity of the use case. For example, the ability to automate or centralize critical algorithms while minimizing manual manipulation is likely to be more appropriate for high volume close to real-time decision making. In contrast, ongoing business *processes* with humans in the loop are more likely to need analytic outputs using visualizations and integrated business intelligence tools. In both scenarios, it is important to ensure the explainability of the *analytics* outputs so that *stakeholders* can gain trust in using the *analytics*.

Questions

→ Do formal *procedures* exist for documenting the purpose, scope, and delivery of *analytics* use case insights?
→ Does the *analytics* specification document identify the specific user community impacted?

→ Does the *analytics* specification document identify how the outputs will be delivered, acted on, or embedded in the business *process*?
→ Have BI visualization tools been deployed for the delivery of *analytics*?
→ Is the *analytics* tooling readily available to all potential beneficiaries in the organization?
→ Has the organization demonstrated the benefits of automating *analytics* that supports business *processes*?

Artifacts

- *Analytics* governance *policies*
- *Analytics* specification documents
- Evidence of alignment with Privacy and ethics *policies*
- Evidence of *analytics* automation
- *Analytics* output visualizations

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
The means of ensuring that *Analytics* support and influence business needs, and are actionable where required, does not exist.	The means of ensuring that *Analytics* support and influence business needs, and are actionable where required, does not exist, but the need is recognized, and the development is being discussed.	The means of ensuring that *Analytics* support and influence business needs, and are actionable where required, is being developed.	The means of ensuring that *Analytics* support and influence business needs, and are actionable where required, has been defined, reviewed, and approved.	The means of ensuring that *Analytics* support and influence business needs, and are actionable where required, is implemented.	The means of ensuring that *Analytics* support and influence business needs, and are actionable where required, is established as part of business-as-usual practice with a continuous improvement routine.

8.2.4 Analytics impact is measured and understood to be driving business value

Description

The objective of *Analytics* is to provide insights that are used to deliver outcomes of business value. The organization must routinely measure the impact and effectiveness of their *analytics* solutions.

Objectives

- Include ways to measure new and additional value created by *analytics* solutions early in the course of their development.
- Include ways to measure new and additional value created by *analytics* solutions post-implementation, as continued ingestion of new data can increase or decrease their value.
- Engage the *stakeholders* to create shared accountability and buy-in to quantified benefits driven by *analytics*.
- Measure the impact of *analytics* use cases, projects, and experiments.
- Communicate the observed and measured business value of *analytics*.

Advice

Organizations should consider in advance how the outcome of any analytical solution will be measured.

A useful starting point is to establish reliable comparison benchmarks from which to measure incremental impacts. Control groups and randomized experimental methods are good ways to establish these benchmarks. These tools become especially important to diagnose cause and effect in situations where actions based on *analytics* insight do not result in the anticipated benefits. For example, the failure of an *analytics*-based marketing campaign that does not deliver expected benefits may not simply result from failed *analytics* *models*. Still, it may arise from other factors such as poor marketing or data.

When the incremental value of *analytics* can be measured, the successful measurement can be applied to a whole portfolio of analytical solutions. The *Analytics* teams quantify the business value of their actions to the organization at large, justifying the initial and additional investments in analytical capabilities.

The systematic quantification and communication of business value support the development of a data-driven culture of inquiry.

There can be *analytics* where the benefits cannot be quantified. In this case, the qualitative benefits of these should be documented and recognized. The value of *analytics* that informs decisions not to change or take action should also be recognized.

Questions

→ Does the *analytics* specification document mention use of KPIs (key performance indicators) to evaluate the *analytics*?
→ Are KPIs, control groups, or other experimental methods routinely used to prove the incremental value of *analytics* against a benchmark?
→ Do the control groups or other experimental methods in use successfully isolate the impact of analytic insights versus the impacts of other related business actions?
→ Are the benefits of *analytics* communicated and understood by senior management?

Artifacts

- *Analytics* specification documents
- Evidence of the return on investment of *analytics* solutions
- Evidence of experimental design to evaluate the incremental value of *analytics*
- Post-implementation benefits evaluations
- Evidence of communication of *analytics* benefits to senior management

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
Analytics usage is not measured or understood to be driving business value.	*Analytics* usage is not measured or understood to be driving business value, but the need is recognized, and the development is being discussed.	Measurement of *analytics* usage and business value is being developed.	Measurement of *analytics* usage and business value has been defined, reviewed, and approved.	Measurement of *analytics* usage and business value is being performed.	Measurement of *analytics* usage and business value is established as part of business-as-usual practice with a continuous improvement routine.

8.3 Analytics is Aligned with Data Architecture

Analytics must be an integrated part of the *data ecosystem* of the organization. It is dependent on data that often originates upstream in the organization and produces *analytical products* and services to support decision making within an organization. Alignment with *Data Architecture* (DA) capabilities is critical to ensure the reliability of data and assure confidence and trust in decisions taken based on them. *Data lineage* must be understood, approved business definitions must be referenced, and identification and classification *standards* must be followed. Consistent *standards* must be applied to the preparation of data for analytics.

8.3.1 Analytics data lineage is understood, and authoritative data sources are used

Description

Knowledge about the lineage of data used in analytics provides a basis of trust for the *analytical products* and services they are used to create as well as the business decisions based on them. Data used in analytics must be clearly defined, referencing the organization's *business glossary* and following the organization's *metadata standards*. *Authoritative data sources* must be used to ensure *completeness* and fit-for-purpose quality. However, there will be cases where analytics will require data for which the authoritative source has yet to be agreed. Recording the use of such new or alternative sources is important to the integrity of Analytics. Data must be traceable from source to final product. Applied transformations or aggregations must be understood. If understanding is not possible, the lack of understanding should be highlighted, for example, in the case of ML. And the use of the transformation or aggregation in a *model*, calculation or visualization must be agreed by *stakeholders* to be reasonable and appropriate.

Objectives

- Document data sources and data analysis outputs by using the *business glossary* and *metadata* descriptions.
- Source data from *authoritative data sources* where relevant to ensure *completeness* and fit-for-purpose Data Quality (DQ).
- Understand *data lineage* from input or *authoritative data source* to final *analytical product*, including any transformations, calculations, and aggregations (or highlight where these are not known).
- Confirm *stakeholder* agreement that data is reasonable and appropriate for use in *analytical products*.

Advice

Throughout data *analytics* tools, *models* and *processes*, *data lineage* must be maintained. Where not possible, that fact must be highlighted. An acceptable *data lineage* includes *metadata* that provides a definition, description, or context for the data, including business, operational and technical characteristics. *Data lineage* must be established across platforms to ensure continuity of proper data usage throughout the *data ecosystem* (landing, staging, sandboxes, reports, and operational and *analytical databases*). *Authoritative data sources* should be leveraged where possible in *data preparation*.

Data lineage provides an audit trail that is key to post-decision defensibility of *analytics* and post-outcome accountability for business decisions. *Data lineage* should be part of any *model*, analytical method, or tool documentation and use the organization's *standards* and *processes*. *Analytics* staff should be well-trained and versed in *data lineage* methods, *processes,* and documentation.

Data lineage solutions for *analytics* should be integrated with the overall lineage approach and tools of the organization that is established by the DA *function*.

Questions

- → Are data sources and data analysis outputs well documented, and do they consistently reference *business glossary* and *metadata* descriptions?
- → Is data sourced from *authoritative data sources* (where relevant)?
- → Is *data lineage* well understood from input or *authoritative data source* to final *analytical product*?
- → Does *data lineage* documentation include any transformations, calculations, and aggregations (or highlight where these are not known)?
- → Are *stakeholders* in agreement that data is reasonable and appropriate for use in *analytical products*?

Artifacts

- *Data lineage policy* and *procedures* and systems that apply to data analytics tools and *processes*
- Analytical *data architecture* and *data flow construct* documentation
- *Analytics* documentation that includes *data lineage*
- Evidence of *stakeholder* agreement including formal sign-offs
- *Data lineage* training materials and evidence of competency

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
Analytics data lineage is not understood, and *authoritative data sources* are not used.	*Analytics data lineage* is not understood, and *authoritative data sources* are not used, but the need is recognized, and the development is being discussed.	The *processes* for understanding *analytics data lineage* and using *authoritative data sources* are being developed.	The *processes* for understanding *analytics data lineage* and using *authoritative data sources* have been defined, reviewed, and approved.	The *processes* for understanding *analytics data lineage* and using *authoritative data sources* have been implemented.	The *processes* for understanding *analytics data lineage* and using *authoritative data sources* are established as part of business-as-usual practice with a continuous improvement routine.

8.3.2 Analytics reference approved business definitions

Description

Clear, unambiguous business definitions are fundamental to trusted decision-making based on *analytical products* and services. Data consumed and produced by *analytics* must be mapped consistently to approved and understood business definitions.

In a mature data environment, business definitions are captured and maintained in authoritative sources (business and semantic glossaries), enriched by *metadata* and *data lineage*, and supported by data management *policy* and *standards* to ensure content quality, controlled access, and appropriate.

Objectives

- Use approved business definitions across all *analytical products* and services.
- Enable all *Analytics processes* and people to retrieve, understand, and trust the data they use based on clear business definitions.
- Drive *accuracy*, effectiveness, and efficiency of reporting by consistent usage of business definitions.
- Ensure *Analytics stakeholder* understanding of business definitions.

Advice

Confirm that the DMP has established authoritative, approved *business glossaries*. Reference to and use of the definitions in these glossaries by *Analytics practitioners* helps ensure that *model* inputs and results are understood consistently across *Analytics* and between *Analytics* and other areas of the organization.

Business glossaries should be accessible to *Analytics practitioners* and *stakeholders* through established data distribution services (see DCAM 4.1.4, Data Management is engaged in the definition and development of the organization-wide data distribution infrastructure). Controlled access to *business glossaries* enables approved definitions to be incorporated in *model* documentation and the visualization of results.

Employ feedback and contribution mechanisms for use by *Analytics stakeholders*. By providing feedback capabilities, *stakeholders* can clarify and improve definitions and propose new definitions for data created by *analytics* applications. Record *Analytics*' interest in definitions as *metadata* to ensure they can be notified of definitional changes and address any impact on *analytics* products.

Discovery and experimentation may commence without a full and formal understanding of all business definitions, particularly where new *data sets* are involved. The organization should be clear where formalization of these is required before adoption into an analytical production environment, *process,* or product.

Questions

→ Is an authoritative glossary of distinct and specific business definitions referenced and applied across all *analytical products* and services?

→ Are all *Analytics processes* and people able to retrieve, understand, and trust the data they use based on clear business definitions?

→ Do *Analytics stakeholders* have a common understanding of business definitions?

Artifacts

- Reference to authoritatively sourced business definitions in design documentation and *analytical products* and outputs
- List of *Analytics stakeholders* and evidence of bi-directional communication on business definitions

- Agreement on business definitions for data produced by *analytics* with verification by *Analytics stakeholders*
- Business definitions readily accessible to *Analytics stakeholders*
- Documentation evidencing linkage of business definitions in *analytics* to the organization's *metadata* repository

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
Analytics do not reference approved business definitions.	*Analytics* do not reference approved business definitions, but the need is recognized, and the development is being discussed.	*Processes* to ensure that *analytics* reference approved business definitions are being developed.	*Processes* to ensure that *analytics* reference approved business definitions are being developed have been defined, reviewed, and approved.	*Processes* to ensure that *analytics* reference approved business definitions are being followed.	*Processes* to ensure that *analytics* reference approved business definitions are established as part of business-as-usual practice with a continuous improvement routine.

8.3.3 Analytics respect the organization's identification and classification standards

Description

Identification and classification *standards* organize data. They increase the ease with which data can be located and retrieved and secure integrity throughout the data analysis life cycle. *Data classification* should follow the organization's data management, information security, and applicable regulatory and compliance *standards*. *Analytics* must respect these *standards* both as a consumer and producer of data.

Objectives

- Define clear roles and responsibilities, as both consumers and producers of data, for the identification and classification of analytical data and products in line with the organization's data management, compliance, and information security *policies*.
- Safeguard confidentiality, integrity, and availability of data and *analytical products* throughout all *analytics data lineage* and *analytics* life cycle *processes* in line with the organization's *policies*.
- Implement and carry *data classifications* throughout data *processes*, systems, and data analysis life cycles.
- Apply the identification and classification *standards* to data used or generated in *analytics* to make it easily searchable and trackable based on categories and criteria set by the organization.

Advice

Data identification and classification facilitates data discovery, querying, and retrieval. *Data classifications* and indexation can significantly improve data extraction *process* performance, storage, and encryption *processes* and facilitate the communication on data value and management within an organization.

The quantity and variety of *data sets* used and generated in *analytics* can create challenges for access and maintenance of data classifications. AI discovery techniques and automated systems that identify and classify data based on content, context, *metadata,* and patterns should be considered. *Policies* and *procedures* should be well defined yet straightforward enough for implementation and compliance by analysts.

The classification of data must include ethical, privacy, and regulatory practices.

Questions

→ Are clear roles and responsibilities defined for both consumers and producers of data and the identification and classification of analytical data and products in line with the organization's data management, *metadata*, compliance, and information security *policies*?
→ Is confidentiality, integrity, and availability of data and *analytical products* safeguarded throughout all *analytics* data life cycle *processes* in line with the organization's *policies*?
→ Are *data classifications* implemented and carried throughout data *processes* (retrieval, transmission, copies), systems, and data analysis life cycles (i.e., sandbox, develop, test, pre-production, production)?
→ Are identification and classification *standards* applied to data used or generated in *analytics*?

Artifacts

- *RACI matrix* that captures roles and responsibilities for data identification and classification in *Analytics*
- Evidence of the application of *data classifications* and indexes in analytical design documentation, testing documentation, and analytical output and products
- Audit trails and audit reports

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
Analytics do not respect the organization's identification and classification *standards*.	*Analytics* do not respect the organization's identification and classification *standards*, but the need is recognized, and the development is being discussed.	The *process* to ensure that *analytics* respect the organization's identification and classification *standards* is being developed.	The *process* to ensure that *analytics* respect the organization's identification and classification *standards* has been defined, reviewed, and approved.	The *process* to ensure that *analytics* respect the organization's identification and classification *standards* is being performed.	The *process* to ensure that *analytics* respect the organization's identification and classification *standards* is established as part of business-as-usual practice with a continuous improvement routine.

8.3.4 Data preparation standards exist and are applied consistently

Description

Data preparation is the *process* of collecting, structuring, cleansing, and transforming data so it can be readily and accurately analyzed for business purposes. The *data preparation process* must be defined, and all *process* steps applied consistently to achieve fit-for-purpose data. *Data preparation* must be subject to the organization's Data Governance (DG) *policy* and *standards,* including the use of the *business glossary* and *metadata*. The data must be from *authoritative data sources,* where relevant. *Data preparation* must be performed in a way that preserves *data lineage* and integrity.

Objectives

- Define *standards* for *data preparation* that include the identification of data required for the analysis, available *authoritative data sources,* and data accessibility. Also define *standards* for specification *of data elements*, DQ, need for data cleansing (including defect tracking and root cause fix) and conformance, transformation, and aggregation.
- Ensure *data preparation* follows the organization's DM *policy* and DM *standards* and preserves *data lineage* and integrity.
- Create and maintain adequate documentation on data definitions, data sources and lineage, data usage, and *data owners*.
- Maximize the re-usability of any prepared data and *data preparation processes* to create efficiencies and time to market improvements for future *data preparation* needs.

Advice

Data preparation processes leverage *authoritative data sources*, *business glossary,* and *metadata* to provide accurate, well-defined data for consumption. When new *data sets* or *data elements* are sourced, documentation should be completed to ease reuse. A well-documented data catalog supports and significantly accelerates future data sourcing and wrangling *processes*.

Understanding DQ and determining whether the data is fit-for-purpose must be a key activity of the *Analytics practitioners* in the *data preparation process*. Data of higher quality can be re-used in other *processes* more readily. Profiling data can support data sourcing decisions. The profiling provides a statistics-based understanding of data content and DQ across multiple data sources.

Data preparation should be a repeatable *process* and a formalized best practice. Preference should be given to the use of self-service *data preparation* solutions versus spreadsheets. Spreadsheets are affordable but error-prone, difficult to control, and maintenance heavy. *Data preparation* tools can provide an effective and controlled *data preparation process* and the ability to integrate structured and unstructured data more efficiently. Strategies for agile creation of re-usable *data sets* should be considered to foster efficiency by reducing the need for internal approvals via obfuscation techniques.

Questions

→ Do *data preparation standards* cover: the identification of data required for the analysis; available (authoritative) data sources; data accessibility; specification of *data elements*; DQ; data cleansing and conformance; transformation and aggregation?

→ Does *data preparation* follow the organization's DM *policy* and DM *standards* and preserve *data lineage* and integrity?

→ Is adequate documentation maintained on data definitions, data sources, *data lineage*, data usage, and *data owners*?

→ Are any prepared data or *data preparation processes* leveraged across multiple analytical *processes*, tools, or teams?

Artifacts

- *Data preparation* and *analytics* guidance and design documentation
- Evidence of adoption of *data preparation* *standards* by *Analytics* teams
- *Data architecture* and *data lineage* documentation
- Data definition, sourcing, lineage, usage and ownership documentation
- Evidence of re-use of *data sets* and *data preparation* *processes*

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No *data preparation standards* exist.	No *data preparation standards* exist, but the need is recognized, and the development is being discussed.	*Data preparation standards* are being developed.	*Data preparation standards* have been defined, reviewed, and approved.	*Data preparation standards* are being applied consistently.	The consistent application of *data preparation standards* is established as part of business-as-usual practice with a continuous improvement routine.

8.4 Analytics is Aligned with Data Quality

Analytics must leverage the resources and activities of the Data Quality Management (DQM) *function* to understand the quality of the data input to *analytics* and ensure it is appropriate for the business purpose. DQM must be applied to data created by *analytics* and to any DQ issues uncovered.

8.4.1 The quality of data both used and created by analytics is fit-for-purpose

Description

The business must define the quality of data required to support each *analytics* business case. The DQM *function* must be leveraged to understand the quality of the *data sets* used and created. *Analytics* must set and achieve the DQ requirements for consumed data.

Objectives

- Understand the quality of data required to meet the needs of the business case.
- Use DQM *profiling* and grading to understand the quality of the data used and created by *analytics*.
- Ensure the quality of data used and created is aligned to the requirements of the business case.
- Ensure prioritized data created by *analytics* is subject to DQM.

Advice

Analytics professionals may identify and accept differing DQ levels. For example, a business analyst that generates a report may not identify nor be concerned with some nulls in a *data set*. In contrast, a data scientist may not only discover them but need to address the nulls before building a *model*.

The acceptable level of DQ will differ by business case. When the decision based on the analytic output is higher risk, higher DQ thresholds may need to be met before proceeding. Analysis conducted on poor quality data is likely to be wrong. Low DQ also increases costs, risks, and the likelihood of ethics issues in the analysis. *Analytics* professionals must coordinate with the *data owners* to discuss requirements and DQ criteria. *Data owners* remain accountable for monitoring DQ and implementing changes, but *Analytics practitioners* should also contribute to improving DQ. There must be consistent communication among *data owners* and *Analytics practitioners* to preserve a consistent DQ framework and a consistent toolset.

Analytics practitioners should document findings, adjustments, assumptions, and decisions. Consideration should also be given to how the data was collected and stored before data analysis took place. *Analytics practitioners* must share their findings with the *data owners* as the findings often warrant a change to business *processes* or business decisions and *business rules*. Suitable *processes* must be in place to ensure that data generated as part of the *Analytics process* comes within the scope of DQM.

Questions

- → Are the DQ requirements of business cases defined?
- → Do *Analytics practitioners* reference DQ metrics?
- → Is the quality of data assessed against the requirements of the business use case?
- → Is business justification provided to support the prioritization of DQ improvement?
- → Are *data quality rules* defined and communicated for data created by *analytics*?
- → How do *Analytics practitioners* assess, document, and communicate DQ findings?

Artifacts

- DQ requirements for *analytics* business cases
- Evidence of assessment of DQ against DQ requirements and criteria
- Documented DQ *processes* for *Analytics practitioners*
- Quality rules for data produced by *analytics*
- Evidence of data produced by *analytics* being in scope of DQM

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
The quality of data used and created by *analytics* is not understood.	The quality of data used and created by *analytics* is not understood, but the need is recognized, and the development is being discussed.	The means of ensuring the quality of data used and created by *analytics* is understood and aligned to the needs of the business case is being developed.	The means of ensuring the quality of data used and created by *analytics* is understood and aligned to the needs of the business case has been defined, reviewed, and approved.	The means of ensuring the quality of data used and created by *analytics* is understood and aligned to the needs of the business case is implemented.	The means of ensuring the quality of data used and created by *analytics* is understood and aligned to the needs of the business case is established as part of business-as-usual practice with a continuous improvement routine.

8.4.2 Issues identified during data preparation are managed via the Data Quality Management framework

Description

The DQM framework must support DQ issue management on *analytics* *data sets* so that the data user can understand the quality of the data being prepared and report DQ issues identified during preparation. Adoption of and integration with the existing DQM framework is required to ensure that all data is managed at its source, regardless of whether it originated inside or outside the data *analytics* environment.

Objectives

- Ensure the DQM framework is understood within *Analytics*.
- Provide DQ metrics for data used in the *analytics* environment.
- Identify the techniques and tooling required for *analytics* DQ issue reporting and management.
- Integrate the reporting and management of *analytics* DQ issues into the DQM framework.

Advice

DQM and *Analytics* must work together to ensure the *analytics* requirements are addressed in the DQ framework. Analytics *stakeholders* play a contributory role in DQM. As consumers and users of data, they are accountable for ensuring that the data being used in *analytics* are of a quality fit for the intended purpose. If data issues are discovered, analytic *stakeholders* must provide feedback to the source systems and contribute to the root-cause analysis. A clearly understood *process* for logging and prioritizing DQ issues must be developed and socialized.

Analytics practitioners leverage a variety of data types with an increased number of variables and a higher volume of data than ever before. This environment creates challenges in determining what data must be monitored and assessed as well as how the DQM framework should be applied. DQM must be able to evaluate the DQ needs of *Analytics* and propose appropriate, cost-effective solutions. DQM professionals should be open to new tools and techniques while also remaining cognizant of industry trends and best practices.

As an organization expands its use of *analytics*, DQM teams need to evolve in support of this. The team should assess its people, *processes,* and technology and share these findings with *stakeholders* to gain support for training, transformation, and tools. This assessment, evaluation, and implementation may need to occur faster than ever before, given the constant advancement within the *analytics* discipline.

Questions

- → Is there a funded education and communication plan to ensure Analytics understands the DQM framework?
- → What are the *data quality dimensions* for data used by *Analytics practitioners*? Do these need to be adjusted?
- → What tools are available to measure DQ?
- → How are DQ issues communicated between *Analytics* and the *data producer*?
- → How has DQ improved over time?
- → Is the *data producer* able to successfully implement changes within the timeframe required by *Analytics*?
- → Is there an assessment of the cost of potential DQ remediation to weigh against the potential benefits to be derived from the *analytics*?

Artifacts

- A learning and development plan and supporting materials to educate *Analytics practitioners* on the DQM framework
- A communication plan and supporting material to convey the DQM framework to *Analytics practitioners*
- Evidence of DQ issues being reported by *Analytics practitioners*
- DQM team skills assessment and training
- DQM tools assessment
- DQM *process* reviews
- Evidence of collaboration between DQM team and *Analytics*

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
Issues identified during *data preparation* are not managed via the DQM framework.	Issues identified during *data preparation* are not managed via the DQM framework, but the need is recognized, and the development is being discussed.	The means of ensuring issues identified during *data preparation* are managed via the DQM framework is being developed.	The means of ensuring issues identified during *data preparation* are managed via the DQM framework has been defined, reviewed, and approved.	Issues identified during *data preparation* are managed via the DQM framework.	The management of issues identified during *data preparation* via the DQM framework is established as part of business-as-usual practice with a continuous improvement routine

8.5 The Analytics Platform is Designed and Operational

For *Analytics* to be effective and efficient, it must be supported by a platform that is designed and implemented to meet its needs. The operating model drives many of these needs. The different requirements of production and non-production environments must be addressed, and there must be a version control regime for *models* that is appropriate for each of these environments. Strategies for obfuscation of sensitive data are required to maximize the reusability of *data sets*. The platform should provide appropriate flexibility to scale up and down.

8.5.1 The platform design meets the needs of the Analytics operating model

Description

The *analytics platform* is a combination of tools, applications, and infrastructure that enables *analytics* to be created and executed in an organization. It must have the necessary capabilities to support the way that *Analytics* teams are structured and operate, and the way they engage with the business and *stakeholders*. It must enable *Analytics* to develop, test, govern, socialize, maintain, and mature analytical *models*. The platform must be flexible enough to allow a variety of individuals to interact appropriately with the system. These interactions must implement any required segregation of duties and the ability to set access rights to different users of the system. Beyond the *Analytics* teams, business users will need to validate and utilize the results of the *analytics* *models*.

Objectives

- Understand the platform requirements of the different roles defined in the *operating model*.
- Obtain *stakeholder* agreement to the requirements to be supported.
- Ensure the platform design takes account of the agreed requirements.
- Ensure the platform supports the necessary and appropriate enforcement of access rights.

Advice

The functional and non-function requirements to support the *Analytics* *operating model* must be understood before the procurement or development of the platform commences. Any pre-existing infrastructure and solutions should not constrain the requirements.

Facilitation of segregation of duties and support and control of different levels of access must be part of the design. The platform design should ensure ease of access and use. It should support the need for platform users to understand both the data being used and the results being produced.

Questions

→ Have the platform requirements of different roles described in the *operating model* been defined?
→ Have platform requirements been discussed and agreed with business and *Analytics* *stakeholders*?
→ Are the requirements understood by those responsible for the procurement and implementation of the platform?
→ Has the platform design been reviewed to confirm it addresses the agreed requirements?
→ Has platform support of enforcement of access rights been validated?
→ Can *stakeholders* easily review the outputs of the platform?
→ Can the platform support users beyond the data analysts who work to produce the outputs?

Artifacts

- Documented *Analytics* *operating model*
- Documented assessment of the platform design support for the *operating model*
- *Policies* relating to segregation of duties and evidence of their review and approval
- Evidence of bi-directional communication with the business on the use of the platform

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
The requirement for the platform design to meet the needs of the *Analytics* *operating model* has not been identified.	The platform design does not meet the needs of the *Analytics* *operating model*, but the need is recognized, and the development is being discussed.	The design of the platform to meet the needs of the *Analytics* *operating model* is being developed.	The design of the platform to meet the needs of the *Analytics* *operating model* has been defined, reviewed, and approved.	The platform design meets the needs of the *Analytics* *operating model*.	Platform design support for the needs of the *Analytics* *operating model* is established as part of business-as-usual practice with a continuous improvement routine.

8.5.2 The platform addresses the separate needs for innovation and production

Description

A separate environment is needed for data discovery and testing before delivering *models* into production. Testing in this low risk, *sandbox environment* ensures appropriate testing takes place before the *model* is used and relied upon by the organization. The sandbox must be provided appropriate capacity for its computational requirements determined by the business needs.

Objectives

- Define and support the requirements for flexibility and agility of innovation environments.
- Define and support the requirements to segregate the development and testing sandbox from the production environment.
- Establish distinct design *processes* for the sandbox, development, and production aspects of the platform.
- Establish distinct change control *processes* for the sandbox, development, and production environments.

Advice

The nature of *analytics* is trial and error, so development environments need a degree of agility that supports the ability to fail fast and fail-safe. Non-production environments, especially a sandbox/innovation environment, need a greater level of flexibility made possible by a sufficiently agile change management *process* that can address multiple scenarios.

There must be clear segregation of environments to ensure the change management process is appropriate for each level. It is change management that enables the *Analytics* teams to be productive. Change management *processes* for sandbox/innovation environments must be flexible enough to support adequate turnaround times for experimentation. A rigidly separate environment must be provided for non-production *model* testing, with clear demarcation separating it from production. The change management *process* for production must be appropriately robust in comparison to the more flexible non-production environments.

Questions

→ Are there documented requirements distinguishing the sandbox/innovation environments and production environments?
→ Are the memory, storage, and processing capabilities of the different environments documented?
→ Has the business approved that the environments meet their requirements?
→ Has the design *process* for each environment been assessed and put into operation?
→ Are different change control *processes* in place for each environment to support the business requirements and risk appetite of the organization?

Artifacts

- Non-production environments strategy
- Documented designs of sandbox, development and production environments
- Documented change control *processes*
- Evidence of engagement with *stakeholders* of *policy* review/implementation
- Evidence of communication of strategy and *policies*

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
The separate needs for innovation and production are not understood.	The separate needs for innovation and production are not understood, but the need is recognized and the development is being discussed.	The separate needs for innovation and production are being defined.	The separate needs for innovation and production have been defined, reviewed and approved.	The platform addresses the separate needs for innovation and production.	The need for the platform to address the separate needs for innovation and production is established as part of business-as-usual practice with a continuous improvement routine.

8.5.3 A version control regime is defined and in place

Description

There must be controlled management of change to *models* developed within the *analytics platform* and a documented *process* for recording the changes. Effective governance and change control of the *models* must be in place and must be auditable.

Objective

- Ensure changes to the *analytics* *models* are made with appropriate authorization.
- Ensure changes are tracked, and the nature of each change is documented and auditable.
- Ensure *models* are released into the production environment only through the controlled *process* of the appropriate test environments release mechanisms.
- Ensure appropriate versioning and archiving are in place for *data sets* used to train and validate *models*.
- Address requirements to re-run *analytics* on historical data with the version of a *model* used at that point in time.

Advice

Analytical *models* should follow the change control *process* and the organization's software development life cycle *process*. Seek concurrence through discussion with the Technology *function*, the change management *function,* and connected business *function*s. Maintain documentation that explains changes made to *models*. This documentation must trace version history to know which versions of which *models* are used in which implementations. Document the *process* for the creation and storage of retrievable backups of previous versions of code and *data sets*. These backup files are needed in the event of errors in the upgrade or in case the analysis provided by previous versions needs to be validated.

Questions

→ Is there a change control *process* for amendments to *analytics* *models* on the platform?
→ Is the defined change control *process* in place and understood by the responsible individuals within the company?
→ Does documentation exist to show and explain the changes made to *models*?
→ Are previous code versions and *data sets* available to ensure understanding of previous *model* analyses?
→ Does the change control *process* ensure appropriate authorization for changes?

Artifacts

- *Model* specifications
- *Policies* and *procedures* associated with version control for *models* and *data sets*
- *Model* testing, approval and review *procedures*
- *Model* release protocols and *policies*
- Evidence of *model* review and approval

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No version control regime exists.	No version control regime exists, but the need is recognized and the development is being discussed.	A version control regime is being developed.	A version control regime has been defined, reviewed, and approved.	The version control regime has been put in place and is being followed.	Adherence to the version control regime is established as part of business-as-usual practice with a continuous improvement routine.

8.5.4 Data obfuscation strategies are defined and supported

Description

Data obfuscation strategies can help to ensure that data used during the development phases of *models* comply with the appropriate regulatory regimes governing the organization and best practices regarding data privacy. *Data obfuscation* also applies to commercially sensitive confidential data, such as company results before their release. The obfuscation capability is important for non-production testing environments and, in some cases, can also be relevant to production environments. Where external professional testers have access to test environments, non-production environments must store the minimum amount of personally identifiable or commercially sensitive information.

Objectives

- Create a strategy or *policy* to understand what data is classed as commercially sensitive or personally identifiable information (PII), using the organization's *data classification* system to help determine the level and sophistication of obfuscation that is required.
- Establish techniques to obfuscate data, to mask or blur it, as required by the business

Advice

Identify the *data classifications* that determine what data is commercially sensitive or classed as PII (according to the relevant regulatory framework). Those classifications will drive the level of obfuscation or deletion that must be applied to the data. Confirm whether *data obfuscation* is needed for production and non-production *models* and whether it applies to *model* outputs as well as inputs. Governance of the appropriate application of obfuscation is covered in DCAM 8.6.3 – "*Model* approval and release is aligned with privacy governance."

Questions

→ Are *data obfuscation* requirements understood?
→ Is there a *process* documented and followed to ensure regulatory compliance in the use of *personal data* and commercially sensitive data?
→ If *personal data* needs to be used to ensure accurate reporting and prediction optimization, is there a *process* to ensure that the *personal data* utilized is obscured or deleted after use?
→ Have obfuscation techniques been established and adopted?

Artifacts

- Evidence of data *anonymization policies*, referencing data types and detailing specific requirements for production and non-production environments
- Non-production environments strategy including the management of data replication across environments
- List of *stakeholders* and evidence of bi-directional communication and recognition of relevant *policies*
- Documentation of obfuscation techniques with guidance on how and when to apply them
- Evidence of *data obfuscation*/*anonymization* of relevant *models*/*model* outputs

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No *data obfuscation* strategies exist.	No *data obfuscation* strategies exist, but the need is recognized and the development is being discussed.	*Data obfuscation* strategies are being developed.	*Data obfuscation* strategies have been defined, reviewed, and approved.	*Data obfuscation* strategies are supported and are being used.	The use and support of *data obfuscation* strategies are established as part of business-as-usual practice with a continuous improvement routine.

8.5.5 Environment scalability requirements are understood and appropriately supported

Description

Business requirements change rapidly. The *analytics model* environment must have the flexibility to cope with new data without requiring significant redesign. The *model* environment needs to be able to provide an increase in data computing power requirements to address forecasted business growth and the growing sophistication of *models*. The costs of computational power need to be tracked. There must be an understanding of how additional requirements could be accommodated, and potential costs must be estimated in advance.

Objectives

- Understand the processing power capacity and flexibility potentially needed for the *analytics model* environment.
- Support the ability to scale up and down to accommodate planned requirements.
- Establish a mechanism for estimating and tracking *model* processing costs.

Advice

A *process* should be established for *stakeholders* to provide direction on what future requirements the *analytics models* may be required to support. These requirements provide input for capacity planning. They should include data volumes and an estimate of the analytical workloads. The scalability requirements should be reflected in the technology roadmap for *Analytics*. They should establish a position for the use of both on-premise and cloud infrastructure as appropriate for the organization.

The costs of the *analytics platform* should be tracked and reviewed against expectations by the *stakeholders* to ensure the environments are sized appropriately. Assess and quantify the business benefits obtained from the *model* outputs to determine if the costs of the *models* and processing and the costs of idle capacity that supports scalability are justified.

Questions

- → Is there business direction on future requirements and strategy that the *analytics models* are required to support?
- → Are future data volumes and workloads understood and quantified?
- → Are volume and workload requirements documented and reviewed by *Analytics*?
- → Can the *models* accommodate increased data volumes or variables without requiring a complete redesign and the related costs?
- → Are the costs of the *analytics platform* tracked and reviewed against expectations to ensure that environments are sized appropriately?
- → Does the assessment include the costs of idle capacity?

Artifacts

- Documentation evidencing production and non-production low-level non-functional requirements
- Non-production environments strategy
- Cost tracking of both non-production and production environments, including any utility computing
- Evidence of spending reviews and ROI for production and non-production environments

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
Environment scalability requirements are not understood.	Environment scalability requirements are not understood, but the need is recognized and the development is being discussed.	Environment scalability requirements are being developed.	Environment scalability requirements have been defined, reviewed, and approved.	Environment scalability requirements are understood and supported.	Understanding and support of environment scalability requirements are established as part of business-as-usual practice with a continuous improvement routine.

8.6 Model Operationalization is Established

While some *analytics* activities will be exploratory or one-off, *Analytics* must be able to deploy *models* into production in a controlled and governed manner. Testing, approval, and release *processes* are central to this, along with *processes* for regular review of deployed *models*. These must be aligned with the organization's governance approaches for data ethics and privacy. Requirements to understand and control *model bias* must be addressed, as must the need to be able to explain how *model* decisions have been reached.

8.6.1 Model testing, approval, release and regular review processes are in place and effective

Description

Before release into an operational environment, *models* must be tested to ensure that they are performing as expected and in consonance with all *model* specifications. The satisfactory outcome of testing and approval for release should be formalized by the official designated for this purpose. The *model* should be released following established release protocols. The performance of the *model* may change as ingested data changes over time, potentially affecting the efficacy of the *model's* original intent. Its performance and its continued adherence to *model* specifications must be reviewed periodically, and any time there is a change to *model* specifications.

Objectives

- Define formal *procedures* for *model* testing, approval, release, and periodic review.
- Ensure the authority for *model* approval is clear and appropriate.
- Obtain *stakeholder* approval and support to enforce testing, approval, release, and review *procedures*.
- Ensure that *models* released or pending release into production environments are performing as expected and in consonance with each *model's* specifications.
- Establish safeguards to ensure *model* release into production minimizes disruption to the organization's operations and protects against data breaches and corruption of data.

Advice

As part of the *model* testing, a wide range of inputs must be run through the *model* to ensure that it is performing as expected. Such inputs sets may include back-test data, current data, and artificially created representative samples of input data that could theoretically, even if rarely, arise in operations. *Model* outputs should be checked for any unintended consequence arising from using the *model,* such as unfair bias or undesirable business outcomes. *Model* testing and controls should be automated where possible.

The individuals authorized to approve a *model* should have input to the specifications of the portfolio of evidence to be provided for review in the approval. They should have an understanding of how the *model* works and the business objectives it is designed to meet. They should also have a broader awareness of the overall business *function* context in which the *model* will operate and of governance and ethical parameters.

Questions

→ Do formal *procedures* exist for *model* testing, approval, release, and periodic review?
→ Is the authority for *model* approval designated at the appropriate level of seniority?
→ Have *stakeholders* approved the enforcement of the *procedures* for testing, approval, release, and review?
→ Do *models* perform as expected and in consonance with their specifications?
→ Have safeguards been established to ensure *model* release minimizes disruption to operations and protects against data corruption and breaches?

Artifacts

- *Model* specifications
- *Model* testing, approval and review *procedures*
- Evidence of automation of *model* tests
- *Model* release protocols and *policies*
- Details of *model* input datasets and outputs
- Schedule of periodic *model* reviews
- Evidence of *model* review and approval
- List of *stakeholders* and evidence of approval of enforcement of *procedures*

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No *model* testing, approval, release, and regular review *processes* exist.	No *model* testing, approval, release, and regular review *processes* exist, but the need is recognized, and the development is being discussed.	*Model* testing, approval, release, and regular review *processes* are being developed.	*Model* testing, approval, release, and regular review *processes* have been defined, reviewed, and approved.	*Model* testing, approval, release, and regular review *processes* are in place and effective.	Effective *model* testing, approval, release, and regular review *processes* are established as part of business-as-usual practice with a continuous improvement routine.

8.6.2 Model approval and release is aligned with data ethics governance

Description

The data ethics oversight *function* should outline guidance and the requirements to operationalize *models*. These requirements and the Code of Data Ethics must be followed and monitored during the *model* development, approval, and release *process*.

Objectives

- Align *model* development, approval, and release *processes* with the data ethics governance structure and guidance.
- Assign data ethics accountability for *models* during approval and release *processes*.
- Review and mitigate any weaknesses in the *model* approval and release *process* with the data ethics oversight *function*
- Involve *analytics* experts in the Code of Data Ethics development and continuous review.
- Execute ethical management.

Advice

Consideration of data ethics needs to be embedded in the end-to-end *process* of *model* development and management. Organizations should have *policies*, *processes,* and controls in place to document the ethical consideration as part of the approval and release *process*. Organizations should retrospectively embed similar *policies* retrospectively in existing *models* and any post operationalization monitoring *process*. This sub-capability is dependent on capability 6.6 – "Govern the Data Ethics."

Model approval and release must take into account the data ethics considerations. These include the credible, responsible, and traceable sourcing and safe sharing of data. The *customer's* best interests are paramount when the organization uses, *models,* and transforms data in ways that make the handling of data processed by the *model* explainable and transparent.

As data ethics is an emerging practice, there should be continuous dialogue between *analytics* experts, associated risk teams, and the data ethics oversight *function* to review and update requirements and guidance to ensure they are up to date.

Questions

- → Has the data ethics oversight *function* provided guidance and requirements outlining how *models* adhere to data ethics?
- → Is there a clear *process* in place for the ongoing monitoring of *models* to ensure they continue to meet the guidelines and requirements of the data ethics oversight *function*?
- → Is the accountability for data ethics during the *model* approval and release *process* at an appropriate level of seniority?
- → Is there a *process* in place to ensure a formal Code of Data Ethics and associated guidance are reviewed and remain up to date?

Artifacts

- Evidence of compliance with data ethics oversight *function*'s guidance
- Evidence of data ethics assessment included in the *model* review and approval *policy*
- Schedule of periodic reviews of the alignment of *model* development, approval and release *processes* with data ethics governance
- List of *stakeholders* and evidence of communication

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
Model approval and release are not aligned with data ethics governance.	*Model* approval and release are not aligned with data ethics governance, but the need is recognized, and the development is being discussed.	Alignment of *model* approval and release with data ethics governance is being developed.	Alignment of *model* approval and release with data ethics governance has been defined, reviewed, and approved.	*Model* approval and release is aligned with data ethics governance.	Alignment of *model* approval and release with data ethics governance is established as part of business-as-usual practice with a continuous improvement routine.

8.6.3 Model approval and release is aligned with privacy governance

Description

The *model* approval and *model* release *processes* should include explicit checks and confirmation that they will be aligned with the organization's privacy governance requirements.

Objectives

- Ensure privacy governance requirements are understood by those involved in *model* approval and release.
- Align *model* development, approval, and release *processes* with the privacy governance structure and guidance.
- Assign accountability for privacy compliance of *models* during approval and release.
- Engage the privacy oversight *function* in regular reviews of the effectiveness of the *model* approval and release *process*.
- Ensure that, where appropriate, *models* allow authorized third-party access to the data used to reverse engineer a decision to original, preprocessing data.

Advice

Compliance of a *model* with privacy requirements must be understood and confirmed before a *model* is released, but should already be taken into account at the design stage. *Models* and the governance *principles* are tightly interrelated, and compliance must be reconfirmed every time a *model* changes or the compliance rules are amended.

Questions

→ Has the data privacy oversight *function* provided guidance and requirements outlining how *models* adhere to privacy requirements?
→ Is there a *model* compliance review *process* that includes the privacy requirements before a *model* is released, or before changes to a *model* are released?
→ Is there a *process* in place to review all *models* in production when the privacy requirements change?

Artifacts

- *Model* specifications with a description of personally identifiable information used and details of how it is protected
- Privacy governance controls and guidance
- Evidence of privacy review and history for all *model* releases and updates
- Evidence of privacy review and history for all updates of privacy requirements

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
Model approval and release are not aligned with privacy governance.	*Model* approval and release are not aligned with privacy governance, but the need is recognized, and its development is being discussed.	Alignment of *model* approval and release with privacy governance is being developed.	Alignment of *model* approval and release with privacy governance has been defined, reviewed, and approved.	*Model* approval and release is aligned with privacy governance.	Alignment of *model* approval and release with privacy governance is established as part of business-as-usual practice with a continuous improvement routine.

8.6.4 Model bias is understood and managed

Description

Analytics stakeholders need to be aware of any prejudices and unfairness of *models* and any unintended consequences they may cause. This type of *model bias* is a governance issue and must be addressed with appropriate checks and balances that include awareness, mitigation, and controls. *Models* must be subject to active management of bias that provides for ongoing assessment and validation of algorithms, *data sets,* and *model* outcomes.

Objectives

- Establish *procedures* and controls to ensure that input *data sets* do not introduce *model bias*.
- Ensure that *model bias* and its effects are identified.
- Establish *procedures* for addressing *model bias* once identified.
- Ensure that the shortcomings of algorithms are understood, and they are not applied to questions where answers will be invalidated by algorithmic bias.

Advice

Analytics model results are based on the data inputs to the *model*, either when created and trained or when running in production. Bias arises when that data is not representative of the real world, whether missing key variables that would lead to different decisions or including human-produced content that incorporates biases of those persons. To minimize the harmful effects of bias, *stakeholders* need to be aware of the possible bias in any *model*. They must understand the various types of bias and how each can impact data, analysis, and decisions. Identification and management of bias must be incorporated in the formal *procedures* for *model* approval and regular review. Regular reviews of *models* must include monitoring for sudden or creeping bias being introduced as the *models* are exposed to new data in production. Many organizations are establishing Data Ethics committees, charged with reviewing *models* and their outcomes, for signs of intended or unintended bias.

Questions

→ Are *stakeholders* aware of the potential sources of *model bias*?
→ Is the analysis of *model bias* included in the *analytics methodology* and *model documentation standard*?
→ Are there *procedures* and controls to address bias in *model* input data?
→ Are *model* shortcomings documented and considered in decisions on the application of the *model*?
→ Is *model bias* addressed in *procedures* of *model* approval and regular review?

→ Are judgment-based decisions on *model bias* risk made at an appropriate level of seniority?

Artifacts

- *Stakeholder* education material on *model bias*
- *Procedures* for the analysis of *model bias*
- *Procedures* and controls for review of bias in input *data sets*
- Documented analysis of *model bias*
- Documentation of *model* shortcomings and limitations
- Evidence of *model bias* and shortcomings being considered in *model* deployment decisions

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
Model bias is not understood or managed.	*Model bias* is not understood or managed, but the need is recognized and the development is being discussed.	*Processes* are being developed to ensure *model bias* is understood and managed.	*Processes* have been defined, reviewed, and approved to ensure *model bias* is understood and managed.	*Processes* are in place and are effective to ensure *model bias* is understood and managed.	Understanding and effective management of *model bias* are established as part of business-as-usual practice with a continuous improvement routine.

8.6.5 Requirements for model explainability are understood and incorporated

Description

Requirements for explaining how a *model* works and reaches its outcomes should be precise and must be incorporated into the modeling *process* and the operational *processes* that support the *models*. Transparency of how a *model* works is critical to recognizing bias and aligning business *processes* and activities to achieve non-biased outcomes.

Objectives

- Define requirements for *model explainability* in business terms.
- Assign accountability for *model explainability* during the requirements phase of the modeling *process*.
- Perform additional steps to mitigate ambiguity in requirements for *model explainability*.
- Incorporate *processes* to ensure requirements for *model explainability* result in a clear understanding of how a *model* works and *model* transparency.

Advice

Model explainability requirements must specify how easily *models* can be understood and how easy it must be to understand the cause of decisions in a *model*. The goal is for *stakeholders* to be able to explain what *models* do and how they do it, both to internal and to external audiences. *Model explainability* must be able to stand up to internal audit or external regulatory reviews. Explanations should be consistent.

Identify and consult the *stakeholders* who can understand and discuss the requirements for *model explainability*.

Include requirements for *model explainability* in the organization's *analytics methodology* and *model documentation standard*. Embedding *standards* into the fabric of an organization is a critical step in achieving *model explainability*.

Update the organization's *processes*, *standards,* and *policies* as the adoption of requirements for *model explainability* increases. Keep documentation and *processes* current. The alternative results in tribal knowledge, operational siloes, and increased risk that intellectual property and knowledge are lost, making *model explainability* more difficult.

Establish methods to monitor and continuously improve *model explainability*. The goal of continuous improvement is to provide opportunities for *stakeholders* to enhance *model explainability* to meet business objectives. Sources of improvement ideas include surveys, interviews and peer reviews, success stories, and analysis of gaps highlighted by monitoring and measurement. Bring new ideas and thinking into the organization by keeping pace with emerging and waning industry *standards*, technology improvements, and by comparing *processes* and outcomes to those of competitors, allies, and other industries.

Questions

→ Are *models* explainable in business terms?
→ Can the requirements for *model explainability* be traced to business *processes*, activities, and outcomes?
→ Is there evidence of roles and responsibilities?
→ Was accountability assigned during the requirements phase of the modeling *process*?
→ Have steps or *processes* been identified to mitigate ambiguity?
→ Are *models* easily interpreted?
→ Is it easy to understand the basis of decisions in the *models*?

Artifacts

- *Policies* for requirements for *model explainability* including the types of *model* acceptable for business use cases
- Steps to mitigate ambiguity, both demonstrated and documented or observed
- Requirements for *model explainability* documented in *analytics methodology* and *model documentation standard*
- Minutes from peer-review sessions of *model explainability*
- Documentation of rationale for *model* choice
- *Analytics practitioner* role definitions that include responsibilities relating to *model explainability*
- Evidence of monitoring and continuous improvement efforts for *model explainability*

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
Requirements for explainability are not understood.	Requirements for explainability are not understood, but the need is recognized and their development is being discussed.	Requirements for explainability are being developed.	Requirements for explainability have been defined, reviewed, and approved.	Requirements for explainability are understood and incorporated.	Understanding and incorporation of explainability are established as part of business-as-usual practice with a continuous improvement routine.

8.7 The Analytics Culture and Education Needs are Managed

Analytics practitioners require specialized skills. An education program must be in place to address any skills gaps identified. Effective *Analytics* will also require behavioral and cultural change across the organization.

8.7.1 The behaviors needed for an Analytics culture are understood and measured

Description

Unique cultural and behavioral changes must be identified and adopted for each organization to produce value from *Analytics* capabilities. Once defined and agreed, culture and behaviors should be measured at specific points throughout the journey.

Objectives

- Identify the behaviors that exist in the organization today that impact the effectiveness of *Analytics*.
- Define the behaviors needed to realize value from data and *analytics*.
- Create a list of behavioral recommendations.
- Develop a backlog of ideas, solutions, and concepts for adoption based on behavioral recommendations.
- Measure the behaviors at key milestones.

Advice

A combination of data, behavioral insights, and human-centered design needs to be applied to determine the behaviors required to enable an *Analytics* culture. Define the future behaviors necessary to enable the business and data strategy of the organization to realize the desired value. Use engaging discovery methods to obtain an understanding of the current behaviors prevalent in the organization.

Validate the observed and desired behaviors needed for the *Analytics* culture with other *stakeholder* groups to ensure they align with the organizational culture. These could include behaviors showing collaboration, positive attitudes towards experimentation and understanding of failure as part of learning, innovation, a culture of inquiry, etc. All of these can potentially be harnessed to deliver business value.

Questions

→ Is there existing data for how people in the organization think, feel, and act in relation to *Analytics*?
→ Does the organization understand how people apply *analytics* in the flow of their daily work?
→ Have the desired behaviors to create value from *analytics* been defined, documented, and communicated?
→ Have *stakeholders* validated the desired behaviors for the *Analytics* culture?

Artifacts

- Evidence of *stakeholder* engagement in the definition of desired behaviors
- Documentation of existing and desired behaviors
- Backlog of potential ideas, solutions, and concepts
- Results of assessments and measurement of behaviors

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
The behaviors needed for an *Analytics* culture are not understood.	The behaviors needed for an *Analytics* culture are not understood, but the need is recognized and the development is being discussed.	The behaviors needed for an *Analytics* culture are being developed.	The behaviors needed for an *Analytics* culture have been defined, reviewed, and approved.	The behaviors needed for an *Analytics* culture are understood and managed.	Understanding and management of the behaviors needed for an *Analytics* culture are established as part of business-as-usual practice with a continuous improvement routine.

8.7.2 Initiatives to address culture gaps are in place

Description

The required behaviors have been identified. The organization must conceive and create prototype initiatives that encourage and drive the desired behaviors. Once the value of those initiatives is proven, they must be embedded and sustained as ongoing activities.

Objectives

- Create prototypes of potential concepts, solutions, and initiatives that address culture gaps.
- Demonstrate business value by documenting that the initiatives lead to desired changes in behavior.
- Enable the adoption of cultural change initiatives as a business-as-usual activity.
- Measure the behaviors at key milestones.
- Ensure the *Analytics* culture takes account of the broader cultural drivers of the organization, such as confidentiality, risk appetite, innovation, and ethics.

Advice

A backlog of potential solutions (as identified in DCAM 8.7.1) is a prerequisite to embed initiatives that will address culture gaps. The solutions must be prototyped and tested to determine if they lead to the desired behavior changes. Those with a positive impact should be prioritized and refined for further rollout.

Work with the *Analytics* community to define the KPIs associated with the behaviors and build the business case for implementation of the solution as a business-as-usual activity. Obtain buy-in and sponsorship to drive the data and *Analytics* culture initiatives. Following implementation, measure the change in behavior and its impact at regular milestones.

Engage HR to ensure the *Analytics* culture is aligned with the broader organization and work with the organization's Learning and Development team, if available, to co-develop education initiatives. Consider how communication of *Analytics* success stories can support cultural objectives.

Questions

→ Have the solutions in the backlog been assigned relative priorities?
→ Are the initiatives embedded in daily activities?
→ Does a *process* exist to determine if new behaviors are sustainable and adding business value?
→ Has a *process* to measure behaviors over time been defined?

Artifacts

- Education and behavioral initiatives roadmaps
- Results and insights from prototype solutions
- A documented *process* to determine sustainability and business value of new behaviors
- Evidence of behavioral measurement

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No education initiatives to address culture gaps exist.	No education initiatives to address culture gaps exist, but the need is recognized, and the development is being discussed.	Education initiatives to address culture gaps are being developed.	Education initiatives to address culture gaps have been defined, reviewed, and approved.	Education initiatives to address culture gaps are in place.	Education initiatives to address culture gaps are established as part of business-as-usual practice with a continuous improvement routine.

8.7.3 The learning experience needs of Analytics practitioners are defined

Description

The skills required to perform the *Analytics practitioner* roles and responsibilities are understood, and a learning map has been created to develop these skills.

Objectives

- Identify and align the skills required to fulfill roles and responsibilities at different levels.
- Create a learning map to provide *Analytics practitioners* with the right skills and tools to carry out their activities.
- Review and confirm that the learning map fulfills role definitions.
- Embed a *process* to refresh the learning map based on industry *standards* and developments.

Advice

Analytics practitioners need to be provided with the right education and training to build their skill set to maximize the value an organization gets from its *Analytics* capability. When developing the learning map, *Analytics practitioners* across all levels should be involved to ensure all the required skills are represented across *Analytics*. Roles and responsibilities should be used as a starting point to identify the skills and tools that will enable practitioners to perform these effectively. A current state assessment of existing skills, tools, and training should be performed to identify any gaps which should then be addressed within the learning map.

The learning map should be designed to provide *Analytics practitioners* with both practical and theoretical experiences. Consider experiences from structured learning, social learning, and experiential learning activities. The various experiences should be built into the learning map with enough time committed to each and delivery roadmap clear with milestones to monitor learning progress. The learning map must be refreshed at intervals aligned with the *operating model* and the learning experiences updated to reflect industry developments and best practices.

Questions

→ Have skills requirements across *Analytics practitioner* levels been defined and agreed?
→ Have *stakeholders* reviewed and approved the learning map?
→ Does the learning map fulfill role definitions across all roles and levels?
→ Do the learning maps cover all types of learning?
→ Has a *process* to refresh the learning map been defined?

Artifacts

- Documented roles and responsibilities of *Analytics practitioners*
- Documented skills requirements and needs of *Analytics practitioners*
- Learning maps for each role type and level
- List of *stakeholders* and evidence of bi-directional communication and approval of the learning map
- Assessment of existing tools and training
- Documented *process* for regular review of learning maps

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
The learning experience needs of Analytics practitioners are not defined.	The learning experience needs of *Analytics practitioners* are not defined, but the need is recognized, and the development is being discussed.	The learning experience needs of *Analytics practitioners* are being developed.	The learning experience needs of *Analytics practitioners* have been defined, reviewed, and approved.	The learning experience needs of *Analytics practitioners* have been defined	*Analytics practitioners* learning experience needs are established as part of business-as-usual practice with a continuous improvement routine.

8.7.4 Education initiatives to address skills gaps are in place

Description

The education initiatives for *Analytics practitioners* are established with a *process* for continuous improvement. The success of the education initiatives must be measurable and monitored to ensure skills gaps are addressed on an ongoing basis.

Objectives

- Perform gap analysis to identify skills gaps for *Analytics practitioners* across levels.
- Design learning experiences based on the defined learning experience needs.
- Define continuously updated roadmaps to deliver ongoing education initiatives across practitioner levels.
- Embed a *process* to monitor learning progress and for continuous improvement.

Advice

The learning maps prescribed in DCAM 8.7.3 should be used as requirements for developing the education initiatives, centering the learning experience around the needs of *Analytics practitioners*. A gap analysis across *Analytics* should be performed to identify existing skills gaps and common challenges, which should drive the focus of the learning experiences. At this point, it should be considered whether hiring may be appropriate to address skills gaps, depending on the size of the gap and the degree of specialization of the skill. For example, it may be optimal to hire practitioners with particular skill sets to meet a new opportunity or act as a catalyst for raising other team members' skill levels.

The learning experiences should be designed to equip *Analytics practitioners* with both practical and theoretical backgrounds, combining structured learning, social learning, and experiential learning. Engaging experiences based on real-life scenarios should be developed to put new skills and tools to use in a simulated, safe environment. An example is to use gamification to motivate and engage practitioners to make the learning experience as useful as possible. The development of non-technical skills should also be considered, along with the rotation of people between *Analytics* and business *functions* to create a well-rounded skill set.

A roadmap to deliver the learning experience should be created within an iterative approach and a feedback mechanism to ensure education initiatives are effective. Embedded measures of success should be tracked with learning experiences that are continuously improved and updated with industry developments.

Questions

- → Has a gap analysis to identify skills gaps been performed?
- → Is a plan to resolve skills gaps in place?
- → Has the education initiative been designed to provide practical and theoretical experiences?
- → Has a learning experience roadmap been defined?
- → Have success metrics been approved?
- → Is a feedback mechanism in place?

Artifacts

- Skills gap analysis
- Backlog of learning experiences
- Learning experience delivery roadmap
- Education initiative success metrics
- Documented *process* for monitoring and improving learning progress

Scoring

Not Initiated	Conceptual	Developmental	Defined	Achieved	Enhanced
No education initiatives to address skills gaps exist.	No education initiatives to address skills gaps exist, but the need is recognized, and the development is being discussed.	Education initiatives to address skills gaps are being developed.	Education initiatives to address skills gaps have been defined, reviewed, and approved.	Education initiatives to address skills gaps are in place.	Education initiatives to address skills gaps are established as part of business-as-usual practice with a continuous improvement routine.